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(12) **United States Patent**
Peyman

(10) **Patent No.:** **US 12,396,889 B2**

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(54) **LAMELLAR CORNEAL AUTOLOGOUS OR
HOMOLOGOUS GRAFT IN REFRACTIVE
SURGERY**

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patent is extended or adjusted under 35
U.S.C. 154(b) by 1247 days.

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Related U.S. Application Data

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filed on Nov. 29, 2020, now Pat. No. 11,666,777,
(Continued)

(51) **Int. Cl.**
A61F 9/008 (2006.01)
A61F 2/14 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **A61F 9/0081** (2013.01); **A61F 2/142**
(2013.01); **A61F 9/0017** (2013.01);
(Continued)

(58) **Field of Classification Search**
CPC A61F 9/0081; A61F 2/142; A61F 9/00827;
A61F 2/147; A61K 33/18; A61K 45/06;
(Continued)

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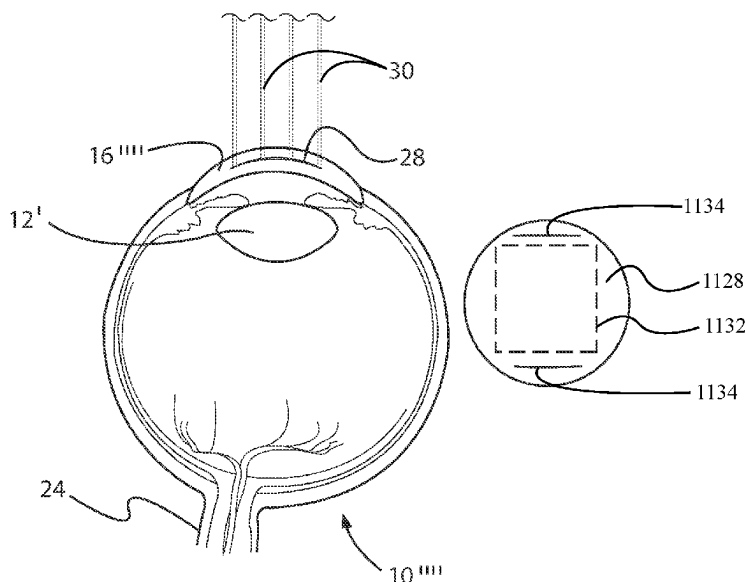
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(57) **ABSTRACT**

A method of lamellar corneal graft implantation is disclosed
herein. The method includes the steps of: (i) forming one or
more intrastromal incisions in a cornea of an eye of a patient;
(ii) removing a cut portion of the cornea defined by the one
or more intrastromal incisions from the eye of the patient;
(iii) applying laser energy to a lamellar corneal graft using
an excimer laser so as to modify the refractive power of the
lamellar corneal graft, the lamellar corneal graft being in the
form of a autologous corneal graft or a homologous corneal
graft; and (iv) implanting the lamellar corneal graft into the
eye of the patient in a location previously occupied by the
cut portion of the cornea so as to correct the refractive power
of the eye of the patient.

20 Claims, 34 Drawing Sheets



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FIG. 1A

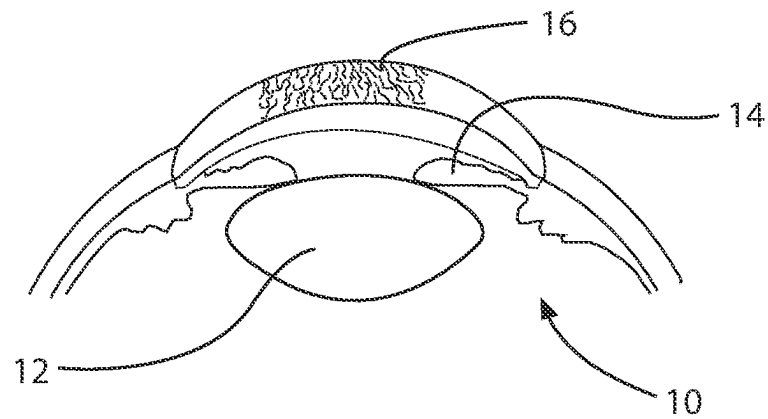


FIG. 1B

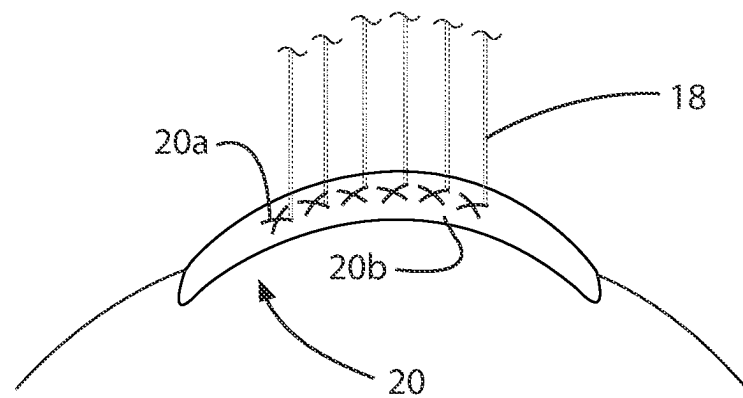


FIG. 1C

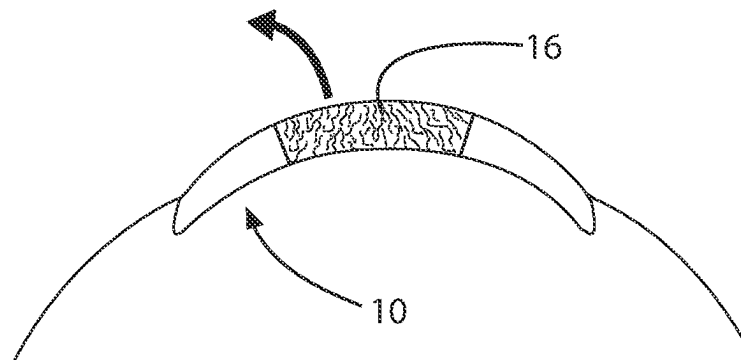


FIG. 1D

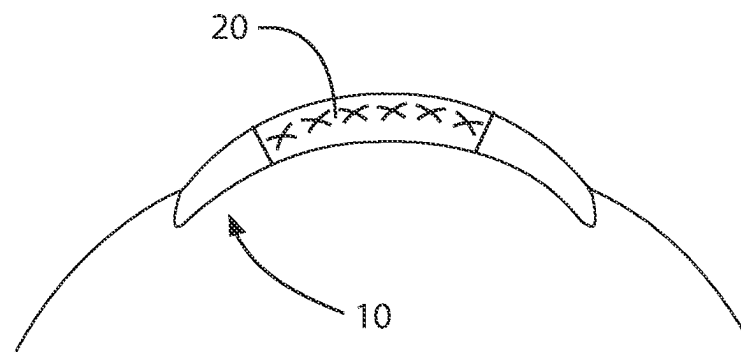


FIG. 2A

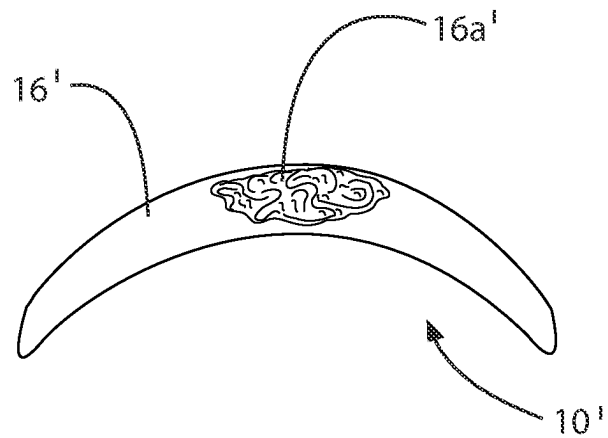


FIG. 2B

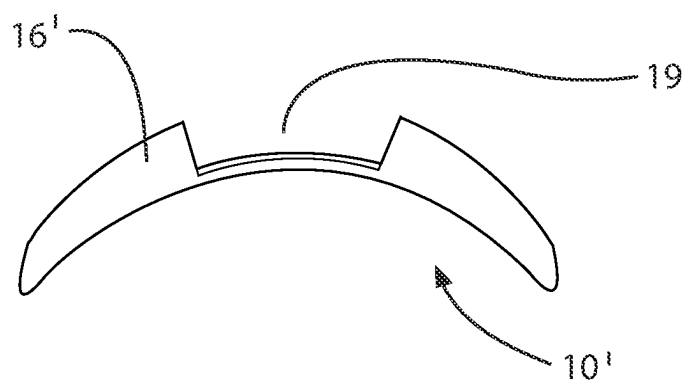
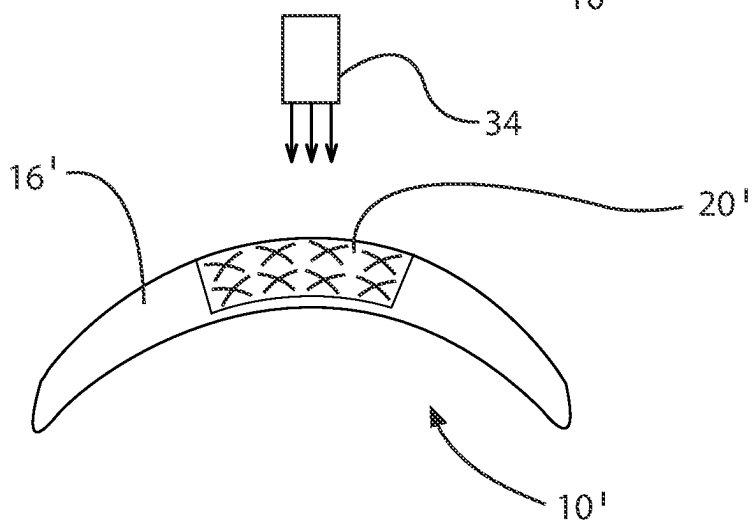


FIG. 2C



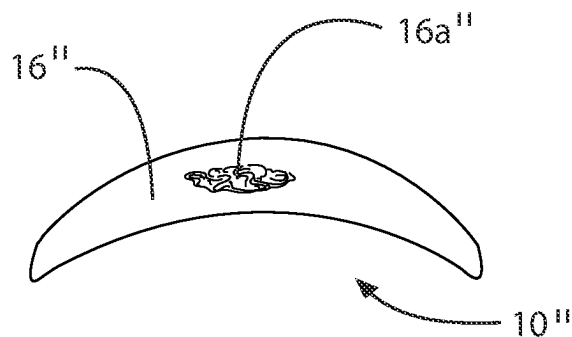


FIG. 3A

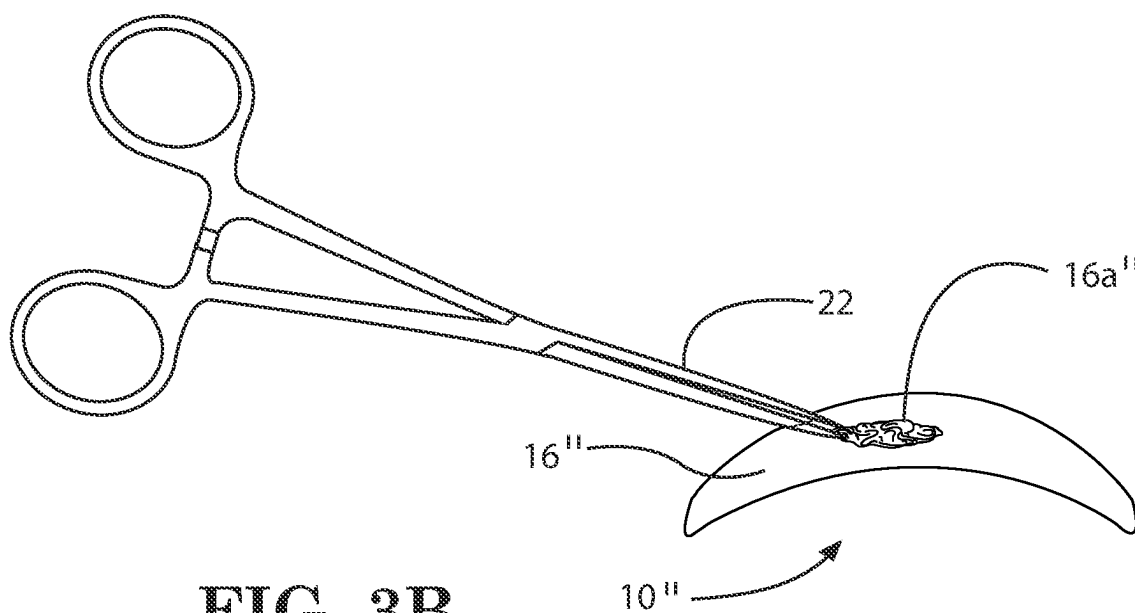


FIG. 3B

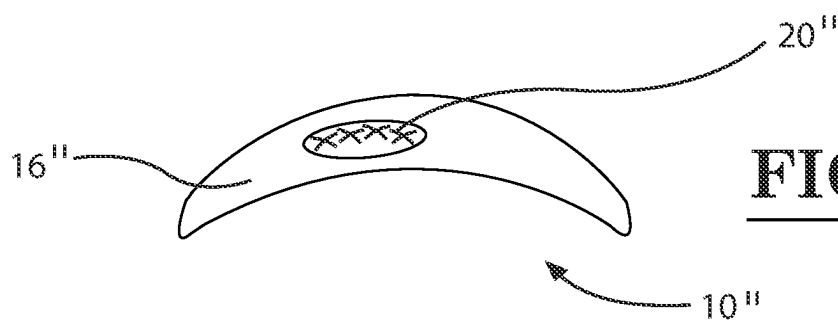


FIG. 3C

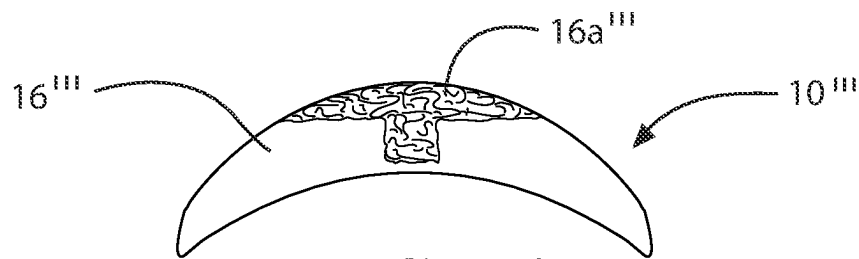


FIG. 4A

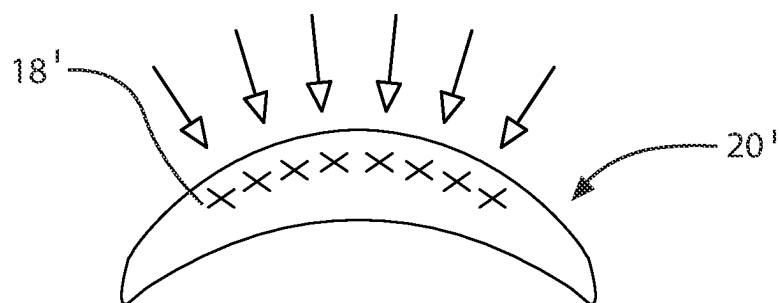


FIG. 4B

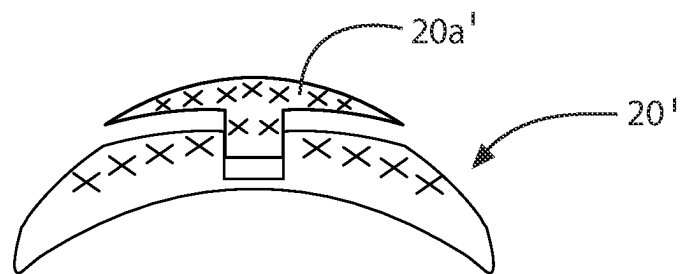


FIG. 4C

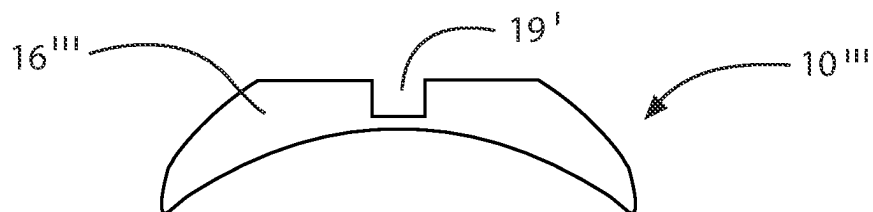


FIG. 4D

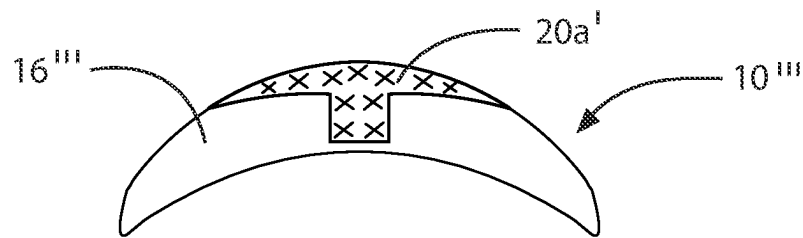


FIG. 4E

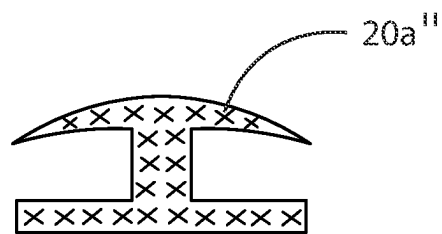


FIG. 5A

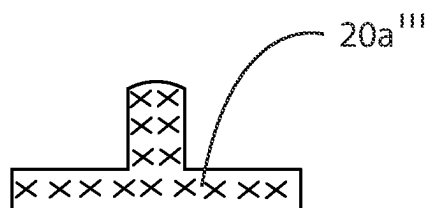


FIG. 5B

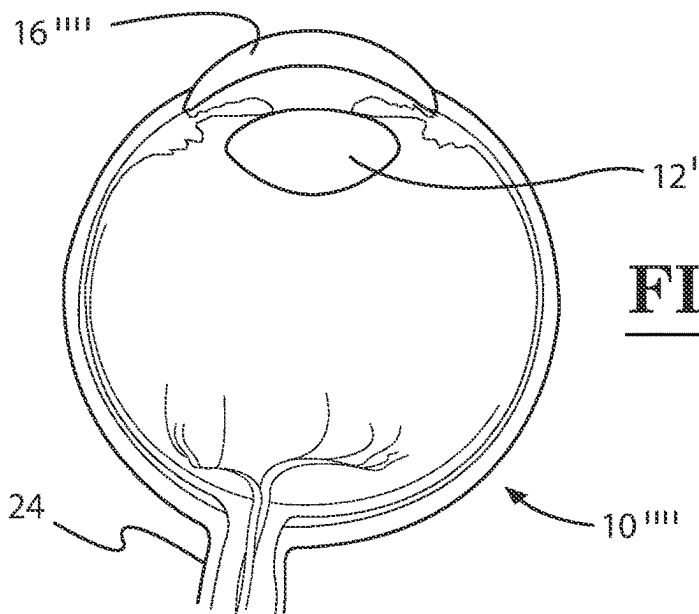


FIG. 6A

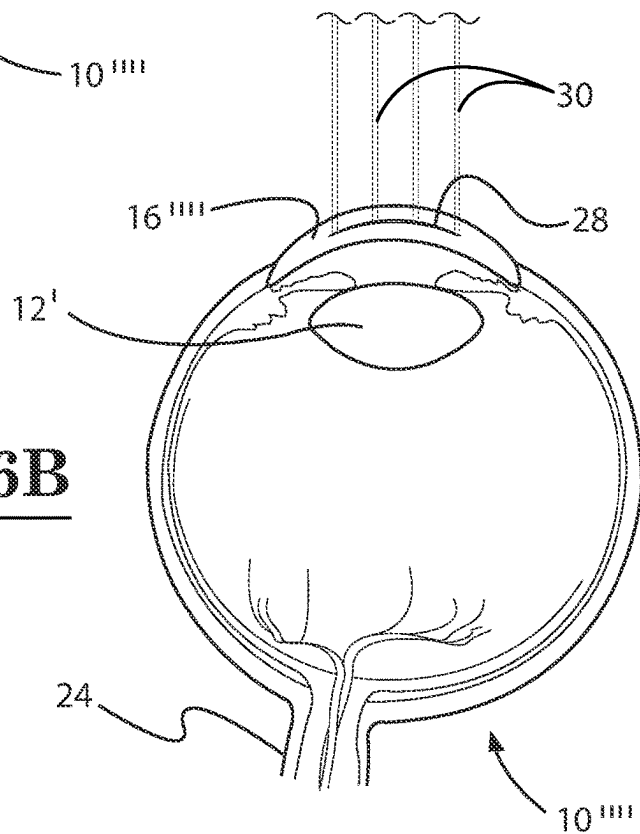


FIG. 6B

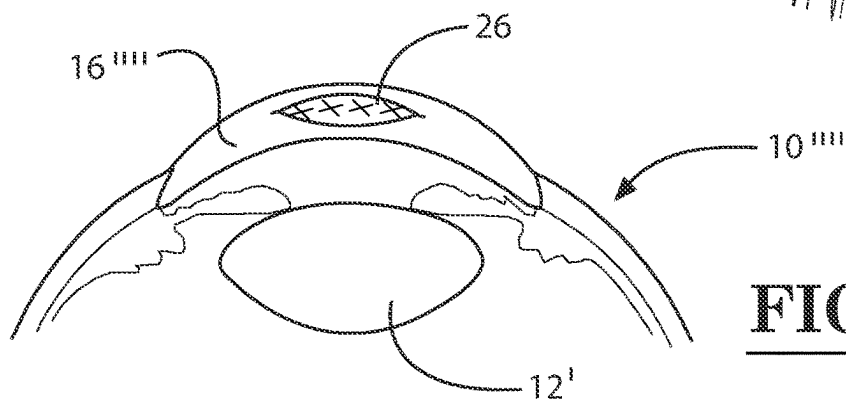


FIG. 6C

FIG. 7A

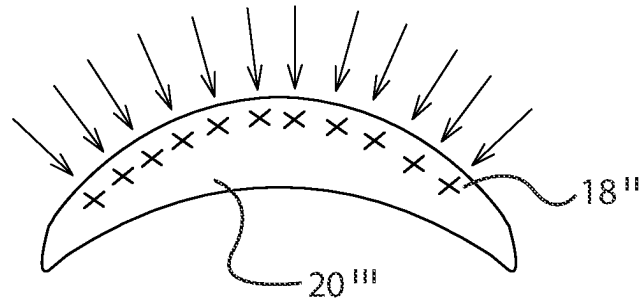


FIG. 7B

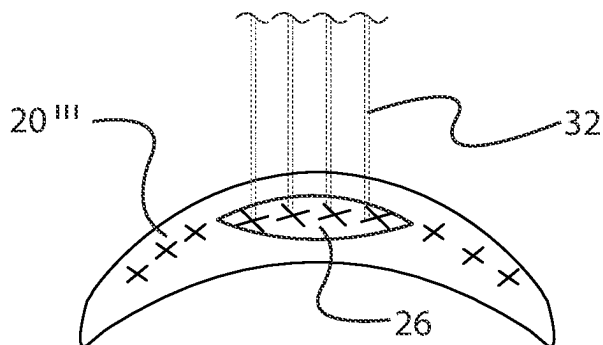
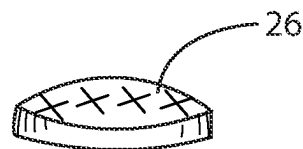
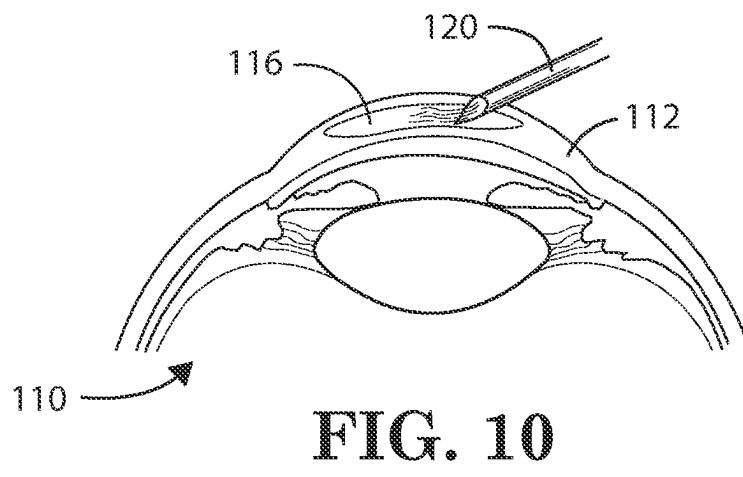
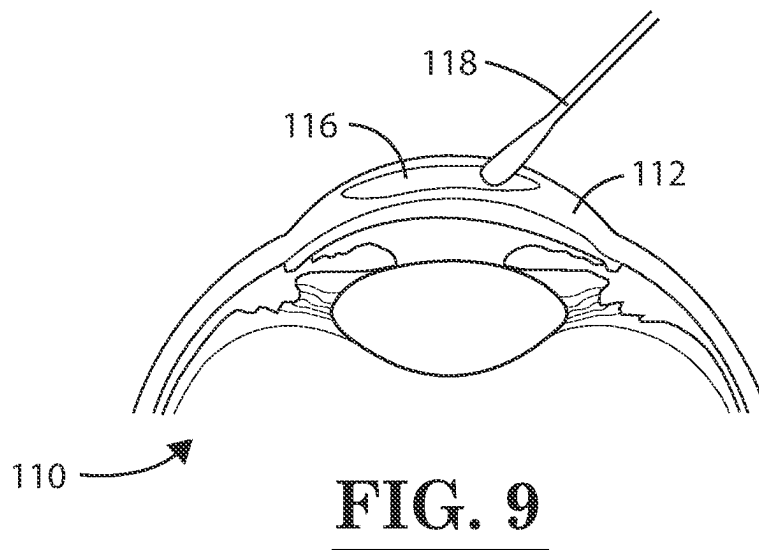
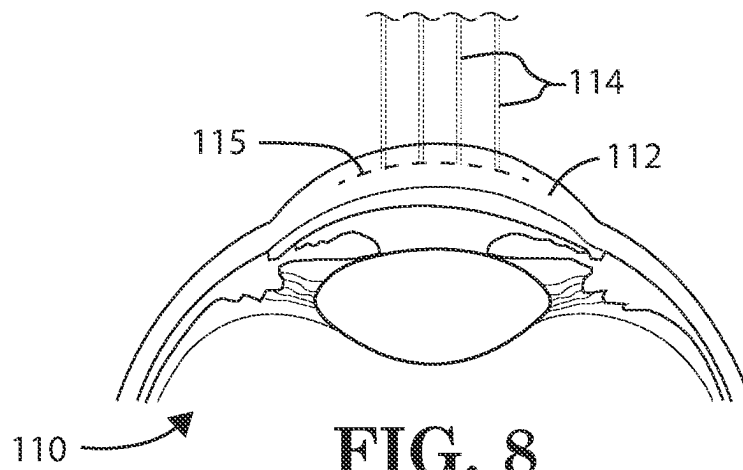


FIG. 7C





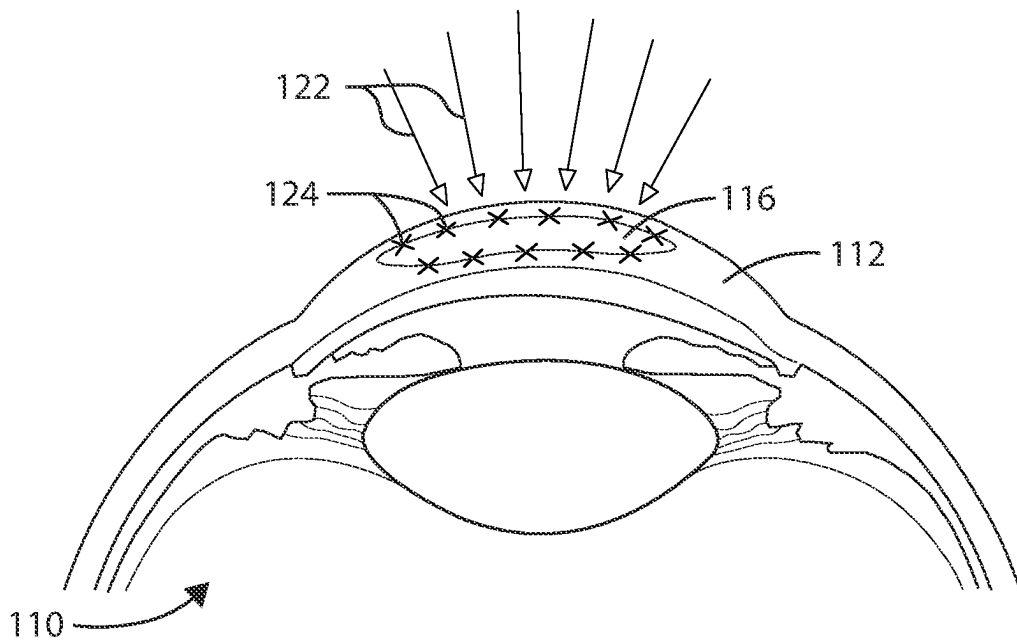


FIG. 11A

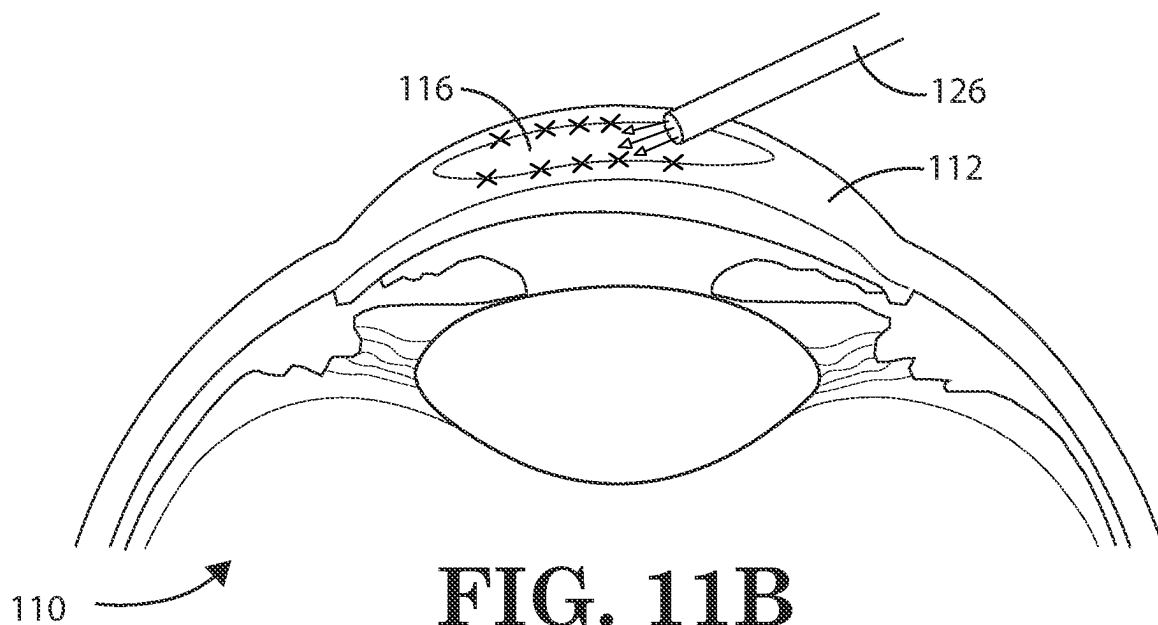


FIG. 11B

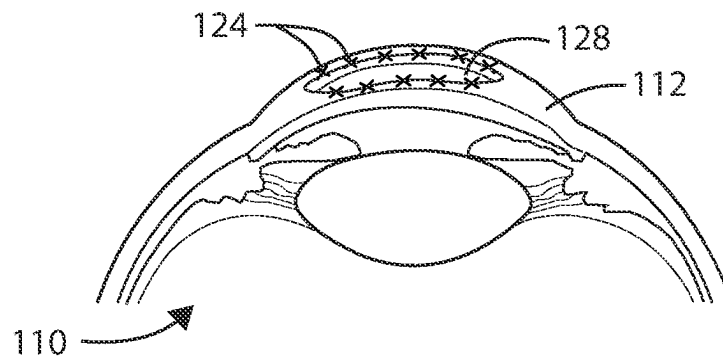


FIG. 12

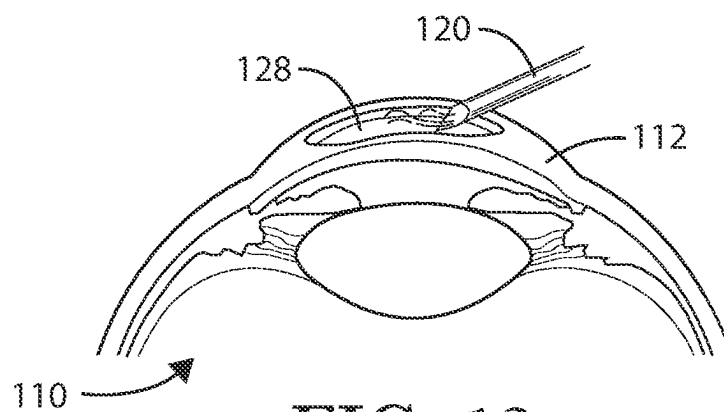


FIG. 13

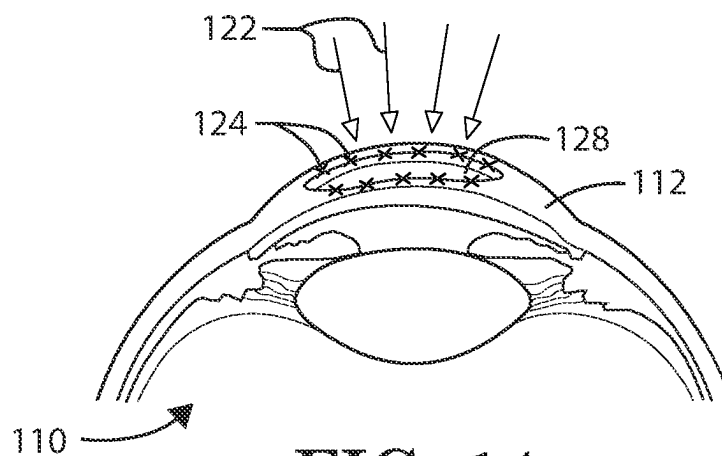


FIG. 14

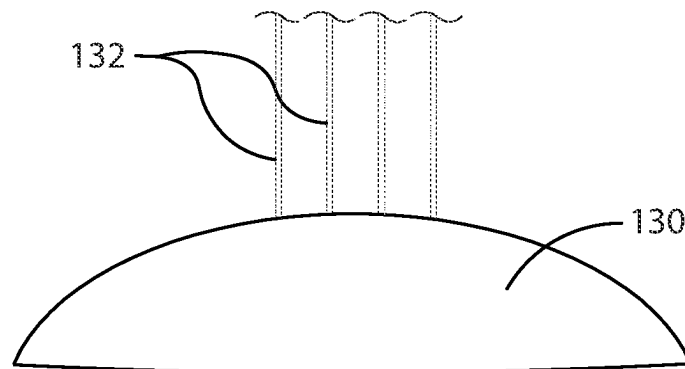


FIG. 15

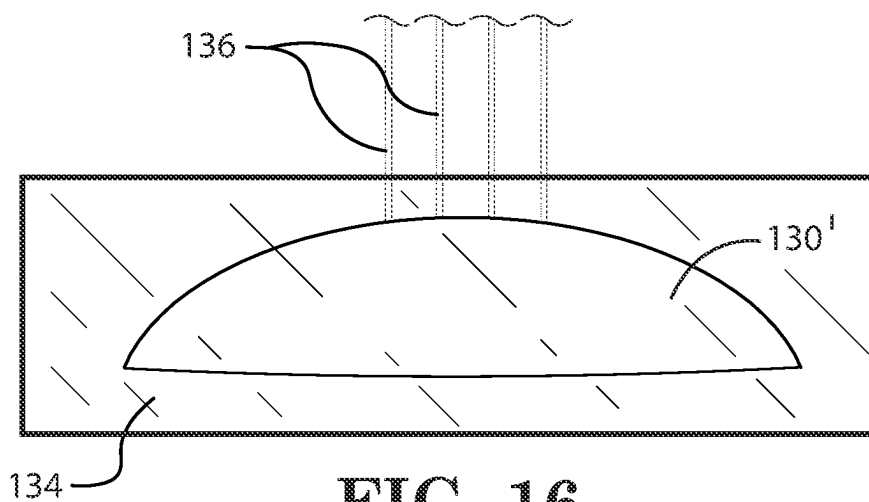


FIG. 16

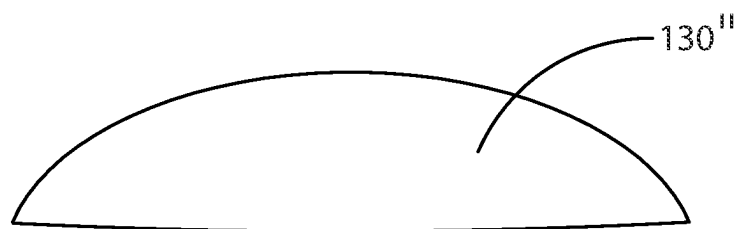


FIG. 17

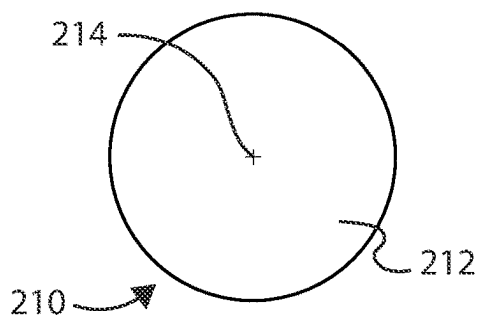


FIG. 18

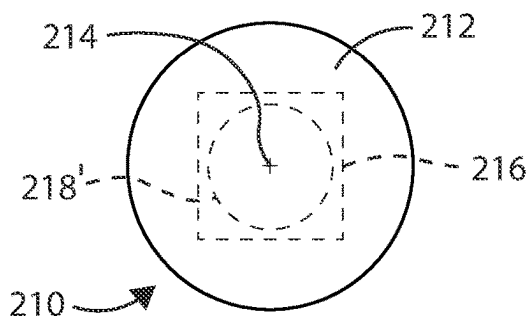


FIG. 21

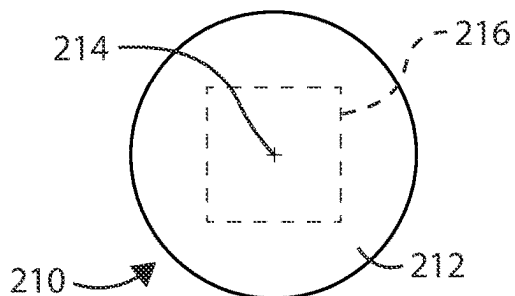


FIG. 19

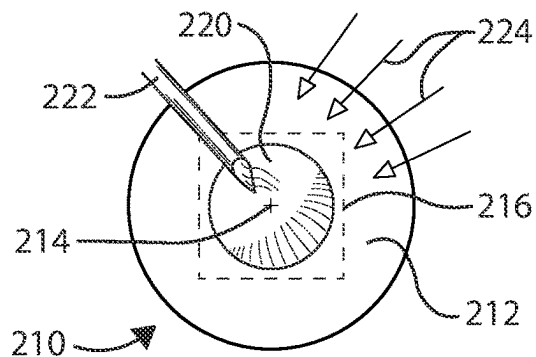


FIG. 22

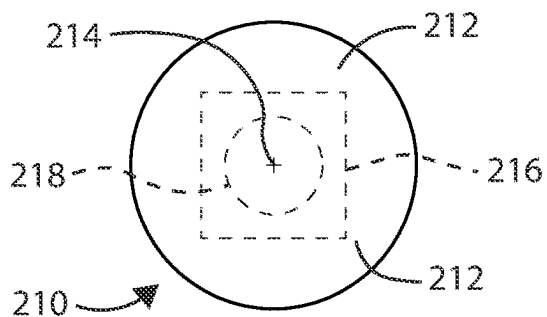


FIG. 20

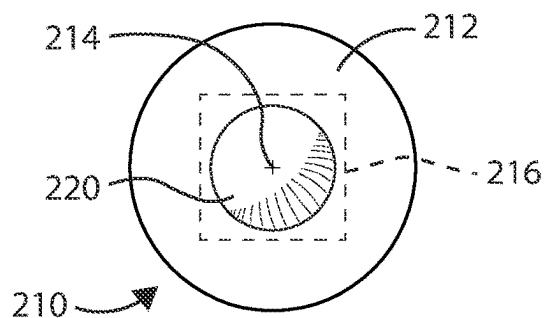


FIG. 23

FIG. 24

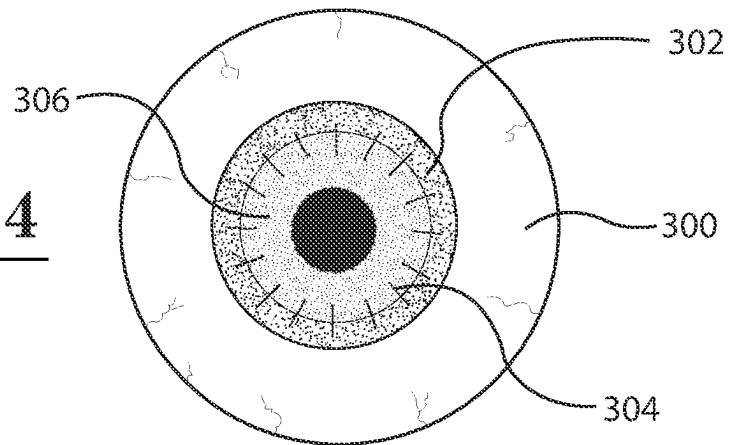


FIG. 25

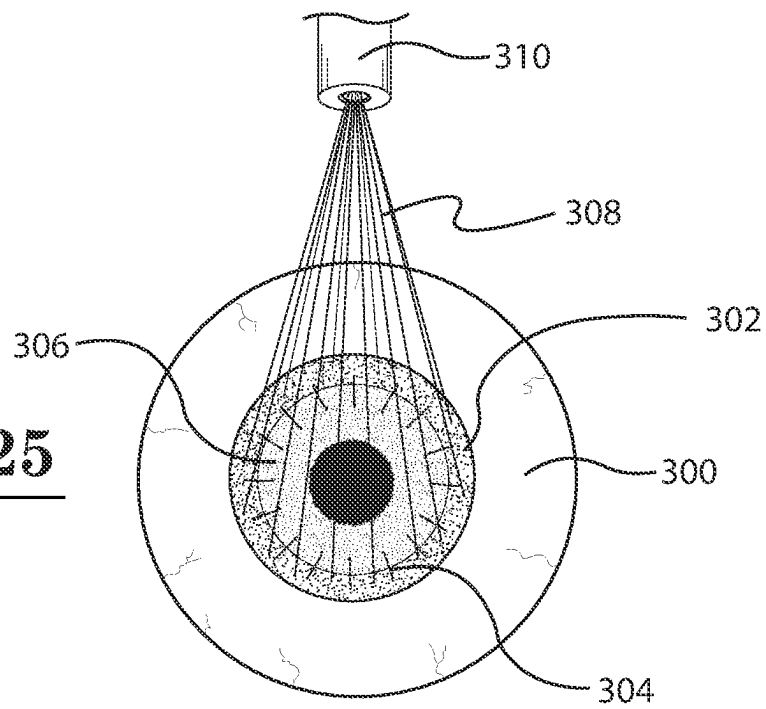


FIG. 26

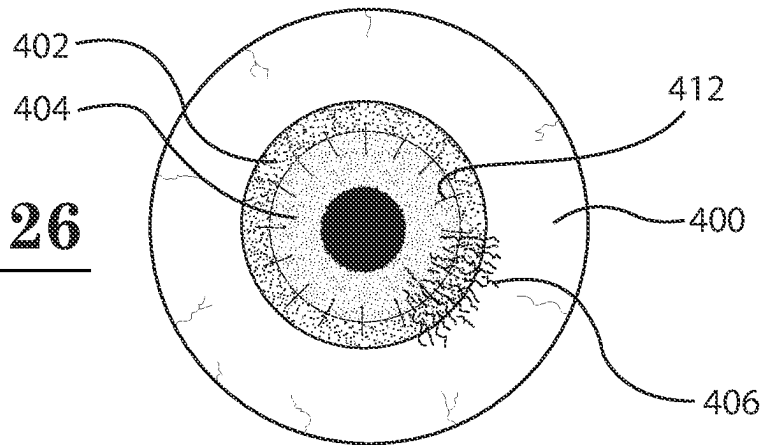
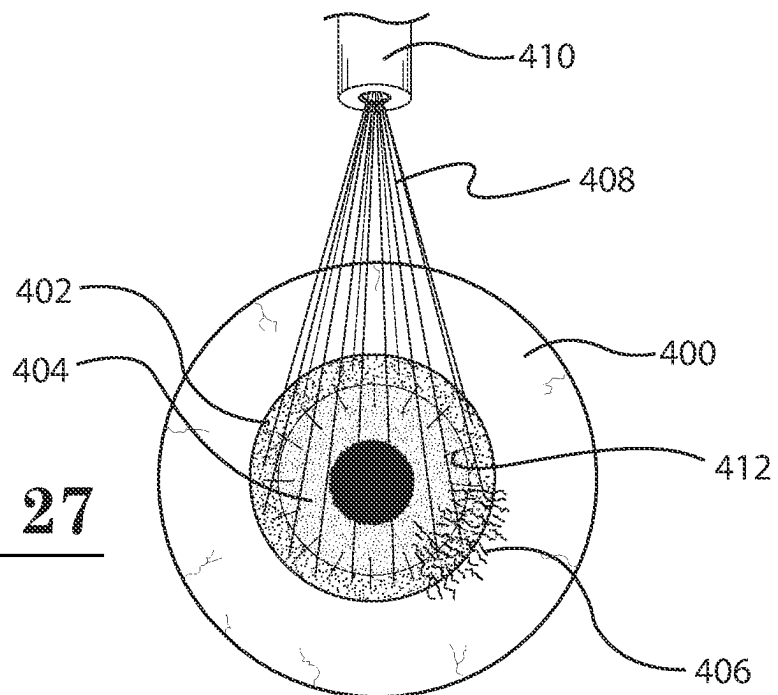


FIG. 27



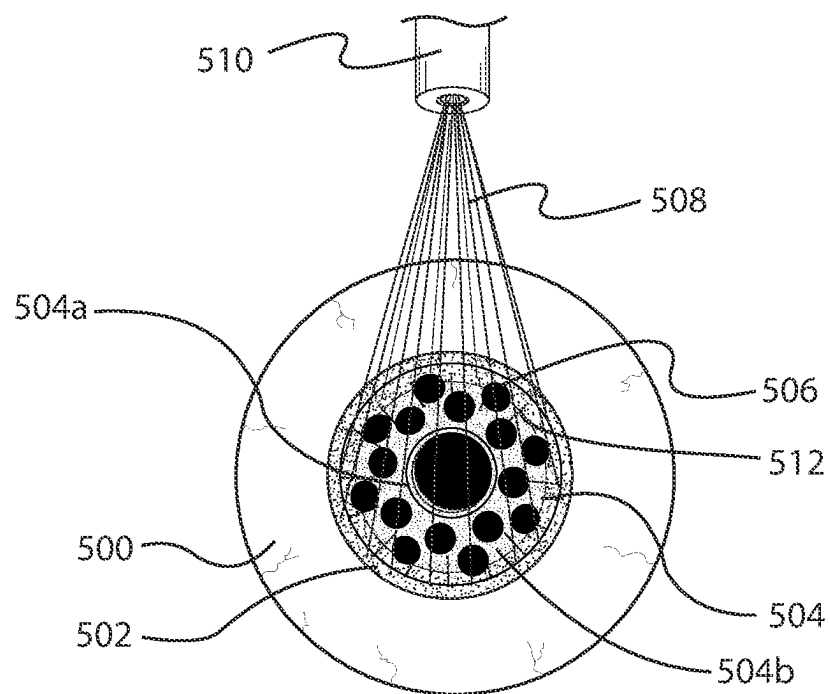


FIG. 28

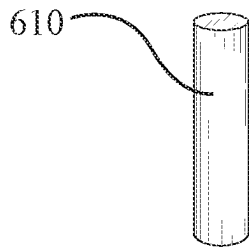


FIG. 29A

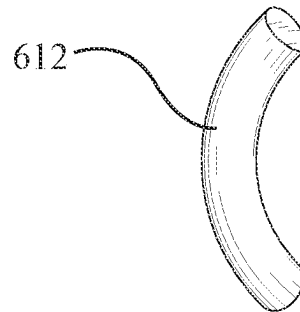


FIG. 29B

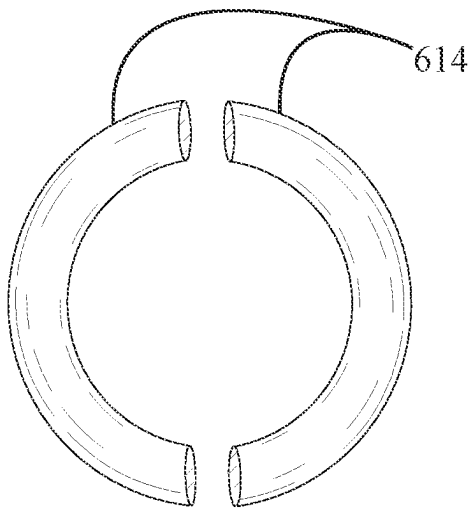


FIG. 29C

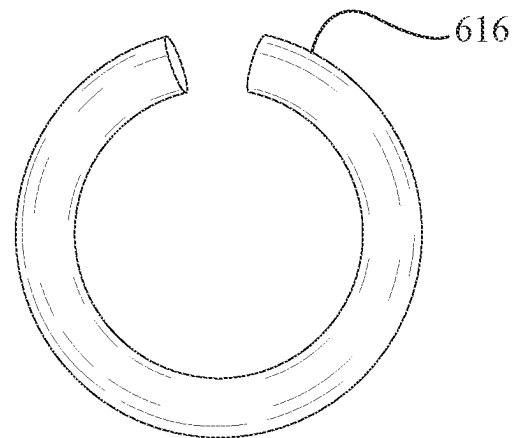


FIG. 29D

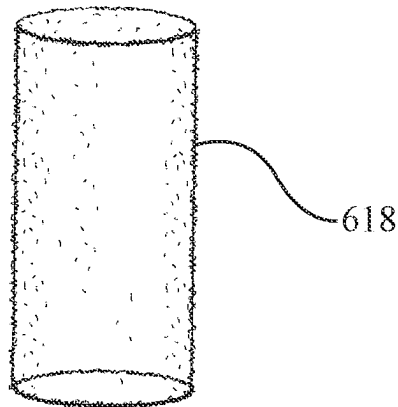


FIG. 30

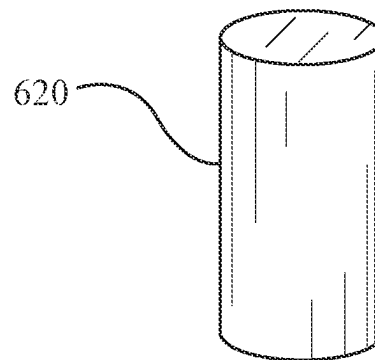


FIG. 31A

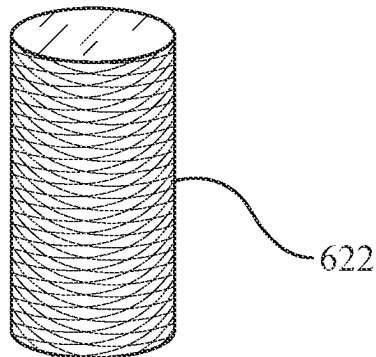


FIG. 31B

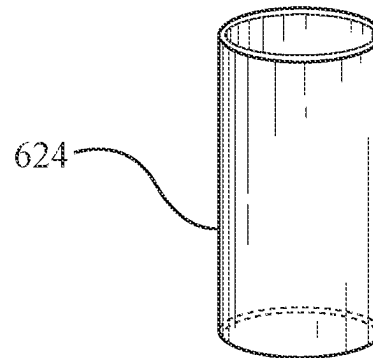


FIG. 31C

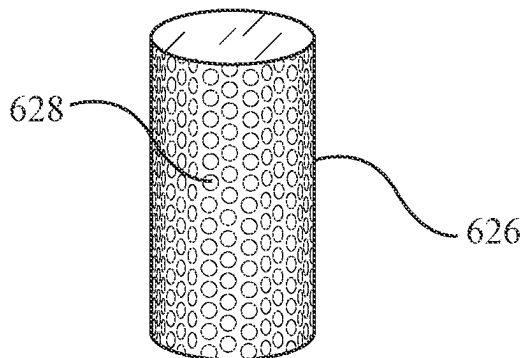


FIG. 32

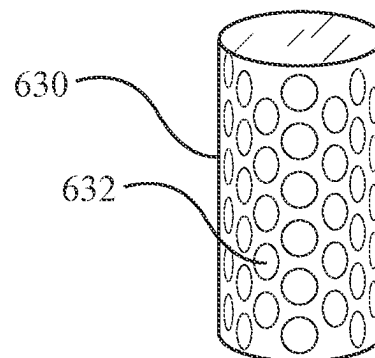


FIG. 33

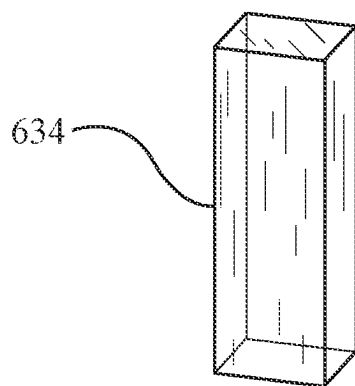


FIG. 34

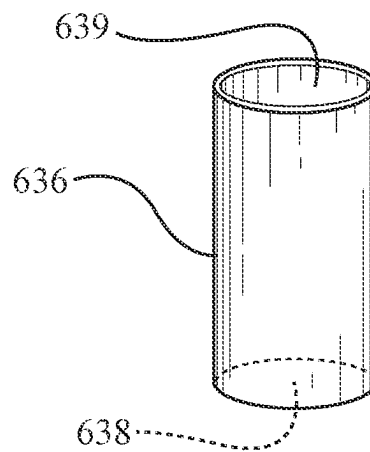


FIG. 35

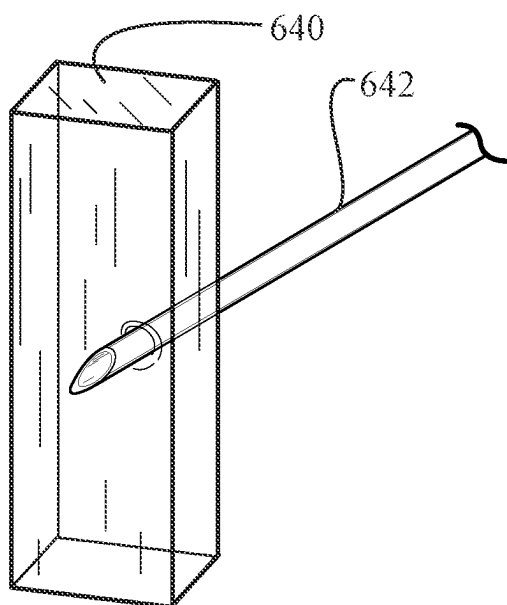


FIG. 36

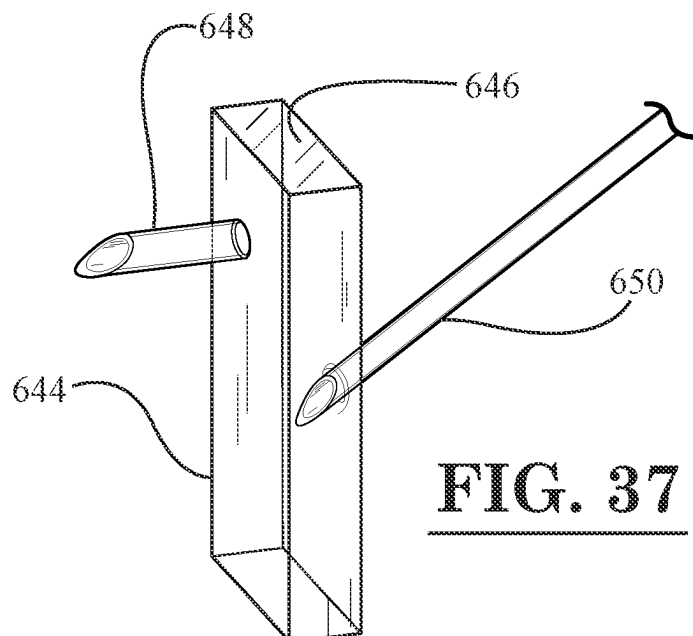


FIG. 37

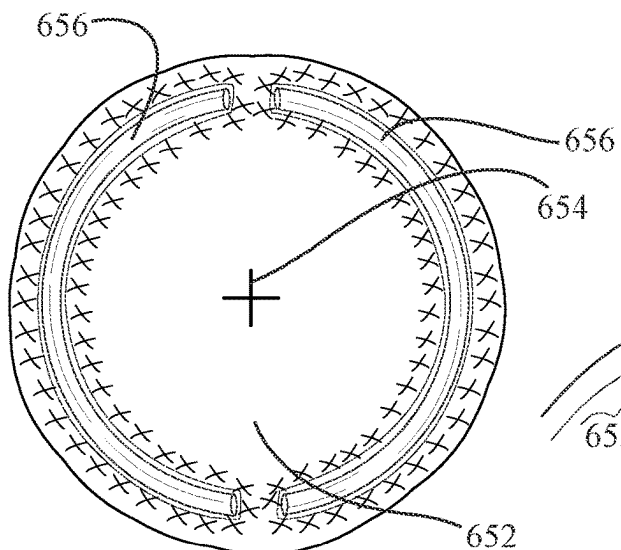


FIG. 38A

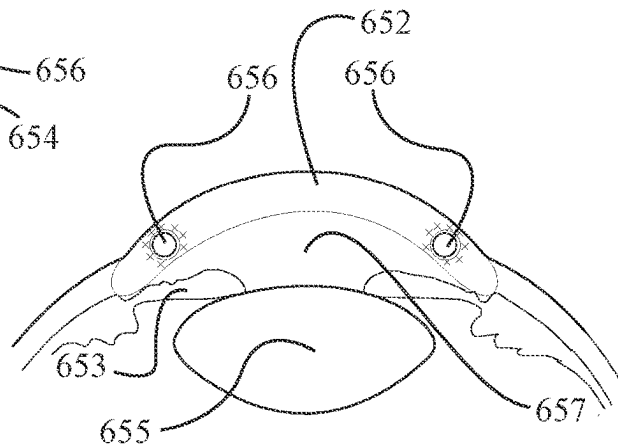


FIG. 38B

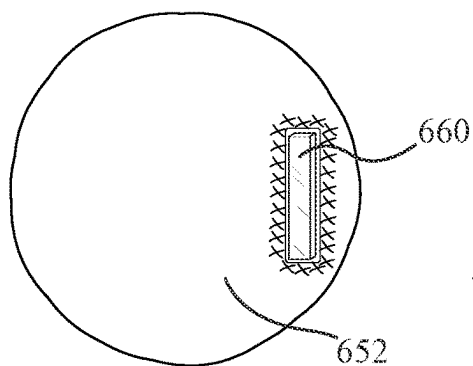


FIG. 39A

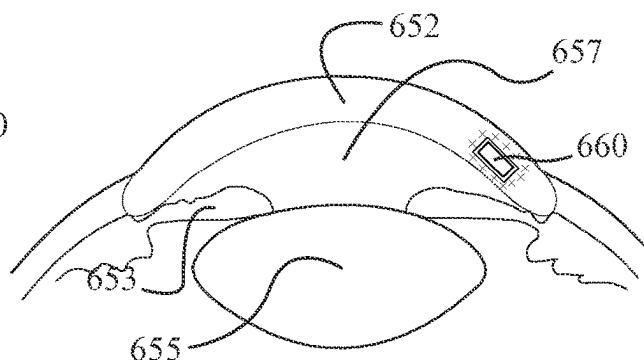


FIG. 39B

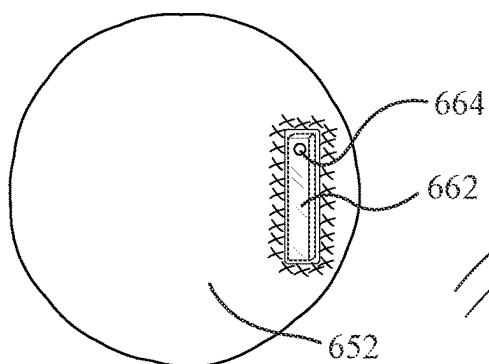


FIG. 40A

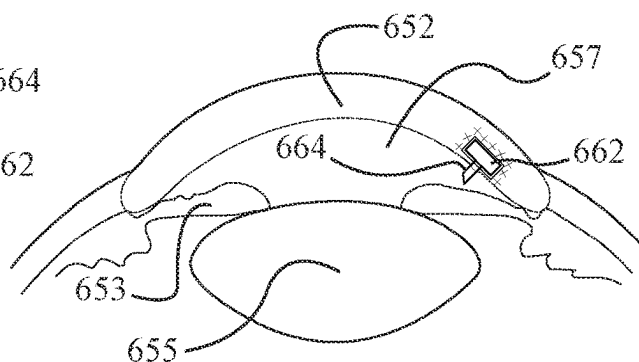


FIG. 40B

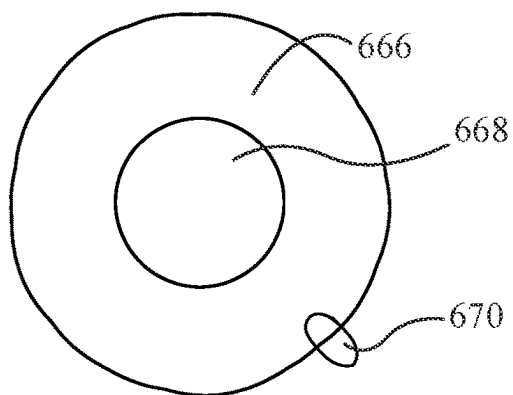


FIG. 41A

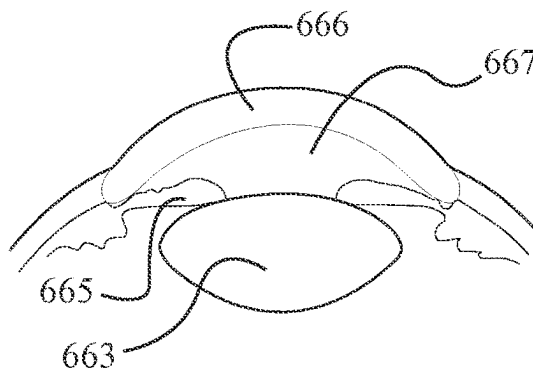


FIG. 41B

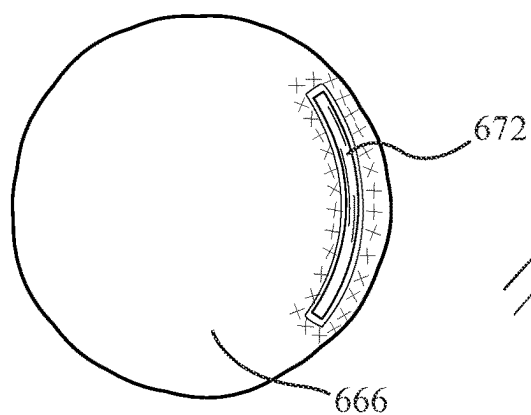


FIG. 42A

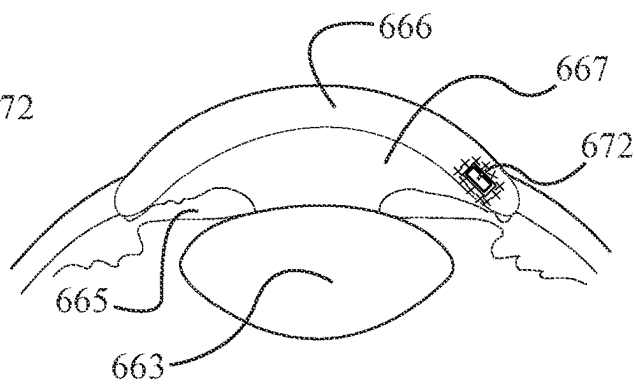


FIG. 42B

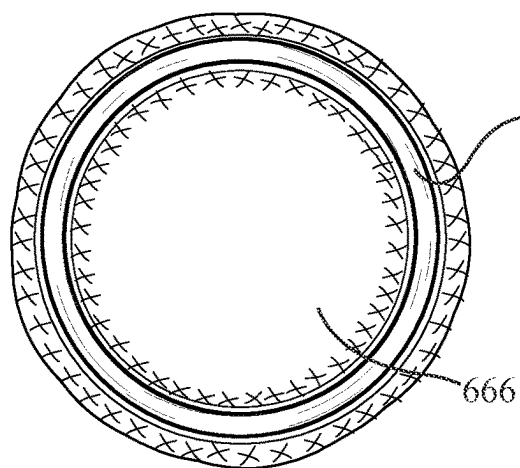


FIG. 43A

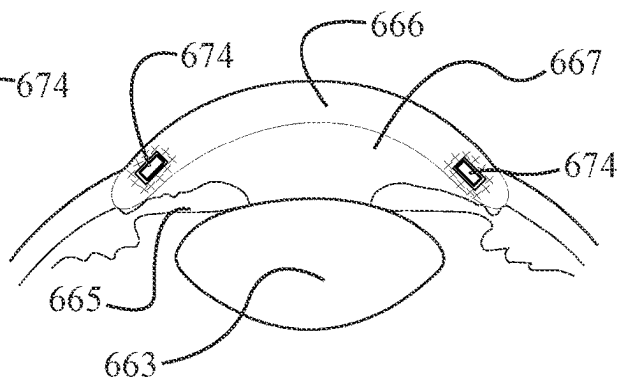


FIG. 43B

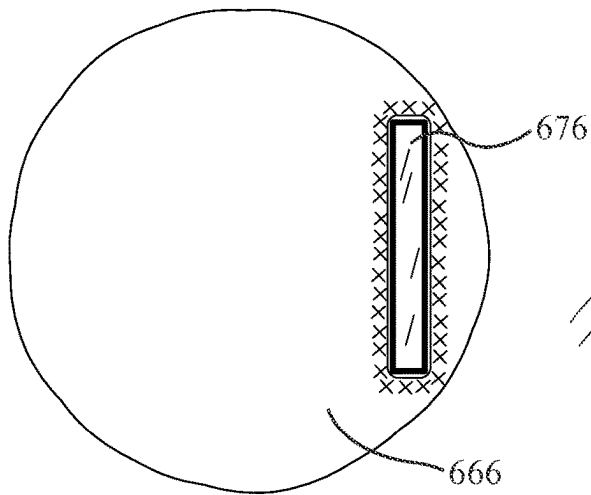


FIG. 44A

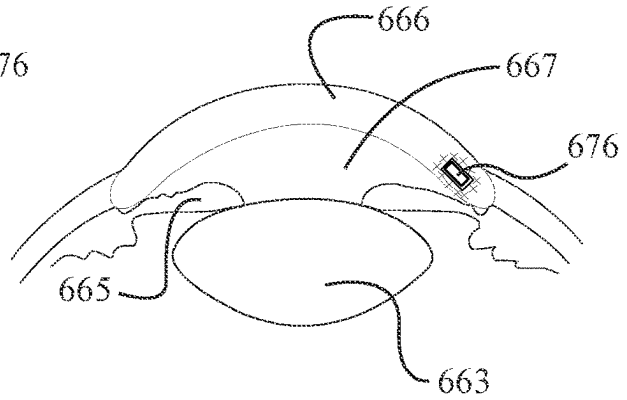


FIG. 44B

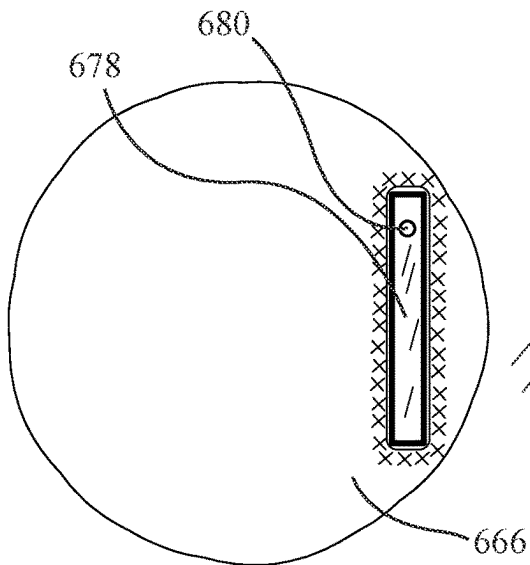


FIG. 45A

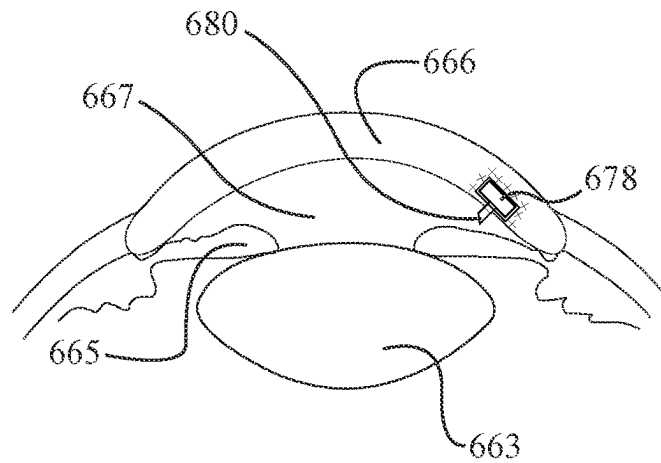
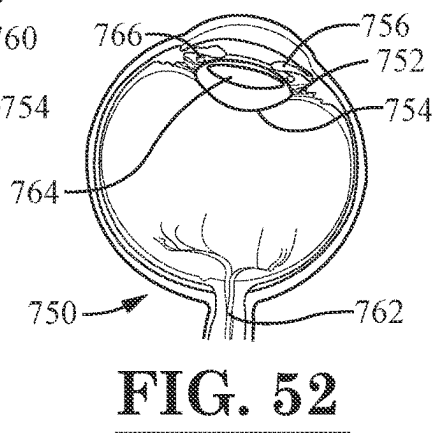
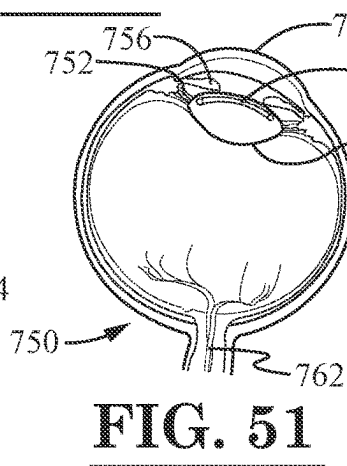
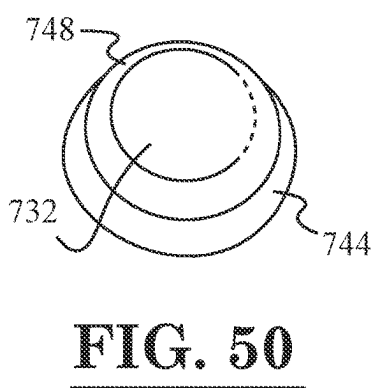
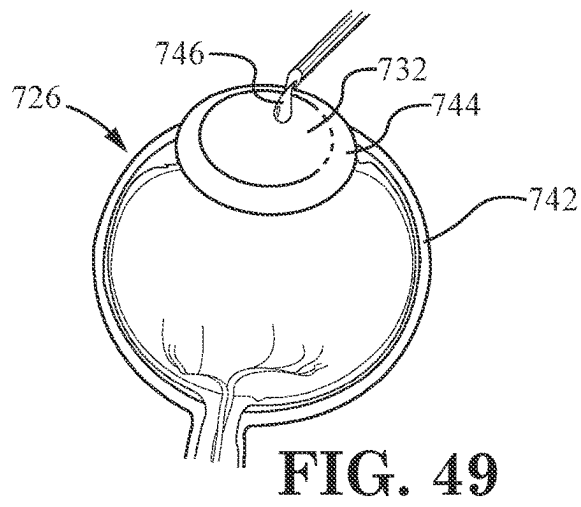
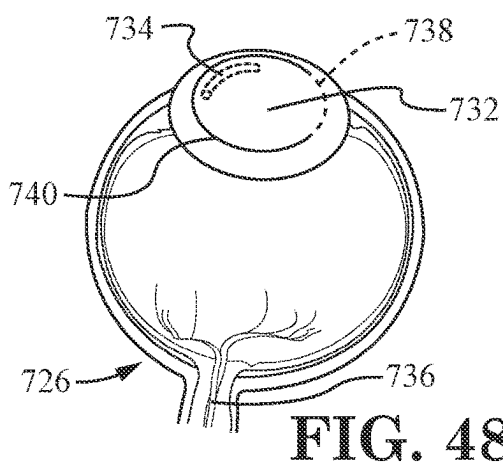
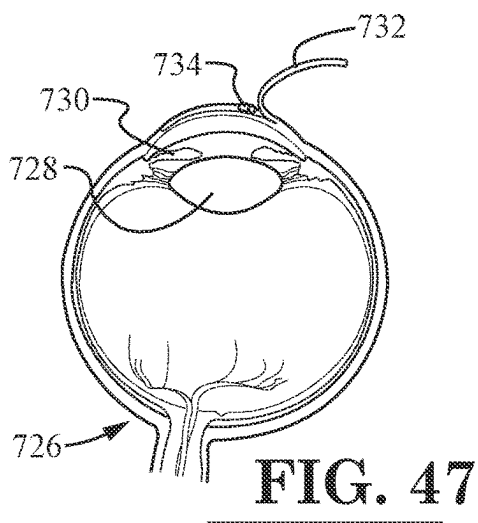
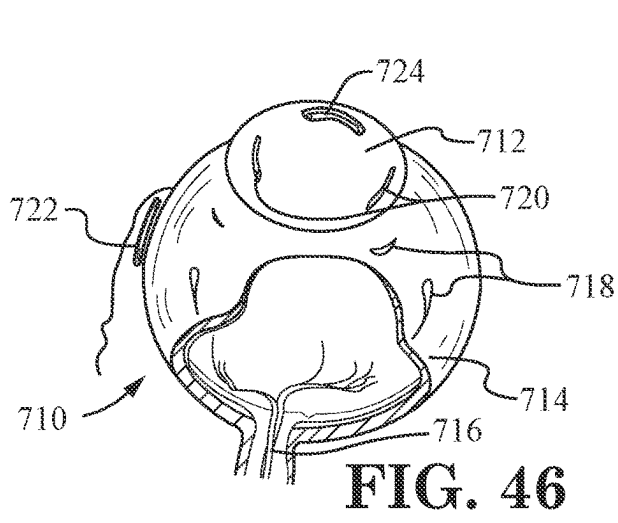


FIG. 45B



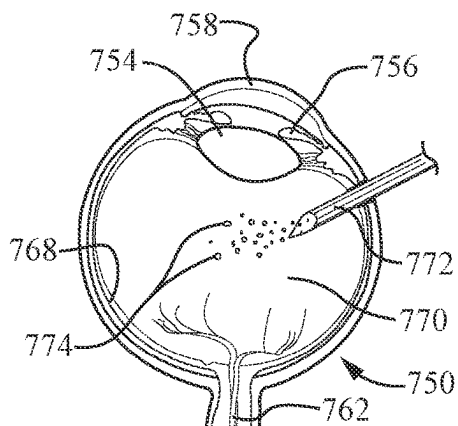


FIG. 53

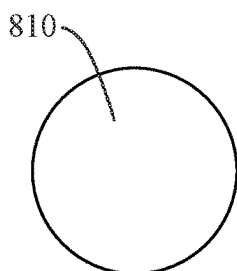


FIG. 54A

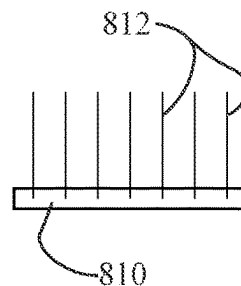


FIG. 54B

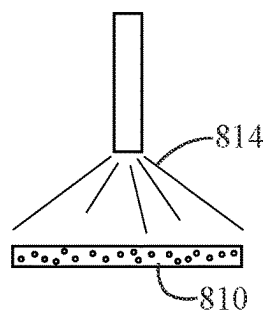


FIG. 55

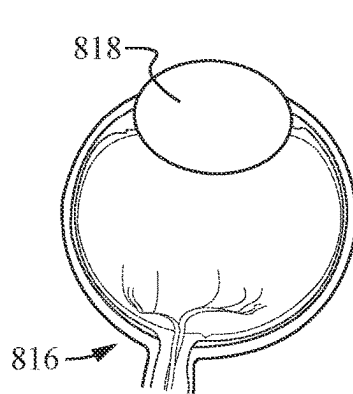


FIG. 56

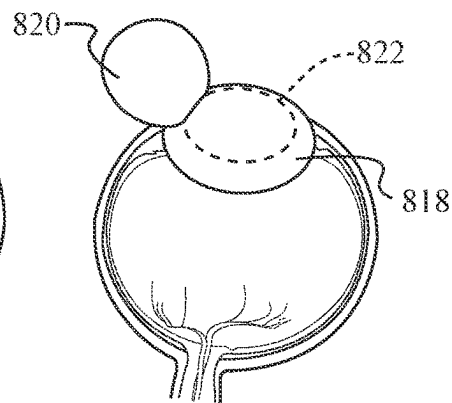


FIG. 57

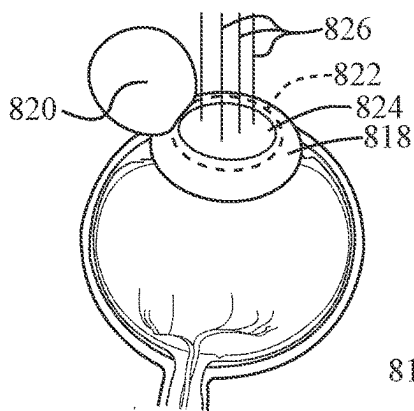


FIG. 58

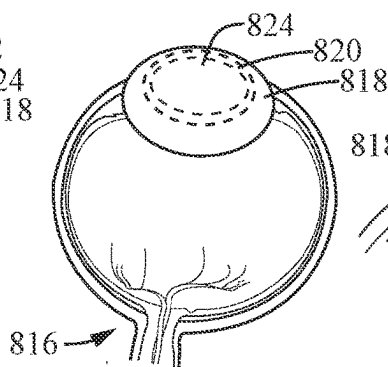


FIG. 59

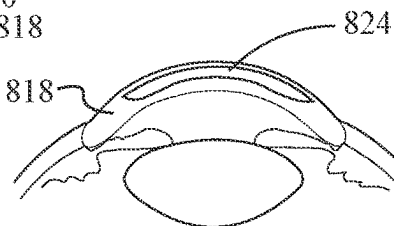


FIG. 60

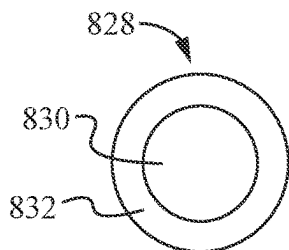


FIG. 61A

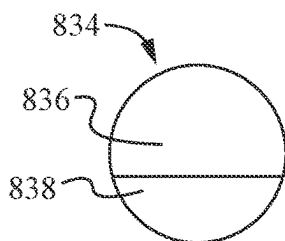


FIG. 61B

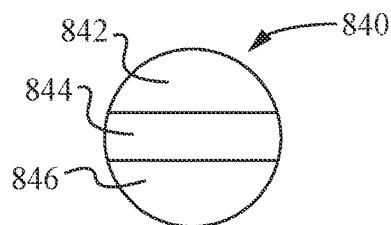


FIG. 61C

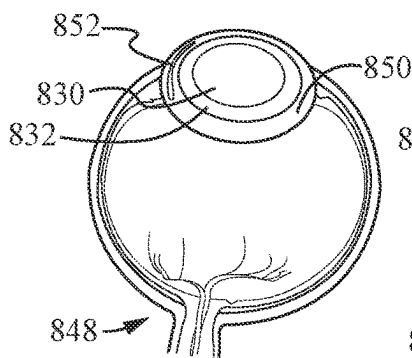


FIG. 62

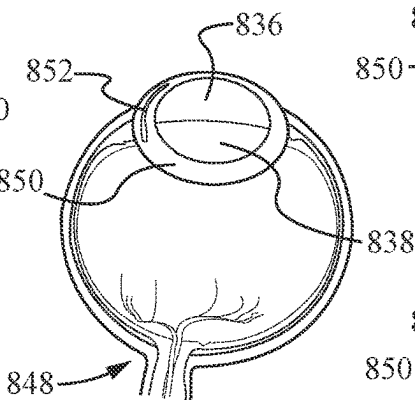


FIG. 63

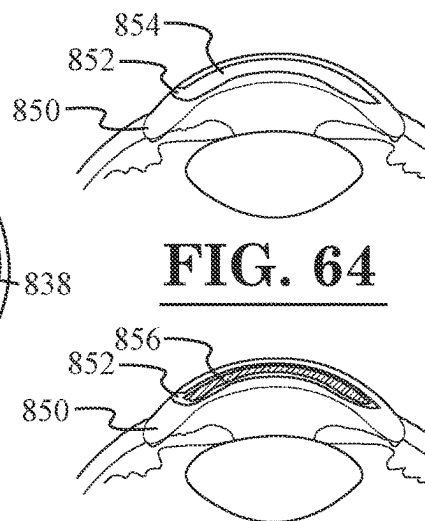


FIG. 64

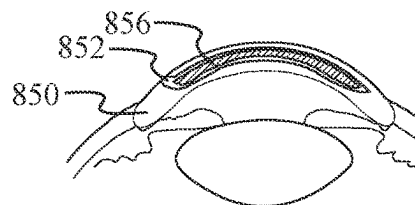


FIG. 65

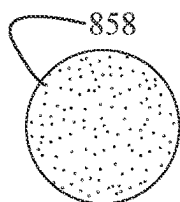


FIG. 66

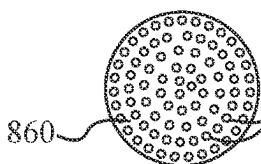


FIG. 67

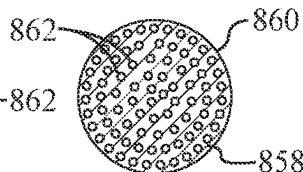


FIG. 68



FIG. 69A



FIG. 69B



FIG. 69C

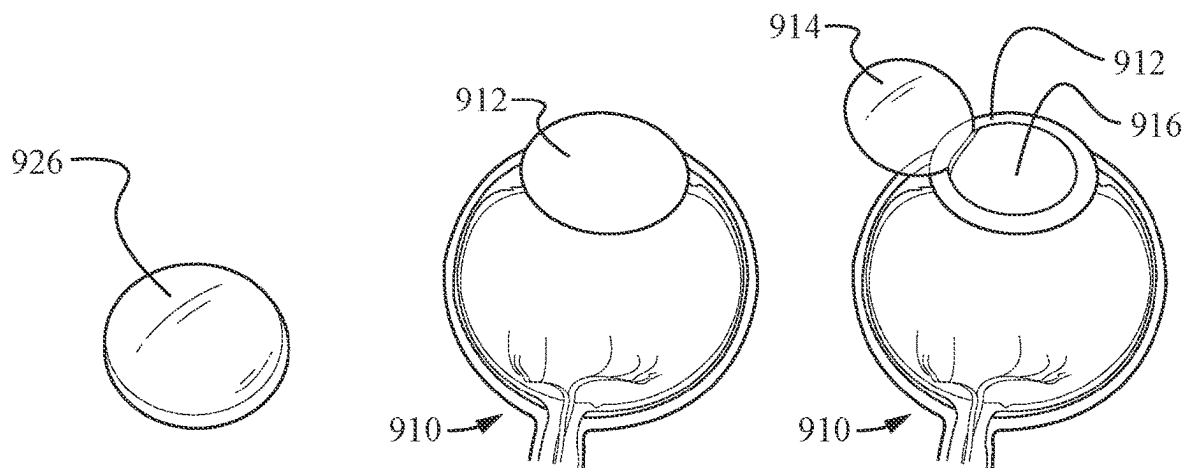


FIG. 70

FIG. 71

FIG. 72

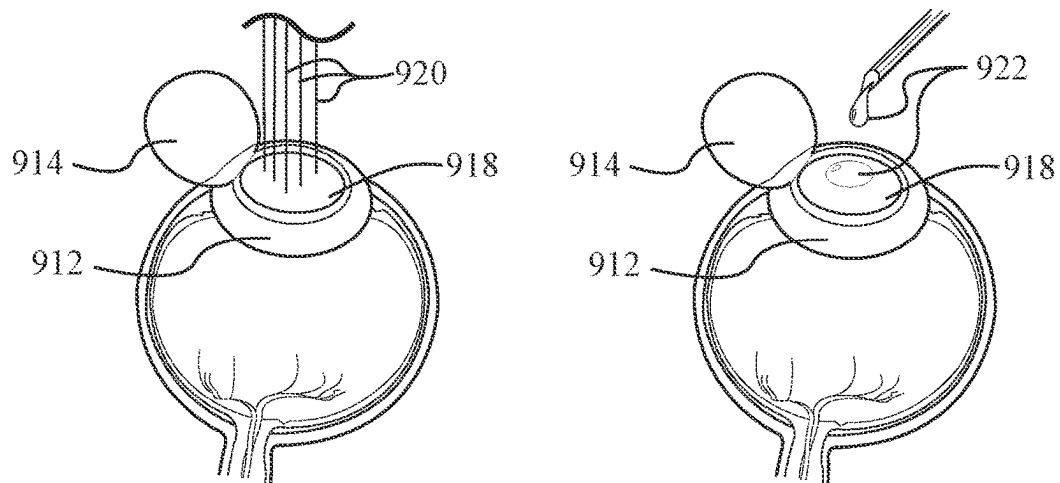


FIG. 73

FIG. 74

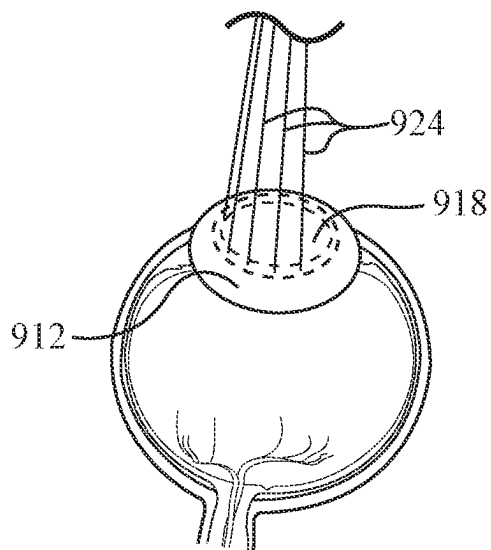


FIG. 75

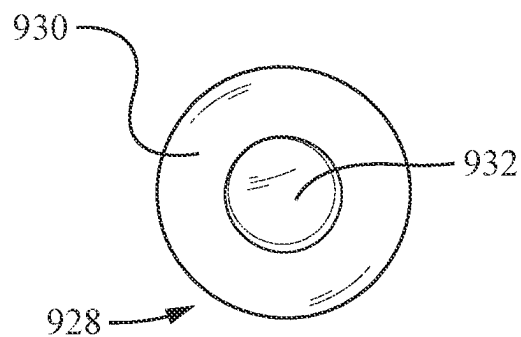


FIG. 76

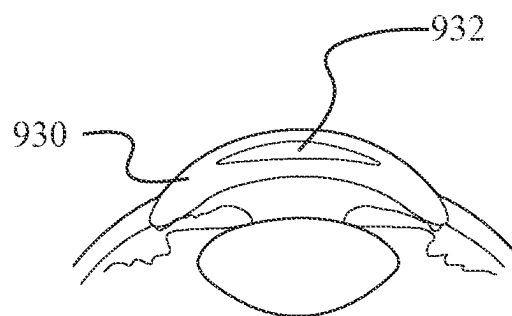


FIG. 77

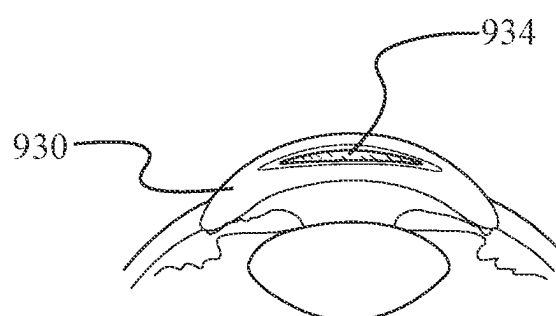


FIG. 78

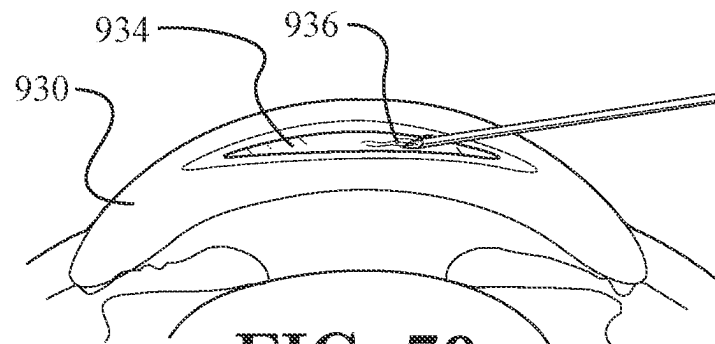


FIG. 79

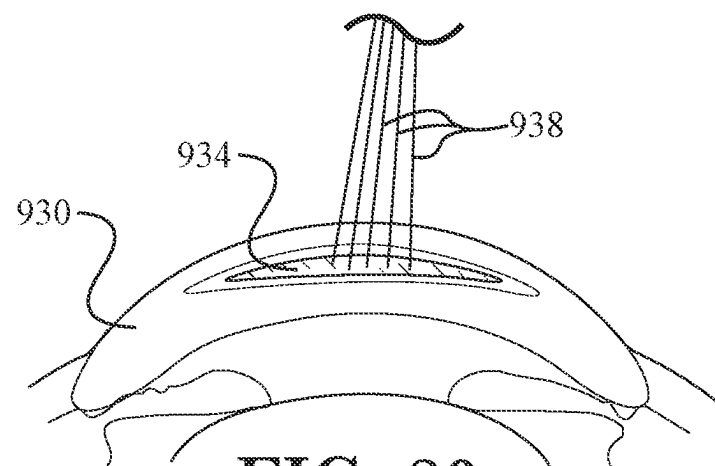


FIG. 80

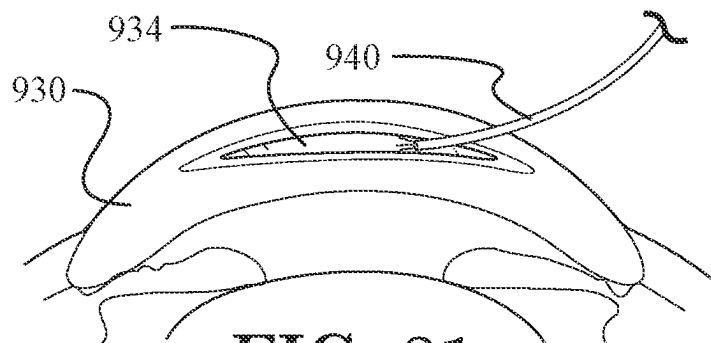


FIG. 81

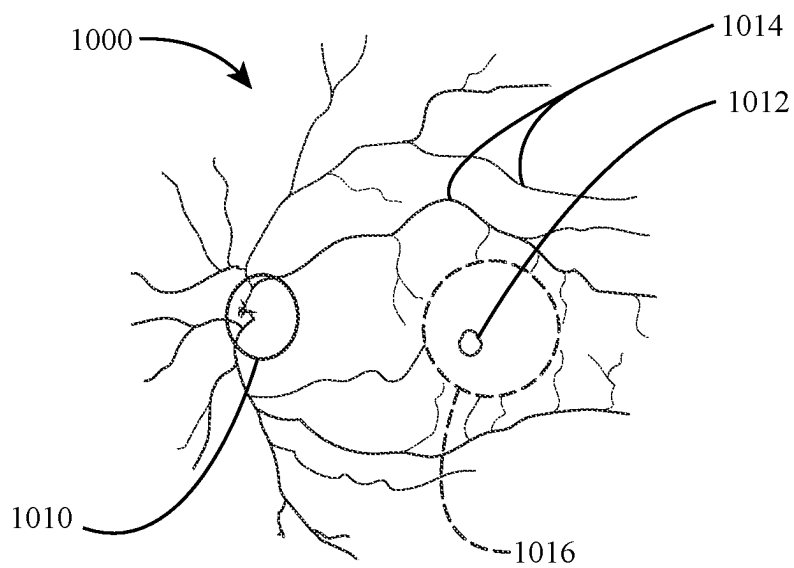


FIG. 82

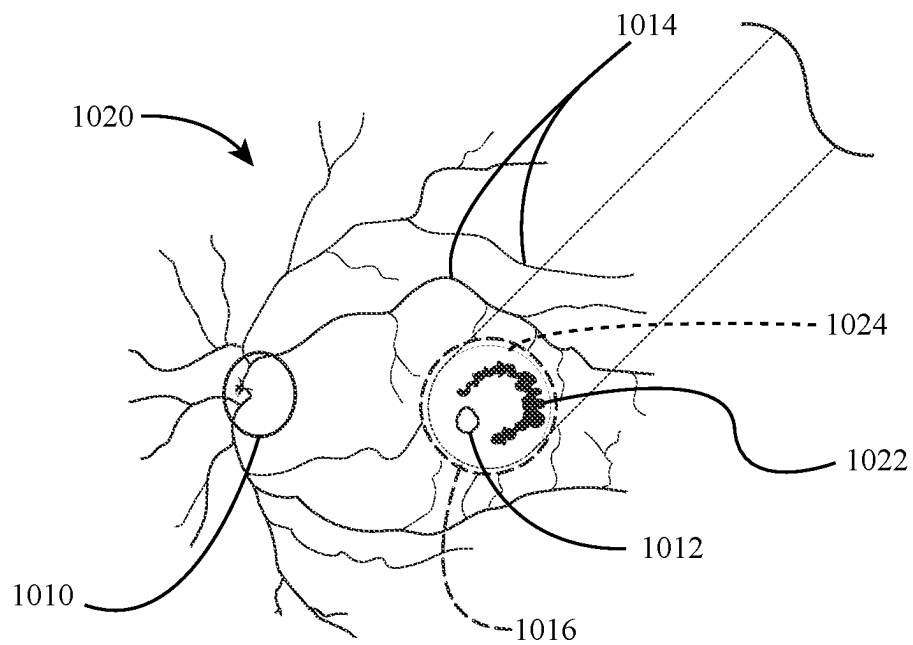


FIG. 83

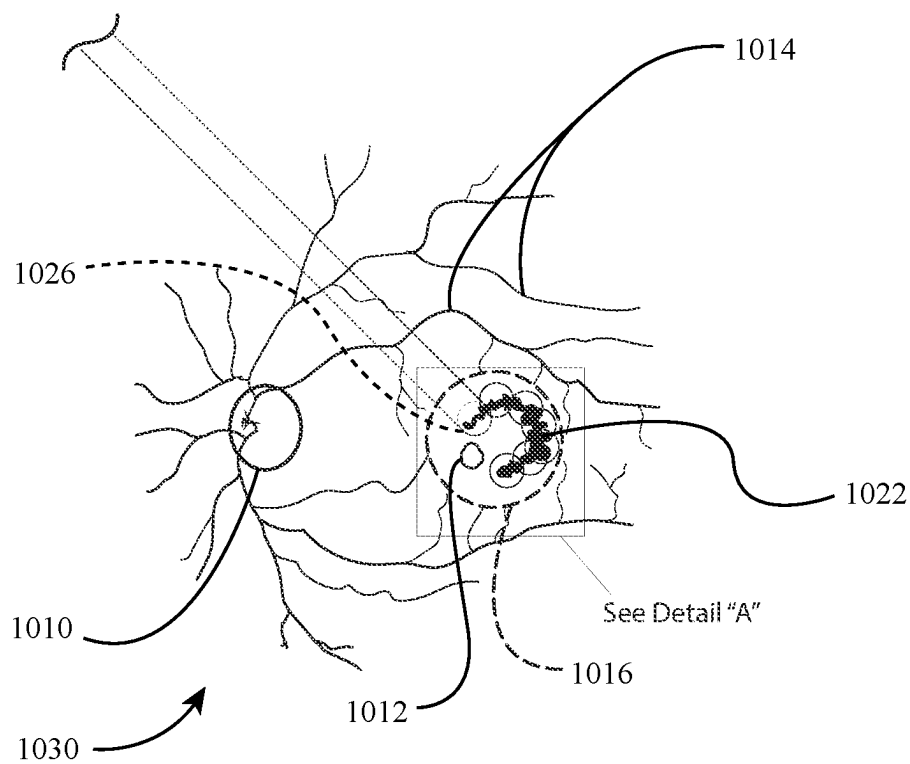
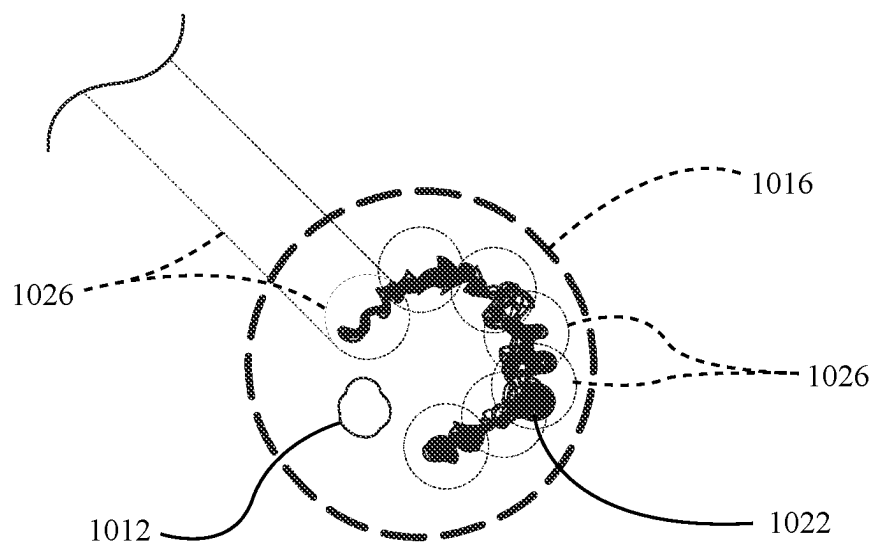


FIG. 84



DETAIL "A"

FIG. 85

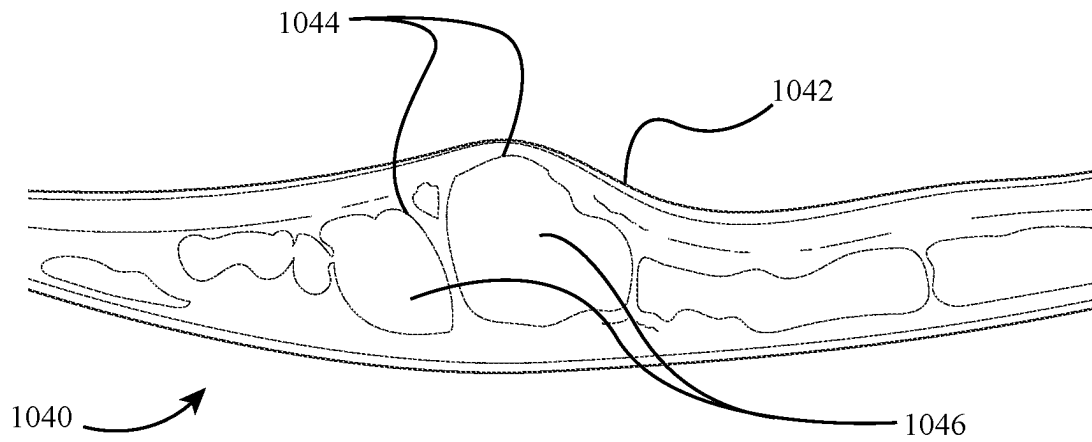


FIG. 86A

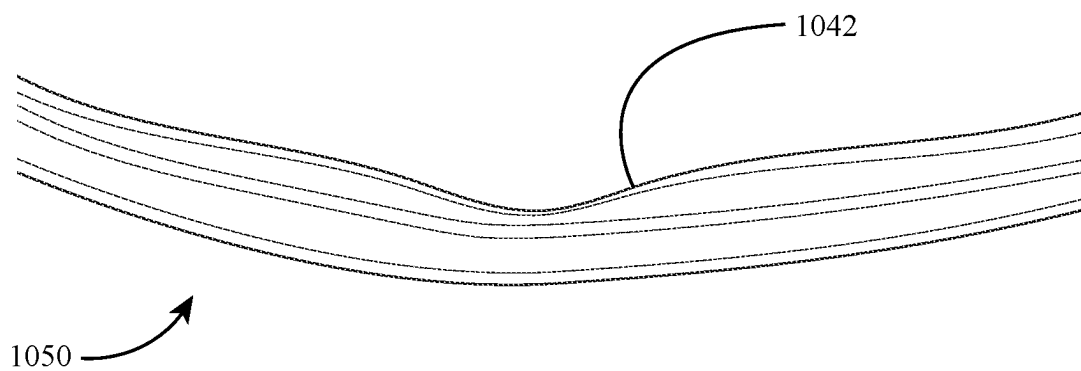


FIG. 86B

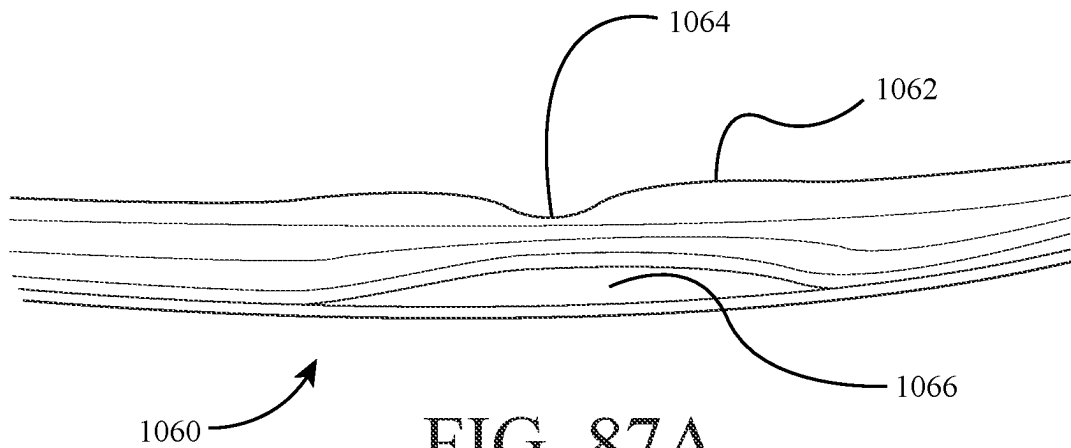


FIG. 87A

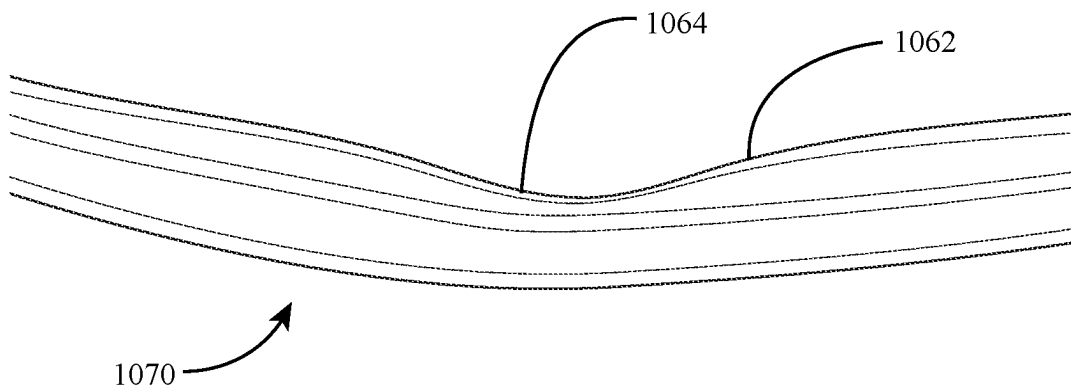


FIG. 87B

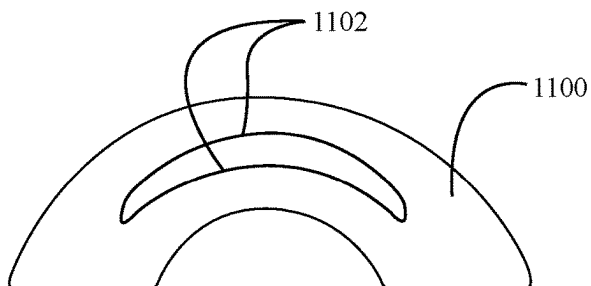


FIG. 88A

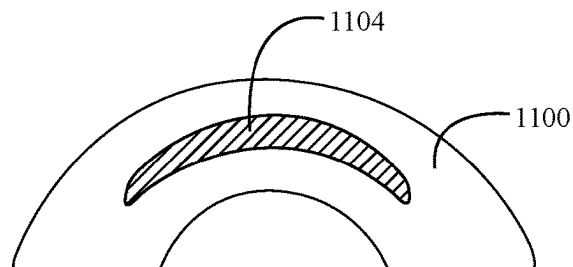


FIG. 88B

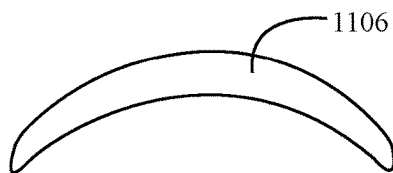


FIG. 88C

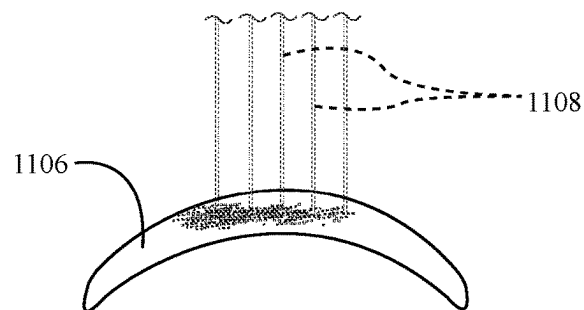


FIG. 88D

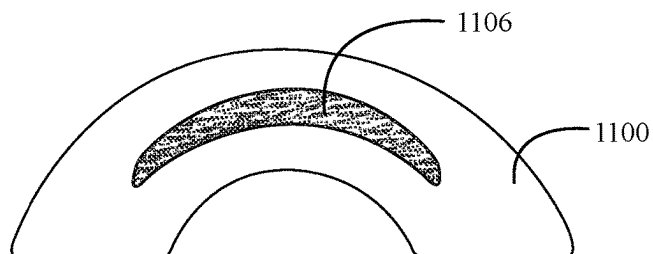


FIG. 88E

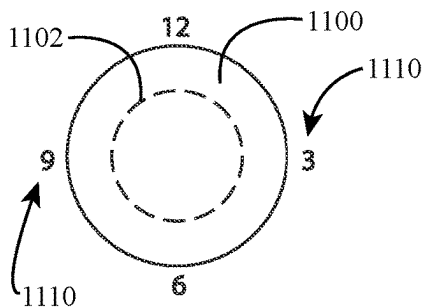


FIG. 89A

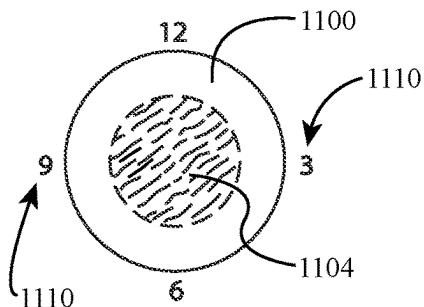


FIG. 89B

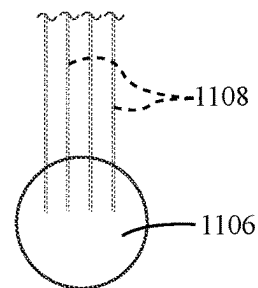


FIG. 89C

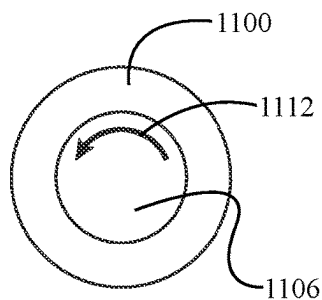


FIG. 89D

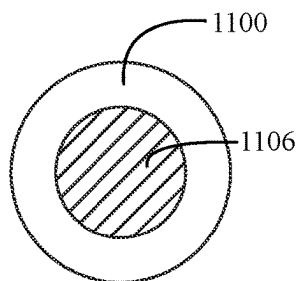


FIG. 89E

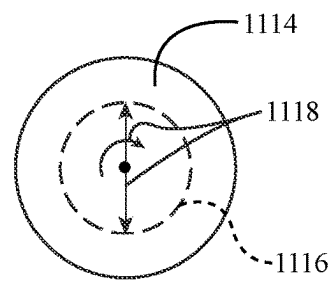


FIG. 89F

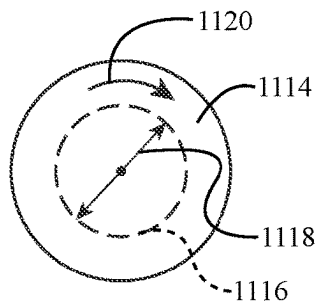


FIG. 89G

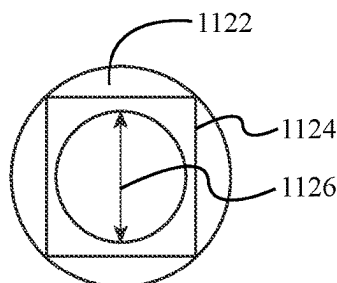


FIG. 89H

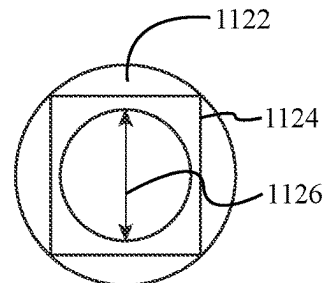


FIG. 89I

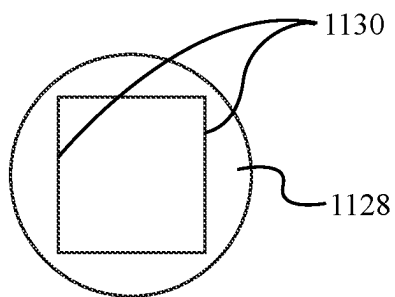


FIG. 90A

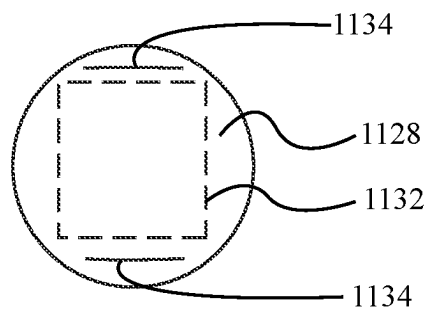


FIG. 90B

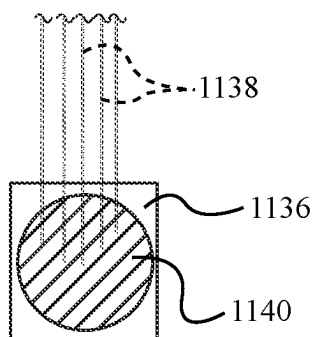


FIG. 90C

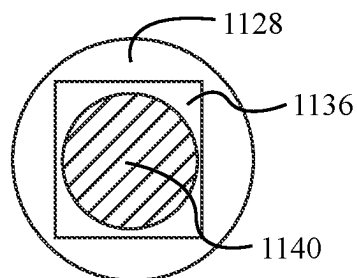


FIG. 90D

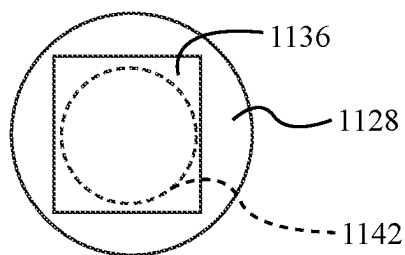


FIG. 90E

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LAMELLAR CORNEAL AUTOLOGOUS OR HOMOLOGOUS GRAFT IN REFRACTIVE SURGERY

CROSS-REFERENCE TO RELATED APPLICATIONS

This patent application claims priority to U.S. Provisional Patent Application No. 62/953,639, entitled “Angulated Lamellar Corneal Autologous Or Homologous Graft In Refractive Surgery”, filed on Dec. 26, 2019, and is a continuation-in-part of application Ser. No. 17/106,167, entitled “Photodynamic Therapy Technique For Preventing Damage To The Fovea Of The Eye Or Another Body Portion Of A Patient”, filed on Nov. 29, 2020, which claims priority to U.S. Provisional Patent Application No. 62/942,061, entitled “Photodynamic Therapy Technique For Preventing Damage To The Fovea Of The Eye Or Another Body Portion Of A Patient”, filed on Nov. 29, 2019, and Ser. No. 17/106,167 is a continuation-in-part of application Ser. No. 16/935,113, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on Jul. 21, 2020, which claims priority to U.S. Provisional Patent Application No. 62/876,736, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on Jul. 21, 2019, and Ser. No. 16/935,113 is a continuation-in-part of application Ser. No. 16/717,705, entitled “Method Of Corneal And Scleral Inlay Crosslinking And Preservation”, filed on Dec. 17, 2019, which claims priority to U.S. Provisional Patent Application No. 62/780,404, entitled “Method Of Corneal And Scleral Inlay Cross-linking And Preservation”, filed on Dec. 17, 2018, and Ser. No. 16/717,705 is a continuation-in-part of application Ser. No. 16/451,247, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on Jun. 25, 2019, which claims priority to U.S. Provisional Patent Application No. 62/849,941, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on May 19, 2019, and Ser. No. 16/451,247 is a continuation-in-part of application Ser. No. 16/414,713, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on May 16, 2019, now U.S. Pat. No. 10,583,221, which claims priority to U.S. Provisional Patent Application No. 62/672,161, entitled “Drug Delivery Implant And A Method Using The Same”, filed on May 16, 2018, and Ser. No. 16/414,713 is a continuation-in-part of application Ser. No. 15/941,641, entitled “Method Of Corneal Transplantation Or Corneal Inlay Implantation With Cross-Linking”, filed on Mar. 30, 2018, now U.S. Pat. No. 10,314,690, which claims priority to U.S. Provisional Patent Application No. 62/478,914, entitled “Method Of Corneal Transplantation With Cross-Linking”, filed on Mar. 30, 2017, and Ser. No. 15/941,641 is a continuation-in-part of application Ser. No. 15/230,445, entitled “Corneal Lenslet Implantation With A Cross-Linked Cornea”, filed on Aug. 7, 2016, now U.S. Pat. No. 9,937,033, which claims priority to U.S. Provisional Patent Application No. 62/360,281, entitled “Method of Altering the Refractive Properties of an Eye”, filed on Jul. 8, 2016, and Ser. No. 15/230,445 is a continuation-in-part of application Ser. No. 14/709,801, entitled “Corneal Transplantation With A Cross-Linked Cornea”, filed on May 12, 2015, now U.S. Pat. No. 9,427,355, which claims priority to U.S. Provisional Patent Application No. 61/991,785, entitled “Corneal Transplantation With A Cross-Linked Cornea”, filed on May 12, 2014, and to U.S. Provisional Patent Application No. 62/065,714, entitled “Corneal Transplantation With A Cross-

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Linked Cornea”, filed on Oct. 19, 2014, the disclosure of each of which is hereby incorporated by reference as if set forth in their entirety herein.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not Applicable.

NAMES OF THE PARTIES TO A JOINT RESEARCH AGREEMENT

Not Applicable.

INCORPORATION BY REFERENCE OF MATERIAL SUBMITTED ON A COMPACT DISK

Not Applicable.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The invention generally relates to angulated lamellar corneal autologous or homologous graft for use in refractive surgery. More particularly, the invention relates to intrastromal corneal modification of refractive error using either an angulated autologous graft or a angulated homologous graft.

2. Background

Corneal scarring is a major cause of blindness, especially in developing countries. There are various causes for corneal scarring, which include: bacterial infections, viral infections, fungal infections, parasitic infections, genetic corneal problems, Fuch’s dystrophy, other corneal dystrophies, corneal trauma, or failed surgery. A corneal transplant is often required if the corneal scarring is extensive, and cannot be corrected by other means. However, there can be major complications associated with a corneal transplant, such as corneal graft rejection wherein the transplanted cornea is rejected by the patient’s immune system.

A normal emmetropic eye includes a cornea, a lens and a retina. The cornea and lens of a normal eye cooperatively focus light entering the eye from a far point, i.e., infinity, onto the retina. However, an eye can have a disorder known as ametropia, which is the inability of the lens and cornea to focus the far point correctly on the retina. Typical types of ametropia are myopia, hypermetropia or hyperopia, and astigmatism.

A myopic eye has either an axial length that is longer than that of a normal emmetropic eye, or a cornea or lens having a refractive power stronger than that of the cornea and lens of an emmetropic eye. This stronger refractive power causes the far point to be projected in front of the retina.

Conversely, a hypermetropic or hyperopic eye has an axial length shorter than that of a normal emmetropic eye, or a lens or cornea having a refractive power less than that of a lens and cornea of an emmetropic eye. This lesser refractive power causes the far point to be focused behind the retina.

An eye suffering from astigmatism has a defect in the lens or shape of the cornea converting an image of the point of light to a line. Therefore, an astigmatic eye is incapable of sharply focusing images on the retina.

While laser surgical techniques, such as laser-assisted in situ keratomileusis (LASIK) and photorefractive keratectomy (PRK) are known for correcting refractive errors of the eye, these laser surgical techniques have complications, such as post-operative pain and dry eye or scarring. Also, these laser surgical techniques cannot be safely used on patients with corneas having certain biomechanical properties. For example, corneal ectasia may occur if these laser surgical techniques are applied to patients having thin corneas (e.g., corneas with thicknesses that are less than 500 microns).

Therefore, what is needed is a method for corneal transplantation that reduces the likelihood that the implanted cornea will be rejected by the patient. Moreover, a method is needed for corneal transplantation that is capable of preserving the clarity of the transplanted cornea. Furthermore, there is a need for a method of corneal transplantation that reduces the likelihood that the transplanted cornea will be invaded by migrating cells. Also, what is needed is a method for corneal lenslet implantation for modifying the cornea to better correct ametropic conditions without removing the tissue that potentially creates a side effect of keratectasia. In addition, a method is needed for corneal lenslet implantation that prevents a lens implant from moving around inside the cornea once implanted so that the lens implant remains centered about the visual axis of the eye, and is not rejected by the host tissue.

Further, a number of the diseases can lead to corneal opacity and loss of sight. Among these are many infectious diseases, caused by bacterial, viral, and fungal or other organisms. Furthermore, penetrating and contusion corneal injury in children and adults can create a corneal scar, which prevents a patient from seeing. Postsurgical procedures, such as cataract surgery, and glaucoma, etc. can also damage corneal endothelial cells with loss of clarity of the corneal tissue. A large number of corneal diseases have a genetic predisposition, and can cloud the cornea over a period of time.

In order to clear the visual axis for light to reach the retina, it often requires performing a corneal transplantation. In this process, a cornea from a recently deceased person is excised and transplanted in place of the diseased cornea in the host.

Often, the first corneal transplantation is successful for about 90% of the cases. However, the rest of the cases require repeated corneal transplantation to replace a rejected one. These cases and all cases that have vascular components to the corneal cloudiness constitute the group of patients in whom the corneal transplant can be rejected in about 30% of the cases. As a general rule, the more corneal transplantations that are performed, the greater are the chances of a graft rejection.

Medical therapy, steroids, immunosuppressants, etc. are often applied to the cornea, but in repeated cases of corneal transplantation and complicated cases, they have limited success with their associated side effects. This process can ultimately lead to corneal edema, cellular immune response, complete corneal cloudiness and vascularization, in addition to dry eye. In these cases, one can remove the center part of the opaque cornea and create a circular pocket in the remaining peripheral cornea horizontally with a knife as known in the art. This produces an anterior and a posterior flap around the central opening and a remaining part of the cornea in which a prosthetic lens with a flange is implanted. However, often these artificial lenses can be rejected because of the body's immune response.

Therefore, it is apparent that a need also exists for a corneal transplantation method with cross-linking that prevents an immune response to a corneal transplant and/or

rejection of the corneal transplant by the patient, and for a corneal transplantation method with cross-linking that prevents vascular and/or fibrous tissue growth on, and surrounding a corneal implant, inlay, or keratoprosthesis lens.

The choroid is the vascular layer of the eye located between the sclera and the retina and is responsible for oxygenation and nutrition of the outer portion of the retina. A number of choroidal and retinal diseases are associated with development of abnormal vessels in the choroidal layer located under the retina.

The fovea is the central part of the sensory retina that receives most of the light coming from the outside world, and is responsible for fine vision. The fovea is made of mostly the cone photoreceptors. The macula is the area of about 5 mm in diameter of the central retina centered around the fovea. The macula is made of a mixture of the cone cells and rod photoreceptors. The cones are involved in fine discrimination of two objects close to each other (e.g., during reading vision), whereas the rods are involved mainly in night vision and motion detection.

Among the most common diseases affecting the choroid-retina are dry and wet forms of age-related macular degeneration, presumed ocular histoplasmosis syndrome, idiopathic choroidal neovascularization, and central serous retinopathy, trauma, and diabetic retinopathy, etc.

The dry form of age-related macular degeneration does not initiate by invasion of the abnormal vessels in the sub-retinal space, but by gradual dry loss of the retinal pigment epithelial (RPE) cells that are providing nutrition to the sensory retina and form a barrier between the choroid and retina. The RPE cells also remove the debris that over time can be collected under the retina.

The wet form of age-related macular degeneration is an aging disease of the eye in which the abnormal vessels protrude under the retina, leak and bleed leading to the loss of sight and scarring of the macula.

The cause of idiopathic choroidal neovascularization is not known, but the patients are generally a young group and the disease is not associated with accumulation of drusen that are more or less small accumulated debris under the retinal pigment epithelium (RPE) that can become confluent, and is found commonly in the wet form of the age-related macular degeneration and the dry form.

The pathogenesis of central serous retinopathy is not well known, but is seen in younger or middle aged patients in which a small or larger amount of fluid leaks through the RPE and elevates the retina. RPE changes may adversely affect photoreceptor function and survival due to disruption of the blood-retinal barrier (BRB) and leakage of fluid into the sub-retinal space with the consequence of loss of sight. The disease can heal by itself or it can be chronic and affect larger areas of the macula.

The initial laser treatment of the abnormal vascular tissue has the side effect that it creates a laser burn which would simultaneously burn the retina and the choroid to cauterize the abnormal vessels located underneath the retina. However, laser treatment has caused immediate blindness in patients.

Central serous retinopathy (CSR) is characterized by serous detachment of the neurosensory retina causing metamorphopsia and visual loss. In CSR, fluid is extruded through a hyperpermeable choroid and passes across a leak in the retinal pigment epithelium (RPE), creating a bullous separation of the RPE and the overlying retinal layers and, at times, with RPE detachment, known risk factors include psychological stress, Type-A personality, elevated levels of cortisol, male sex, and pregnancy.

Similarly, contusion and penetrating traumatic injuries can damage the barrier between the retina and choroid (Bruch's membrane) and permit abnormal vessels to invade the subretinal space leak and bleed damaging the sensory retinal area.

Photodynamic therapy (PDT) was developed so that the retinal vessels would not be damaged by thermal energy since the effect of photodynamic therapy is to reduce the thermal damage while damaging the abnormal vessels. In PDT, a photosensitizer (e.g., Verteporfin or Visudyne), is a benzoporphyrin derivative) where verteporfin 6 mg/m² body surface area, is injected intravenously. Then, the abnormal vessels are exposed to a non-thermal wavelength of laser that damages the vessels that contain a dye (e.g., verteporfin or Riboflavin 0.1-1 mg/ml or more) through a mechanism called the photodynamic effect. The photosensitizer absorbs the light energy after its intravenous injection or topical administration. The absorption of light has two synergistic mechanisms: (1) converting energy level of the photosensitizer molecules to a higher level of energy that upon return to its base creates triplet oxygen and reactive toxic species that damage the wall of the endothelial cells of the capillary vessels, and (2) the damaged endothelial cells initiate a platelet aggregation and blood clot formation that block the blood flow in the abnormal vessels.

The most important factor for photodynamic therapy (PDT) includes the coexistence of a sensitizer, light, and oxygen. The main mechanism of action of PDT is vascular occlusion due to damage to endothelial cells and subsequent thrombosis of both neovascular and normal choriocapillaris. The response to PDT appears to be caused by a combination of direct cytotoxicity to vascular endothelial cells, subsequent platelet adhesion and degranulation, thrombosis, and vasoconstriction, leading to blood flow stasis and vaso-occlusion of the choriocapillaris.

However, in PDT, the standard practice recommended method is that the amount of energy delivered requires that the size of the "non-thermal" beam should cover the entire lesion and slightly beyond it (refer to FIG. 2), while using continuous radiation with a laser beam, e.g., at a 689 nm wavelength for a period of 80 seconds continuously, at the described circular spot size, and with a power of 600 mW or (600 mW/cm² fluence rate and 50 J/cm² dose) at all times for 80 seconds each case. However, in real life, a single PDT treatment does not eliminate all the sub-retinal abnormal vessels, and recurrences are the norm. In experimental studies, the initial application of the laser and the dye in animal eyes had minimal damage to the retina, but by the second and third treatment, the treatment area of the retina became non-viable retinal scar tissue that could not function and effectively had the same results as the thermal damage seen by the standard laser treatment (see Peyman et al., ref. 1). Reducing the amount of the energy by 50% reduces the damage; however, it did not eliminate the side effects of scar formation when the treatment is repeated in the subsequent sessions.

PDT has been shown to induce structural changes in the RPE and choriocapillaris, abnormal vessels depending on the light intensity, duration of exposure time and concentration of the photosensitizer, in addition the time between dye administration of the photosensitizer or riboflavin and the laser therapy. A waiting time of more than 10 minutes, from its intravenous injection, or topical administration reduces the circulating photosensitizer in the blood, which is eliminated through the kidney and liver, thus, reduces the effect of PDT, since the photosensitizer is removed rapidly from the body. Other factors influencing the results of the

PDT are media opacity reducing the amount of light reaching the retina. Increased intraocular pressure causes reduction of the dye in the choriocapillaries, location of treatment within the fundus, such as central areas respond more than the peripheral area, equivalent fluence, lower energy and longer duration causes increases in the size of the lesion, and increases in fundus pigmentation reduces the effect since the light is absorbed by the retinal pigment epithelium and reduces the light reaching the choriocapillaries.

Therefore, what is also needed is a photodynamic therapy technique for preventing damage to the fovea of the eye and other normal tissue of the eye, or for preventing damage to another body portion of a patient being irradiated during photodynamic therapy thereof.

Further, in the past, corneal transplantation and partial lamellar transplantation has been performed to remove the diseased cornea of a person or animal and replace it with a donor cornea of another animal or person as a homologous or allogeneic graft. Among the indications for these procedures include corneal ulcers caused by infection, cornea scarring, thinned progressed keratoconus, corneal swelling in Fuchs' dystrophy, etc. Reported complications of the cornea after transplantation are rejection of the donor cornea, neovascularization of the graft with subsequent perforation of the corneal tissue, increased intraocular pressure, astigmatism produced by the need for fixing the corneal graft with suture, or infected suture and scarring, endothelial cell damage and subsequent graft failure requiring repeat grafting with an increased chance of immune response to the corneal graft.

The full corneal transplantation requires a functioning corneal endothelium to maintain the corneal transparency. The lamellar corneas are similarly used as a corneal inlay to augment or replace a thin or opaque part of the cornea.

In all these cases, the full thickness cornea or the corneal stroma is obtained from the eye bank (i.e., a removed cornea from a deceased patient). The main problem with corneal transplantation has been that the graft is rejected by the host immune response. The cornea can also be crosslinked to prevent the corneal rejection by the host (e.g., see U.S. Pat. No. 9,427,355).

Unfortunately there not enough corneas obtained from deceased persons to fulfill the need of the people with the corneal injury. In addition, one cannot eliminate, at present, certain viral pathogens and prions that can cause serious central nervous system (CNS) disease.

Specifically, in the developing countries, there is a shortage of corneas to treat everyone needing them. At times, religious beliefs or superstitions have been the reasons for not accepting the tissue from a dead human. In addition, there is an additional need for corneal tissue for some of the corneal procedures such as lamellar anterior or posterior replacement of the cornea after cataract surgery when the corneal endothelial cells are damaged, etc.

The present refractive surgical procedures do not require lamellar transplantation. These procedures are based on removing or ablating a part of the cornea with an excimer laser such as LASIK and photorefractive keratectomy (PRK). Briefly, these procedures are laser in-situ keratomileusis (LASIK) (see e.g., U.S. Pat. No. 4,840,175), where the refractive errors of the eye are corrected by creating a corneal flap using a femtosecond laser or a microkeratome, ablating the corneal stroma with an excimer laser using a wavefront technology and repositioning the flap after the surgery, and photorefractive keratectomy (PRK) where the surface of the cornea is ablated with the excimer laser. The newly approved small incision lenticule

extraction (SMILE) procedure utilizes a femtosecond laser to cut away a part of the cornea with a femtosecond laser and scraping it out the corneal stroma which causes roughness in the surrounding stromal tissue. All these procedures have the limitation of not correcting higher myopic refractive errors or high hyperopia or hyperopia. In another refractive procedure (Intacs), one implants an acrylic ring in the corneal periphery which expands the tissue and flattens the central cornea in myopia, but does not have the precision of correction of the refractive errors of the eye compared with LASIK, SMILE or PRK procedures.

Therefore, there is also a need to expand the refractive surgery either with the use of a homologous lamellar transplant (allograft or homograft) or an autologous lamellar transplant in which the refractive correction is done on the lamellar transplant which is then inserted in a pocket inside the cornea.

BRIEF SUMMARY OF EMBODIMENTS OF THE INVENTION

Accordingly, the present invention is directed to a method of lamellar corneal graft implantation that substantially obviates one or more problems resulting from the limitations and deficiencies of the related art.

In accordance with one or more embodiments of the present invention, there is provided a method of lamellar corneal graft implantation. The method comprises the steps of: (i) forming one or more intrastromal incisions in a cornea of an eye of a patient; (ii) removing a cut portion of the cornea defined by the one or more intrastromal incisions from the eye of the patient; (iii) applying laser energy to a lamellar corneal graft using an excimer laser so as to modify the refractive power of the lamellar corneal graft, the lamellar corneal graft being in the form of an autologous corneal graft or a homologous corneal graft; and (iv) implanting the lamellar corneal graft into the eye of the patient in a location previously occupied by the cut portion of the cornea so as to correct the refractive power of the eye of the patient.

In a further embodiment of the present invention, the step of forming one or more intrastromal incisions in the cornea of the eye of the patient further comprises: (a) forming two generally parallel cuts in the cornea of the eye using a femtosecond laser; and (b) forming two generally parallel corneal access incisions in the cornea outwardly from the two generally parallel cuts.

In yet a further embodiment, the lamellar corneal graft is in the form of an autologous corneal graft, and wherein the cut portion of the cornea that is removed from the eye of the patient forms the autologous corneal graft.

In still a further embodiment, the lamellar corneal graft is in the form of an homologous corneal graft, and wherein a donor corneal portion that is removed from an eye bank cornea forms the homologous corneal graft.

In yet a further embodiment, prior to implantation of the homologous corneal graft or after implantation of the homologous corneal graft, the method further comprises the steps of: (v) treating the homologous corneal graft with a solution of riboflavin or benzalkonium chloride (BAK) for making the homologous corneal graft acellular, and for killing any bacteria, viruses, or parasites prior to implantation; and (vi) irradiating the homologous corneal graft so as to cross-link the homologous corneal graft, thereby killing keratocytes and decellularizing the homologous corneal graft and preventing graft rejection without affecting the transparency of the homologous corneal graft.

In still a further embodiment, the lamellar corneal graft is circular in shape such that the lamellar corneal graft is capable of being rotated about an astigmatic axis of the eye in order to permit adjustment after the lamellar corneal graft has been implanted into the eye.

In yet a further embodiment, the lamellar corneal graft is rectangular in shape such that the lamellar corneal graft is not able to be rotated about an astigmatic axis of the eye, thus preventing movement of the lamellar corneal graft after the lamellar corneal graft has been implanted into the eye.

In still a further embodiment, the rectangular lamellar corneal graft comprises a central refractive section surrounded by a peripheral non-refractive section.

In yet a further embodiment, the method further comprises the step of: (v) disinfecting the eye of the patient using a solution of povidone iodine prior to surgery to prevent infection.

In still a further embodiment, the step of forming the one or more intrastromal incisions in the cornea of the eye of the patient further comprises cutting out a three-dimensional portion of stromal tissue from the cornea of the eye using a femtosecond laser; and the step of removing the cut portion of the cornea further comprises removing the three-dimensional cut portion of the cornea using forceps so as to create a three-dimensional pocket for receiving the lamellar corneal graft.

In yet a further embodiment, the method further comprises the step of: (v) treating the cornea of the patient with one or more topical anti-inflammatory agents in combination with one or more slow release nanoparticle polymers so as to release the one or more anti-inflammatory agents in the cornea for a long time after implantation of the lamellar corneal graft.

In still a further embodiment, the one or more topical anti-inflammatory agents are selected from the group consisting of steroids, NSAIDs, ROCK inhibitors, Wnt inhibitors, integrin inhibitors, GSK inhibitors, and combinations thereof, and wherein the slow release nanoparticle polymers are selected from the group consisting of polylactic or polyglycolic acid, micelles, liposomes, porous silicon, polyester, and combinations thereof.

In yet a further embodiment, the lamellar corneal graft implantation is performed on a human patient, and the lamellar corneal graft implanted into the human patient is formed from a human cornea.

In still a further embodiment, the lamellar corneal graft implantation is performed on an animal, and the lamellar corneal graft implanted into the animal is formed from an animal cornea.

In yet a further embodiment, the method further comprises the step of: (v) measuring the refractive error of the animal cornea with a handheld phoropter.

In still a further embodiment, the lamellar corneal graft implantation is performed on an animal, and the lamellar corneal graft implanted into the animal is formed from a human cornea.

In yet a further embodiment, the lamellar corneal graft implantation is performed on a human patient, and the lamellar corneal graft implanted into the human patient is formed from an animal cornea.

In still a further embodiment, the lamellar corneal graft implantation is performed on a human patient, the lamellar corneal graft is decellularized and crosslinked, and the lamellar corneal graft implanted into the human patient is formed from an animal cornea.

In yet a further embodiment, the lamellar corneal graft implantation is performed on an animal, the lamellar corneal

graft is decellularized and crosslinked, and the lamellar corneal graft implanted into the animal is formed from a human cornea.

In still a further embodiment, the lamellar corneal graft is decellularized using chemical means, the chemical means for destroying the cellular elements in the lamellar corneal graft are selected from the group consisting of ethanol, glycerol, acids, alkalis, peracetic acid, ammonium hydroxide ionic detergents, sodium dodecyl sulfate, sodium deoxycholate non-ionic detergents, zwitterionic detergents, Triton X-100, benzalkonium chloride, Igepal, genipin, and combinations thereof.

It is to be understood that the foregoing general description and the following detailed description of the present invention are merely exemplary and explanatory in nature. As such, the foregoing general description and the following detailed description of the invention should not be construed to limit the scope of the appended claims in any sense.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

The invention will now be described, by way of example, with reference to the accompanying drawings, in which:

FIG. 1A is a partial side cross-sectional view of an eye having a scarred cornea, wherein substantially the entire thickness of the cornea is scarred;

FIG. 1B is a partial side cross-sectional view of a donor cornea undergoing cross-linking;

FIG. 1C is a partial side cross-sectional view of the eye of FIG. 1A, wherein the scarred cornea is shown being removed;

FIG. 1D is a partial side cross-sectional view of the eye of FIG. 1A, wherein the cross-linked donor cornea is shown being implanted in the location previously occupied by the scarred cornea;

FIG. 2A is a partial side cross-sectional view of an eye having internal corneal scar tissue;

FIG. 2B is a partial side cross-sectional view of the eye of FIG. 2A, wherein the scarred corneal tissue has been externally removed from the eye;

FIG. 2C is a partial side cross-sectional view of the eye of FIG. 2A, wherein a cross-linked donor cornea is shown being implanted in the location previously occupied by the scarred corneal tissue;

FIG. 3A is a partial side cross-sectional view of an eye having internal corneal scar tissue;

FIG. 3B is a partial side cross-sectional view of the eye of FIG. 3A, wherein the scarred corneal tissue is shown being internally removed from the eye;

FIG. 3C is a partial side cross-sectional view of the eye of FIG. 3A, wherein a cross-linked donor cornea is shown being implanted in the location previously occupied by the scarred corneal tissue;

FIG. 4A is a partial side cross-sectional view of an eye having a T-shaped corneal scar and/or diseased tissue portion;

FIG. 4B is another partial side cross-sectional view of a donor cornea undergoing cross-linking;

FIG. 4C is a partial side cross-sectional view illustrating a T-shaped portion of the cross-linked donor cornea being cut out from a remainder of the donor cornea;

FIG. 4D is a partial side cross-sectional view of the eye of FIG. 4A, wherein the T-shaped scarred and/or diseased portion of corneal tissue has been removed from the eye;

FIG. 4E is a partial side cross-sectional view of the eye of FIG. 4A, wherein the cross-linked T-shaped donor cornea

portion is shown being implanted in the location previously occupied by the scarred and/or diseased corneal tissue portion;

FIG. 5A illustrates an alternative configuration for the cross-linked donor cornea implant, wherein the donor cornea implant has a dumbbell shape;

FIG. 5B illustrates another alternative configuration for the cross-linked donor cornea implant, wherein the donor cornea implant has a reversed or upside down T-shape;

FIG. 6A is a side cross-sectional view of a host eye prior to an transplant procedure;

FIG. 6B is another side cross-sectional view of the host eye of FIG. 6A, which illustrates a creation of a corneal pocket therein;

FIG. 6C is another side cross-sectional view of the host eye of FIG. 6A, which illustrates an implantation of the cross-linked lamellar lenslet into the host eye;

FIG. 7A is a partial side cross-sectional view of a donor cornea being cross-linked prior to being shaped for use in a transplant procedure;

FIG. 7B is another partial side cross-sectional view of the donor cornea of FIG. 7A, which illustrates the cutting of a cross-linked lamellar lenslet from a remainder of the cross-linked donor cornea;

FIG. 7C is a side cross-sectional view of the cross-linked lamellar lenslet after it has been appropriately shaped and removed from the donor cornea of FIGS. 7A and 7B;

FIG. 8 is a partial side cross-sectional view illustrating the formation of a two-dimensional cut into a cornea of an eye, according to another embodiment of the invention;

FIG. 9 is another partial side cross-sectional view of the eye of FIG. 8, which illustrates the creation of a three-dimensional pocket in the cornea of the eye;

FIG. 10 is yet another partial side cross-sectional view of the eye of FIG. 8, which illustrates the injection of a photosensitizer into the three-dimensional pocket in the cornea of the eye;

FIG. 11A is still another partial side cross-sectional view of the eye of FIG. 8, which illustrates the irradiation of the stromal tissue surrounding the three-dimensional pocket of the eye using ultraviolet radiation delivered from outside of the cornea;

FIG. 11B is yet another partial side cross-sectional view of the eye of FIG. 8, which illustrates the irradiation of the stromal tissue surrounding the three-dimensional pocket of the eye using a fiber optic delivering ultraviolet radiation inside the three-dimensional pocket, according to an alternative embodiment of the invention;

FIG. 12 is still another partial side cross-sectional view of the eye of FIG. 8, which illustrates a lens implant inserted into the pocket so as to change the refractive properties of the eye;

FIG. 13 is yet another partial side cross-sectional view of the eye of FIG. 8, which illustrates the reinjection of a photosensitizer into the three-dimensional pocket with the lens implant disposed therein so that the cross-linking procedure may be repeated;

FIG. 14 is still another partial side cross-sectional view of the eye of FIG. 8, which illustrates the re-irradiation of the stromal tissue surrounding the three-dimensional pocket of the eye during the repetition of the cross-linking procedure;

FIG. 15 is a side cross-sectional view illustrating the creation of a lens implant from an organic block of polymer using an excimer laser;

FIG. 16 is a side cross-sectional view illustrating the cutting of a lens implant from an organic block of polymer using a femtosecond laser;

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FIG. 17 is a side cross-sectional view illustrating a lens implant that has been formed using a three-dimensional printing technique or a molding technique;

FIG. 18 is a front view of a cornea of an eye, according to yet another embodiment of the invention;

FIG. 19 is another front view of the cornea of the eye of FIG. 18, wherein a square-shaped intrastromal pocket has been formed in the cornea of the eye;

FIG. 20 is yet another front view of the cornea of the eye of FIG. 18, wherein a circular three-dimensional portion of tissue having a first diameter has been removed from the area within the square-shaped intrastromal pocket;

FIG. 21 is still another front view of the cornea of the eye of FIG. 18, wherein a circular three-dimensional portion of tissue having second diameter has been removed from the area within the square-shaped intrastromal pocket, the second diameter of the circular three-dimensional portion of tissue in FIG. 21 being larger than the first diameter of the circular three-dimensional portion of tissue in FIG. 20;

FIG. 22 is yet another front view of the cornea of the eye of FIG. 18, wherein a circular lens implant has been implanted in the area where the circular three-dimensional portion of tissue has been removed, and wherein a photosensitizer is being injected into the pocket in the cornea of the eye;

FIG. 23 is still another front view of the cornea of the eye of FIG. 18, wherein the circular lens implant is shown in the area where the circular three-dimensional portion of tissue was removed;

FIG. 24 is a front view of an eye where a corneal graft has been implanted in the cornea of the eye, according to still another embodiment of the invention;

FIG. 25 is another front view of the eye of FIG. 24, wherein the application of ultraviolet radiation is being illustrated following the administration of a cross-linking agent or photosensitizer to the cornea of the eye;

FIG. 26 is a front view of an eye where a corneal graft has been implanted in the cornea of the eye and neovascularization has occurred, according to yet another embodiment of the invention;

FIG. 27 is another front view of the eye of FIG. 26, wherein the application of ultraviolet radiation is being illustrated following the administration of a cross-linking agent or photosensitizer to the cornea of the eye;

FIG. 28 is a front view of an eye where a keratoprosthesis lens has been implanted in the cornea of the eye and the cornea of the eye is being irradiated with ultraviolet radiation following the implantation, according to still another embodiment of the invention;

FIG. 29A illustrates a first exemplary shape for the drug delivery implant described herein, which is in the form of a rod-shaped implant;

FIG. 29B illustrates a second exemplary shape for the drug delivery implant described herein, which is in the form of a curved implant;

FIG. 29C illustrates a third exemplary shape for the drug delivery implant described herein, which is in the form of a two-part semi-circular implant;

FIG. 29D illustrates a fourth exemplary shape for the drug delivery implant described herein, which is in the form of a one-part semi-circular implant;

FIG. 30 illustrates an exemplary coated drug delivery implant, wherein the drug delivery implant is coated with a polymer and a photosensitizer;

FIG. 31A illustrates a first exemplary form of the drug delivery implant described herein, which is in the form of a solid tubular implant;

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FIG. 31B illustrates a second exemplary form of the drug delivery implant described herein, which is in the form of a porous tubular implant;

FIG. 31C illustrates a third exemplary form of the drug delivery implant described herein, which is in the form of a tubular implant with open ends;

FIG. 32 illustrates another exemplary form of the drug delivery implant described herein, wherein the implant is tubular-shaped with holes formed in the side thereof;

FIG. 33 illustrates yet another exemplary form of the drug delivery implant that is similar to that which is depicted in FIG. 32, except that the tubular-shaped implant of FIG. 33 has larger-sized holes formed in the side thereof;

FIG. 34 illustrates still another exemplary form of the drug delivery implant described herein, wherein the implant is in the form of a rectangular flat tube;

FIG. 35 illustrates yet another exemplary form of the drug delivery implant described herein, wherein the implant is in the form of a semi-solid or silicone tubular implant with one closed end and one open end;

FIG. 36 illustrates still another exemplary form of the drug delivery implant described herein, wherein the implant is in the form of a rectangular tube that is refillable by injection;

FIG. 37 illustrates yet another exemplary form of the drug delivery implant described herein, wherein the tubular implant comprises a needle for tissue penetration and the tubular implant is capable of being penetrating by a needle for taking liquid biopsies;

FIG. 38A is a front view of a cornea of an eye illustrating a two-part semi-circular drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea that is spaced apart from the central visual axis of the eye so as not to obstruct the central portion of the eye;

FIG. 38B is a partial side cross-sectional view of the eye of FIG. 38A illustrating the two-part semi-circular drug delivery implant disposed in the cross-linked pocket in the peripheral portion of the cornea;

FIG. 39A is a front view of a cornea of an eye illustrating a generally linear drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea that is spaced apart from the central visual axis of the eye so as not to obstruct the central portion of the eye;

FIG. 39B is a partial side cross-sectional view of the eye of FIG. 39A illustrating the generally linear drug delivery implant disposed in the cross-linked pocket in the peripheral portion of the cornea;

FIG. 40A is a front view of a cornea of an eye illustrating a tubular drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea, wherein the implant comprises a needle fluidly coupling the implant to the anterior chamber of the eye;

FIG. 40B is a partial side cross-sectional view of the eye of FIG. 40A illustrating the tubular drug delivery implant with the needle extending into the anterior chamber of the eye;

FIG. 41A is a front view of a cornea of an eye illustrating a pupil, cornea, sclera, and limbus of the eye;

FIG. 41B is a partial side cross-sectional view of the eye of FIG. 41A illustrating an anterior chamber, iris, and lens of the eye;

FIG. 42A is a front view of a cornea of an eye illustrating a one-part semi-circular drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea;

FIG. 42B is a partial side cross-sectional view of the eye of FIG. 42A illustrating the one-part semi-circular drug

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delivery implant disposed in the cross-linked pocket in the peripheral portion of the cornea;

FIG. 43A is a front view of a cornea of an eye illustrating a doughnut-shaped drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea;

FIG. 43B is a partial side cross-sectional view of the eye of FIG. 43A illustrating the doughnut-shaped drug delivery implant disposed in the cross-linked pocket in the peripheral portion of the cornea;

FIG. 44A is a front view of a cornea of an eye illustrating a generally linear drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea;

FIG. 44B is a partial side cross-sectional view of the eye of FIG. 44A illustrating the generally linear drug delivery implant disposed in the cross-linked pocket in the peripheral portion of the cornea;

FIG. 45A is a front view of a cornea of an eye illustrating a drug delivery implant disposed in a cross-linked pocket in the peripheral portion of the cornea, wherein the implant comprises a needle fluidly coupling the implant to the anterior chamber of the eye;

FIG. 45B is a partial side cross-sectional view of the eye of FIG. 45A illustrating the tubular drug delivery implant with the needle extending into the anterior chamber of the eye with the aqueous fluid of the eye;

FIG. 46 is a cutaway perspective view of an eye with a drug delivery implant disposed in a pocket formed in the corneal stroma of the eye;

FIG. 47 is a cross-sectional view of an eye illustrating the formation of a corneal LASIK flap in the eye;

FIG. 48 is a cutaway perspective view of the eye of FIG. 47, wherein a drug delivery implant has been placed under the corneal LASIK flap;

FIG. 49 is a cutaway perspective view of an eye illustrating the application of a medication to the cornea of the eye by drops;

FIG. 50 is a front view of a cornea of an eye illustrating a polymeric contact lens containing a medication disposed on the cornea of the eye after the eye has undergone refractive surgery or a traumatic corneal injury;

FIG. 51 is a cross-sectional view of an eye illustrating an intracapsular drug delivery implant after the eye has undergone cataract surgery;

FIG. 52 is a cross-sectional view of an eye illustrating a drug delivery implant that has been placed on the lens capsule of the eye after the eye has undergone cataract surgery;

FIG. 53 is a cross-sectional view of an eye illustrating intravitreal administration of a medication by injecting the medication along with microparticles or nanoparticles;

FIG. 54A is a front view of a corneal inlay or implant;

FIG. 54B is a side view of the corneal inlay or implant of FIG. 54A illustrating the formation of small holes in the inlay or implant made using a laser;

FIG. 55 is a side view of the corneal inlay or implant of FIG. 54A illustrating the cross-linking of the inlay or implant using ultraviolet radiation after the inlay or implant has been soaked in a photosensitizer solution;

FIG. 56 is a cutaway perspective view of an eye before a corneal flap has been formed in the eye;

FIG. 57 is another cutaway perspective view of the eye of FIG. 56 illustrating the formation of a corneal flap in the eye using a laser;

FIG. 58 is yet another cutaway perspective view of the eye of FIG. 56 illustrating the insertion of a corneal inlay under the flap of the eye, wherein corneal inlay is undergoing refractive correction;

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FIG. 59 is still another cutaway perspective view of the eye of FIG. 56 illustrating the eye after the corneal flap has been replaced;

FIG. 60 is a partial side cross-sectional view of the eye of FIG. 56 illustrating the corneal inlay surrounded by the stromal tissue of the cornea;

FIG. 61A is a front view of a composite two-piece inlay or implant with a central section and a peripheral section;

FIG. 61B is a front view of another composite two-piece inlay or implant that is divided transversely into two sections;

FIG. 61C is a front view of a composite three-piece inlay or implant that is divided transversely into three sections;

FIG. 62 is a cutaway perspective view of an eye illustrating the cornea of the eye after the two-piece inlay or implant of FIG. 61A has been inserted into a pocket in the cornea through a small incision;

FIG. 63 is a cutaway perspective view of an eye illustrating the cornea of the eye after the two-piece inlay or implant of FIG. 61B has been inserted into a pocket in the cornea through a small incision;

FIG. 64 is a partial side cross-sectional view of the eye of FIGS. 62 and 63 illustrating the cavity of the corneal pocket that is surrounded by stromal tissue;

FIG. 65 is a partial side cross-sectional view of the eye of FIGS. 62 and 63 illustrating an inlay or implant that is disposed in the corneal pocket;

FIG. 66 is a front view of a cross-linked corneal implant or inlay;

FIG. 67 is a front view of a polymeric back sheet used for supporting the corneal implant or inlay of FIG. 66;

FIG. 68 is a front view illustrating the corneal implant or inlay of FIG. 66 disposed on the polymeric back sheet of FIG. 67;

FIG. 69A is a side view of a polymeric back sheet used for supporting a corneal implant or inlay that has a flat surface;

FIG. 69B is a side view of a polymeric back sheet used for supporting a corneal implant or inlay that has a convex surface;

FIG. 69C is a side view of a polymeric back sheet used for supporting a corneal implant or inlay that has a concave surface;

FIG. 70 is a front view illustrating donor corneal stromal tissue, according to one embodiment of the invention;

FIG. 71 is a cutaway perspective view of an eye before a corneal flap has been formed in the eye;

FIG. 72 is another cutaway perspective view of the eye of FIG. 71 illustrating the formation of a corneal flap in the eye;

FIG. 73 is yet another cutaway perspective view of the eye of FIG. 71 illustrating the insertion of a corneal inlay under the flap of the eye, wherein corneal inlay is undergoing refractive correction;

FIG. 74 is still another cutaway perspective view of the eye of FIG. 71 illustrating the application of a photosensitizer solution to the corneal inlay by drops;

FIG. 75 is yet another cutaway perspective view of the eye of FIG. 71 illustrating the cross-linking of the corneal inlay using ultraviolet radiation after the photo sensitizer solution has been applied to the corneal inlay;

FIG. 76 is a front view of an eye illustrating an intrastromal cavity that has been created in the eye using a laser, according to another embodiment of the invention;

FIG. 77 is a partial side cross-sectional view of the eye of FIG. 76 illustrating the intrastromal cavity;

FIG. 78 is another partial side cross-sectional view of the eye of FIG. 76 illustrating a shaped donor cornea that is disposed in the intrastromal cavity;

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FIG. 79 is yet another partial side cross-sectional view of the eye of FIG. 76 illustrating the injection of a photosensitizer into the shaped donor cornea inside the intrastromal cavity;

FIG. 80 is still another partial side cross-sectional view of the eye of FIG. 76 illustrating the cross-linking of the shaped donor cornea using externally applied ultraviolet radiation;

FIG. 81 is yet another partial side cross-sectional view of the eye of FIG. 76 illustrating the cross-linking of the shaped donor cornea using internally applied ultraviolet radiation delivered using a fiber optic;

FIG. 82 is a front view of the normal retina with its retinal vessels, optic nerve head, and the circular area is the macular area, the center of the macula is the fovea;

FIG. 83 is a front view of a retinal area with subretina neovascular tissue seen in age-related macular degeneration (ARMD) demonstrated as a kidney-shaped lesion area behind the retina adjacent to the fovea, where a standard PDT treatment with a circular laser spot size with the described parameters is being applied to this area, which unfortunately can damage the fovea simultaneously;

FIG. 84 is a front view illustrating the kidney-shaped lesion being treated with a paint brush technique using a small laser spot which moves over the lesion and its border while sparing the fovea from laser damage (the treatment with the laser can be repeated up to 7 times depending on the thickness of the lesion without damaging either the fovea or even the retina overlying the lesion);

FIG. 85 is an enlarged front view of the kidney-shaped lesion area in FIG. 84 that is being treated with the paint brush technique using the small laser spot (Detail "A");

FIG. 86A depicts an optical coherence tomography (OCT) image of the central retina showing an elevated sensory retina by sub-retinal fluid; the retinal area shows numerous lacunae filled with fluid while the structure of the sensory retina is being pulled and separated by the pressure of the fluid; and the underlying structure shows scarred retinal pigment epithelium (RPE), and beneath it are varicose-like abnormal choroidal vessels which leak fluid;

FIG. 86B depicts an optical coherence tomography (OCT) image after PPDt treatment demonstrating disappearance of most of the fluid and re-establishment of a somewhat normal relationship of the sensory retina with its underlying congested choroid;

FIG. 87A depicts an optical coherence tomography (OCT) image of the retina and fovea overlying a separated fluid area (i.e., a dark area) where the fluid caused the separation of the sensory retina from the choroid in central serous retinopathy that prevents proper nutrition of the retinal cells in that area with reduced vision;

FIG. 87B depicts an optical coherence tomography (OCT) image after PPDt treatment of the area, but avoiding the central or foveal area directly (the retina is now reattached after therapy and the normal anatomical connection between the retina and the choroid is re-established which leads to improvement of the vision in the patient);

FIG. 88A is a partial side cross-sectional view of a cornea of an eye illustrating an intrastromal cut being formed in the cornea;

FIG. 88B is another partial side cross-sectional view of the cornea of the eye illustrating the cut portion of the cornea stroma that is to be removed;

FIG. 88C is a side cross-sectional view of the removed cut portion of the cornea stroma;

FIG. 88D is a side cross-sectional view of the removed cut portion of the cornea stroma being ablated by an excimer laser;

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FIG. 88E is yet another partial side cross-sectional view of the cornea of the eye illustrating the re-implantation of the modified cut portion of the cornea stroma;

FIG. 89A is a top view of the cornea of the eye in FIG. 88A illustrating the intrastromal cut being formed in the cornea;

FIG. 89B is a top view of the cornea of the eye in FIG. 88B illustrating the removal of the cut portion of the cornea stroma;

FIG. 89C is a top view of the removed cut portion of the cornea stroma of FIGS. 88C and 88D being ablated by an excimer laser;

FIG. 89D is a top view of the cornea of the eye in FIG. 88E illustrating the rotation of the removed cut portion of the cornea stroma;

FIG. 89E is a top view of the cornea of the eye in FIG. 88E illustrating the modified cut portion of the cornea stroma after it has been re-implanted and positioned in the cornea of the eye;

FIG. 89F is a top view of a cornea of an eye illustrating a circular implant with a rotatable astigmatic axis;

FIG. 89G is a top view of a cornea of an eye illustrating a circular implant with a movable graft and astigmatic axis;

FIG. 89H is a top view of a cornea of an eye illustrating a rectangular implant with a non-movable graft and axis;

FIG. 89I is another top view of the cornea of the eye illustrating the rectangular implant with the non-movable graft and astigmatic axis;

FIG. 90A is a top view of a cornea of an eye illustrating two parallel cuts that are formed using a femtosecond laser;

FIG. 90B is another top view of the cornea of the eye of FIG. 90A illustrating two parallel corneal access incisions outward from the two parallel cuts;

FIG. 90C is a top view of the rectangular removed cut portion of the cornea stroma being ablated by an excimer laser;

FIG. 90D is yet another top view of the cornea of the eye of FIG. 90A illustrating the re-implantation of the modified rectangular removed cut portion of the cornea stroma; and

FIG. 90E is yet another top view of the cornea of the eye of FIG. 90A illustrating the modified intrastromal area of the implant.

Throughout the figures, the same elements are always denoted using the same reference characters so that, as a general rule, they will only be described once.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

A first illustrative embodiment of a corneal transplant procedure with a cross-linked cornea is shown in FIGS. 1A-1D. The corneal transplant procedure illustrated in FIGS. 1A-1D involves full corneal replacement of the scarred or diseased cornea by the donor cornea. In other words, FIGS. 1A-1D illustrate a penetrating keratoplasty procedure wherein the full thickness of the scarred or diseased cornea is replaced with a cross-linked donor cornea (i.e., a full-thickness corneal transplant).

Referring initially to FIG. 1A, it can be seen that substantially the entire thickness of the cornea 16 of the eye 10 is scarred and/or diseased (i.e., scarred, diseased, or scarred and diseased). FIG. 1A also illustrates the lens 12 and iris 14 of the eye 10, which are located posteriorly of the cornea 16. In this embodiment, it is necessary to replace substantially the entire thickness of the cornea 16 with a donor cornea.

In FIG. 1B, the cross-linking 18 of the clear donor cornea 20 is diagrammatically illustrated. As depicted in FIG. 1B,

only the front portion **20a** of the donor cornea **20** is cross-linked. That is, the cross-linking does not extend all the way to the rear portion **20b** of the donor cornea **20**. It is to be understood that the cross-linking **18** of the donor cornea **20** may also be done after implanting the donor cornea into the eye of the patient, rather than before implantation as shown in the illustrative example of FIGS. 1A-1D. Also, it is to be understood that all or just a part of the donor cornea **20** may be cross-linked.

In the illustrative embodiments described herein (i.e., as depicted in FIGS. 1A-1D, 2A-2C, and 3A-3C), the cross-linking of the clear donor cornea may comprise the steps of: (i) applying a photosensitizer to the donor cornea, the photosensitizer facilitating cross-linking of the donor cornea; and (ii) irradiating the donor cornea with ultraviolet light so as to activate cross-linkers in the donor cornea and thereby strengthen the donor cornea. The photosensitizer may comprise riboflavin or a solution comprising a liquid suspension having nanoparticles of riboflavin. The cross-linker may have between about 0.1% Riboflavin to about 100% Riboflavin or any other suitable range or specific percentage therein. The ultraviolet radiation or rays used to irradiate the donor cornea may be between about 370 nanometers and about 380 nanometers (or between 370 nanometers and 380 nanometers). The radiation is preferably about 3 mW or more as needed and emanates from a laser source at about a 3 cm distance from the donor cornea for about 30 minutes or less. The time of the exposure can vary depending on the light intensity, focus, and the concentration of riboflavin. However, the ultraviolet radiation can be applied at any suitable distance, time or wavelength. Preferably, cross-linking the donor cornea does not significantly change the refractive power of the donor cornea; however, if desired, cross-linking can change the refractive power of the donor cornea to any suitable degree.

In addition to Riboflavin, other suitable cross linking agents are low carbon carbohydrates, such as pentose sugar (e.g., ribose) or hexose sugar (e.g., glucose), or complex carbohydrates. Other crosslinking agents may include Transaminidases, transglutaminases or a naturally-derived cross-linker named malic acid derivative (MAD) concentrations higher than 30 mM, commercially available cross-linkers such as 1-ethyl-3-(3'-dimethylaminopropyl) carbodiimide (EDC), or ethyl-3(3-dimethylamino) propyl carbodiimide (EDC), etc. The cross-linking may also be done postoperatively by the application of other crosslinking agents, such as Triglycidylamine (TGA) synthesized via reacting epichlorhydrin and a carbodiimide, or the oxidized glycogen hexoses. The ribose, glucose and similar agents may penetrate the cornea easily using drops, gel, or the slow release mechanisms, nanoparticle, microspares, liposome sets. In addition, the crosslinkers may be delivered with Mucoadhesives.

In one or more embodiments, all or part of the donor cornea is cross-linked. Also, in one or more embodiments, a very high concentration of Riboflavin may be used because the in vitro cross-linking process may be stopped whenever needed prior to the transplantation of the donor cornea in the host eye. In addition, the power of the ultraviolet (UV) laser may also be increased so as to cross-link the tissue of the donor cornea faster. The use of a high concentration of Riboflavin, and the increasing of the ultraviolet (UV) laser power, are not possible during an in vivo cross-linking procedure because the aim of such an in vivo procedure is to protect the cells of the host cornea. Also, the in vivo process cannot be controlled as efficiently as in the vitro crosslinking of the corneal transplant.

In one or more embodiments, the donor cornea may be extracted from a human cadaver, or the cornea may be reconstructed as known in tissue engineering in vitro and three-dimensionally (3D) printed. Cross-linking of a culture-grown cornea eliminates the cellular structure inside the cornea. If needed again, the healthy corneal endothelium of the patient may be grown in vitro for these tissues by placing them on the concave surface of the cornea and encouraging their growth under laboratory control conditions prior to the transplantation.

In the embodiments where the donor cornea is tissue culture grown, the cornea may be formed from mesenchymal fibroblast stem cells, embryonic stem cells, or cells derived from epithelial stem cells extracted from the same patient, or a mixture of these cells. Using known tissue culture techniques, the cells may produce a transparent corneal stroma. This culture-grown corneal stroma will not have a corneal epithelium or a corneal endothelium. Thus, it eliminates the complexity of developing a full thickness cornea in the tissue culture. This stromal transplant may be used as a lamellar or partial thickness replacement of the existing host cornea. This transplant may also be used to augment or add to the thickness of the host cornea. This transparent corneal stroma may be transplanted either prior to, or after being cross-linked using various cross-linking methods.

In one or more embodiments, the cross-linked donor cornea may be sized and precisely cut with a femtosecond laser to the desired shape and curvature to replace the removed host cornea so that the refractive errors of the recipient are also automatically corrected with the cross-linked cornea.

Now, referring to FIG. 1C, it can be seen that the scarred and/or diseased cornea **16** is shown being removed from the eye **10**. The scarred and/or diseased cornea **16** may be removed from the eye **10** by using various suitable means, such as mechanical means or cutting using a laser. When mechanical means are used to remove the scarred and/or diseased cornea **16** from the eye **10**, the scarred and/or diseased cornea **16** may initially be cut away or dissected from the remainder of the eye **10** using a sharp mechanical instrument (e.g., a surgical micro-knife, a needle, a sharp spatula, a pair of micro-scissors), and then subsequently removed or extracted with a pair of micro-forceps. When laser cutting is used to remove the scarred and/or diseased cornea **16** from the eye **10**, the scarred and/or diseased cornea **16** may be cut away using a suitable laser, such as a femtosecond laser. Also, in some embodiments, the mechanical means for cutting and extraction (e.g., the surgical micro-knife and/or pair of micro-scissors) may be used in combination with the laser means (e.g., the femtosecond laser).

In one or more embodiments, the donor cornea may be shaped and cut with the femtosecond laser prior to the cross-linking thereof so as to replace part or all of the recipient cornea which is cut with the femtosecond laser. In these one or more embodiments, the entire donor and host cornea together may be cross-linked with Riboflavin and UV radiation. These procedures may also be performed on a culture-grown transplant cornea.

Then, as shown in FIG. 1D, after the scarred and/or diseased cornea **16** has been removed from the eye **10**, the cross-linked donor cornea **20** is implanted into the eye **10** of the patient in the location previously occupied by the scarred and/or diseased cornea **16**. After implantation of the cross-linked donor cornea **20**, sutures or a suitable adhesive may be utilized to secure the cross-linked donor cornea **20** in

place on the eye 10. When sutures are used for holding the donor cornea 20 in place, the sutures may comprise nylon sutures, steel sutures, or another suitable type of non-absorbable suture. When the cornea 16 is subsequently ablated after the implantation of the donor cornea, as will be described hereinafter, additional sutures may be required after ablation.

In one or more embodiments, a biodegradable adhesive is used in a corneal transplantation procedure with the cross-linked donor cornea 20 described above, or with a non-cross-linked corneal transplant. In these one or more embodiments, the biodegradable adhesive obviates the need for a suture in the corneal transplant procedure. Sutures generally distort the surface of the cornea and can produce an optically unacceptable corneal surface. Also, the use of the biodegradable adhesive obviates the need for glues requiring exothermic energy. Glues that use an exothermic effect, such as Fibronectin, need thermal energy to activate their adhesive properties. This thermal energy, such as that delivered by a high-powered laser, produces sufficient heat to coagulate the Fibronectin and the tissue that it contacts. Any thermal effect on the cornea produces: (i) corneal opacity, (ii) tissue contraction, and (iii) distortion of the optical surface of the cornea. The tissue adhesion created by these glues, including Fibronectin or fibrinogen, is flimsy and cannot withstand the intraocular pressure of the eye.

In fact, sutures are superior to these types of adhesives because the wound becomes immediately strong with sutures, thereby supporting the normal intraocular pressure of between 18 and 35 mmHg. In contrast to the use of a suture in which distortion that is caused by suture placement can be managed by cutting and removing the suture, the distortion caused by the coagulated corneal tissue cannot be corrected.

Other glues, such as cyanoacrylate, become immediately solid after coming into contact with the tissue or water. These glues produce a rock-hard polymer, the shape of which cannot be controlled after administration. Also, the surface of the polymer created by these glues is not smooth. Thus, the eyelid will rub on this uneven surface, and the uneven surface scratches the undersurface of the eyelid when the eyelid moves over it. In addition, the cyanoacrylate is not biodegradable or biocompatible. As such, it causes an inflammatory response if applied to the tissue, thereby causing undesirable cell migration and vascularization of the cornea.

Thus, by using a biocompatible and absorbable acrylate or other biodegradable glues that do not need exothermic energy for the process of adhesion (i.e., like fibronectin or fibrinogen), one is able to maintain the integrity of the smooth corneal surface. In one or more embodiments, the biocompatible and biodegradable adhesive may be painted only at the edges of the transplant prior to placing it in the host or diseased cornea. In these embodiments, the biocompatible and biodegradable adhesive only comes into contact with the host tissue at the desired predetermined surface to create a strong adhesion. The adhesion may last a few hours to several months depending on the composition of the molecule chosen and the concentration of the active component.

Other suitable biodegradable adhesives or glues that may be used in conjunction with the transplant include combinations of gallic acid, gallic tannic acid, Chitosan, gelatin, polyphenyl compound, Tannic Acid (N-isopropylacrylamide (PNIPAM), and/or Poly(N-vinylpyrrolidone) with polyethylene glycol (PEG). That is, polyethylene glycol (PEG) may be mixed with any one or plurality of gallic acid, gallic

tannic acid, Chitosan, gelatin, polyphenyl compound, Tannic Acid (N-isopropylacrylamide (PNIPAM), and Poly(N-vinylpyrrolidone), so as to form a molecular glue. These adhesives are suitable for the use on the cornea because they create a tight wound that prevents leakage from the corneal wound and maintain the normal intraocular pressure shortly after their application and also do not distort the wound by causing traction on the tissue.

In one or more embodiments, the donor cornea may be temporarily sutured to the host cornea by only a few single sutures to the host cornea. Then, the sutures may be removed immediately after donor cornea is fixed to the host cornea with a suitable adhesive.

A second illustrative embodiment of a corneal transplant procedure with a cross-linked cornea is shown in FIGS. 2A-2C. Unlike the first embodiment described above, the corneal transplant procedure illustrated in FIGS. 2A-2C does not involve full corneal replacement of the scarred or diseased cornea by the donor cornea. Rather, FIGS. 2A-2C illustrate a lamellar keratoplasty procedure wherein only a portion of the cornea 16' of the eye 10' contains scarred and/or diseased tissue (i.e., a full-thickness corneal section is not removed). In the procedure of FIGS. 2A-2C, an internal scarred and/or diseased portion 16a' of the cornea 16' is externally removed from the eye 10' of a patient.

Referring initially to FIG. 2A, it can be seen that only an internal portion 16a' of the cornea 16' is scarred and/or diseased. As such, in this embodiment, it is not necessary to replace the entire thickness of the cornea 16 with a donor cornea as was described above in conjunction with FIGS. 1A-1D, but rather just a portion of the cornea 16'.

Next, referring to FIG. 2B, it can be seen that the scarred and/or diseased portion 16a' has been externally removed from the cornea 16' of the eye 10' such that the cornea 16' comprises a cavity 19 disposed therein for receiving the donor cornea. Because an external approach was utilized for removing the scarred and/or diseased portion 16a' of the cornea 16', the cavity 19 comprises a notch-like void in the outside or anterior surface of the cornea 16'. As described above for the first embodiment, the scarred and/or diseased corneal portion 16a' may be removed from the remainder of the cornea 16' using various suitable means, such as mechanical means or the laser cutting means (e.g., femto-second laser) described above.

Finally, as shown in FIG. 2C, after the scarred and/or diseased portion 16a' has been removed from the remainder of the cornea 16' of the eye 10', the cross-linked donor cornea or cross-linked donor corneal portion 20' is implanted into the eye 10' of the patient in the location previously occupied by the scarred and/or diseased corneal portion 16a'. As described above, after implantation of the cross-linked donor corneal portion 20' into the eye 10', sutures or a suitable adhesive (e.g., the biocompatible and biodegradable adhesive described above) may be utilized to secure the cross-linked donor corneal portion 20' in place on the host cornea of the eye 10'.

After the cross-linked donor corneal portion 20' is implanted into the eye 10' of the patient, a portion of the cornea 16' may be ablated so as to change the refractive properties of the eye (e.g., to give the patient perfect or near perfect refraction). The ablation of the portion of the cornea 16' may be performed using a suitable laser 34, such as an excimer laser. The ablation by the laser causes the ablated tissue to essentially evaporate into the air. Also, the ablation of the portion of the cornea 16' may be done intrastromally, as with LASIK (laser-assisted in situ keratomileusis), or on the surface of the cornea, as with PRK (photorefractive

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keratectomy). The ablation may be performed a predetermined time period after the corneal transplantation so as to enable the wound healing process of the recipient's cornea to be completed. It is to be understood that the ablation, which follows the corneal transplantation, may be performed in conjunction with any of the embodiments described herein.

It is also to be understood that, in some alternative embodiments, the ablation may be performed prior to the transplantation of the donor cornea, rather than after the transplantation of the donor cornea. For example, in one or more alternative embodiments, a lenticle may be precisely cut in the tissue of a culture-grown stroma of a donor cornea by using a femtosecond laser so that when implanted into the host cornea, it corrects the residual host eye's refractive error.

A third illustrative embodiment of a corneal transplant procedure with a cross-linked cornea is shown in FIGS. 3A-3C. Like the second embodiment described above, the corneal transplant procedure illustrated in FIGS. 3A-3C only involves replacing a scarred and/or diseased portion 16a" of the cornea 16" with a donor corneal portion. Thus, similar to the second embodiment explained above, FIGS. 3A-3C illustrate a lamellar keratoplasty procedure wherein only a portion of the cornea 16" of the eye 10" contains scarred and/or diseased tissue (i.e., a full-thickness corneal section is not removed). Although, in the procedure of FIGS. 3A-3C, an internal scarred and/or diseased portion 16a" of the cornea 16" is internally removed from the eye 10" of a patient, rather than being externally removed as in the second embodiment of FIGS. 2A-2C.

Referring initially to FIG. 3A, it can be seen that only an internal portion 16a" of the cornea 16" of the eye 10" is scarred and/or diseased. As such, in this embodiment, like the preceding second embodiment, it is not necessary to replace the entire thickness of the cornea 16" with a donor cornea, but rather just a portion of the cornea 16".

Next, referring to FIG. 3B, it can be seen that the scarred and/or diseased portion 16a" is being internally removed from the remainder of the cornea 16" using a pair of forceps 22 (i.e., mechanical means of removal are illustrated in FIG. 3B). Advantageously, because an internal approach is being utilized for removing the scarred and/or diseased portion 16a" of the cornea 16", the cornea 16" will not comprise the notch-like cavity 19 disposed in the outside or anterior surface of the cornea, which was described in conjunction with the preceding second embodiment. As described above for the first and second embodiments, the scarred and/or diseased corneal portion 16a" may be removed from the remainder of the cornea 16" using other suitable alternative means, such as laser cutting techniques (e.g., using a femtosecond laser). Advantageously, the femtosecond laser is capable of cutting inside the tissue without involving the surface of the tissue. The cut part of the tissue can then be removed by other means (e.g., micro-forceps).

Finally, as shown in FIG. 3C, after the scarred and/or diseased corneal portion 16a" has been removed from the remainder of the cornea 16" of the eye 10", the cross-linked donor cornea or cross-linked donor corneal portion 20" is implanted into the eye 10" of the patient in the location previously occupied by the scarred and/or diseased corneal portion 16a". After implantation of the cross-linked donor corneal portion 20", sutures or a suitable adhesive (e.g., the biocompatible and biodegradable adhesive described above) may be utilized to secure the cross-linked donor corneal portion 20" in place on the host cornea of the eye 10". Advantageously, the cross-linked donor corneal portion 20",

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which is strengthened by the cross-linking performed thereon, reinforces the cornea 16" and greatly reduces the likelihood of corneal graft rejection.

It is to be understood that the scarred and/or diseased corneal portion 16a" that is removed from the cornea 16" may also be replaced with stroma stem cells or mesenchymal stem cells, which can be contained in a medium, and then injected in the internal cavity previously occupied by the scarred and/or diseased corneal tissue 16a".

In one or more embodiments, mesenchymal stem cells also may be injected inside the donor cornea before or after transplantation. In addition, in one or more embodiments, daily drops of a Rho Kinase inhibitor may be added to the host eye after the surgery. The use of a medication, such as a Rho Kinase inhibitor, with the stem cells will encourage stem cell proliferation.

A fourth illustrative embodiment of a corneal transplant procedure with a cross-linked cornea is shown in FIGS. 4A-4E. Like the second and third embodiments described above, the corneal transplant procedure illustrated in FIGS. 4A-4E only involves replacing a scarred and/or diseased portion 16a'" of the cornea 16'" with a donor corneal portion. Thus, similar to the second and third embodiments explained above, FIGS. 4A-4E illustrate a lamellar keratoplasty procedure wherein only a portion of the cornea 16'" of the eye 10'" contains scarred and/or diseased tissue (i.e., a full-thickness corneal section is not removed). Although, in the procedure of FIGS. 4A-4E, a different-shaped scarred and/or diseased portion 16a'" of the cornea 16'" is removed.

Referring initially to FIG. 4A, it can be seen that only a portion 16a'" of the cornea 16'" having a T-shape or "top hut" shape is scarred and/or diseased. As such, in this embodiment, it is not necessary to replace the entire thickness of the cornea 16'" with a donor cornea as was described above in conjunction with FIGS. 1A-1D, but rather just a portion 16a'" of the cornea 16'"'. In this illustrative embodiment, the back side of the cornea 16'" is maintained (see e.g., FIG. 4D).

In FIG. 4B, the cross-linking 18' of the clear donor cornea 20' is diagrammatically illustrated. As mentioned above, it is to be understood that all or just a part of the donor cornea 20' may be cross-linked. Then, in FIG. 4C, it can be seen that a portion 20a' of the clear donor cornea 20', which has a T-shape or "top hut" shape that matches the shape of the scarred and/or diseased portion 16a'" of the cornea 16'", is cut out from the remainder of the clear donor cornea 20' such that it has the necessary shape. In one or more embodiments, the portion 20a' may be cut from the clear donor cornea 20' and appropriately shaped using a femtosecond laser. As shown in FIGS. 5A and 5B, other suitably shaped cross-linked corneal portions may be cut from the clear donor cornea 20', such as a dumbbell-shaped corneal portion 20a" (see FIG. 5A) or a corneal portion 20a'" having a reversed T-shape or "reversed top hut" shape (see FIG. 5B), in order to accommodate correspondingly shaped scarred and/or diseased areas in the host cornea.

Next, referring to FIG. 4D, it can be seen that the scarred and/or diseased portion 16a'" having the T-shape or "top hut" shape has been removed from the cornea 16'" of the eye 10'" such that the cornea 16'" comprises a cavity 19' disposed therein for receiving the donor cornea. As described above for the first three embodiments, the scarred and/or diseased corneal portion 16a'" may be removed from the remainder of the cornea 16'" using various suitable means, such as mechanical means or the laser cutting means (e.g., femtosecond laser) described above.

Finally, as shown in FIG. 4E, after the scarred and/or diseased portion 16a''' has been removed from the remainder of the cornea 16''' of the eye 10''', the cross-linked donor corneal portion 20a' is implanted into the eye 10''' of the patient in the location previously occupied by the scarred and/or diseased corneal portion 16a'''. Because the shape of the transplant corresponds to that of the removed portion 16a''' of the cornea 16''', the transplant sits comfortably in its position in the host cornea. As described above, after implantation of the cross-linked donor corneal portion 20a' into the eye 10''', sutures or a suitable adhesive (e.g., the biocompatible and biodegradable adhesive described above) may be utilized to secure the cross-linked donor corneal portion 20a' in place on the host cornea 16''' of the eye 10'''. For example, if a biocompatible and biodegradable adhesive is used to secure the cross-linked donor corneal portion 20a' in place in the cornea 16''' of the eye 10''', the edges of the donor corneal portion 20a' are coated with the biocompatible and biodegradable adhesive so as to give the transplant a reliable stability. In this case, it is desirable to have the attachment of the transplant maintained by the biocompatible and biodegradable adhesive for a period of months (i.e., it is desirable for the transplant to be secured in place by the biocompatible and biodegradable adhesive for as long as possible).

An illustrative embodiment of a corneal lenslet implantation procedure with a cross-linked cornea is shown in FIGS. 6A-6C and 7A-7C. Similar to the second, third, and fourth embodiments described above, FIGS. 6A-6C and 7A-7C illustrate a lamellar keratoplasty procedure wherein only a portion of the cornea 16''' of the host eye 10''' is removed during the procedure (i.e., a full-thickness corneal section is not removed). Although, the procedure of FIGS. 6A-6C and 7A-7C differs in several important respects from the abovedescribed procedures. In this embodiment, the corneal transplant is cross-linked in vitro. Then, using a femtosecond laser or an excimer laser, the surgeon carves out or ablates a three-dimensional (3D) corneal cross-linked augment from the donor cornea 20''' that exactly compensates for the refractive error of the recipient of the transplant. That is, the corneal cross-linked augment or inlay may be cut to the desired shape using a femtosecond laser, or the inlay may be shaped in vitro using an excimer laser prior to its implantation in the cornea 16''' of the host eye 10'''. After making an internal pocket 28 in the recipient cornea 16''' of the host eye 10''' with a femtosecond laser, the cross-linked transplant is folded and implanted in a predetermined fashion inside the host's corneal pocket 28 to provide stability to the eye 10''' having keratoconus, keratoglobus, a thin cornea or abnormal corneal curvature, thereby preventing future corneal ectasia in this eye 10''' and correcting its refractive errors. Advantageously, the procedure of this embodiment comprises a lamellar cross-linked corneal transplantation, which additionally results in simultaneous correction of the refractive error of the eye 10''' of the patient. As used herein, the term "lenslet" refers to a lens implant configured to be implanted in a cornea of an eye. The lens implant may be formed from an organic material, a synthetic material, or a combination of organic and synthetic materials.

Now, with reference to FIGS. 6A-6C and 7A-7C, the illustrative embodiment will be described in further detail. The host eye 10''' with lens 12', cornea 16''', and optic nerve 24 is shown in FIG. 6A, while the donor cornea 20''' is depicted in FIG. 7A. The donor cornea 20''' of FIG. 7A may be a cross-linked cornea of a cadaver or a tissue culture-grown cornea that has been cross-linked. Turning to FIG. 6B, it can be seen that an internal corneal pocket 28 is

created in the cornea 16''' of the host eye 10''' (e.g., by using a suitable laser, which is indicated diagrammatically in FIG. 6B by lines 30).

In FIG. 7A, the cross-linking 18'' of the donor cornea 20''' is diagrammatically illustrated. As mentioned in the preceding embodiments, it is to be understood that all or just a part of the donor cornea 20''' may be cross-linked. Then, after the donor cornea 20''' of FIG. 7A has been cross-linked (e.g., by using a photosensitizer in the form of riboflavin and UV radiation as described above), it can be seen that a cross-linked lamellar lenslet 26 is cut out from the remainder of the donor cornea 20''' (e.g., by using a suitable laser, which is indicated diagrammatically in FIG. 7B by lines 32) such that it has the necessary shape for implantation into the host eye 10'''. As explained above, the cross-linked lamellar lenslet 26 may be cut from the donor cornea 20''' and appropriately shaped using a femtosecond laser or an excimer laser. The cross-linked lamellar lenslet 26 is capable of being prepared to any requisite shape using either the femtosecond laser or the excimer laser. FIG. 7C illustrates the shaped cross-linked lamellar lenslet 26 after it has been removed from the remainder of the donor cornea 20'''.

Finally, as shown in FIG. 6C, the cross-linked lamellar lenslet 26 is implanted into the cornea 16''' of the host eye 10''' of the patient in the location where the pocket 28 was previously formed. Because the shape of the transplant corresponds to that of the pocket 28 formed in the eye 10''', the transplant sits comfortably in its position in the host cornea 16'''. As described above, after implantation of the cross-linked lamellar lenslet 26 into the eye 10''', the refractive errors of the eye 10''' have been corrected because the cross-linked lamellar lenslet 26 has been appropriately shaped to compensate for the specific refractive errors of the host eye 10''' prior to its implantation into the eye 10'''. In addition, as explained above, the implantation of the cross-linked lamellar lenslet 26 provides additional stability to an eye having keratoconus, keratoglobus, a thin cornea, or abnormal corneal curvature.

Another illustrative embodiment of a corneal lenslet implantation procedure with a cross-linked cornea is shown in FIGS. 8-14. In general, the procedure illustrated in these figures involves forming a two-dimensional cut into a cornea of an eye; creating a three-dimensional pocket in the cornea of the eye, cross-linking the interior stroma, and inserting a lenslet or lens implant into the three-dimensional pocket after the internal stromal tissue has been cross-linked.

Initially, in FIG. 8, the forming of a two-dimensional cut 115 into the cornea 112 of the eye 110 is diagrammatically illustrated. In particular, as shown in the illustrative embodiment of FIG. 8, the two-dimensional cut 115 is formed by making an intrastromal incision in the cornea 112 of the eye 110 using a femtosecond laser (i.e., the incision is cut in the cornea 112 using the laser beam(s) 114 emitted from the femtosecond laser). Alternatively, the two-dimensional cut 115 may be formed in the cornea 112 of the eye 110 using a knife.

Then, in FIG. 9, the forming of a three-dimensional corneal pocket 116 in the cornea 112 of the eye 110 is diagrammatically illustrated. In particular, as shown in the illustrative embodiment of FIG. 9, the three-dimensional corneal pocket 116 is formed by using a spatula 118. The formation of the intracorneal pocket 116 in the cornea 112 of the eye 110 allows one to gain access to the tissue surrounding the pocket 116 (i.e., the interior stromal tissue surrounding the pocket 116).

Turning again to FIGS. 8 and 9, in the illustrative embodiment, the corneal pocket 116 formed in the cornea 112 of the

eye 110 may be in the form of an intrastromal corneal pocket cut into the corneal stroma. A femtosecond laser may be used to form a 2-dimensional cut into the cornea 112, which is then opened with a spatula 118 to create a 3-dimensional pocket 116. In one embodiment, a piece of the cornea 112 or a cornea which has a scar tissue is first cut with the femtosecond laser. Then, the cavity is cross-linked before filling it with an implant or inlay 128 to replace the lost tissue with a clear flexible inlay or implant 128 (see FIG. 12).

In one embodiment, a three-dimensional (3D) uniform circular, oval, or squared-shaped corneal pocket 116 is cut with a femtosecond laser and the tissue inside the pocket is removed to produce a three-dimensional (3D) pocket 116 to be cross-linked with riboflavin and implanted with a prepared implant.

After the pocket 116 is formed using the spatula 118, a photosensitizer is applied inside the three-dimensional pocket 116 so that the photosensitizer permeates the tissue surrounding the pocket 116 (see FIG. 10). The photosensitizer facilitates the cross-linking of the tissue surrounding the pocket 116. In the illustrative embodiment, the photosensitizer is injected with a needle 120 inside the stromal pocket 116 without lifting the anterior corneal stroma so as to cover the internal surface of the corneal pocket 116. In one or more embodiments, the photosensitizer or cross-linker that is injected through the needle 120 inside the stromal pocket comprises riboflavin, and/or a liquid suspension having nanoparticles of riboflavin disposed therein. Preferably, the cross-linker has between about 0.1% riboflavin to about 100% riboflavin therein (or between 0.1% and 100% riboflavin therein). Also, in one or more embodiments, an excess portion of the photosensitizer in the pocket 116 may be aspirated through the needle 120 until all, or substantially all, of the excess portion of the photosensitizer is removed from the pocket 116 (i.e., the excess cross-linker may be aspirated through the same needle so that the pocket 116 may be completely emptied or substantially emptied).

Next, turning to the illustrative embodiment of FIG. 11A, shortly after the photosensitizer is applied inside the pocket 116, the cornea 112 of the eye 110 is irradiated from the outside using ultraviolet (UV) radiation 122 so as to activate cross-linkers in the portion of the tissue surrounding the three-dimensional pocket 116, and thereby stiffen the cornea 112, prevent corneal ectasia of the cornea 112, and kill cells in the portion of the tissue surrounding the pocket 116. In the illustrative embodiment, the ultraviolet light used to irradiate the cornea 112 may have a wavelength between about 370 nanometers and about 380 nanometers (or between 370 nanometers and 380 nanometers). Also, in the illustrative embodiment, only a predetermined anterior stromal portion 124 of the cornea 112 to which the photosensitizer was applied is cross-linked (i.e., the surrounding wall of the corneal pocket 116), thereby leaving an anterior portion of the cornea 112 and a posterior stromal portion of the cornea 112 uncross-linked. That is, in the illustrative embodiment, the entire corneal area inside the cornea 112 exposed to the cross-linker is selectively cross-linked, thereby leaving the anterior part of the cornea 112 and the posterior part of the stroma uncross-linked. The portion of the cornea 112 without the cross-linker is not cross-linked when exposed to the UV radiation. In an alternative embodiment, the cornea 112 may be irradiated using wavelengths of light other than UV light as an alternative to, or in addition to being irradiated using the ultraviolet (UV) radiation 122 depicted in FIG.

11A. Also, microwave radiation may be used synergistically or additively to correct non-invasively the remaining refractive error(s) of the cornea.

Alternatively, as shown in FIG. 11B, a fiber optic 126 may be inserted into the corneal pocket 116 so as to apply the ultraviolet radiation and activate the photosensitizer in the wall of the corneal pocket 116. When the fiber optic 126 is used to irradiate the wall of the pocket 116, the ultraviolet radiation is applied internally, rather than externally as depicted in FIG. 11A.

Now, with reference to FIG. 12, it can be seen that, after the wall of the corneal pocket 116 has been stiffened and is devoid of cellular elements by the activation of the cross-linkers, a lens implant 128 is inserted into the corneal pocket 116 in order to change the refractive properties of the eye. In particular, in the illustrated embodiment, the lens implant 128 is inserted through a small incision, and into the corneal pocket 116, using forceps or microforceps. In one or more embodiments, the lens implant 128 that is inserted inside the pocket 116 in the cornea 112 is flexible and porous. Also, in one or more embodiments, the lens implant 128 may comprise a hybrid lens implant with an organic outer portion and a synthetic inner portion. The organic outer portion of the hybrid lens implant may be made from a transparent, hydrophilic organic polymer, while the synthetic inner portion of the hybrid lens implant may be made from a transparent, gas permeable, porous flexible polymer. For example, the transparent, hydrophilic polymer forming the organic outer portion may be formed from collagen, chitosan, poloxamer, polyethylene glycol, or a combination thereof (or any other transparent hydrophilic coating which can be deposited over the entire lens surface), while the flexible polymer forming the synthetic inner portion of the hybrid lens implant may be formed from silicone, acrylic, polymetacrylate, hydrogel, or a combination thereof. The surface of the lens implant 128 may have the appropriate shape to reshape the cornea 112 or the dioptric power to nullify the remaining spheric or astigmatic error of the eye. More particularly, in one or more embodiments, the lens implant 128 may have one of: (i) a concave surface to correct myopic refractive errors (i.e., a minus lens for correcting nearsightedness), (ii) a convex surface to correct hyperopic refractive errors (i.e., a plus lens for correcting farsightedness), or (iii) a toric shape to correct astigmatic refractive errors.

In the illustrative embodiment, the irradiation of the cornea 112 using the ultraviolet (UV) radiation 122 only activates cross-linkers in the portion of the stromal tissue surrounding the three-dimensional pocket 116, and only kills the cells in the portion of the tissue surrounding the pocket 116, so as to leave only a thin layer of cross-linked collagen to prevent an immune response and rejection of the lens implant 128 and/or encapsulation by fibrocytes, while preventing post-operative dry eye formation. In addition to preventing encapsulation of the lens implant 128 by fibrocytes, the cross-linking of the stromal tissue surrounding the pocket 116 also advantageously prevents corneal haze formation around the lens implant 128. That is, the cross-linking of the stromal tissue surrounding the lens implant 128 prevents formation of myofibroblast from surrounding keratocytes, which then convert gradually to fibrocytes that appear as a haze, and then white encapsulation inside the cornea, thereby causing light scattering in front of the patient's eye.

As shown in FIGS. 13 and 14, the crosslinking procedure described above may be repeated after the lens implant 128 is implanted so as to prevent any cellular invasion in the area surrounding the implant 128. Initially, with reference to FIG.

13, the photosensitizer is reinjected inside the space between the lens implant 128 and the surrounding corneal tissue using a needle 120. In one or more embodiments, the needle 120 for injecting the photosensitizer may comprise a 30-32 gauge needle. Then, after the reinjection of the cross-linker, the cornea 112 is re-irradiated with ultraviolet radiation 122 to cross-link the tissue surrounding the lens implant 128 so as to prevent cellular migration towards the lens implant 128 (see FIG. 14).

In one or more embodiments, the lens implant or inlay 128 may be prepared ahead of time with known techniques, wherein the inlay 128 may be coated with a biocompatible material, such as collagen, elastin, polyethylene glycol, biotin, streptavidin, etc., or a combination thereof. The inlay 128 and the coating may be cross-linked with a photosensitizer or cross-linker, such as riboflavin, prior to being implanted into the pocket 116 in the cornea 112 of the eye.

In another embodiment, the lens implant or inlay 128 may be silicone, methacrylate, hydroxyethylmethacrylate (HEMA), or any other biocompatible transparent material, or a mixture thereof. The lens implant or inlay 128 also may be coated with materials, such as collagen or elastin, and may have a desired thickness of from 2 microns to 70 microns or more.

In yet another embodiment, the lens implant or inlay 128 is formed from an eye bank cornea, or a cross-linked eye bank cornea, etc. In general, there is a tremendous paucity of normal cadaver corneas for total or partial implants, such as for a corneal transplant of a corneal inlay. Because all the cellular elements are killed during the crosslinking of the corneal inlay, and because the corneal collagen is cross-linked and denatured, the remaining collagenous elements are not immunogenic when implanted inside the body or in the cornea of a patient. Advantageously, the prior cross-linking of the organic material, such as in the cadaver cornea, permits transplantation of the corneal inlay from an animal or human cornea or any species of animal to another animal or human for the first time without inciting a cellular or humoral response by the body, which rejects the inlay. Thus, cross-linking transparent cadaveric tissue for corneal transplantation, or as an inlay to modify of the refractive power of the eye, is highly beneficial to many patients who are on the waiting list for a corneal surgery. In addition, the surgery may be planned ahead of time without necessitating the urgency of the surgery when a fresh cadaver eye becomes available. In one or more embodiments, the collagens may be driven from the animal cornea, and cross-linked. Also, in one or more embodiments, the implant or inlay 128 may be made of cross-linked animal cornea or human cornea that is cut using a femtosecond laser to any desired shape and size, and then ablated with an excimer laser or cut with a femtosecond laser to have a desired refractive power.

For example, as shown in FIG. 15, the lens implant or inlay 130 may be formed from an organic block of a polymer (e.g., donor cornea) by cutting the lens implant 130 using an excimer laser (e.g., by using the laser beam(s) 132 emitted from the excimer laser). Alternatively, referring to FIG. 16, the lens implant or inlay 130' may be formed from an organic block 134 of a polymer (e.g., donor cornea) by cutting the lens implant 130' from the block 134 using a femtosecond laser or a computerized femto-system (e.g., by using the laser beam(s) 136 emitted from the femtosecond laser).

In still another embodiment, as depicted in FIG. 17, the lens implant or inlay 130" is made using three-dimensional (3D) printing technology or a molding technique in order to

form the lens implant or inlay 130" into the desired shape, size or thickness. The transparent material of the 3D-printed implant or inlay 130" may be coated with one or more biocompatible polymers and cross-linked prior to the implantation.

In yet another embodiment, after the implantation of an intraocular lens, the remaining refractive error of the eye may be corrected by the implantation of a lens implant or inlay 128 in the cross-linked pocket 116 of the cornea 112, thereby eliminating the need for entering the eye cavity to replace the original intraocular lens.

In still another embodiment, the remaining refractive error of the eye is corrected after an intraocular lens implantation by placing an inlay 128 on the surface of the cornea 112 of the patient while the shape of the cornea 112 is corrected with an excimer laser and wavefront optimized technology so that the patient is provided instant input on its effect on his or her vision. In this embodiment, an inlay similar to a contact lens is placed on the cornea 112 that, after correction, matches the desired refractive correction of the eye, and then, subsequently, the inlay 128 is implanted inside the cross-linked corneal pocket 116.

In yet another embodiment, the implant or inlay 128 may be ablated with an excimer laser for implantation in the cross-linked pocket 116, or after cross-linking the exposed corneal stroma in LASIK surgery.

In still another embodiment, a small amount of hyaluronic acid or a viscous fluid is injected into the pocket 116 prior to the implantation of the implant or inlay 128 so as to simplify the insertion of the implant or inlay 128 in the corneal pocket 116.

In yet another embodiment, the implant or inlay 128 is prepared having four marking holes of 0.1-2 millimeter (mm) in diameter in the inlay periphery at an equally sized distances so that the implant 128 may be rotated with a hook, if desired, after the implantation as needed to match the axis of an astigmatic error of the eye during the surgery as measured simultaneously with a wavefront technology system, such as an Optiwave Refractive Analysis (ORA) system or Holos® system, which are commercially available for measurement of astigmatism or its axis.

In still another embodiment, the implant or inlay 128 is located on the visual axis and may provide 1 to 3 times magnification for patients whose macula is affected by a disease process needing magnifying glasses for reading, such as in age-related macular degeneration, macular edema, degenerative diseases of the retina, etc. Because these eyes cannot be used normally for reading without external magnifier glasses, providing magnification by a corneal implant to one eye assists the patients in being able to read with one eye and navigate the familiar environment with their other eye.

In yet another embodiment, the surface of the cornea 112 is treated after surgery in all cases daily with an anti-inflammatory agent, such as steroids, nonsteroidal anti-inflammatory drugs (NSAIDs), immune-suppressants, such as cyclosporine A or mycophenolic acid, anti-proliferative agents, antimetabolite agents, or anti-inflammatory agents (e.g., steroids, NSAIDs, or antibiotics etc.) to prevent inflammatory processes after the corneal surgery, inlay implantation or crosslinking, while stabilizing the integrity of the implant 128 and preventing future cell growth in the organic implant or the adjacent acellular corneal tissue. In this embodiment, the medication is injected in the corneal pocket 116 along with the implantation or the implant 128 is dipped in the medication first, and then implanted in the cross-linked corneal pocket 116.

In still another embodiment, a cross-linked corneal inlay is placed over the cross-linked corneal stroma after a LASIK incision, and is abated to the desired size with an excimer laser using a topography guided ablation. By means of this procedure, the refractive power of the eye is corrected, while simultaneously providing stability to an eye prone to conceal ectasia postoperatively after a LASIK surgery. Then, the LASIK flap is placed back over the implant.

Yet another illustrative embodiment of a corneal lenslet implantation procedure with a cross-linked cornea is shown in FIGS. 18-23. In general, the procedure illustrated in these figures involves initially making an intrastromal square pocket surrounding the visual axis of the eye, and then, after forming the initial square pocket, a three-dimensional circular portion of diseased or weak stromal tissue is cut, removed, and replaced with a circular implant which fits into the circle that borders the four sides of the square. A front view of the cornea 212 of the eye 210 with the centrally-located visual axis 214 is illustrated in FIG. 18. Advantageously, in the illustrative embodiment of FIGS. 18-23, corneal tissue removal around the visual axis is greatly facilitated, and nearly perfect centration of the lens implant or inlay 220 about the visual axis is possible because the lens implant 220 fits within a depressed circular recess at the bottom of the pocket 216. As such, the undesirable decentering of the lens implant is prevented.

Initially, in FIG. 19, the forming of an intrastromal square-shaped pocket 216 surrounding the visual axis 214 (represented by a plus sign) in the cornea 212 of the eye 210 is diagrammatically illustrated. In particular, as shown in the illustrative embodiment of FIG. 19, the square-shaped pocket 216 is formed by making a two-dimensional intrastromal incision in the cornea 212 of the eye 210 using a femtosecond laser (i.e., the incision is cut in the cornea 212 using the laser beam(s) emitted from the femtosecond laser).

Then, in FIG. 20, the removal of a three-dimensional circular portion 218 of diseased or weak stromal tissue in the cornea 212 of the eye 210 is diagrammatically illustrated. In particular, as shown in the illustrative embodiment of FIG. 20, the three-dimensional circular stromal tissue portion 218 has a first diameter, which is less than a width of the square-shaped pocket 216 so that the three-dimensional circular stromal tissue portion 218 is disposed within the boundaries of the square-shaped pocket 216. The three-dimensional circular stromal tissue portion 218' depicted in FIG. 21 is generally similar to that illustrated in FIG. 20, except that the three-dimensional circular stromal tissue portion 218' depicted in FIG. 21 has a second diameter that is slightly larger than the first diameter of the three-dimensional circular stromal tissue portion 218 in FIG. 20. As such, the periphery of the three-dimensional circular stromal tissue portion 218' depicted in FIG. 21 is disposed closer to the square-shaped pocket 216, but still within the confines of the square-shaped pocket 216. In the illustrative embodiment, the three-dimensional circular stromal tissue portion 218, 218' may be removed using forceps or micro-forceps. In an exemplary embodiment, the diameter of the circular stromal tissue portion 218, 218' that is removed from the cornea 212 is between approximately 5 millimeters and approximately 8 millimeters, inclusive (or between 5 millimeters and 8 millimeters, inclusive).

In an alternative embodiment of the corneal lenslet implantation procedure, three (3) sequential cuts may be made in the stromal portion of the cornea 212 of the eye 210 using a femtosecond laser in order to form the pocket. First, a lower circular cut or incision centered about the visual axis (i.e., a lower incision with the patient in a supine position)

is made using the femtosecond laser. Then, a second vertical cut is made above the lower incision using the femtosecond laser to form the side(s) of a circular cutout portion. Finally, a third square or circular cut (i.e., an upper incision) is made above the vertical cut using the femtosecond laser. In the illustrative embodiment, the lower incision is parallel to the upper incision, and the vertical cut extends between lower incision and the upper incision. In this alternative embodiment, the three-dimensional circular stromal tissue cutout portion bounded by the lower incision on the bottom thereof, the vertical cut on the side(s) thereof, and the upper incision on the top thereof is removed from the cornea 212 of the eye 210 using a pair of forceps. A cavity formed by the upper incision facilitates the removal of the three-dimensional circular stromal tissue cutout portion. As described above, the third cut or incision formed using the femtosecond laser may be an upper circular cut that is larger than the lower circular cut, rather than an upper square cut that is larger than the lower circular cut.

Turning to FIG. 22, after the three-dimensional circular stromal tissue portion 218, 218' is removed, a photosensitizer is applied inside the pocket 216 so that the photosensitizer permeates the tissue surrounding the pocket 216. The photosensitizer facilitates the cross-linking of the tissue surrounding the pocket 216. In the illustrative embodiment, the photosensitizer is injected with a needle 222 inside the stromal pocket 216. In one or more embodiments, the photosensitizer or cross-linker that is injected through the needle 222 inside the stromal pocket 216 comprises riboflavin, and/or a liquid suspension having nanoparticles of riboflavin disposed therein. Preferably, the cross-linker has between about 0.1% riboflavin to about 100% riboflavin therein (or between 0.1% and 100% riboflavin therein). Also, in one or more embodiments, an excess portion of the photosensitizer in the pocket 216 may be aspirated through the needle 222 until all, or substantially all, of the excess portion of the photosensitizer is removed from the pocket 216 (i.e., the excess cross-linker may be aspirated through the same needle 222 so that the pocket 216 may be completely emptied or substantially emptied).

Next, turning again to the illustrative embodiment of FIG. 22, shortly after the photosensitizer is applied inside the pocket 216, the cornea 212 of the eye 210 is irradiated from the outside using ultraviolet (UV) radiation 224 so as to activate cross-linkers in the portion of the tissue surrounding the three-dimensional pocket 216, and thereby stiffen the cornea 212, prevent corneal ectasia of the cornea 212, and kill cells in the portion of the tissue surrounding the pocket 216. In the illustrative embodiment, the ultraviolet light used to irradiate the cornea 212 may have a wavelength between about 370 nanometers and about 380 nanometers (or between 370 nanometers and 380 nanometers). Also, in the illustrative embodiment, only a predetermined anterior stromal portion of the cornea 212 to which the photosensitizer was applied is cross-linked (i.e., the surrounding wall of the corneal pocket 216), thereby leaving an anterior portion of the cornea 212 and a posterior stromal portion of the cornea 212 uncross-linked. That is, in the illustrative embodiment, the entire corneal area inside the cornea 212 exposed to the cross-linker is selectively cross-linked, thereby leaving the anterior part of the cornea 212 and the posterior part of the stroma uncross-linked. The portion of the cornea 212 without the cross-linker is not cross-linked when exposed to the UV radiation. In an alternative embodiment, the cornea 212 may be irradiated using wavelengths of light other than UV light as an alternative to, or in addition to being irradiated using the ultraviolet (UV) radiation 224

depicted in FIG. 22. Also, microwave radiation may be used synergistically or additively to correct non-invasively the remaining refractive error(s) of the cornea. In addition, in an alternative embodiment, the ultraviolet (UV) radiation may be applied after the implantation of the lens implant 220 to perform the crosslinking, rather than before the implantation of the lens implant 220 as described above. Further, rather than applying the ultraviolet (UV) radiation from outside the cornea 212, the stromal tissue of the pocket 216 may be irradiated from inside by means of a fiber optic, before or after the implantation of the lens implant 220.

Now, with combined reference to FIGS. 22 and 23, it can be seen that, before or after the wall of the corneal pocket 216 has been stiffened and is devoid of cellular elements by the activation of the cross-linkers, a circular lens implant 220 is inserted into the circular recess at the bottom of the pocket 216 formed by the three-dimensional circular stromal tissue cutout portion 218, 218' that was removed. That is, the circular lens implant 220 fits within the periphery of the circular recess that borders the four sides of the squared-shaped pocket 216. In particular, in the illustrated embodiment, the circular lens implant 220 is inserted through a small incision, and into the circular recess at the bottom of the pocket 216 using forceps or microforceps. In the illustrative embodiment, the flexible lens implant 220 may be folded, inserted through the small incision, placed inside the circular recess at the bottom of the pocket 216, and finally unfolded through then small incision. In one or more embodiments, the lens implant 220 that is inserted inside the pocket 216 in the cornea 212 is flexible and porous. Also, in one or more embodiments, the lens implant 220 may comprise a hybrid lens implant with an organic outer portion and a synthetic inner portion. The organic outer portion of the hybrid lens implant may be made from a transparent, hydrophilic organic polymer, while the synthetic inner portion of the hybrid lens implant may be made from a transparent, gas permeable, porous flexible polymer. For example, the transparent, hydrophilic polymer forming the organic outer portion may be formed from collagen, chitosan, poloxamer, polyethylene glycol, or a combination thereof (or any other transparent hydrophilic coating which can be deposited over the entire lens surface), while the flexible polymer forming the synthetic inner portion of the hybrid lens implant may be formed from silicone, acrylic, polymetacrylate, hydrogel, or a combination thereof.

Advantageously, the lens implant 220 of the aforescribed illustrative embodiment always remains perfectly centered around the visual axis 214 of the eye 210, and will not move because it is disposed within the circular recess at the bottom of the pocket 216. As explained above, the lens implant 220 may be formed from an organic material, synthetic material, polymeric material, and combinations thereof. The lens implant 220 may replace either a diseased tissue or create a new refractive power for the eye 210, as explained hereinafter.

In the illustrative embodiment, the lens implant 220 may correct the refractive errors of the eye 210. The refractive error correction may be done by the lens implant 220 having a curvature that changes the corneal surface of the cornea 212. Alternatively, the lens implant 220 may have a different index of refraction that corrects the refractive power of the cornea 212. In the illustrative embodiment, the lens implant 220 may have the appropriate shape to reshape the cornea 212 or the dioptric power to nullify the remaining spheric or astigmatic error of the eye. More particularly, in one or more embodiments, the lens implant 220 may have one of: (i) a concave anterior surface to correct myopic refractive errors

(i.e., a minus lens for correcting nearsightedness), (ii) a convex anterior surface to correct hyperopic refractive errors (i.e., a plus lens for correcting farsightedness), or (iii) a toric shape to correct astigmatic refractive errors.

In the illustrative embodiment, the irradiation of the cornea 212 using the ultraviolet (UV) radiation 224 only activates cross-linkers in the portion of the stromal tissue surrounding the three-dimensional pocket 216, and only kills the cells in the portion of the tissue surrounding the pocket 216, so as to leave only a thin layer of cross-linked collagen to prevent an immune response and rejection of the lens implant 220 and/or encapsulation by fibrocytes, while preventing post-operative dry eye formation. In addition to preventing encapsulation of the lens implant 220 by fibrocytes, the cross-linking of the stromal tissue surrounding the pocket 216 also advantageously prevents corneal haze formation around the lens implant 220. That is, the cross-linking of the stromal tissue surrounding the lens implant 220 prevents formation of myofibroblast from surrounding keratocytes, which then convert gradually to fibrocytes that appear as a haze, and then white encapsulation inside the cornea, thereby causing light scattering in front of the patient's eye.

In the further illustrative embodiments described hereinafter, the cornea is cross-linked soon after the initial corneal transplant where a suture is placed around the implant to keep the implant in place until the tissue heals between the host and the corneal transplant, before the graft and often the suture itself can act as a foreign body inducing an immune response with neovascularization of the corneal transplant which will be rejected and become scarred. For example, in the aforescribed embodiments, the cross-linking may be performed between two and three months after transplantation of the corneal graft or prosthesis in order to allow the transplantation area time to heal before the cross-linking procedure is performed.

A fifth illustrative embodiment is depicted in FIGS. 24 and 25. In this illustrative embodiment, shortly after corneal transplantation (see FIG. 24) the remaining peripheral host cornea 302 of the eye 300, the suture 304 and the transplant (i.e., the corneal graft 306) is treated with drops of 0.05-1% or more of riboflavin, CPP/nanoparticles, physiological solution or the use of another photosensitizer or cross-linking agent, that is applied to the cornea 302 for a period of a few minutes to 30 minutes. Then, the entire cornea 302 is irradiated with a UV laser 310 emitting ultraviolet radiation 308 of 370-400 nm wavelength, 1-30 mw/cm² for a desired time of 1-30 minutes, depending on the concentration of the photosensitizer and the power of the laser light to cross-link the corneal collagen and damage the keratocytes in it, in at least one-third of the front thickness of the host cornea and the transplant to prevent an immune response to the transplant and its rejection (refer to FIG. 25).

A sixth illustrative embodiment is depicted in FIGS. 26 and 27. In this illustrative embodiment, the neovascularization 406 has started or has reached the edges of the transplanted cornea 404 with sutures 412 (see FIG. 26), one applies riboflavin or another photosensitizer to the anterior part of the cornea 402 of the eye 400 prior to the cross-linker molecule diffusing to the posterior corneal layers and the corneal endothelial cells under observation with a slit lamp, then the cornea 402 is irradiated with UV light 408 from a UV laser 410 as described previously to cross-link at least one half of the anterior thickness of the cornea 402.

A seventh illustrative embodiment is depicted in FIG. 28. In this illustrative embodiment, the repeated corneal transplant has led to a cloudy vascularized cornea 502 in which

no corneal transplantation can be considered, the cornea 502 of the eye 500 is prepared for implantation of a keratoprosthesis lens 504 by removing the center part of the opaque cornea 502 by a trephine, thus creating a central opening in the cornea 502, and then a circular pocket in the remaining peripheral cornea is produced horizontally with a knife or laser so as to produce an anterior and a posterior flap around the central opening in the remaining part of the peripheral cornea, in which a prosthetic lens 504 with a central portion 504a and a peripheral flange 504b is implanted. As shown in FIG. 28, the peripheral flange 504b of the prosthetic lens 504 is provided with a plurality of apertures or holes 506 disposed therethrough for allowing the aqueous humour fluids of the eye to provide nutrients to the donor graft stroma. The riboflavin drops or nanoparticle suspension of riboflavin with or without CPP are applied to penetrate the front of the cornea 502 and the posterior corneal flap, followed a few anchoring 8-0 or 10-0 nylon sutures 512 to hold the prosthetic lens 504 in place, then followed by UV radiation 508 from a UV laser 510 of the entire cornea 502, except the lens 504 which is covered with a piece of tissue paper to prevent the light from getting inside the eye 500, thereby preventing vascular growth in the prosthesis 504 from the front and its side or fibrous tissue growth to the back side of the prosthesis 504 that could block the light reaching the retina (refer to FIG. 28).

Next, illustrative corneal crosslinking procedures with enhanced penetration of the crosslinking agent will be described. The process of crosslinking can be very time consuming in which the penetration of the cross-linker (e.g., riboflavin or any other cross-linker) is applied as drops to the corneal tissue for about 15-30 minutes to penetrate typically a depth of 150-200 microns of the anterior corneal stroma. Considering that corneal crosslinking often requires UV radiation for a period of 30 minutes, the operation takes at least between 30-60 minutes to perform. Also, riboflavin does not penetrate the cell membrane well. In the above-described embodiments, because the purpose is to reduce and eliminate the potential of corneal transplant rejection by the survived host corneal epithelial cells or the host stromal cells that produce cytokines against the transplanted cornea and vice versa, it is desirable to eliminate the cellular component of the host cornea or the transplanted cornea, or both, while maintaining the host corneal endothelial cells intact. Since the corneal endothelial cells, in general, are not affected if the riboflavin is applied over the corneal surface through which it penetrates inside the stroma, it is desirable to enhance cellular penetration of the cross-linker, and simultaneously penetration through the corneal stroma, which is made of collagen. Although the procedures described above can be used with standard riboflavin or any other crosslinking solution, a technique and formulation for expediting the cell penetration of riboflavin and the crosslinking of the cornea for corneal crosslinking and elsewhere in the body is very desirable.

In one or more embodiments, cell penetrating peptides are used that comprise the short peptide lysine or arginine, which are known as cell penetrating peptides (CPP) and activatable-cell penetrating peptides (ACPP). CPP and ACPP may be conjugated to dendrimers (ACPPDs) or other nanoparticles (e.g., riboflavin) or any other cross-linkers. The ACPP may be labeled with a polycationic CPP. ACPP and CPP may be naturally-occurring or artificially constructed protein segments (<30 amino acids) rich in arginine, lysine, cysteine, histidine, ornithine, etc.; preferably .alpha.-helices and about 17-amino acids. The ACPP and CPP may include a penetration accelerating peptide sequence (Pas) or

an INF7 fusion peptide sequence. CPP and/or ACPP can be linked to cargoes either covalently or non-covalently. Nanoparticles may be delivered by cell-penetrating peptides comprised of nona-arginine and a penetration accelerating sequence. Also, nona-arginine may be used to facilitate the delivery of the riboflavin cross-linker into cells via multiple pathways. Exemplary, but not limiting ACPP and CPP may include transportan, penetratin, TAT, VP22, MAP, KALA, ppTG20, proline-rich peptides, MPG-derived peptides, Pep-1, nona-arginine, and the carboxy-terminal tail of TFPI-2, polyproline helices having cationic amino acids and/or cationic-functionalized amino acids within the helix). Nanoparticles may be coated or otherwise associated with organic or non-organic biodegradable compounds, aliphatic biodegradable polymers, as needed. The nanoparticles may comprise organic nanoparticles, non-organic nanoparticles, synthetic nanoparticles, or non-synthetic nanoparticles.

In one embodiment, riboflavin or other cross-linkers may be linked to, associated with, complexed or conjugated with nanoparticles using linking agents and methods including but not limited to the following: amino groups, carboxyl groups, S—S deprotected sulfhydryl groups in biomolecules, carbodiimide conjugation, sulfosuccinimidylsuberyl linkage, synthetic tripyrrole-peptide linkage, NHS-esters and other esters, etc.

In one embodiment, the riboflavin or a cross-linker conjugated to dendrimers (ACPPDs) is applied to a cornea which has developed neovascular tissue, where the penetration of the riboflavin is very minimal in the endothelial cells or the neovascular tissue, so that subsequent UV radiation will not damage these cells or close the neovascular tissue of the cornea. Dendrimers and other types of nanoparticles do not need to have an antibody attached to them as long as they are applied topically for the crosslinking of the cornea. Although, nanoparticles do need to have a specific antibody attached to them if they are injected in the circulation of the patient for targeting a tumor.

In one embodiment, the cross-linker is conjugated with nanoparticles in the form of dendrimers or functionalized dendrimers conjugated with CPP or ACPP and administered locally, topically or injected in body cavity, to be absorbed by normal or abnormal tissue or tumor with their neovascular tissue and subsequently cross-linked by UV radiation or other laser wavelength absorbed by the cross-linker, not only damaging the neovascular tissue, but also the tumor cells, as an example. Also, the cross-linker conjugated with the dendrimers or functionalized dendrimers conjugated with CPP or ACPP may be applied to a surface lesion/tumor on the skin, mucosa or conjunctiva of the eye or inside a body cavity.

In one embodiment, the nanoparticles carrying riboflavin or another cross-linker are made of lactic acid, glycolic acid, or polycaprolactone and conjugated with a cell specific or organism specific antibody to attach to their cell membrane receptors.

In one embodiment, the nanoparticles are made of lactic acid, glycolic acid, polycaprolactone, or chitosan, or are in the form of dendrimers, and are conjugated with riboflavin, CPP and antibodies to target specific cells, such as tumors of neovascular cells, and to attach to the cell membrane receptors of these cells or to one or more organisms and to penetrate the cell walls of specific cells or organisms, thus carrying the photosensitizer or riboflavin inside the cell, while activated with a light or UV light, thereby crosslinking the cytoplasmic proteins and kill the cells or the organism.

In one embodiment, the photosensitizer is in the form of drops, and the drop solution or suspension of nanoparticles

or dendrimers comprises between about 0.05% and about 1% riboflavin or a photosensitizer therein.

In one embodiment, the photosensitizer or riboflavin is in the form of drops or suspension of nanoparticles, and the drop solution comprises between about 0.0005% and about 5% riboflavin therein.

In one embodiment, riboflavin or the cross-linker solution may have ethylenediaminetetraacetic acid (EDTA) or Disodium Edetate Dihydrate 0.1 w/w %, Sodium Chloride 0.4 w/w %, Polycarbophil 0.95 w/w %, Octoxynol 40 (70% Solution), 2N NaOH (active dissolution @ pH 7.9), q.s. to adjust pH to about 7.4-7.7 as desired, Mannitol 0.15 w/w %, Sodium thiosulfate 0.3 w/w %, Water q.s. to 100%. The riboflavin or the cross-linker solution may have an osmolarity of about 300 mOsm/L. The riboflavin or the cross-linker solution may also contain a cell penetrating agent (e.g., CPPs or ACPPs). In addition, the riboflavin or the cross-linker solution may be provided with or without a poloxamer and with or without dextran.

In one embodiment, the cross-linker is applied through a circular gel with the desired diameter of 1 millimeters (mm) to 15 millimeters (mm) or more to be placed on the surface of the cornea or other surfaces so that the cross-linker penetrates the desired area of the cornea selectively, etc.

In one embodiment, the gel has a thickness of 0.01 mm to 3 mm or more.

In one embodiment, the diameter of the gel is 5-10 mm.

In one embodiment, the gel is circular with an opening of 1-8 mm.

In one embodiment, the gel is made of an organic or synthetic material.

In one embodiment, the gel is made from cellulose derivatives.

In one embodiment, the gel is soaked with the photosensitizer, such as riboflavin or another chemical, at a desired concentration that can diffuse from it inside the tissue within the desired time.

In one embodiment, the circular gel fits on the surface of the cornea or another desired surface and has strategically made holes to selectively release the photosensitizer in that area, so that during the crosslinking, only specific areas are selectively cross-linked, while the other areas are left alone to have specific effect either in the refractive power, elasticity on the cornea or for specific selective implantation of an implant.

In one embodiment, the cross-linker or riboflavin/CPP are conjugated with functionalized nanoparticles, such as acrylic or acrylic derivatives or crystalline silicon or other organic transparent nanoparticles, to enhance cell penetration and penetration of the nanoparticles through the corneal stroma to kill the stromal cells and entrap the acrylic or crystalline silicon nanoparticle in the cross-linked stromal collagen and inside the cells after irradiation with UV radiation applied externally of through a fiber optic and crosslinking their cytoplasmic proteins and the collagen. Additionally, the corneal stroma and nanoparticles's index of refraction can be changed in the same session or later, using a femtosecond or multi-photon laser and a Shack-Hartmann unit to achieve a perfect refractive power for the cornea.

In one embodiment, a corneal inlay is soaked in a solution having a cross-linker or riboflavin/CPP are conjugated with coated nanoparticles, such as acrylic or acrylic derivatives or crystalline silicon nanostructure or organic transparent nanoparticles, to enhance cell penetration and penetration of the nanoparticles through the corneal stroma to kill the stromal cells and entrap the acrylic or crystalline silicon nanoparticle

in the cross-linked stromal collagen and inside the cells after irradiation with UV radiation applied externally before implanting inside a corneal pocket or after implantation inside the stromal pocket using a fiber optic and crosslinking their cytoplasmic proteins where the corneal stroma and subsequently the refractive index of the inlay and the cornea with the nanoparticles can be changed using a femtosecond or multi-photon laser and a Shack-Hartmann unit to achieve a perfect refractive power for the cornea or modifying it for astigmatic, presbyopic, myopia, or hyperopia correction.

In one embodiment, the corneal inlay is prepared from a 3-D stromal tissue culture or an eye bank eye cornea where the inlay is soaked in a solution having a cross-linker or riboflavin/CPP are conjugated with PEGylated nanoparticles, such as acrylic or acrylic derivatives or crystalline silicon nanostructure or organic transparent, organic nanoparticles to enhance cell penetration and penetration of the nanoparticles through the corneal stroma to kill the stromal cells and entrap the acrylic nanoparticle or crystalline silicon in the cross-linked stromal collagen and inside the cells after irradiation with UV radiation applied externally before implanting inside a corneal pocket or after implantation inside the stromal pocket using a fiber optic and crosslinking their cytoplasmic proteins where the corneal stroma and subsequently the refractive index of the inlay or the cornea with the nanoparticles can be changed using a femtosecond or multi-photon laser and a Shack-Hartmann unit to achieve a perfect refractive power for the cornea.

In one embodiment, the cross-linker or riboflavin/CPP are conjugated with antibody coated nanoparticles or dendrimers, to reach specific cells targeted to kill after irradiation with UV radiation applied externally of through a fiber optic and crosslinking their cytoplasmic proteins, etc.

In one embodiment, the targeted cells are tumor cells.

In one embodiment, the tumor cells are located on the skin or mucosa.

In one embodiment, the riboflavin/CPP conjugated with antibody coated nanoparticles are injected inside the body cavity where the nanoparticles are attached to the tumor cells, then treated with UV radiation brought in using a fiber optic crosslinking their cytoplasmic proteins, etc.

In one embodiment, the riboflavin/CPP conjugated with antibody coated nanoparticles are administered intravenously to reach an internally located tumor and to attach to their cell membranes and kill them after locally administered UV radiation.

In one embodiment, the tumor is in the mouth, nose, throat, eye, conjunctiva, or lid of the eye, or lung and can be reached with a UV laser with a fine fiber optic to irradiate the tumor and kill them by crosslinking their cytoplasmic proteins, etc.

In one embodiment, the lesion is an infected ulcer of the cornea, skin or mucosa, nasal, throat, etc. that can be treated by topical application of the riboflavin/CPP or ACPP conjugated with antibody coated nanoparticles to kill the bacteria, viruses, fungi, protozoal infections treated with UV radiation applied through a fiber optic and crosslinking their cytoplasmic proteins, etc.

In one embodiment, the lesion is an infected ulcer of the cornea, skin or mucosa, nasal, throat, etc. that can be treated by topical application of the riboflavin/CPP or ACPP conjugated with antibody coated nanoparticles or dendrimers to kill the bacteria, viruses, fungi, protozoal infections treated with UV radiation applied through a fiber optic and crosslinking their cytoplasmic proteins, etc.

In one embodiment, the ulcer is an infected corneal ulcer treated by topical application of the riboflavin/CPP or ACPP

conjugated with antibody coated nanoparticles or dendrimers to kill the bacteria with UV radiation applied through a fiber optic or an external UV light and crosslinking their cytoplasmic proteins, etc. as described before.

In one embodiment, the ulcer is an infected skin or mucosal ulcer treated by topical application of the riboflavin/CPP or ACPP conjugated with PEGylated nanoparticles/dendrimers to kill the bacteria, with UV radiation applied through a fiber optic or an external UV light by crosslinking their cytoplasmic proteins, etc., as described before.

In one embodiment, the keratoprosthesis **504** described above is coated with biocompatible nanoparticles, such as gold or silica, or a combination thereof, etc. in a manner that does not cover the central optical lens to prevent rejection.

In one embodiment, the riboflavin nanoparticle/dendrimer and CPP is used as a surface coating for any intracorneal implantation followed with UV radiation to kill the cells surrounding it by crosslinking their cytoplasmic proteins etc.

In one embodiment, the riboflavin nanoparticle or dendrimers and CPP is administered after extracapsular lens removal followed by administration of CPP/cross-linker conjugated with polyethylene glycol (PEG) coated nanoparticles in the capsular bag, then followed by lens implantation where the CPP enhances the penetration of the riboflavin into the lens epithelial cells and kills them by crosslinking their cytoplasmic proteins etc., thereby preventing capsular opacification.

In one embodiment, a glaucoma stent surface is coated with biocompatible nanoparticles, such as gold or silica, or a combination thereof, etc. to prevent rejection of the stent, then coated with CPP/riboflavin nanoparticles or dendrimers prior to the implantation of the glaucoma stent and then irradiated with UV radiation after the implantation to kill the cells by crosslinking their cytoplasmic proteins, etc. around the stent and prevent ingrowth of the cells blocking the stent and prevent ingrowth of the cells blocking the stent using a fiber optic.

In one embodiment, a cardiac/vascular stent is coated with biocompatible nanoparticles, such as gold or silica, or a combination thereof, etc. to prevent rejection, then coated with CPP/riboflavin nanoparticles or dendrimers prior to the implantation of the vascular stent and irradiated with UV radiation after the implantation to kill the cells by crosslinking their cytoplasmic proteins, etc. around the stent and preventing ingrowth of the cells blocking the stent using a fiber optic.

It is readily apparent that the aforescribed corneal transplant procedures and inlay implantation procedures offer numerous advantages. First, the implementation of the aforescribed corneal transplant procedures reduces the likelihood that the implanted cornea will be rejected by the patient. Secondly, the aforescribed corneal transplant procedures enable the clarity of the transplanted cornea to be preserved. Finally, the aforescribed corneal transplant procedures reduce the likelihood that the transplanted cornea will be invaded by migrating cells, such as migrating cells that might initiate an immune response such as macrophage, lymphocytes or leucocytes or vascular endothelial cells. These types of migrating cells are discouraged by the cross-linked corneal collagen which does not provide an easily accessible tissue to invade. In addition, the use of abovescribed tissue adhesives reduces the surgical procedure significantly. Moreover, the aforescribed corneal lenslet implantation procedures modify the cornea so as to better correct ametropic conditions. Furthermore, the corneal lenslet implantation procedures described above prevent the lens implant from moving around inside the cornea

once implanted, thereby ensuring that the lens implant remains centered about the visual axis of the eye. In addition, the aforescribed inlay implantation procedures prevent an immune response to the corneal inlay and to prevent a rejection of the corneal inlay by the patient.

Illustrative embodiments of a drug delivery implant and methods using the same will now be described hereinafter. In accordance with the various embodiments described herein, in order to provide the medication to the anterior and posterior part of the eye with a slow release drug system, it is required to create an immune privileged space inside the cornea to keep the cellular response away and prevent production of cytokine by them, and position the device outside the central visual axis so that the device would not interfere with the patient's vision.

In the embodiments described herein, the device is placed in the far corneal periphery so that it will not affect the vision or visual field of the patient, and so that it has created a so-called artificial "immune-privilege" which does not generate an immune response from the body while fluid, soluble medications or nano-particulates and micro-particulates can travel through it. See, for example, FIGS. **38A-45B**.

Because of the location of the implant inside the cornea, the released medication bypasses the epithelial barrier of the cornea, while providing medication in a slow manner by diffusion to the anterior part of the cornea, to the sclera, to the conjunctival tissue, and to the posterior segment of the eye including the retina, choroid, and the optic nerve head. This technique can provide similar immune-privileged spaces in other part of the body so that devices implanted there are not encapsulated.

The drug delivery system of the embodiments described herein may be constructed so that it can have direct access to the anterior chamber, if needed, for both obtaining repeatedly a fluid biopsy from the eye or deliver medication(s) directly inside the eye in a fast or slow release manner, or for reducing the intraocular pressure of the eye by creating a minor flow through a porous implanted stent or tube through the corneal limbus without inducing a fibrous encapsulation of the stent. The stent can ameliorate also corneal dryness caused by dry eye syndrome. The stent can also be equipped with a pressure sensor indicating directly the intraocular pressure and communicating it with a radiofrequency device to outside the eye to a receiver or a processor. As one example, as shown in FIG. **37**, the implant **644** may comprise a closed end **646** and a needle **648** for tissue penetration so that the implant **644** is capable of being used for taking liquid biopsies. In addition, stem cells or other cells can reside in, for example, a tubular implant, while having access to the oxygen and nutrients through the artificial barrier in an appropriately prepared corneal pocket. However the porous tubular implant permits these cells to migrate elsewhere in the eye or remain in place without being attacked by body's cellular response. Because the cellular body immune response is dependent on the production of the cells close to the implant or a foreign body to be taken up by the dendritic cells of the body at that location by creating a cell free space around the implant made of transparent amorphous cross-linked collagen. The invention of the embodiments described herein has eliminated the incentive for a Major Histocompatibility Complex (MHC) to occur. Because these MHC are present on the cell surface of the body cells to be activated in the production of an immune response. The release of these cytokines activates the cellular immune system of the body to either eliminate the threat or isolate the device from the body completely by fibrocytes, thereby building a dense membrane (i.e., scar)

around the implant. However, the cross-linked collagen permits the diffusion of water and small molecules permitting the needed growth factors from the incorporated stem cells placed inside the tubular implant needed for survival and the health of the cornea, retina etc.

Though this mechanism is very effective and useful, it affects the function of an implant that usually either releases a needed medication or measures or controls the release of a medication (e.g., measuring the blood glucose level and/or releasing insulin according to the glucose level found in the blood, etc.).

In order to isolate an implant in the body while preventing the immune cell to gain access to the device or build a membranous scar tissue around it, a method has been developed to isolate the implant in the body by killing all the cells adjacent to an implant, while maintaining a fluid-filled area around the implant or creating a barrier out of the surrounding tissue containing collagen and cross-linking the tissue in vivo. This barrier protects the implant from the antigen presenting dendritic cells in the tissue, while permitting the soluble medication or nano-sized particulate material to pass through the barrier so as to treat a pathological process in the body. One can also monitor the level of the analytes in the tissue fluid (e.g., aqueous fluid levels of glucose), which is a representative of the blood glucose level in the blood, from which it is originated. Aqueous level of most if not analytes found in the blood and could be used effectively to provide information on the health or disease processes affecting the eye or the body as a whole.

In the embodiments herein, implantation of a drug delivery device is described for the release or monitoring and controlling of a disease process in the eye, while crosslinking the tissue around the implant or implants (if more than one implant is provided). In any of the embodiments described herein, a plurality of drug delivery implants may be used (e.g., for delivering different medications), rather than a single drug delivery implant.

The technology described herein may be applied for any other device implantation in the body regardless of the location in the body. One of the benefits of the technology is that, if the device needs to be replaced, it can be done easily without dealing with the scar tissue formation that otherwise forms and makes the removal or replacement of the implant very complex because the tissue adhesions that usually forms between the tissue and the device.

One can use this concept described herein for diagnosis or therapy in diseases affecting the cornea, a metabolic disorder, genetic disorder, glaucoma, an infection affecting the eye or another portion of the body, a disease or disorder affecting the front or the back part of the eye or the conjunctiva or lens, an aging process, such as dry eye formation, retinal diseases including infective processes, genetic diseases requiring gene therapy (e.g., retinitis pigmentosa, etc. or metabolic disorders such as diabetes, etc.).

In one embodiment, if the media is clear, a two dimensional intrastromal corneal incision is created that is subsequently converted into a pocket in the corneal stroma using a femtosecond laser or a mechanical cutting system. The femtosecond laser passes through the clear media of the cornea. When the laser beam is focused inside the cornea, one can produce a two-dimensional cut or a three-dimensional cut around a thin part of the tissue that is removed to desired shape, shape, depth, and location.

In another embodiment, in opaque elastic tissue (e.g., skin), one can use a knife or a syringe needle ending in a sharp cutting tip to cut a pocket in the tissue. If needed, the incision simultaneously involves removal of a three-dimen-

sional tissue surrounding the surgical pocket to create some additional space for the implant using a similar cutting instrument, in the skin or soft tissue. In general, a cut creates a flexible three-dimensional space that can be filled with an implant. The implant is placed inside the needle and can be expelled from the needle by the syringe into the space created by knife.

In one or more embodiments, an injectable anesthetic (e.g., lidocaine or Bupivacaine) in a desired non-toxic preparation or concentration of 0.1-2% or more in a physiologic solution with, but preferably without, a preservatives, is injected in the corneal pocket to anesthetize the cornea postoperatively for a period up to 8-12 hours (e.g., if a PRK procedure is contemplated or after a corneal inlay implantation to prevent pain sensation completely in the postoperative period). This eliminates subjecting the entire corneal epithelium or the conjunctival epithelial cells to the damaging effect of topical anesthesia, which delays corneal re-epithelialization or conjunctival epithelial cells. Generally, the topical preservatives present in the topical anesthesia damages the cells that are bathed in them, and at times affects the regeneration of these cells (i.e., corneal epithelial or conjunctival cells) if applied frequently. Also, it may produce addiction to the topical anesthesia for eliminating the pain sensation caused by the loss of the corneal epithelial cells, whereas the injectable anesthetic does not damage the epithelial cells, including the nerve cells or their axons, except for blocking temporarily the neuronal transmission.

In one or more embodiments, the collagen cross-linker is mixed with the intracorneal locally injectable anesthetic, and injected simultaneously or sequentially in the corneal pocket.

In one or more embodiments, the pocket is filled with a biocompatible implant or implants (if more than one implant is provided) made of organic or non-organic material, or a mixture of it, and the implant is used for drug delivery. The implant may further be coated with a biocompatible material, such as collagen, elastin, polyethylene glycol, biotin and streptavidin, etc., as known in the art, or a composition thereof, to make the implant more biocompatible. The implant and/or the coating can be cross-linked with a cross-linker with the desired thickness and shape before or after implantation.

In one or more embodiments, the diameter of the corneal pocket can be 0.1 to 4 millimeters (mm), as needed. Only flat implants need a larger space with more than 0.2 mm. As shown in FIG. 38A-45B, the pocket can be circular, semi-circular, C-shaped, doughnut-shaped, rectangular, or any other shape.

In one or more embodiments, the implant or implants (if more than one implant is provided) can be located at a desired distance from the Bowman's membrane or from the corneal periphery, that is located away from the center of the visual axis (i.e., the implant may be off-centered, or ring-shaped in the peripheral cornea). See, for example, FIGS. 38A-45B.

In one or more embodiments, the implant or implants (if more than one implant is provided) is made to the desired shape, and size in diameter and length that fits with ease inside the corneal pocket without exerting pressure on the corneal tissue (i.e. without bulging it).

In one or more embodiments, a photosensitizer or cross-linker, such as riboflavin, is injected at the desired concentration in a biocompatible fluid or a viscous fluid prior to the implantation of the implant. However, it can be also administered simultaneously with the implant in the corneal pocket sufficiently to cover the internal wall of the pocket for a

desired duration so that it penetrates at least 20 micron or wider, taking 5-30 seconds after injection prior to the cross-linking of the cornea, which prevents cell proliferation, encapsulation, or rejection of the implant while preserving an acellular barrier.

In one or more embodiments, ultraviolet (UV) radiation at the desired power (e.g., 1 to 4 mW/mm²) and duration of 1-15 minutes, as needed, depending on the concentration of the photosensitizer or other radiation if another cross-linker is used (e.g., visible or infrared (IR) or another wave length) is applied externally to activate the photosensitizer in the corneal pocket, and to cross-link the collagen of the corneal stroma surrounding the corneal pocket, thereby killing only the cells located within the cross-linked cornea while preventing encapsulation of the drug implant while providing a physical stability to the cornea and preventing the wall of the pocket from adhering together or to the implant. This permits the implant to be replaced, if needed, with another implant with ease.

In one or more embodiments, the implant is coated with an organic material, such as collagen, dipped in a photosensitizer, or the implant can be coated with nanoparticles of the photosensitizer and implanted in the corneal pocket and ultraviolet (UV) radiation is applied with the desired power and duration using a painting technique using a small diameter fiber optic or other radiation with another wave length is applied if another cross-linker is used, externally or internally inside the pocket via a fiber optic to activate the photosensitizer in the corneal pocket and to cross-link the collagen of the corneal stroma surrounding the corneal pocket, thereby killing all cells located within the cross-linked cornea and cross-link the implant simultaneously. The corneal cross-linking prevents implant encapsulation with fibrous tissue, but provides a physical stability to the cornea without gluing the wall of the pocket together or to the implant.

In one or more embodiments, an injection of a small amount of hyaluronic acid in the pocket simplifies insertion of the drug implant in the corneal pocket.

In one or more embodiments, the drug implant has a tube-like structure with a size of 0.01 to 3 micron diameter holes in its wall, or having one micron or larger-sized holes for diffusion of fluid across it.

In one or more embodiments, the implant can be silicone, acrylic, methacrylate, hydroxyethyl methacrylate (HEMA), cross-linked organic or any other biocompatible transparent or non-transparent material, metallic or non-metallic, or a mixture thereof or coating other polymers, such as collagen or elastin with the desired thickness of 2 microns or more, as needed.

In one or more embodiments, the implant is made of various drug delivery polymers, such as polylactic acid or polyglycolic acid, or a combination thereof or polycaprolactone, or chitosan or other organic materials that can deliver the medication at a certain concentrations and dissolve within time ranging from 3-12 months or more.

In one or more embodiments, the biodegradable or non-biodegradable implant can be replaced with another one as before or a non-biodegradable material, but having biocompatible material or coating where the drug release occurs either through the small holes in the body of the implant at a certain rates depending on the size of the holes, or from one or both ends of the implant for drug delivery, as needed.

In one or more embodiments, the implant is a porous biodegradable polymer.

In one or more embodiments, the material inside the tubular implant is liquid, nanoparticles, suspension, powder, porous polymeric drug, etc.

In one or more embodiments, the implant is made using 3-D printing technology to the desired shape, size and/or coated with more biocompatible polymer(s) and cross-linked prior to the implantation, or it is implanted in a cross-linked pocket.

In one or more embodiments, the cross-linked corneal implant can be loaded with one or multiple medications needed for a short biocompatible drug delivery, or prophylactically to prevent an infection, or other used therapeutically medications to treat a disease process (e.g., inflammation, intraocular pressure (TOP), neovascularization, infection, or a cytokine, etc.).

In one or more embodiments, an organic cross-linked material can be used as above for a short term drug delivery of 1 to 4 weeks.

In one or more embodiments, an organic cross-linked material can be used as above for a short term drug delivery of 5 to 50 weeks or longer.

In one or more embodiments, the implant is a C-shaped flexible or semi-flexible structure, and can be implanted in the prepared corneal pocket according to the size or the shape of the implant (e.g., centered around the visual axis having a string shape, rod-like shape, or flat shape), while removing a small 3-D tissue from the stroma for the pocket formation to provide space for the implant for drug delivery to the cornea or the anterior chamber, trabecular meshwork, conjunctiva, or diffusing toward the posterior segment, such as the retina, choroid or the optic nerve of the eye. As shown in FIGS. 29A-29D, the drug delivery implant may be rod-shaped 610, C-shaped 612, two-part semi-circular 614, or one-part semi-circular 616. Also, as illustrated in FIG. 34, the implant may also be in the form of a rectangular flat tube 634. In FIGS. 38A and 38B, a two-part semi-circular drug delivery implant 656 disposed in a cross-linked pocket in the peripheral portion of the cornea 652 that is spaced apart from the central visual axis 654 of the eye so as not to obstruct the central portion of the eye. As shown in FIG. 38B, the two-part semi-circular drug delivery implant 656 is disposed adjacent to the anterior chamber 657 of the eye, and anteriorly with respect to the iris 653 and lens 655 of the eye. In FIGS. 39A and 39B, a generally linear drug delivery implant 660 is disposed in a cross-linked pocket in the peripheral portion of the cornea 652.

Turning to FIGS. 41A and 41B, it can be seen that an eye generally includes a lens 663, an iris 665, cornea 666, an anterior chamber 667, a pupil 668, and a limbus 670. In FIGS. 42A and 42B, a one-part semi-circular drug delivery implant 672 is shown disposed in a cross-linked pocket in the peripheral portion of the cornea 666. In FIGS. 43A and 43B, a doughnut-shaped or ring-shaped drug delivery implant 674 is shown disposed in a cross-linked pocket in the peripheral portion of the cornea 666. In FIGS. 44A and 44B, a generally linear drug delivery implant 676 is shown disposed in a cross-linked pocket in the peripheral portion of the cornea 666.

In one or more embodiments, the implant is inserted in the corneal pocket through a small external incision made into the corneal pocket.

In one or more embodiments, the pocket itself can be filled with biodegradable nanoparticles for drug delivery to the entire ocular structures from the cornea to the optic nerve, and all tissues in between. The medication(s) can be anti-inflammatory, anti-infective, immune-suppressants, AntiVEGFs, biologics, Anti-PDGF, Anti IL-6, Rho kinase

inhibitors, nerve growth factors, anti-glaucoma medications, gene(s) delivery in conjugation with viral or non-viral nanoparticles, such as nanoparticles, quantum dots, biodegradable polymers, etc. coated with polyethylene glycol (PEG) or cell penetrating agents along with an antibody to the specific tissue. This permits the genes or medications to be delivered after their migration out of the implant and the corneal pocket and to attach to the targeted cells in the cornea, conjunctiva, trabecular meshwork, retinal ganglion cells or photoreceptors, retinal and optic glial or nerve cells or their axons etc.

In one or more embodiments, the one or more medications in the drug implant may be anti-inflammatory agents, such as steroids, Dexamethasone, NSAIDs, Anti IL-17, Anti IL-6 and other Anti-ILs or antibiotics, fluoroquinolones, macrolides, cephalosporin A, vancomycin, aminoglycosides, penicillin and its derivatives or combination of antibiotics, etc., anti-virals, ganciclovir, valganciclovir, etc., anti-fungals, amphotericin B, etc., Anti-VEGFs, Avastin, lucentis, Aflibercept, Anti-IL-6, anti-parasitic, etc., or other anti-inflammatory agents, such as NSAIDs after any corneal surgery and act therapeutically to various diseases affecting the conjunctiva (e.g., dry eye), immune-suppressants, such as cyclosporine A, Mycophenolic acid, anti-proliferative agents, anti-metabolite agents, in uveitis, choroiditis or other medications, such as anti-glaucoma medication or combination of medications, gene delivery, DNA, RNA, siRNA etc. along with viral or non-viral delivery vehicles and CRISPR-cas9 mediated homology-independent targeted integration (HITI) or homology directed repair (HDR) to modify the genetic components of various diseases of the eye.

In one or more embodiments, repeated crosslinking of the pocket can be performed as needed to prevent new cellular ingrowth and adhesion around the implant from the corneal tissue so that the implant's barrier is maintained, and the implant can be removed or replaced as needed (e.g., if the eye needs another or a combined medication to regulate disease process, such as age related macular degeneration, glaucoma, uveitis, choroiditis or an infectious process of any origin).

In one or more embodiments, the peripheral cross-linked pocket is used to insert or inject medications needed to treat a corneal disease or glaucoma or a disease of the posterior segment. The medication can be in a form of nanoparticles, microspheres, lipid coating or PEG, streptavidin, biotin coating, etc., micelles, liposomes, thermosensitive chitosans, etc.

In one or more embodiments, one can inject or implant in the peripheral corneal pocket large-sized flexible, semi-solid or porous or solid rod, flat or tube or any shape and size polymeric material that can be absorbed with time and the medication is released slowly to the cornea or the anterior chamber of the eye or diffuses to the back of the choroid or retina and optic nerve.

In one or more embodiments, the diameter of these rod or flat-shaped shape implants can vary between 10 microns to 1 millimeter (mm) in diameter or larger with a length of 1 to 50 mm or longer.

In one or more embodiments, the porous tube can be made of semi-permeable non-biodegradable material that permits only the diffusion of the fluid/medication, etc. in and out of the tube, implanted in the peripheral cross-linked pocket. In these one or more embodiments, the tube can be refilled with medication as needed. For example, as shown in FIG. 36, the implant 640 in the form of a rectangular tube is refillable by injection with a needle 642.

In one or more embodiments, the drug implant tube contains stem cells, embryonic stem cells, ciliary hormone producing cells, or other hormone or factors producing stem cells, neuronal or glial stem cells, Mesenchymal stem cells, trabecular meshwork stem cells, limbal stem cells, modified skin stem cells, etc. in a biocompatible fluid that permits nutrition to reach the cells injected in the tube where the cells are immortalized to produce one or the other medication, growth factors, such as ciliary neurotrophic growth factor, RPE growth factor, nerve growth factors, anti-VEGFs, or other medications needed.

In one or more embodiments, the non-biodegradable tube with pores for drug and cell delivery is implanted in a cross-linked pocket with an implant in any part of the body for medication and cell delivery for various medications and functions.

In one or more embodiments, the implant is coated with biocompatible polymer(s) that is used for delivery of stem cells with medication in a corneal pocket. The implant has larger diameter holes of 5 microns and more in its wall permitting the cells to escape from the tubular implant into any tissue (e.g., corneal pocket containing stem cells, embryonic stem cells, ciliary body factor producing stem cells, neuronal or glial stem cells, Mesenchymal stem cells, trabecular meshwork stem cells, Limbal stem cells, modified skin stem cells, etc. in a biocompatible fluid) that permits nutrition to reach the cells injected in the tube where these cells can grow and pass through the holes of the implant and move toward a tissue. In FIG. 32, the tubular implant 626 has small holes 628 disposed in the circular peripheral side thereof, whereas the tubular implant 630 in FIG. 33 has large holes 632 disposed in the circular peripheral side thereof.

In one or more embodiments, the implant contains stem cells, embryonic stem cells, ciliary body hormone producing stem cells, neuronal or glial stem cells, Mesenchymal stem cells, trabecular meshwork stem cells, limbal stem cells, modified skin stem cells, etc. in a biocompatible fluid that permits nutrition to reach the cells injected in the tube along with Rho kinase inhibitors or Nerve growth factors to stimulate their regeneration and migration of these cell and repair the pathology in the tissue.

In one or more embodiments, genetically modified cells are used to produce needed enzymes and medications. The combination of cross-linking of the cornea and killing the corneal cells and lack of vessels in the cornea makes it a suitable place for these cells in the tube implant to survive inside the tube without being attacked by the cellular body's response, thereby creating an immune privileged space for these cells to survive and produce medications needed locally or systemically (e.g. in many genetic diseases of the cornea such as Fuchs dystrophy, etc.).

In one or more embodiments, the pocket can be filled with a polymeric material that can become more semisolid, or becomes a gel, and contains medication for slow release of medication.

In one or more embodiments, the medication can be injected in the peripheral pocket along with corneal stem cells from the limbus or genetically modified skin stem cells, embryonic or pluripotent stem cells, or mesenchymal stem cells grown in the culture for implantation, in cases of cell loss of endothelium, or in genetically caused corneal opacification, such as macular dystrophy or trauma.

In one or more embodiments, the stem cells are mesenchymal stem cells injected in the corneal pocket along with ROCK inhibitors to replace or repair a cloudiness of the cornea.

In one or more embodiments, the stem cells are nerve cells to induce regeneration of the damaged corneal nerve (e.g., in diabetic patient) and after traumatic corneal injuries or after LASIK surgery.

In one or more embodiments, all tubular drug implants or devices are replaceable with ease.

In one or more embodiments, the tube can be refilled with medication to be used as slow release drug delivery that releases the drug in the cornea and anterior of the posterior segment of the eye.

In one or more embodiments, the tube is used for taking fluid samples from the eye.

In one or more embodiments, one creates an intrastromal corneal pocket in the peripheral cornea involving 2-4 mm 4-8 mm in width areas involving the cornea and the anterior sclera after bleaching out the peripheral conjunctival capillaries with a low dose of vasoconstrictive medication such as 0.5%-4% phenylephrine applied locally with a Q-tipped applicator using a femtosecond laser.

In one or more embodiments, a small knife can be used to create a pocket in the cornea or elsewhere under the skin etc. if needed.

In one or more embodiments, the pocket width is more toward the corneal side than the scleral side or vice versa. The circumferential extent of the pocket can be 1 degree to 360 degrees of the corneal periphery (see FIGS. 38A, 38B, 42A, 42B, 43A, and 43B).

In one or more embodiments, using a small incision to access the intrastromal incision, one uses a curved probe to separate the corneal adhesion for injection of a photosensitizer cross-linker, such as riboflavin, at the desired concentration of 0.5%-4% in a biocompatible fluid, such as a physiological saline solution, etc. or suspension of particulates in a volume of 0.01 milliliters (ml) to 1 milliliter (mm) as needed only for the extent of the corneal pocket to cover the internal walls of the pocket for a desired duration that the photosensitizer penetrates at least 20 microns and beyond the corneal pocket in the corneal stroma to isolate that localized area of the cornea from the rest so that it does not respond with cell migration into the surrounding implant and so that it avoids tissue bounding together or to the implant.

In one or more embodiments, 0.01 ml to 0.1 ml of 0.02-2% lidocaine or bupivacaine solution can be injected alone or along with the photosensitizer in the corneal pocket to anesthetize the cornea for the next 1-15 hours, thereby eliminating pain sensation or discomfort of the surgery, and dry eye after surgery.

In one or more embodiments, the width of the corneal pocket can be 1-3 mm as needed. The peripheral corneal pocket can be circular, semi-circular, C-shaped, doughnut-shaped, straight, curved, or any other shape.

In one or more embodiments, the cross-linked pocket can be located at a desired distance from the Bowman's membrane in the corneal periphery.

In one or more embodiments, the ultraviolet (UV) radiation or other appropriate wavelength of light at the desired power 0.5-50 mW/Cm² and duration of 1-15 minutes, or other radiation with another wave length is applied externally in a stationary pattern or as a continuous painting/oscillatory technique with a focused small sized spot of 1-4 mm and a high energy to cover the width of the pocket, or on a painting oscillatory fashion entering the corneal pocket with a small diameter fiber optic and to activate the photosensitizer in the corneal pocket and crosslink the collagen of the corneal stroma surrounding the corneal pocket, and preventing the wall from adhering to itself or to a future

implant, while providing a physical stability to the wall of the corneal pocket and preventing cell migration and rejection of an implant.

In one or more embodiments, ultraviolet (UV) radiation at the desired power in a stationary or focused light for a duration of 10 seconds to 15 minutes for the stationary radiation, or for a duration of 10 seconds to 20 minutes for the painting approach, depending on the power of the radiation and the length of the pocket used (or other radiation with another photosensitizer and wave length) is applied externally or via a fiber optic inserted inside the pocket to activate the photosensitizer and cross-link the collagen of the corneal stroma surrounding the corneal pocket while preventing cell migration, encapsulation, or rejection of the implant and protecting the anterior corneal stroma and the stem cells.

In one or more embodiments, the corneal pocket is three-dimensionally cut in order to remove a part of the stroma and create a space for the implant.

In one or more embodiments, the wall of the corneal pocket can absorb the photosensitizer from the implant after it is dipped in a photosensitizer solution or the implant is coated with nanoparticles of the cross-linker and placed in the corneal pocket to leak out, which is then followed by UV radiation at the desired power and duration or other radiation with another wave length applied externally or internally via a fiber optic to activate the photosensitizer in the corneal pocket and cross-link the collagen surrounding the implant. This technique provides a physical stability to the cornea preventing adhesion or gluing the implant to the surrounding tissue and preventing fibrous ingrowth or encapsulation or rejection of the implant, which can lead to implant rejection. This makes it possible to exchange the implant when needed without much trauma to the cornea surrounding the implant.

In one or more embodiments, the photosensitizer is conjugated to the surface of the implant having a polymeric coating, such as collagen, that releases the photosensitizer (e.g., riboflavin) from the implant once it is exposed to the water content of the tissue in the corneal pocket surrounding it. The riboflavin is released and stains the wall of the implant which is subsequently cross-linked with UV light. This prevents tissue adhesion between the implant and the corneal tissue and maintains a potential space between the corneal wall and the implant, thereby preventing activation of an immunologic response or neovascular tissue response by releasing from the tissue vascular endothelial cell factors (VEGF) in response to a foreign implant. The cross-linking process can be repeated as needed every 6 months to a year or more as needed. The cross-linking of the collagen protects the implant containing particulate medication(s), which releases the drug for a long time, and prevents the pocket from being invaded by the immune cellular elements and keeps the lumen of the tube shaped implant open.

In one or more embodiments, during the cross-linking, the corneal pocket remains pristine not allowing cell traffic or access to the pocket surrounded by the cross-linked amorphous collagen or other cross-linked tissues located in that area.

In one or more embodiments, the crosslinking can be repeated again in the postoperative period after implantation by injecting a cross-linker in the corneal pocket through a needle inside the wall of the pocket, which diffuses readily through the potential space around the implant and the wall of the pocket, and then is irradiated with UV light from the outside.

In one or more embodiments, the implant can be made of silicone, acrylic, methacrylate, HEMA, metallic or non-

metallic, synthetic, organic, polymeric biodegradable, etc., coated with another or a biocompatible polymeric materials or a mixture thereof or coated with, for example, collagen or elastin, formed with a desired thickness of 2 microns to 100 microns, and conjugated with a cross-linker or the cross-linker is injected in the potential pocket space in the tissue and is cross-linked.

In one or more embodiments, the implant is made by the use of 3-D printing technology with the desired material, shape, size or thickness, transparent or non-transparent organic or non-organic or a mixture of them, a material such as collagen elastin, synthetic polymers can be coated again with riboflavin nanoparticles with one or more biocompatible polymer(s), and cross-linked with UV light prior to or preferably after implantation.

In one or more embodiments, the implant is coated with a collagen polymer to a desired thickness or in combination with another polymer, such as polyvinyl alcohol, chitosan, polycaprolactone, etc., conjugated with riboflavin or another cross-linker and cross-linked before or after implantation in a preformed pocket with an appropriate wavelength of light or UV radiation to cross-link the polymeric coating inside the body allowing the cross-linker to be released in the tissue, and then cross-link the tissue surrounding the implant in order to, after implantation, release the incorporated medication from the implant slowly without inciting cellular attraction or encapsulation of the implant which inhibits a release of the medication(s) from the implant that is unpredictable. For example, as shown in FIG. 30, the implant 618 is coated with a polymer and/or a photosensitizer.

In one or more embodiments, the non-biodegradable flexible porous tube made of polymeric material or a non-organic compound in combination with cross-linked organic polymer coating is filled with microspheres, drug nanoparticles incorporated in a polymeric material, such as polylactic glycolic acid, chitosan, liposomes, polycaprolactone, or lipid-coated nanoparticles, etc. containing the medication so as to release the medication slowly from the tube implant.

In one or more embodiments, the implant can serve as a reservoir that releases the medications through the pores of 1 to 3 microns in diameter in its wall, and then can be refilled repeatedly by injecting in the tubular implant the medication through a 33-34 gauge needle through the cornea surrounding the tube.

In one or more embodiments, the implant releases immunosuppressive agents, such as cyclosporine, calcineurin inhibitors, mycophenolic acid, tacrolimus, siraliums, steroids, MPP inhibitors, NSAIDs, antimetabolites, polyclonal antibodies, monoclonal antibodies, TNF inhibitors, Fingolimod, antibiotics, intraocular pressure (TOP) lowering agents, such as Rho kinase inhibitors, Fasudil, and other agents, pilocarpine, prostaglandin analogues, Brimonidine, etc., anti-virals, Anti-VEGFs, biologics, or neuroprotective releasing medication. The medications being released as needed at concentrations of nanograms or micrograms or mg/per hour depending on the polymeric material size of the holes, length of the polymer, etc.

In one or more embodiments, the implant can be positioned at any place in the body to control a function or release a medication without being encapsulated by the surrounding tissue, due to the cross-linking of the polymeric coating or the pocket being cross-linked prior to the implantation, while the medication can be an anti-VEGF, neuroprotective agents, such as nerve growth factors, Rho kinase inhibitor such as Fasudil, antibiotics, antiproliferative agents, anti-inflammatory agents, etc. at a non-toxic, beneficial concentration.

In one or more embodiments, the implant is made using 3-D printing technology to the desired shape, size or thickness from any material coated with collagen, elastin, or made of collagen, elastin, etc. or synthetic polymers which are further coated with more biocompatible polymer(s), such as acrylic, organic, etc., which are cross-linked prior to the implantation or coated with a cross-linker or the crosslinking nanoparticles are done subsequent to its release in the tissue prior to radiation with the UV light. In another embodiment, the implant is formed from glass using 3-D printing technology (i.e., the implant is 3-D printed glass).

In one or more embodiments, the implant is implanted in another part of the eye, such as under the conjunctiva, under the sclera, in the retina or sub-retinal space, under the skin using an implant containing medications such as Botox, or in other parts of the body using an implant which is coated with collagen to a desired thickness, dipped in a photosensitizer or has photosensitizer nanoparticles, such as riboflavin, etc. or the photosensitizer is injected surrounding the implant and implanted in desired location, such as under or over the sclera in the choroid, under the conjunctiva, etc. Then, ultraviolet (UV) radiation or another wavelength of light is used to cross-link the tissue at the desired power and duration depending on what technique is used. In these conditions, a focused UV or another wavelength of light is applied externally, in a painting oscillatory fashion only to the desired areas or internally through a fiber optic, etc. to activate the photosensitizer in the surrounding tissue where the implant is located. The cross-linked collagenous tissues surrounding the implant prevent creating an adhesion between the tissue and the implant or gluing the wall of the pocket together or to the implant. The cross-linked collagenous tissues surrounding the implant also have these additional benefits: (1) it is easier to replace the implant if needed, (2) fibrous ingrowth or encapsulation is prevented, (3) it permits injection of the cross-linker again to repeat the cross-linking process if needed, and (4) it prevents rejection of the implant and contributes to the slow release of the medication from the implant. Also, these implants can act as a shunt for glaucoma, or drainage shunt for cerebrospinal fluid, or other part of the body, such as a bladder neck for urine if the drainage system is provided with a unilateral valve that only opens when the bladder pressure increases to certain level, etc.

In one or more embodiments, the injection of a small amount of hyaluronic acid or other viscous fluid in the pocket simplifies the inserting of the implant in the peripheral corneal pocket or a pocket created in another tissue.

In one or more embodiments, the implant can be a biodegradable polymer carrying various medications and can be replaced.

In one or more embodiments, the implant is a tube-like structure having a thickness or diameter of 0.02 millimeters (mm) to 0.4 millimeters (mm) in one direction and up to 8 mm in another (flat) width, and being 1-60 mm long covering the entire corneal periphery without pressing the corneal tissue in any direction. The implant may be filled with a medication(s), a fluid, or a combination of them.

In one or more embodiments, the tube is not biodegradable having holes made in the wall of the tube with 0.2 to 3 microns in diameter, or 5 microns to 500 microns in diameter, to permit diffusion of the medications or cells placed in it to produce desired needed proteins, hormones, nerve growth factors, or other products needed for other body cell survival, such as cornea, retina, brain, etc.

In one or more embodiments, the tube has holes that are 5 to 15 microns in diameter so as to permit stem cells to exit

the tube. The tube can be biodegradable implanted in a lightly cross-linked corneal pocket permitting, for example, stem cells to proliferate and/or migrate to the cornea. The stem cells can be obtained from limbal stem cells or mesenchymal stem or skin and cultured cells prior to the injection in the cornea or in another part of the body.

In one or more embodiments, the device is implanted in the wall of the vitreous cavity with one end closed and one end open to the vitreous cavity, or the implant can be under the retina or it can penetrate both the retina and the choroid and permit release of medication or the cells.

In one or more embodiments, the implant is implanted in the tissue surrounding the eye, on the face, etc. with one end closed and one end open to the tissue. The implant can be removed after the drug is released, and then replaced.

In one or more embodiments, the repeated crosslinking of the tissue surrounding the pocket can be performed as needed to prevent cellular ingrowth, and the implant can be removed and replaced as needed (e.g. in age related macular degeneration) to maintain delivery of the anti-glaucoma medication, anti-VEGFs, immunosuppressive or anti-inflammatory agents, or nerve growth factors, or Rho kinase inhibitors.

In one or more embodiments, the peripheral cross-linked pocket is used to insert or inject medications needed to treat a corneal disease, glaucoma, or a disease of the posterior segment. The medication can be in a form of nanoparticles, microparticles, micelles, liposomes, chitosans, polycaprolactone as nanoparticles, dendrimers, etc.

In one or more embodiments, one can inject or insert an implant in the peripheral corneal pocket that is in the form of a large-sized flexible, semi-solid or solid, porous or solid rod-shaped implant, a flat implant, or tube-shaped implant that contains medication, or any shape and size polymeric material that can be absorbed with time and the medication is released slowly to the cornea or the anterior chamber of the eye or diffuses through the anterior chamber or through the sclera to the back of the eye, for treatment of the choroidal or retina and optic nerve diseases. As shown in FIGS. 31A-31C, the implant may be in the form of a solid implant 620, a porous implant 622, or a solid tubular implant 624 with an open end. Also, as shown in FIG. 35, the implant may be in the form of a semi-solid or silicone tubular implant 636 with one closed end 638 and one open end 639.

In one or more embodiments, the diameter of the rod or flat-shaped implant can have a length of 1 microns to a few millimeters (mm), or the length can be 1 to 40 millimeters (mm) or longer.

In one or more embodiments, the non-biodegradable tube is open-ended so that the medication exits only from one or both ends of the tube.

In one or more embodiments, the medication can be released for a duration of from 3 months to 3 or more years, such as when containing nanoparticles of fluoroquinolone dexamethasone, diclofenac, etc., and the implant can be replaced or removed if the desired effect has been achieved or reinserted in the corneal pocket.

In one or more embodiments, the tube is closed ended, but has pores for diffusion of the medication. For example, refer to the implants 626, 630 in FIGS. 32 and 33.

In one or more embodiments, the implant can be placed near any joint in the body and the cross-linking is done using ultraviolet (UV) radiation through the skin or through the fiber optic as described for localized drug delivery.

In one or more embodiments, the porous tube can be made of semipermeable non-biodegradable material that permits only the diffusion of fluid/medication, etc. in and out of the

tube, and the tube is implanted in the peripheral cross-linked corneal pocket, wherein the tube can be refilled with medication as needed via an injection using a 33-34 gauge needle. For example, refer to FIG. 36.

In one or more embodiments, the tube contains cells in a biocompatible fluid that permits nutrition to reach the cells which are injected in the tube where the cells are immortalized to produce one or more medications, growth factors, such as a ciliary neurotrophic growth factor, RPE growth factor, nerve growth factors, anti-VEGFs, or other medications needed.

In one or more embodiments, the implant contains genetically modified cells producing other needed enzymes and medications. The combination of crosslinking of the cornea produces a wall of amorphous, acellular collagen and the corneal location that lacks vessels provides a suitable place for these cells in the tube implant to survive and produce medications as needed, which otherwise would have to be given repeatedly either locally or systemically, and in many genetic diseases of the cornea, such as Fuchs dystrophy, the cells have to be injected in the subconjunctival space where the cells could be attacked by the normal cellular body's immune response. The cross-linked pocket with the implant creates an immune-privileged space in the cornea or elsewhere for these cells to survive. For example, refer to FIGS. 38A-45B.

In one or more embodiments, the medication in the implant can be in any form or composition, such as antibiotics, anti-inflammatory, immune suppressants, anti-glaucoma medication, anti-vascular proliferation, stimulatory, such as Rho inhibitors. The polymers can be made of bio-degradable compounds, such as polylactic, polyglycolic acid or a combination of them, polycaprolactone, etc.

In one or more embodiments, the corneal cross-linked pocket contains a tubular implant filled with particulate immunosuppressive agents, such as cyclosporine etc., that release the medication at a constant, but low concentration of micrograms as needed. The medication diffuses in the cornea, sclera, and/or conjunctiva, thus eliminating the burning sensation of topical cyclosporine drops and the need for daily drop administration in dry eye syndromes, or after refractive surgery or other diseases.

In one or more embodiments, the medication can be injected in the peripheral pocket along with corneal stem cells taken from the limbus or genetically modified stem cells and grown in the culture for implantation.

In one or more embodiments, as shown in FIGS. 40A and 40B, a non-biodegradable implant tube 662 as described herein is implanted in the cross-linked corneal pocket of the cornea 652 of the eye with iris 653 and lens 655, and the implant tube 662 is connected to the anterior chamber 657 with the aqueous fluid via a thin 23-34 gauge needle 664, where biomarkers such as VEGFs, glucose, and analytes, etc. are present both inside the aqueous and the tube system made of soft silicone. Similarly, as depicted in FIGS. 45A and 45B, an implant 678 is implanted in the cross-linked corneal pocket of the cornea 666 of the eye with iris 665, and the implant 678 is connected to the anterior chamber 667 with the aqueous fluid via a needle 680. The implants 662, 678 can be penetrated with a 30-34 gauge needle from outside and the aqueous can be aspirated in a volume of less than 0.50 microliters repeatedly over a long period of time without causing a collapse of the anterior chamber. The volume of the anterior chamber is 25 times more than the sample fluid taken. The minimal amount of aqueous fluid withdrawn will be replaced by the eye in less than 10 minutes. This provides a means of obtaining easily a fluid

biopsy repeatedly from the eye without penetrating the entire cornea or the eye wall directly with the complication of iris or lens injury and retinal injury. The fluid sample can be examined in chronic disease processes, such as uveitis for biomarker of a disease, viral infection that persist in the eye long after the body has healed, such as Ebola, Zika, Herpes viruses or other viral diseases or non-viral infections that can be detected and treated appropriately. The biomarkers can be obtained from the implanted tube, and can provide valuable information on many metabolic diseases of the body or the eye, a systemic disease (e.g., Alzheimer disease), age related macular degeneration, glucose level, or other analytes (e.g., diabetes) in diabetic retinopathy and other slow progressive degenerative eye diseases, tumors, infection, uveitis, poisoning or drug overdose, etc.

In one or more embodiments, a plurality of implants are implanted in the cornea of the eye. In these one or more embodiments, each of the implants is used for a different purpose. For example, a first one of the implants may be in form of a corneal drug delivery implant used for delivering one or more medications to the eye, as described above. A second one of the implants may be used for taking liquid biopsies from a portion of the eye, as described herein (e.g., extracting a liquid biopsy of the aqueous fluid from the anterior chamber of the eye). A third one of the implants may be used for stem cell delivery and/or gene therapy in the manner described above. A fourth one of the implants may be used for measuring the intraocular pressure of the eye of the patient (e.g., intracorneal implant comprising a pressure sensor). That is, the fourth implant may contain a pressure sensor configured to measure an intraocular pressure of an eye and to output a signal based on the measured intraocular pressure of the eye, the pressure sensor configured to be implanted in a cornea of the eye; a processor operatively coupled to the pressure sensor, the processor configured to generate intraocular pressure data based upon the signal outputted by the pressure sensor; and a transmitter device operatively coupled to the processor, the transmitter device configured to transmit the intraocular pressure data generated by the processor to a remote receiver located outside of the eye, the transmitter device configured to be implanted in the cornea of the eye. In addition to the pressure sensor, the third implant may further comprise a needle configured to penetrate a posterior portion of the cornea of the eye, the needle configured to open into the anterior chamber of the eye so as to measure the intraocular pressure of the eye without obstructing vision through the central cornea.

In one or more embodiments, one can measure the amount of VEGF present in the aqueous providing information on the disease progression requiring treatment (e.g., anti-VEGFs or no treatment). Anti-VEGFs or another medication can be administered directly in the tube to reach the posterior segment avoiding repeated intraocular injection through the sclera, without having the risk of retinal detachment or lens injury. As another example, liquid biopsy of aqueous in a patient with diabetic retinopathy, where the retina is in need of treatment with the laser coagulation, provides the information regarding whether the disease process is under the control or not.

In one or more embodiments, for the first time one can obtain from the aqueous biopsy, instant information needed for the doctor to diagnose a disease process at the bedside and be able to follow the process over a long period of time with ease.

In one or more embodiments, nanoparticles carrying other medications can be delivered as slow release nanoparticles from the tube in the anterior chamber to treat glaucoma for

a long period of time, thereby eliminating the need for repeat therapy. These medications may include pilocarpine, prostaglandin analogues for treatment of glaucoma, Rho kinase inhibitors, or neuroprotective agents or Brimonidine, etc.

In one or more embodiments, the implanted tube is filled with desired medications, as described above, and is coated with collagen or albumin loaded with riboflavin particles that are diffused after implantation in the pocket. The ultraviolet (UV) radiation used for cross-linking permits the diffusing of the medication from the implant as a slow release device, and prevents vascular growth around the implant containing the medication.

In one or more embodiments, the implanted tube can be 100 microns to 1 millimeters (mm) in diameter and 4 mm to 40 mm long, or less than 100 micron in diameter and no longer than a few millimeters in length. The implanted tube maybe filled with any desired medication to be implanted in any tissue and cross-linked after implantation.

Methods disclosed hereinafter include administering Wnt inhibitors either alone, or in combination with Rho inhibitors, or one or more Wnt inhibitors with the dendrimers or liposomes, or (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) that are useful for alleviating the effects of conditions that are caused by acute or chronic inflammatory processes, such as chronic inflammatory dry eye disease, diabetes, optic nerve neuritis, scleritis, keratitis, chronic Meibomian gland inflammation, and uveitis.

In one embodiment, Wnt inhibitors or Rho kinase inhibitors with the dendrimers or liposomes, or as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) are used as topical drops, spray, ointment, gel, or a non-toxic injectable formulation under the conjunctiva, in the corneal pocket, or in the vitreous to treat the dry eye syndrome or mucosal inflammatory diseases, such as uveitis, optic nerve neuritis, diabetic macular edema, cystoid macular edema after cataract surgery, macular edema in uveitis, vasculitis, Behçet's disease, sarcoidosis, multiple sclerosis, lichen planus, chronic joint disease arthritis, chronic chorioiditis, plantar fasciitis, pars planitis, scleritis, iritis, and/or scleritis gingivitis.

A method of treating dry eye with deficiency of aqueous production which is associated often with the Meibomian gland disease, affecting about 7% to 34% of all Americans, pathophysiology of chronic dry eye disease including a cycle of inflammation involving both innate and adaptive immune responses is also disclosed herein.

In one embodiment, dry eye syndrome (DES) or keratoconjunctivitis sicca, a disease affecting tear production leading to damage to the corneal surface, associated often with disturbance of Meibomian gland, lachrymal gland, conjunctival goblet cells, nasolacrimal duct and pain sensation is treated by Wnt inhibitors or Rho kinase inhibitors used as topical drops, ointment, gel, non-toxic injectable formulation such as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the method used for treatment of the eye utilizes over the counter physiological saline solutions with some other components to affect the inflammatory component of the dry eye or improve on the composing of the tear film, such as tear film osmolarity, or adding lipids, mucin, etc. Other topical medication include TheraTears® (Advanced Vision Research), Refresh® and Celluvisc® (Allergan), Tears Natural® and Bion Tears® (Alcon), GenTeal® and HypoTears® (CIBA Vision), each of which

contain electrolytes and has varying pH levels, osmolarities, Restasis® (0.05% cyclosporine, Allergan), and Xiidra® (5% lifitegrast, Shire), which attacks the inflammatory process by a different mechanism than cyclosporine. Most of these medications are applied as a drop to maintain the conjunctival wetness as needed usually 1-3 drops during the day or ointment at night most of these medications may be used in combination with Rock inhibitors, such as Fasudil, or Wnt inhibitors.

In one embodiment, the Rock inhibitors not only reestablishes the tear production by reducing the conjunctival inflammatory cytokines and inflammatory response, but enhances the nerve fibers to grow and reestablish the function of conjunctival goblet cells to produce mucin, which is essential for tear film lubrication. RHO associated protein Kinase (Rock) is a kinase belonging to the family of serine-threonine Kinase involved in regulating the shape and the cytoskeleton of the cells, it is an important regulator of cell migration, stimulates PTEN phosphatase activity, leading to uncontrolled cell division in cancer. Rock is active in inflammatory processes, cancer, diabetes, and many neurodegenerative diseases such as dry or wet form of age related macular degeneration. Therefore, Rock inhibitors inhibit inflammatory processes.

In one embodiment, Rock inhibitors may be used in combination with functionalized nanoparticles of polycaprolactone, polylactic or polyglycolic acid, as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) etc. to reduce the inflammation during immune therapy or thermoimmune therapy.

In one embodiment, a potent ROCK inhibitor administered systemically, locally, as an implant, orally, topically as functionalized nanoparticles, microparticles, dendrimers or such as bioavailable Fasudil hydrochloride, inhibitor of cyclic nucleotide dependent- and Rho-kinases GSK 269962 is used. In one embodiment, potent and selective ROCK inhibitor GSK 429286, Selective Rho-kinase (ROCK) inhibitor H1152 dihydrochloride, or Botox, Wnt inhibitors, and brimonidine are used as an implant for slow release of the medication.

In one embodiment, selective Rho-kinase (ROCK) inhibitor Glycyl H 1152 dihydrochloride, or another selective Rho-kinase (ROCK) inhibitor is administered as a topical ointment, drop, or gel. Also, a more selective analogue of H1152, that is cell-permeable, a selective Rho-kinase inhibitor OXA 06 dihydrochloride, a potent ROCK inhibitor PKI1447 dihydrochloride, potent and selective ROCK inhibitor antitumor SB 772077B, a potent Rho-kinase inhibitor, vasodilator SR 3677 dihydrochloride, a potent, selective Rho-kinase (ROCK) inhibitor TC-S7001, a potent and highly selective ROCK inhibitor, orally active Y-27632 dihydrochloride or Botox also may be administered as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, ROCK inhibitors include, but are not limited to, Fasudil, Ripasudil, RKI-1447, Y-27632, GSK429286A, Y-30141, etc. They may be administered in a polymer, an implant, microparticles or nanoparticles, or porous silicone implant, etc. They may also contain poly(amidoamine) (PAMAM), poly(amidoamine-organosilicon) (PAMAMOS), poly(propyleneimine) (PPIO), poly(caprolactone), poly(lactic acid) (PLA), polylactic-co-glycolic acid (PLGA); may be tecto, multilingual, chiral, hybrid, amphiphilic, micellar, multiple antipen peptide, and Frechet-type dendrimers; may be functionalized microparticles or nan-

oparticles with an antibody and/or a ligand for a receptor or covalent coupling to one or more of cell penetrating peptides (CPP), arginine-CPP, cysteine-CPP, polyethylene glycol (PEG), biotin-streptavidin, and/or acetyl cysteine or as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, ROCK inhibitors include, but are not limited to, Fasudil, Ripasudil, RKI-1447, Y-27632, GSK429286A, Y-30141, etc. They may be administered in a suspension, or solution over the cornea or injected inside the corneal channel provided with a femtosecond laser as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the implant may be a C-shaped or ring-shaped flexible implant with a diameter of the implant of 20 microns to 200 microns or more, with a length of 1-12 mm and if it is circular with the diameter of 3-8 mm or less than the diameter of the lens capsule after cataract extraction without exerting any pressure or tension on the lens capsule or expanding it. In one embodiment, the polymeric material can be or porous silicone implant etc. They may also contain poly(amidoamine) (PAMAM), poly(amidoamine-organosilicon) (PAMAMOS), poly(propyleneimine) (PPIO), poly(caprolactone), poly(lactic acid) (PLA), polylactic-co-glycolic acid (PLGA); may be tecto, multilingual, chiral, hybrid, amphiphilic, micellar, multiple antipen peptide, and Frechet-type dendrimers; may be functionalized microparticles or nanoparticles with an antibody and/or a ligand for a receptor or covalent coupling to one or more of cell penetrating peptides (CPP), arginine-CPP, cysteine-CPP, polyethylene glycol (PEG), biotin-streptavidin, and/or acetyl cysteine or as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the implant is injected in the cornea or scleral pocket using a 20 gauge or 25 gauge or 27 gauge or 30 gauge needle.

In one embodiment, one or more corneal pockets 720 are produced in the corneal stroma 712 of an eye 710 with a femtosecond laser or a small knife (see FIG. 46) or one or more scleral pockets 718 are formed in the sclera 714 of the eye 710 or under the conjunctiva and the implant 724 carrying Rho inhibitors, Wnt inhibitors, Fingolimod, or antibiotics, is placed in the pocket 718 or 720. As shown in FIG. 46, a subconjunctival implant 722 may also be implanted in the eye 710 with optic nerve 716. In another embodiment, a femtosecond laser flap or a flap formed by a knife is created in the corneal stroma of an eye 726 as a LASIK flap 732 and after correcting the refractive error either with an excimer laser or an inlay, then an implant 734 is placed under the corneal flap 732 (see FIGS. 47-48). As shown in FIGS. 47 and 48, the eye 726 with lens 728, iris 730, and optic nerve 736 comprises a corneal drug delivery implant 734 disposed in the stromal tissue underneath the LASIK flap 732. FIG. 48 illustrates the eye 726 after the flap 732 has been replaced. In FIG. 48, the attached area of the flap 732 is represented by the dashed line 738, while the flap incision is represented by the solid line 740. In yet another embodiment, a corneal pocket is created with femtosecond laser and an inlay is placed inside the pocket to correct refractive error of the eye followed with implantation of an implant in the pocket for drug delivery and the implant carrying Rho inhibitors, Wnt inhibitors, Fingolimod, or antibiotics as nano- or microparticles with (alpha)-cyclodex-

trin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), etc.

In another embodiment, after refractive surgery, the polymeric medication is applied either as a drop **746** or spraying to the cornea **744** of eye **726** with sclera **742** (see FIG. **49**). In yet another embodiment, after refractive surgery or traumatic corneal injury, a soft polymeric contact lens **748** carrying medication is placed over the cornea **744** to protect it and treat it with medication (see FIG. **50**), and the implant carrying Rho inhibitors, Wnt inhibitors, Fingolimod, or antibiotics, Fasudil, brimonidine, etc. with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In another embodiment, the polymeric drug delivery implant **760** has a C-shape or has a ring or a curved shape and is placed in the lens capsule **754** of an eye **750** without expanding or putting tension or pressure on the capsule and the implant **760** carrying Rho inhibitors, Fasudil, or Wnt inhibitors, Fingolimod, or antibiotics, an agent to lower the intraocular pressure (TOP), NSAIDs, with as nano- or microparticles and/or with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), etc. The polymeric implant **760** can be placed, either before or after the intraocular lens (IOL) implantation. The drug delivery implant **760** never covers the view of the patient (see FIG. **51**). In FIG. **51**, it can be seen that eye **750** with lens zonules **752**, lens capsule **754**, iris **756**, and cornea **758** comprises an intracapsular C-shaped or ring-shaped drug delivery implant **760** that has been inserted into the lens capsule **754** after cataract surgery on the eye **750**.

In one embodiment, an implant **766** (e.g., a C-shaped or ring-shaped drug delivery implant **766**) is placed over the lens capsule **754** before the crystalline lens is removed or can be placed on the lens zonules **752** after cataract extraction and intraocular (IOL) lens **764** implantation (see FIG. **52**), and the implant carrying Rho inhibitors, or Wnt inhibitors, or Fingolimod, or antibiotics or an agent to lower IOP, such as prostaglandin analogues, NSAIDs, etc.). In FIG. **52**, it can be seen that eye **750** with lens zonules **752**, lens capsule **754**, iris **756**, optic nerve **762**, and retina **768** comprises an intracapsular C-shaped or ring-shaped drug delivery implant **766** disposed over the lens zonules **752**.

In one embodiment, the polymeric material is administered to the vitreous before or after removal of the vitreous for both visualization and delivery of medication to the eye, and the implant carrying either Rho inhibitors, or Wnt inhibitors, or Fingolimod, or antibiotics, NSAIDs, etc. as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the polymeric drug delivery carrier can be nanoparticles or microparticles or up to one mm in diameter injected as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) compound in the eye cavity or in the pocket created in the cornea or sclera or under the conjunctiva to place an implant. For example, as shown in FIG. **53**, nanoparticles or microparticles **774** carrying one or more medications may be injected into the vitreous cavity **770** of an eye **750** using a needle **772**. In FIG. **53**, it can be seen that eye **750** with lens capsule **754**, iris **756**, cornea **758**, optic nerve **762**, and retina **768** is being treated with nanoparticles or microparticles **774** carrying one or more medications that are injected into the vitreous cavity **770**.

In one embodiment, the implant is visible by the nature of the medication it carries or may be transplanted such as implants (if more than one implant is provided) and comprises one or more medications selected from the group consisting of immunosuppressive agents, calcineurin inhibitors, mycophenolic acid, tacrolimus, siraliums, steroids, MPP inhibitors, NSAIDs, antimetabolites, polyclonal antibodies, monoclonal antibodies, TNF inhibitors, Rho inhibitors, Wnt inhibitors, Fingolimod, antibiotics, Brimonidine, an intraocular pressure (TOP) lowering agents, pilocarpine, prostaglandin analogues, anti-virals, anti-VEGFs, biologics, neuroprotective medications, and combinations thereof as a preparation to penetrate the tissue and the cells using as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, aqueous tear-deficient dry eye, occurring as a result of not enough tears being produced due to a dysfunction of the lacrimal glands, is treated with Wnt inhibitors or Rho kinase inhibitors as topical drops, ointment, gel, or a non-toxic injectable formulations as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for tissue and cell penetration and slow release.

In one embodiment, the Wnt inhibitors compound that is used includes FH535, IWP-2, PNU-74654, IWR-1endo, IWR-exo, demethoxycurcumin, sulforaphane and vitamin D, CCT036477, KY02111, WAY-316606, SFRP, IWP, LGK974, C59, Ant1.4Br/Ant 1.4C1, ivermectin, niclosamide, apicularen and bafilomycin, XAV939, XAV939, G007-LK and G244-LM, NSC668036, SB-216763, gemtuzumab, and akinumab as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) because of their hydrophilic and hydrophobic nature.

In one embodiment, patients with moderate-to-severe dry eye having both elements of evaporative dry eye and aqueous tear-deficient dry eye, and that are on topical medications for other diseases, such as glaucoma, drops, or antibiotics containing preservative that over time damage the conjunctival goblet cells and other cells and induce dry eye syndrome, are treated with Wnt inhibitors or Rho kinase inhibitors as topical drops, ointment, gel, or a non-toxic injectable formulation or as nano- or microparticles with (alpha)-cyclodextrin, (beta)-cyclodextrin, (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) to release medication slowly without causing irritation of the conjunctiva or the cornea.

In one embodiment, administration of Wnt inhibitors, such as demethoxycurcumin, sulforaphane and vitamin D, or Rho kinase inhibitors, such as Fasudil derivatives, is done as topical drops, a gel, a non-toxic injectable formulation, or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or injectable Botox, 1-100 units as needed, administered locally at multiple locations with the same formulation having nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or Rock inhibitors molecules at doses of 1 picogram (pg) to nanograms to a few micrograms as a slow release delivery system.

In one embodiment, patients who are on topical medications for other diseases, such as glaucoma, drops or antibiotics containing preservatives and over time damage the conjunctival goblet cells and other cells and induce dry eye syndrome, or patients with dry eye and glaucoma are treated

either by implanting matrices polylactic acid or poly glycolic acid, polyanhydride, or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or chitosan polymers under the conjunctiva with slow release polymers containing either Wnt inhibitors or Rock inhibitors, such as Botox or Fasudil derivatives, releasing the medication over months or years locally at multiple locations to release the non-toxic doses of the medications from 1 picogram (pg) to 1 nanograms (ng) or more each day.

In one embodiment, patients who develop dry eye as a result of systemic medication, such as in cancer patients developing dry eye after administration of checkpoint inhibitors in cancer immune therapy, are treated either by Wnt inhibitors or Rock inhibitors with slow release polymers containing either Wnt inhibitors, such as demethoxycurcumin, CCTO36477, KY02111, WAY-316606, SFRP, IWP, LGK974, C59, Ant1.4Br/Ant 1.4Cl, nitazoxanide (NTZ) ivermectin, niclosamide, sulforaphane and vitamin D, or Rock inhibitors, such as Botox or Fasudil derivatives, etc., releasing the medication over months or years locally at multiple locations to release the non-toxic doses slow release medications from 1 picogram to 10 nanograms each day or more using nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the Sjogren syndrome is associated with low salivary flow, lymphocytic infiltration of the lacrimal gland and salivary gland auto antibodies in serum, rheumatoid factor, connective tissue diseases, such as Sjogren's syndrome, to the list of immune-related adverse events that can develop during cancer treatment with immune checkpoint inhibitors are treated with Rock inhibitors and Wnt inhibitors at non-toxic concentrations of sulforaphane and vitamin D, CCTO36477, KY02111, WAY-316606, SFRP, IWP, LGK974, C59, Ant1.4Br/Ant 1.4Cl, nitazoxanide (NTZ), ivermectin, niclosamide, or Rock inhibitors such as Botox or Fasudil etc., releasing the medication slowly over months or years locally at multiple locations to release the non-toxic doses slow release medications from 1 pg to 10 ng each day locally using as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for slow release and tissue penetration.

In one embodiment, the patients being treated have a dry eye syndrome unassociated with SS (non-SS KCS) with a sensation of foreign body in the eyes, photophobia, excessive tearing, ocular irritation and pain. Other symptoms are increased tear film osmolality, decrease in tear breakup time, increase in the conjunctival enzymes metalloproteinase 9 and 17, and changes in impression cytology of the conjunctival cells. These patients are treated with Rock inhibitors, such as injectable Botox, 1-10 units or in combination with metalloproteinase inhibitors doxycycline, low molecular weight heparin, lovenox, and dexamethasone at concentration of 0.1%-5% as polymeric implant or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, when inflammation is one of the mechanisms that causes damage to the ocular surface in dry eye disease seen in autoimmune diseases such as Sjogren's syndrome, and rheumatoid arthritis and neuropathic disorders, optic nerve neuritis, papillitis, scleritis, uveitis, inflammatory, infectious, chemical, traumatic diseases, etc., the patients are treated with injectable Rock inhibitors, such as Botox or Fasudil derivatives, conjugated with slow release

polymer, such as porous silicone, etc. releasing the medication over months or years locally at multiple locations as the non-toxic doses slow release medications months to a year at concentration of 1 picogram to 10 nanograms each day.

In one embodiment, the pathological conditions resulting in dry eye include pemphigus and Sjogren's syndrome, which affect the eye by either damaging the conjunctival cells responsible for maintaining the wetness of the cornea and the conjunctiva, or by damaging the lacrimal glands of the eye and/or the meibomian glands of the eye lid or other pathological conditions resulting in dry eye include hypolacrimation, alacrima, Stevens-Johnson syndrome, marginal blepharitis pemphigus, ocular pemphigoid, scleritis, or diabetes are treated with Rock inhibitor Fasudil, Botox, etc. at a picogram to nanogram concentration or in combination with metalloproteinase inhibitors, doxycycline 0.1%-5% solution, low molecular weight heparin 0.1%-5% solution, or dexamethasone 0.1-2% solution in combination with MTOR inhibitors at 0.1%-5% solution or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) solution or suspension.

In one embodiment, the dry eye of patients occurring in post-corneal surgery (including but not limited to post-LASIK surgery or PRK) with surgical damage to the corneal nerves, other conditions resulting in dry eye including the aging process, environmental factors (e.g., dry home and/or work environments), and extended use of visual display terminals (e.g., employment, recreation) are treated with Rock inhibitors, or in combination with metalloproteinase inhibitors, low molecular weight heparin, or Wnt inhibitors or Rock inhibitor, such as Botox, 1-100 units administered locally at multiple locations, small doses or Rock inhibitors molecule, Fasudil and its derivatives, etc., at doses of 1 nanogram (ng) to a few micrograms (μ g) as slow release polymer or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the dry eye can also occur after cataract surgery and refractive surgery (i.e., the LASIK procedure or PRK) and photorefractive keratectomy, smile procedure, partial or complete corneal transplants, which are the majorities of present refractive surgery where these procedures are performed, but dry eye is more common with LASIK where the superficial nerves are cut, and where the eye dries out because the corneal reflex is affected and the eye subsequent to these surgeries becomes dry while many eyes experience regeneration of the nerves, but it takes about one year or more to achieve it all. Patients with these conditions are treated with Rock inhibitors, or in combination with metalloproteinase inhibitors, low molecular weight heparin, or Wnt inhibitors or Rock inhibitors, such as Botox, 1-100 units administered locally over the cornea as drops 1-4 times daily or injectable preparation at multiple locations, small doses or Rock inhibitors molecule, such as Fasudil or its derivatives, etc., at doses of 1 nanogram or a few micrograms as slow release non-toxic preparation or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, patients with paresis or paralysis of the fifth or seventh cranial nerves causing dry eye as a result of interfering with proper lid closure are treated with Rock inhibitors, such as Botox, 1-10 units administered topically over the cornea at multiple locations, small doses or Rock

inhibitors molecule Fasudil, or its derivatives, etc. at doses of 1 nanogram to a few micrograms as topical ointment, drop, gel, etc.

In one embodiment, Wnt signaling is involved in the control of stem cell proliferation. Wnt mutation causes developmental defects in many disease processes including inflammation and cancer, wet form of age related macular degeneration, laser surgery, diabetic macular edema retinal vascular occlusion.

In one embodiment, the Wnt inhibitors compounds used are: FH535, IWP-2, PNU-74654, IWR-1endo, IWR-exo, nitazoxanide (NTZ), demethoxycurcumin, sulforaphane and vitamin D, CCTO36477, KY02111, WAY-316606, SFRP, IWP, LGK974, C59, Ant1.4Br/Ant 1.4Cl, Ivermectin, Niclosamide, apicularen and bafilomycin, XAV939, XAV939, G007-LK and G244-LM, NSC668036, SB-216763, gemtuzumab, akimab Wnt inhibitors administered as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, the oral doses for the Wnt inhibitor niclosamide is 1 to 2 g tablet once, or to repeat in 7 days, if needed.

In one embodiment, the small molecule Fasudil, a rock inhibitor Fasudil (HA-1077), a selective RhoA/Rho kinase (ROCK) inhibitor, or Y-27632, small molecule inhibitor of ROCK1 and ROCK2, Botulinum toxin a is rock inhibitor marketed under the brand names Botox, Dysport, Myobloc, Xeomin, etc. Botulinum toxin, all having good penetration into the cornea, and do not increase intraocular pressure or cause cataracts and may be dissolved in an organic solvent such as DMSO or alcohol or sterol, lanosterol, squalene, and/or squalamine, or containing a polyanhydride, poly(glycolic) acid, poly(lactic) acid, or polycaprolactone polymer or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for the treatment of the lid, conjunctiva, lacrimal gland corneal diseases and glaucoma.

In one embodiment, Fasudil is used as a single, oral 40-80 milligram (mg) dose orally as two 40 mg Fasudil tablets are administered.

In one embodiment, the methods include administering Wnt inhibitors, either alone or in combination with Rho inhibitors, orally, locally by injection or drops, spray or ointment for alleviating the effects of conditions that result in lack of moisture or wetness in the eye.

In one embodiment, Rho inhibitors, may be administered orally, locally by injection or drops, spray or ointment for alleviating the effects of conditions that result in lack of moisture or wetness in the eye, such as the inflammatory conditions resulting in dry eye including pemphigus and Sjogren's syndrome, which affect the eye by either damaging the conjunctival cells, or by damaging the lacrimal glands of the eye and/or the meibomian glands of the eye lid.

In one embodiment, the required treatment of Rho inhibitors such as Botox in 1-2 units, may be administered locally by injection or drops, spray or ointment or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for inflammatory processes resulting in dry eye including hypolacrimation, alacrima, xerophthalmia, Stevens-Johnson syndrome, pemphigus, ocular pemphigoid, marginal blepharitis, nerve pain, diabetes, and/or post-corneal surgery after cutting the corneal nerves (including but not limited to post-LASIK surgery or PRK or corneal inlay). Other conditions resulting in dry eye include the aging process, environmental factors (e.g., dry home and/or work

environments), and extended use of visual display terminals (e.g., employment, recreation, etc.).

In one embodiment, the required treatment of Rho inhibitors, such as Botox in 1-2 units, may be administered locally by injection or drops, spray or ointment or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for chronic inflammatory processes, lichen planus, arthritis, psoriasis, and plantar fasciitis.

In one embodiment, inhibition of Wnt signaling or ABC transporters by RNA interference may be a valuable therapeutic strategy in dry eye including hypolacrimation, alacrima, xerophthalmia, and Stevens-Johnson syndrome, pemphigus, where the Wnt inhibitors administered as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, a number of Rock inhibitors are used in non-toxic doses in combination with functionalized nanoparticles, conjugated with polymeric coating, such as chitosan, polyanhydride, cyclodextrin as a potent ROCK inhibitor; bioavailable Fasudil hydrochloride, inhibitor of cyclic nucleotide dependent- and Rho-kinases GSK 269962, potent and selective ROCK inhibitor GSK 429286, selective Rho-kinase (ROCK) inhibitor H1152 dihydrochloride, selective Rho-kinase (ROCK) inhibitor Glycyl H 1152 dihydrochloride, selective Rho-kinase (ROCK) inhibitor, more selective analogue of H1152, Cell-permeable, selective Rho-kinase inhibitor OXA 06 dihydrochloride, potent ROCK inhibitor PKI1447 dihydrochloride, potent and selective ROCK inhibitor, antitumor SB 772077B, potent Rho-kinase inhibitor, vasodilator SR 3677 dihydrochloride, potent, selective Rho-kinase (ROCK) inhibitor TC-S7001, potent and highly selective ROCK inhibitor, orally active Y-27632 dihydrochloride and may be dissolved in an organic solvent such as DMSO or alcohol or sterol, lanosterol, squalene, and/or squalamine, or in nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, or hydroxypropyl-b-cyclodextrin (bHPCD) or containing a polyanhydride, poly(glycolic) acid, poly(lactic) acid, or polycaprolactone polymer to reduce the inflammation processes in the eye, retina, choroid, sclera, lid, conjunctiva, or other mucosal diseases, or in the mouth.

In one embodiment, small molecule Wnt inhibitor PKF118-310, the Wnt/ β -catenin pathway inhibitor and Fasudil, a rock inhibitor Fasudil (HA-1077), a selective RhoA/Rho kinase (ROCK) inhibitor, or Y-27632, small molecule inhibitor of ROCK1 and ROCK2, etc. may be dissolved in an organic solvent such as DMSO or alcohol or sterol, lanosterol, squalene, and/or squalamine, or containing a polyanhydride, poly(glycolic) acid, poly(lactic) acid, or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or polycaprolactone polymer to release non-toxic medication slowly at desired concentration to the eye.

In one embodiment, early management includes, the use of lubricants, artificial tear substitutes, ointment, gel, or emulsion. Topical anti-inflammatory agents, topical rock inhibitors, anti-interleukin (IL1) TNF- α , TNF- α , hyaluronic acid, low molecular heparin 0.1-5% solution alone or in combination with, metalloproteinase inhibitors doxycycline 0.1-5% solution immunosuppressive agent or inhibitor mycophenyllic acid as local or systemic therapy.

In one embodiment, topical Rock inhibitors are applied to the cornea as drops or spray or subconjunctival injection as a slow release compound combined with chitosans or in

nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) in 0.1 microgram/ml to 40 microgram/ml or more for topical application.

In another embodiment, the Rock inhibitors are coated with the slow release polymers, such as lactic acid and/or glycolic acid at a concentration of 200 nanograms to 1 micrograms/ml or more and administered, topically, subconjunctival or inside the eye subcutaneously inside the plantar fascia in the joint, etc. using nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) for slow release and tissue penetration with or without cell penetrating agents.

In another embodiment, the Rock inhibitors are released from a polymeric explant or implant either placed over or under the conjunctiva and sutured to the sclera to release, e.g., Fasudil, etc. at concentrations of 0.01 microgram/ml to 40.0 microgram/ml or more per day.

In one embodiment, the Rock inhibitors release, after placement in the upper or lower cul-de-sack of the conjunctiva or as a slow release punctal plaque or implanted subconjunctivally, at a rate of 1 picogram to a 10 nanograms/day of the medication or delivered as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) injection.

In one embodiment, the Rock inhibitors release, after placement of an implant or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) in the suprachoroidal space, inside the eye, behind the eye, inside the gingiva, subcutaneously in plantar fascia, or as a slow release polymeric plaque or implanted to release medication at a rate of 1 picogram to a 10 nanograms/day of the non-toxic medication.

In another embodiment, the nanoparticles or dendrimers are conjugated with Rock inhibitors and chitosan delivered as a slow release system that can be released as a temperature sensitive polymer that melts at 42-43 degrees used with a warm compressor over or under the lid, or light thermal application, or the use of a compressive focused ultrasound applied to lid, conjunctiva or cornea or the lid releasing 1 picogram to a 10 nanograms/day of the medication.

In one embodiment, the Rock inhibitors or Wnt inhibitors are delivered as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) with simultaneous application of amniotic membrane and slow release nanoparticles applied post corneal surgery, such as LASIK, or corneal inlay cataract corneal transplant, or cataract surgery and lens implantation or any other corneal surgical intervention, scleritis, retinitis, vitreoretinal surgery, diabetic retinopathy, macular edema, at 10 picograms to 20 nanograms of medication per day.

In one embodiment, the Rock inhibitors or Wnt inhibitors are delivered with simultaneous application of amniotic membrane and low molecular weight heparin slow release nanoparticles applied post corneal surgery, such as LASIK, cataract, or cataract surgery and lens implantation, after retinal surgery, vitreous surgery, corneal transplant uveitis scleritis or chemical injury to the cornea or conjunctiva at concentrations of 0.001 micrograms/ml to 40 micrograms/ml or more or topical or subconjunctival Botox, at 1-100 units or topical at 1-5 units or more in a physiological solution of Botox, or similar preparations.

In one embodiment, the Rock inhibitors or Wnt inhibitors are delivered with simultaneous application of low molecular weight heparin (levonox) with other medications, such as tetracycline, Doxycycline or metalloproteinase inhibitors, dexamethasone 0.1%-1% concentration as slow release polymeric nanoparticles or liposomes applied post corneal surgery such as LASIK, PRK, cataract, corneal transplant, uveitis, scleritis or thermal or chemical injury to the cornea or conjunctiva, e.g., Fasudil derivatives, etc., at 0.1 micrograms/ml to 40 micrograms/ml or more, or Botox at 1-3 units as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) solution or suspension.

In one embodiment, after LASIK or cataract surgery and lens implantation or any refractive surgery or cataract surgery, Rock inhibitors at doses of 0.1 micrograms/ml to 40 micrograms/ml or more for topical application or Wnt inhibitor can be injected as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) in the corneal pocket or in the anterior chamber, or applied as drops in the post-operative period to replace prednisolone or other steroids, or NASIDs, and encourage regrowth of the cut neurons in the cornea.

In one embodiment, after LASIK or any refractive surgery or cataract surgery, Wnt inhibitors or, Rock inhibitors such as botulinum toxin (Botox) can be injected under the conjunctiva or applied as drops in the post-operative period to encourage regrowth of the cut neurons in the cornea after LASIK or other corneal surgery at doses of 1 to 10 units of Botox injected under the conjunctiva or 1-2 drops daily at concentration of 10 picograms to 500 picograms of Botox in physiological solution or topically as drops.

In one embodiment, in dry eye syndrome, Rock inhibitors or Wnt inhibitor, such as botulinum toxin (Botox) can be applied as drops or injected subconjunctivally to eliminate the inflammatory component of dry eye at doses of 1-10 units once a month or once every 2 to 3 months with slow release nanoparticle conjugates in biodegradable polymers or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, in dry eye syndrome, Rock inhibitors such as botulinum toxin (Botox), Fasudil, etc. or Wnt inhibitors, such as niclosamide, nitazoxanide (NTZ), ivermectin, FH535, IWP-2, PNU-74654, IWR-1endo. IWR-exo, Demethoxy Curcumin, sulforaphane as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) and vitamin D can be given orally at the tolerated dose or 40 mg Fasudil or 1 gram niclosamide or 10-100 units of Botox to eliminate the inflammatory component of dry eye, sulforaphane at 400 micrograms and Vitamin D 3000-5000 IU.

In one embodiment, the Rock inhibitors, such as Fasudil derivatives at concentrations or 10 picograms to 10 nanograms to 1 microgram per drop Botox solution of 0.1 units of Botox can be administered with small molecule WNT inhibitors or a low concentration 1-10 micrograms.

In one embodiment, a topical or subconjunctival or intraocular administration of the Rock inhibitors, such as Fasudil derivatives, etc., at concentrations or 10 picograms to 100 nanograms/0.25 ml or Botox solution of 0.1-1 units can be administered with small molecule WNT inhibitors or a low concentration of sulforaphane and vitamin D to inhibit the inflammatory processes or auto-immune response as

nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, Rock inhibitors are administered with antibody coated nanoparticles conjugated with thermo-sensitive nanoparticles and Adalimumab, a humanized antibody administered topically or subcutaneously at a non-toxic dose.

In one embodiment, Rock inhibitors are administered with antibody coated nanoparticles, dendrimers, liposomes, etc. to the conjunctiva as liposomes or ointment in Meibomian gland inflammation to release medication at a concentration of 1 picogram to 100 units or more picograms/0.25 ml to 0.5 ml along with an antibiotic with nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, Wnt inhibitors or Rock inhibitors, such as Fasudil derivatives, etc. are administered with nanoparticles, dendrimers, thermosensitive polymers conjugated with polylactic or polyglycolic acid or chitosan, microspheres, liposomes, dendrimers, microparticles or nanoparticles, or porous silicone implant and combinations thereof, and they are administered as drops, or injected in the conjunctival or lacrimal glands along with immunosuppressive agents, such as mycophenolic acid, etc. as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, topical administrations, subconjunctival injections, sub-tenon injections, suprachroideal injections, intravitreal injections can be combined with small molecule Wnt inhibitors or standard anti-inflammatory agents (e.g., steroids, dexamethasone, etc.) depot, nanoparticle implants, biodegradable or non-biodegradable polymers, NASIDs, Diclofenac, immunotherapy immunosuppressants, etc. to treat inflammatory processes of the lid conjunctiva or the cornea and the lid or throughout the day. For injection, a dose of about 50 picograms/ml to about 200 micrograms/ml may be used as a surgical implant, for example, in a diffusible walled reservoir sutured to the wall of the sclera, or may be contained within an inert carrier, such as microspheres dendrimers, or liposomes, porous silicon oxide microparticles or as nano or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) to provide a slow-release drug delivery system.

In one embodiment, a formulation of Wnt or Rock inhibitors is used from the group consisting of topical administration at a concentration of about 50 picograms/ml to less than 1 micrograms/ml, subconjunctival injection at a dose in the range of about 1 picogram/ml to about 200 micrograms/ml, intravitreal injection at a dose in the range of about 0.1 picogram/ml to about 20 micrograms/ml, or retrobulbar injection at a dose in the range of about 2 micrograms/ml to about 200 micrograms/ml in slow release microspheres or dendrimers. In one embodiment, a formulation of Wnt or Rock inhibitors is used comprising intraocularly administering to a patient after corneal surgery at picogram to nanogram concentrations or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, a formulation of Wnt or Rock inhibitors is used as a composition consisting essentially of Rock inhibitors in a pharmaceutically acceptable formulation and in an amount effective to enhance post-surgical to enhance

ocular moisture, nerve regeneration in the patient wherein the composition is administered at a concentrations up to about 10 micrograms/ml by at least one of slow release polycaprolactone, polylactic, or polyglycolic acid, etc. over many months, intraocular administration of the composition or is administered topically at a concentration in the range between about 10 picograms/ml to less than 1 microgram/ml depending on the composition of the medication or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, wherein the polymeric composition is administered by subconjunctival injection at a dose in the range of about 1 picogram/ml to about 20 micrograms/ml, intravitreal injection at a dose in the range of about 1 picogram/0.1 ml to about 20 nanograms/ml, or retrobulbar injection at a dose in the range of about 20 nanograms/ml to about 2 micrograms/ml, as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, a formulation of Wnt or Rock inhibitors is used to enhance post-surgical ocular moisture or in papillitis, optic nerve neuritis, uveitis, or scleritis in the patient wherein the composition is administered at a concentration up to about 50 picograms/ml by at least one of intraocular injection, or the composition is administered topically at a concentration in the range between about 50 picograms/ml to less than 1 micrograms/ml as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, a formulation of Wnt or Rock inhibitors is used wherein the composition is administered by subconjunctival injection at a dose in the range of about 1 picogram/ml to about 2 micrograms/ml, intravitreal injection at a dose in the range of about 1 nanogram/0.1 ml to about 20 nanograms/ml, or retrobulbar injection at a dose in the range of about 200 nanograms/ml to about 2 micrograms/ml as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, a method to treat an ocular condition in a patient comprises intraocularly administering to the patient a pharmaceutically acceptable formulation of a drug selected from the group consisting of Rock inhibitors, such Rock inhibitors as Fasudil or derivatives in nanogram to microgram concentrations in microspheres, dendrimers, physiological solution, botulinum toxin in picogram concentrations in polymeric microspheres or 0.3-5 units injectable, or Wnt inhibitors, such as niclosamide, nitazoxanide (NTZ), ivermectin, nanogram to microgram concentration in microspheres, dendrimers, suspension or another polymer, sulforaphane 10-400 nanograms in microspheres, dendrimers, or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or another polymer and Vitamin D taken orally in 1000-5000 IU etc., Fasudil derivatives taken orally 1-40 mg, niclosamide orally in 10-500 mg tablets, sulforaphane in capsule 10-40 mg or more ivermectin taken orally 1-400 mg or more and topical formulation as drops, ointment, or gel in a non-toxic formulation for the patient undergo surgery in the eye for refractive errors, diabetic retinopathy, retinal detachment, or after cataract surgery or refractive surgery for the duration until the eye is free of inflammation and has recovered from the surgery.

In one embodiment, non-toxic doses of Rock inhibitors in an amount up to about 1-200 micrograms/ml effective to treat dry eye or another ocular condition selected from diabetic retinopathy, retinitis pigmentosa, or age related macular degeneration without substantial toxicity and at least one Wnt inhibitor or Rock inhibitor, wherein the composition is administered by at least one of intraocular injection at a concentration up to about 2 picograms/ml, or the composition is administered topically at a concentration in the range between about 1 picograms/ml to less than 10 nanograms/ml as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, a formulation of Wnt or Rock inhibitors is used as topical administration at a concentration between about 50 picograms/ml to 200 nanograms/ml, subconjunctival injection at a dose in the range of about 1 picograms/ml to about 20 micrograms/ml in slow release polymer, intravitreal injection at a dose in the range of about 1 picogram/0.1 ml to about 2 micrograms/ml, or retrobulbar injection at a dose in the range of about 1 picograms/ml to about 200 nanograms/ml suspension in slow release polymer depending on the composition of the medication as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, a method to treat an ocular condition in a patient by intraocularly administering a pharmaceutically acceptable formulation of Wnt inhibitors or rock inhibitors in an amount effective to treat the condition. The method provides treatment while avoiding systemic administration of systemic medication. In one embodiment, a sustained release pharmaceutically acceptable formulation is implanted intraocularly in polymeric slow release compound having about 20 nanograms to 1 microgram or more of Fasudil to about 0.1 micrograms to 40 micrograms or 1 milligram of Fasudil or other rock inhibitors implanted in or on the eye and may continuously deliver Fasudil for five or more years.

In another embodiment, a concentration up to about 10 or more micrograms of Rock inhibitors is administered intraocularly without substantial toxicity.

In another embodiment, Fasudil derivatives are taken orally 1-40 mg, niclosamide is taken orally in 10-500 mg tablets, sulforaphane is taken orally in capsule 10-40 mg or more, ivermectin is taken orally 1-400 mg or more, and topical formulations may be administered as drops, ointment, gel in a non-toxic formulation.

In another embodiment, Rock inhibitors at a concentration in the range of about 1 picogram/ml (0.000000001%) to less than 0.1 micrograms/ml (less than 0.001%) is administered topically. In other embodiments, Fasudil or another Rock inhibitor at a concentration in the range of about 1 nanogram/ml to about 200 micrograms/ml is injected under the conjunctiva, or a concentration in the range of about 1 picogram/0.1 ml to about 200 micrograms/ml is injected in the vitreous, or a concentration in the range of about 20 picograms/ml to about 200 nanograms/ml is injected behind the eyeball as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, the Rock inhibitors, such as Fasudil, etc., or Wnt inhibitors, such as niclosamide, are administered as topical or a spray at non-toxic concentrations of 1 picogram/ml to 20 nanograms/ml in a physiological pH balanced, with osmolarity of 310 to prevent and treat, decrease the time of onset, or lessen the severity of a wide

variety of diseases such as lichen planus, ocular conditions, such as retinitis pigmentosa, ocular irritation following corneal surgery (e.g., LASIK surgery), age related macular degeneration, diabetic retinopathy, dry eye disease, scleritis, papillitis, and uveitis, scleritis pars planatis, or vogt-koyanagii syndrome.

In one embodiment, the Rock inhibitors, such as Fasudil, etc., or Wnt inhibitors, such as niclosamide, nitazoxanide (NTZ), are administered as topical or a spray at non-toxic concentrations of 1 picogram/ml to 20 nanograms/ml or in an ointment or cream or suspension of microspheres and dendrimers in meibomian gland disease as nano or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, the Rock inhibitors, such as Botulinum toxins are administered as topical or a spray at non-toxic concentrations of 1 picogram/ml to 1 nanograms in a cream, ointment, suspension of microspheres or dendrimers, etc. for topical application in lichen planus, nerve damage after LASIK or refractive surgery procedures, or diabetes or wrinkle treatment.

In one embodiment, the Rock inhibitors, such as Fasudil 40-80 mg/kg, etc., or Wnt inhibitors, such as niclosamide, 100-500 mg or ivermectin, 250 mg to 2000 mg are administered orally to prevent and treat, decrease the time of onset, or lessen the severity of a wide variety of diseases, such as optic nerve neuritis, papillitis, variety of idiopathic uveitis, scleritis, or ocular conditions, such as retinitis pigmentosa, ocular irritation following corneal surgery (e.g., LASIK surgery), age related macular degeneration, diabetic retinopathy, dry eye disease, papillitis, or uveitis.

In one embodiment, the Rock inhibitors or Wnt inhibitors are administered as topical or a spray at non-toxic concentrations of 1 picogram/ml to 20 nanograms/ml in a physiological pH balanced solution with osmolality of 310 to treat the corneal nerve cuts after LASIK surgery to decrease inflammatory process and encourage fast regrowth of neurons from the cut end of the corneal nerves and enhance corneal sensation recovery time and prevent dry eye formation.

In one embodiment, the Rock inhibitors (40-80 mg/Kg) or Wnt inhibitors are administered orally after LASIK surgery to decrease inflammatory process and to encourage fast regrowth of neurons from the cut end of the corneal nerves and enhance corneal sensation recovery time and prevent dry eye formation.

Another embodiment of the invention is a method to treat ocular conditions including ocular irritation following corneal surgery, conjunctivitis, canaliculitis or schlemm's canal of the eye, iritis, lacrimal and Meibomian glands are treated with Rock inhibitors, such as Fasudil or its derivatives in nanogram to microgram concentrations in microspheres, dendrimers, physiological solution, Botulinum toxin in picogram concentrations in polymeric microspheres dendrimers, or 0.3-5 unit injectable, or Wnt inhibitors, such as niclosamide, nitazoxanide (NTZ), ivermectin, nanogram to microgram concentration in microspheres suspension or another polymer, sulforaphane 10-400 nanogram in microspheres, dendrimers, or another polymer and Vitamin D taken orally in 1000-5000 IU, etc.

In one embodiment, a sustained release pharmaceutically acceptable formulation is implanted intraocularly. For example, a matrix containing in the range of between about 0.4 to 1 mg Fasudil can last for ten or more years. In another embodiment, a concentration up to about 1 microgram Fasudil or others Rock inhibitors is administered intraocu-

larly, inside the joint in arthritis, or subcutaneously or sub-gingival injection in lichen planus without substantial toxicity as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, or hydroxypropyl-b-cyclodextrin (bHPCD).

In another embodiment, Rock inhibitors at a concentration in the range of about 1 nanogram/ml (0.0000001%) to less than 1 microgram/ml (less than 0.0001%) are administered topically. In other embodiments, Fasudil at a concentration in the range of about 1 nanogram/ml to about 20 microgram/ml is injected under the conjunctiva, or a concentration in the range of about 1 nanogram/0.1 ml to about 200 micrograms/ml is injected in the vitreous, or a concentration in the range of about 20 nanograms/ml to about 20 micrograms/ml is injected in a slow release polymer, such as polycaprolactone or polylactic or glycolic, in the vitreous cavity or behind the eyeball or other part of the body as needed.

In another embodiment, Rock inhibitors at a concentration in the range of about 1 nanogram/ml (0.0000001%) to less than 1 microgram/ml (less than 0.0001%) are administered topically. In other embodiments, Fasudil at a concentration in the range of about 1 nanogram/ml to about 20 micrograms/ml is injected under the conjunctiva, or a concentration in the range of about 1 nanogram/0.1 ml to about 200 microgram/ml is injected in the vitreous, or a concentration in the range of about 20 nanogram/ml to about 20 micrograms/ml is injected in a slow release polymer, such as polycaprolactone or polylactic or glycolic, in the vitreous cavity or behind the eyeball in subconjunctival space, or subcutaneously as needed as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, the non-toxic doses of Wnt inhibitors, Rock inhibitors, or Botox, act as an anti-inflammatory agent. The botulinum toxin or botox preparation may be administered topically to the eye or eye lid, forehead skin at 1 pictogram to 1 nanogram concentrations, 1 pictogram to 5 nanogram concentrations, for example, using drops, an ointment, a cream, a gel, a suspension of microsphere, dendrimers, etc. The agent(s) may be formulated with excipients such as methylcellulose, hydroxypropyl methylcellulose, hydroxypropyl cellulose, the LD50s of any naturally occurring botulinum toxin protein is at 1.3 nanograms per kilogram (abbreviated ng/kg). In a 75 kg (165 lbs.) subjects, the LD50 for botulinum toxin would be 97.5 nanograms if injected directly into a vein or artery. 100 unit vials contains 0.75 nanograms=750 picograms of botulinum toxin A in the entire vial.

In one embodiment, a dose of botulinum toxin in 100-2000 picograms will not be toxic if injected subcutaneously, or 750 picograms (100 units) 1-2 times a month will not be toxic. Higher doses can be used with caution and it would be desirable not to exceed these levels to prevent an immune response to the medication.

In one embodiment, a dose of botulinum toxin in 100-2000 picograms will not be toxic if injected subcutaneously, or 750 picograms (100 units) 1-2 times a month will not be toxic. Higher doses can be used with caution and it would be desirable not to exceed these levels to prevent an immune response to the medication.

In one embodiment, the concentrations 1-20 picograms/Botox in a physiological solution, or up to 30 picograms conjugated with antibody coated nanoparticles would be non-toxic to the body or when conjugated with thermosensitive polymeric coating of the nanoparticles in a physi-

ologic solution or used as drops or injectable or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) preparation.

In one embodiment, the Wnt inhibitors or Rock inhibitors may be injected into the eye, for example, injection under the conjunctiva or tenon capsule, intravitreal injection, or retrobulbar injection as a slow release nanoparticle. The agent(s) may be administered with a slow release drug delivery system, such as polymers, matrices, microcapsules, nanoparticles or microparticles of porous silicone or other delivery systems formulated from, for example, glycolic acid, lactic acid, combinations of glycolic and lactic acid, liposomes, silicone, as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) or polyanhydride polyvinyl acetate alone or in combination with polyethylene glycol, etc. The delivery device can be implanted intraocularly, in the lens capsule, over the lens in the choroid for example, implanted under the conjunctiva, implanted in the wall of the eye, sutured to the sclera, for long-term drug delivery or injected in the vitreous cavity.

In one embodiment, one uses a composition containing Rock inhibitors, such as Fasudil etc., at a concentration in the range of about 50 picogram/ml (0.000000005%) to about 50 micrograms/ml (0.005%), niclosamide at a concentration in the range of about 50 picograms/ml to about 50 micrograms/ml, or a combination of Fasudil or an immune suppressive agent, such as mycophenolic acid, to achieve a total concentration of both agents of about 50 picogram/ml to about 50 microgram/ml as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD). Within this range, the agent(s) has wide safety and efficacy, permitting specific doses or administration protocols to be formulated for specific applications. For example, some patients may prefer once a day administration compared to administration more than once a day, so a higher concentration of agent(s) may be used for these patients.

In another embodiment, Rock inhibitors, such as Fasudil may also be administered by injection. Intraocular injection may be desirable or necessary, for example, for conditions in which topical administration is either not advised or is inadequate, for patients who have difficulty self-administering medications, etc. In one embodiment, the volume injected is less than 0.3 ml. In another embodiment, the volume injected is in the range of about 0.01 ml to about 0.3 ml. For intravitreal administration (injection into the vitreous), Rock inhibitor concentrations in the range of about 1 nanogram/0.1 ml to about 20 microgram/ml (0.002%) may be used without toxicity or adverse side effects.

In another embodiment, niclosamide used in amounts ranging from about 1 nanogram to about 10 micrograms is contained in an aqueous-based cream excipient. In another embodiment, the amount of Fasudil, etc. or other Rock inhibitors ranges from about 1 nanogram to about 10 micrograms, and is contained in an aqueous-based cream excipient. In another embodiment, Fasudil and niclosamide or mycophenolic acid are present in an aqueous-based cream excipient in various proportions. In another embodiment, to achieve a total amount of combined agents of about 1 nanogram to about 10 micrograms, the drug(s) may be incorporated directly into the cream in solution, or may be contained in liposomes or microspheres, dendrimers, either in solution or in an anhydrous form as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin

(bHPCD). The cream formulation is usually applied to the eye at bedtime, but it may be applied any time throughout the day if the cream does not cause blurred vision. In another embodiment, the agent(s) is formulated as a solution or suspension and is applied topically in the form of eye drops.

In another embodiment, for long term delivery of a Rock inhibitor or a Wnt inhibitor, either alone or in combination, and/or for sustained release, a matrix housing or containing the agent(s) may be implanted into the eye. For example, a reservoir containing in the range of about 1 milligram to about 5 milligrams of agent(s) is estimated to be able to release about 1 microgram agent(s) per day. At such a release rate, continuous, sustained dosing may occur over 1000 to 5000 days. If less than 1 microgram of agent(s) per day is released, sustained dosing may last up to or more than a decade. In one embodiment, less than 50 micrograms/day of agent(s) is released from the matrix. In another embodiment, agent(s) is released from the matrix at a rate in the range of about 50 picogram/day to about 50 micrograms/day. In another embodiment, agent(s) is released from the matrix at a rate in the range of about 1 micrograms/day to about 5 micrograms/day.

In another embodiment, a surgically implanted intraocular device or matrix may be provided with a reservoir container having a diffusible wall of polyvinyl alcohol or polyvinyl acetate or polycaprolactone and containing milligram quantities of a Rock inhibitor or Wnt inhibitor, or a combination of them may be implanted in the sclera. As another example, milligram quantities of agent(s) may be incorporated into a polymeric matrix having dimensions of about 1 millimeter (mm) by 2 millimeter (mm), and made of a polymer such as polycaprolactone, poly(glycolic) acid, poly(lactic) acid, or a polyanhydride, or a lipid such as sebatic acid, and may be implanted on the sclera or in the eye.

In another embodiment, as one example of an inert matrix, liposomes may be prepared from dipalmitoyl phosphatidylcholine (DPPC), preferably prepared from egg phosphatidylcholine (PC) since this lipid has a low heat transition. Liposomes are made using standard procedures as known to one skilled in the art. The agent(s), in amounts ranging from picogram to microgram quantities, is added to a solution of egg PC, and the lipophilic drug binds to the liposome.

In another embodiment, the implantable formation may be in the form of a capsule of any of the polymers previously disclosed (e.g., polycaprolactone, poly(glycolic) acid, poly(lactic) acid, polyanhydride) or lipids that may be formulated as microspheres or dendrimers. As an illustrative example, Fasudil may be mixed with polyvinyl alcohol (PVA), the mixture then dried and coated with ethylene vinyl acetate, then cooled again with PVA. Niclosamide bound with liposomes may be applied topically, either in the form of drops or as an aqueous based cream, or may be injected intraocularly as nano or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD). In a formulation for topical application, the drug is slowly released over time as the liposome capsule degrades due to wear and tear from the eye surface. In a formulation for intraocular injection, the liposome capsule degrades due to cellular digestion, other slow release polymers such as PLA, PGA, Polycaprolactone, microsphere, dendrimers) or as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) are also utilized.

In another embodiment, the time-release administration, however, is formulated so that the concentration released at

any period of time does not exceed a toxic amount. This is accomplished, for example, through various formulations of the vehicle (coated or uncoated microspheres, coated or uncoated capsule, lipids, dendrimers, or polymer components, unilamellar or multilamellar structure, and combinations of the above, etc.). Other variables may include the patient's pharmacokinetic-pharmacodynamic parameters (e.g., body mass, gender, plasma clearance rate, hepatic function, etc.). The formation and loading of microspheres, dendrimers, microcapsules, liposomes, etc. and their ocular implantation are standard techniques known by one skilled in the art.

In one embodiment, Rock inhibitors, such as Fasudil, or Botox, etc. or Wnt inhibitors such as niclosamide, alone or in combination with low molecular weight heparin and metalloproteinase inhibitors, such as doxycycline, tetracycline, etc. can be used at non-toxic concentrations with or without dexamethasone, for dry eye or lichen planus or pemphigus or Stevens-Johnson syndrome, lesions of the mucosa, or skin or other inflammatory diseases of the retina, cornea, conjunctival sclera or optic nerve neuritis, scleritis, uveitis in an appropriate physiological solution or ointment, as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), etc.

In one embodiment, the intravenous solution form of Rock inhibitors or Wnt inhibitors may be diluted to achieve the indicated concentration using 0.9% NaCl or 5% dextrose, or an organic solvent such as dimethyl sulfoxide (DMSO) or sterol, lanosterol, squalene, and/or squalamine. Intraocular administration may be any of the routes and formulations previously described. For injection, either a solution, emulsion, suspension of a liquid, capsular formulation of microspheres, dendrimers, or liposomes, as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), etc. may be used.

In one embodiment, Rock inhibitors or Wnt inhibitors or Botox may be injected subconjunctivally to treat uveitis at a dose in the range of about 1 picogram/ml to about 200 picograms/ml, or intravitreally at a dose of about 1 gram/0.1 ml to about 200 picograms/ml. In one embodiment, the dose is about 50 picograms/0.1 ml. To treat scleritis involving the anterior sclera, Rock inhibitors or Wnt inhibitors or Botox may be administered topically as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, or hydroxypropyl-b-cyclodextrin (bHPCD).

In one embodiment, Rock inhibitors or Wnt inhibitors or Botox may be injected to treat scleritis involving the posterior sclera, may be administered by retrobulbar injection at a dose in the range of about 20 picogram/ml to about 800 picograms/ml or more and dissolved in DMSO or a very low concentration of alcohol or sterol, lanosterol, squalene, and/or squalamine.

In one embodiment, to treat neuritis or papillitis, Rock inhibitors may be administered by retrobulbar injection at a dose in the range of about 200 picogram/ml to about 800 nanograms/ml of Fasudil and its derivatives, etc.

In one embodiment, to treat neuritis or papillitis, Rock inhibitors (e.g., Fasudil) may be administered orally at a dose in the range of about 40-80 milligrams of Fasudil tablets, etc. or one time niclosamide 1-2 grams orally.

In one embodiment, the ocular solutions contain at least one Rock inhibitor or Wnt inhibitor such as sulforaphane and provide anti-inflammatory, anti-cell proliferation, anti-

cell migration effects if given orally with Vitamin D, topically as dendrimer or microsphere delivery or an injectable non-toxic preparation.

In one embodiment, the solution or suspension is administered intraocularly after cataract surgery before insertion of a replacement intraocular lens, resulting in reduced post-operative inflammation, which may eliminate the need for a steroid therapy.

In one embodiment, the solution may be one that is invasively administered, for example, an irrigation or volume replacement solution containing at least one Rock inhibitor, such as Botox, or Wnt inhibitor.

In one embodiment, the solution may be one that is non-invasively or topically administered in the form of drops, ointments, gels, creams, etc. and may include eye lubricants and contact lens solutions. The solution may contain a supratherapeutic concentration of agent(s), such as 40 microgram/ml or to 80 micrograms/ml or more for topical application ranges 40 nanograms/ml to 4 micrograms/ml Fasudil and its derivatives, etc. as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD) so that a therapeutic concentration of a topically administered solution accumulates in a diseased ocular structure sufficient to treat the disease.

In one embodiment, medications are administered with antibody coated nanoparticles, dendrimers, or as nano or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), thermosensitive polymers, nanoparticles, dendrimers, lactic or glycolic acid, chitosan or combinations, etc. Immunosuppressives are all conjugated with the antibody coated nanoparticles for slow release as topical drops or an injectable preparation for dry eye after LASIK, meibomian gland inflammation, optic nerve neuritis, uveitis, scleritis, etc.

In one embodiment, Rock inhibitor or Wnt inhibitors are administered by topical drops, spray, subconjunctival injection, subtenon injection, suprachroideal injection, intravitreal injection in combination with standard anti-inflammatory agents etc. and steroids, dexamethasone, etc. as depot, nanoparticles implant biodegradable or non-biodegradable polymers.

In one embodiment, a method of using Rock inhibitors or Wnt inhibitors is disclosed where Rock inhibitors or Wnt inhibitors are administered at non-toxic doses to the skin or mucosa.

In one embodiment, in treating lichen planus, for example, a topical administration may contain between about 10 picogram/ml drug to about 50 micrograms/ml of Fasudil, etc. or other Rock inhibitors in a formulation which may be applied at bedtime or throughout the day or as an injection, a dose of about 50 picograms/ml to about 200 micrograms/ml around or inside the lesion. In one embodiment, the medication may be used as a surgical implant, for example, in a diffusible walled reservoir sutured to the surrounding tissue, or may be contained within an inert carrier, such as microspheres, dendrimers, or liposomes, microparticles or nanoparticles, or porous silicone implant to provide a slow-release drug delivery system to release the medication at 1 picogram to 100 picograms (e.g., Fasudil, etc.) per day.

In one embodiment, a formulation of Wnt or Rock inhibitors is used to treat ocular conditions, such as dry eye disease, as well as other conditions, is disclosed. Rock inhibitors and Wnt inhibitors are used as a topical drop spray application or injection into the eye, or implantation in or on

the eye. For example, a topical administration may contain between about 10 picograms/ml drug to about 50 micrograms/ml drug in a formulation which may be applied at bedtime.

In one embodiment, the patient is administered with Rock inhibitors or Wnt inhibitors as nano- or microparticles with (alpha)-cyclodextrin, or (beta)-cyclodextrin, or (gamma)-cyclodextrin, hydroxypropyl-b-cyclodextrin (bHPCD), microparticles or nanoparticles, or porous silicone implant alone or in combination with NSAIDs or to treat associated chronic pain, with more or less inflammatory processes as seen in the eye after refractive surgery or vitreoretinal surgery or ocular hypotony caused either by surgery and associated systemic disease such as diabetes or autoimmune uveitis.

In one embodiment, non-toxic doses of Rock inhibitors, such as Fasudil, etc., 200 picograms to 2 nanograms or as Botox (10-100 units) are administered locally at multiple locations in treatment or pain or chronic inflammation.

In one embodiment, an intrastromal corneal implant is prepared for implantation inside a patient's corneal stroma, and is comprised of an organic stroma, a synthetic 3-D printed corneal stroma, a hybrid polymeric/organic stroma, a genetically modified animal cornea stroma, an eye bank corneal stroma, or tissue culture grown corneal stroma, which are made non-immunogenic to the host cornea by cross-linking the intrastromal corneal implant with riboflavin solution and UV radiation.

In one embodiment, after the enucleated eyes are obtained from the cornea bank, the eyes are dipped in povidone-iodine to sterilize the surface of the tissue, rinsed with saline solution, and then, using a Q-tipped applicator dipped in a saline solution. The corneal epithelium is removed from the donor cornea, then a standard 6-7 mm circular trephine cuts though the remaining cornea and a circular remaining corneal tissue is removed from the eye, cleansed again and the endothelial cells of the cornea are removed while leaving the corneal stromal with its cells.

In one embodiment, using a femtosecond laser or microkeratome, a circular piece of the donor corneal stroma is cut with the thickness of 50, 100, or 150 microns or more thickness that serves for correcting the refractive error of the human recipient cornea from $<+1.00$ D to $+15$ D or <-1.00 D to -15.00 power and up to ± 6.00 D power of astigmatism including lower and higher order of aberrations.

In one embodiment, the donor stromal inlay is sterilized by dipping it in a solution of 0.1-5% photosensitizer or riboflavin, or riboflavin nanoparticles conjugated with cell penetrating peptides (CPP) or activatable cell penetrating peptides (ACPP) to penetrate the entire stromal block. In one embodiment, the riboflavin solution contains other medications such as NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors or GSK inhibitors in a polymeric slow release polylactic, glycolic, polycaprolactone, micelles, porous silicon, etc. to release for 3-4 weeks or more. In one embodiment, the riboflavin has the osmolality of about 300 mOsmol, pH of about 7, with or without dextran or low molecular heparin or hyaluronic acid, and polymeric nanoparticles.

In one embodiment, the corneal inlay containing riboflavin is irradiated with the UV light of a laser with the wavelength of 350-390 nm with a power of 3-30 mW/cm² for a period of 1-10 minutes or more to cross-link the inlay by a combination effect of riboflavin and UV radiation creating singlet oxygen and reactive species that cross-link proteins making the inlay non-immunogenic regardless of its origin, such as human or animal of genetically modified

animals, such as pig for organ transplant, etc., and the cross-linking simultaneously sterilizes the implant by killing all the potentially existing pathogens.

In one embodiment, the implant is made more permeable to a riboflavin solution by applying femtosecond laser pulses to it (see FIGS. 54A and 54B), then cross-linking by a combination effect of riboflavin and UV radiation (see FIG. 55) creating singlet oxygen and reactive species that cross-link proteins making the inlay non-immunogenic and simultaneously sterilizing the implant by killing all the potentially existing pathogens. The femtosecond laser will create some air bubbles initially, which rapidly dissipate in the air, and do not change the thickness of the inlay. As shown in FIGS. 54A and 54B, small cavities or holes may be formed in the corneal inlay or implant 810 using a femtosecond laser (i.e., the small cavities or holes are formed in the corneal inlay or implant 810 using the laser beam(s) 812 emitted from the femtosecond laser) so as to make the corneal inlay or implant 810 more permeable to a photosensitizer solution (e.g., riboflavin solution), thereby allowing more of the photosensitizer solution to get inside the corneal inlay or implant 810. In the illustrative embodiment, the corneal inlay or implant 810 of FIGS. 54A and 54B is formed from a donor human cornea, and has an outside diameter between 6 and 7 millimeters. As shown in FIG. 55, after the corneal inlay or implant 810 has been soaked in the photosensitizer solution (e.g., the riboflavin solution), the corneal inlay or implant 810 is irradiated using ultraviolet (UV) radiation 814 so as to cross-link the corneal inlay or implant 810. The photosensitizer solution (e.g., the riboflavin solution) in which the corneal inlay or implant 810 is soaked may also contain one or more medications for treating the recipient eye of the patient (e.g., one or more medications described above, such as Rock inhibitors, Wnt inhibitors, integrin inhibitors, GSK inhibitors, etc.).

In another embodiment, the donor cornea, genetically modified cornea, or humanized cornea is prepared by removing it from the donor eye using a 6 to 8 millimeter diameter trephine and scissors to separate it from the eye. The corneal epithelium and endothelium is removed chemically, mechanically, or combination thereof using a Q-tipped applicator dipped either in saline or a solution of 1% or more in water or saline solution and subsequently rinsed immediately with saline to remove the remaining alcohol and/or the loose cells to create a corneal stromal tissue by using a standard femtosecond laser or a mechanical microkeratome to create two or three stromal implant circles of about 100-150 microns in thickness (e.g., see FIG. 61A), when exposed to a solution of 1-5% riboflavin nanoparticles and cell penetrating peptides (CPP), and if needed, an appropriate medication and/or another photosensitizer to cross-link the implant rapidly in less than 3 minutes. Then, the implant is radiated using either a laser with UV radiation of wavelength of between 380 nm to 390 nm, or other wavelength when another photosensitizer rather than riboflavin is used, thereby crosslinking the corneal proteins, including collagen, elastin, etc., and stiffening the corneal collagen and simultaneously killing corneal stromal cells and all pathogens present in the implant and making the implant non-immunogenic without affecting its transparency.

In another embodiment, a 3-D printed cornea is prepared without having the epithelial cells or the endothelial cells, or having only the stromal fibrils, and other proteins, such as collagen and elastin, etc. and mesenchymal stem cells to create a corneal stromal tissue, and by using a standard femtosecond laser or a mechanical microkeratome, create two or three stromal implant circles of about 100-150

microns in thickness, exposed to a solution of 1-5% riboflavin or riboflavin nanoparticles and cell penetrating peptides (CPP) or another photosensitizer to cross-link the implant rapidly in less than 3 minutes. Then, the implant is radiated using a laser with UV radiation of wavelength of 380 nm to 390 nm, a power of 3 milliW/cm² to 10 milliW/cm² or more for a time of 1-10 minutes as needed for cross-linking the corneal proteins, including collagen, elastin, etc. to stiffen the corneal collagen and simultaneously kill corneal stromal cells and all pathogens if present in the implant, and make the implant non-immunogenic by cross-linking it, without affecting its transparency.

In one embodiment, the implant is modified to be able to pass through a small incision so as to be placed with an instrument under a flap of the cornea, such as after a LASIK procedure or a procedure that does not require a large corneal incision cutting more corneal nerves. By inserting the implant through a small incision into a stromal pocket, the potential side effects of corneal anesthesia and dry eye are prevented, such as in the small incision lenticule extraction (SMILE) procedure. However in the SMILE procedure, the removed stromal lenticule requires scratching the bordering stroma to the explant, the previously cut stromal lenticule with femtosecond laser inside the stroma for correction of low myopia. However in this typical SMILE procedure, the removed tissue (stromal lenticule) is not replaced with an appropriate lenticule, which would provide a smooth border and shape that reduces or prevents the post-operative glare experienced by majority of the patients.

In one embodiment, in the SMILE procedure, the stromal pocket is not left empty, but is filled with a small porous human corneal, cross-linked implant with an appropriate size and shape to achieve the required refractive correction, while preventing the post-operative glare in this patient.

In one embodiment, the surrounding tissue of the inlay is also cross-linked when the implant has some crosslinker in it, wherein the cross-linker diffuses out of the implant and penetrates the surrounding tissue when the eye is exposed to an external UV radiation for a short period of time after implantation.

In one embodiment, the inlay is placed after a LASIK flap is prepared, and the refractive error of the eye regardless of being myopic, hyperopic, astigmatic, or having any other combined refractive errors, is corrected using a standard excimer laser equipped with wavefront technology and Shack-Hartmann wavefront sensor during in vivo surgery objectively and corrected with the excimer laser immediately, then the corneal flap is repositioned, and the appropriate antibiotic and other medications are applied to the eye.

In one or more embodiments, referring to FIGS. 56-59, a LASIK flap 820 is formed in a cornea 818 of an eye 816 using a femtosecond laser cut 822 (see FIG. 57). Then, a corneal inlay or implant 824 is inserted under the flap 820. The inlay or implant 824 is cross-linked so as to prevent an immune response to the implant and/or rejection of the implant 824 by the patient. Then, as shown in FIG. 58, laser energy 826 is applied to the implant 824 in the eye using an excimer laser so as to modify the refractive power of the implant 824 while being monitored using a Shack-Hartmann wavefront system so as to achieve a desired refractive power for the implant. Finally, as shown in FIG. 59 the cross-linked implant 824 is covered with the flap 820, the cross-linked implant being surrounded entirely by the stromal tissue of the cornea.

In one embodiment, the stromal inlay is a single piece and does not have any curvature to correct the refractive error and as such it is "refractively" neutral with an index of

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refraction of the normal cornea of 1.33 which similar to the water inside the stroma, and the inlay does not change the refractive power of the cornea, but can increase the thickness of the cornea. In one embodiment, a larger diameter inlay of 8 to 12 mm in diameter and a thickness of 50 to 150 microns can be used in a patient with keratoconus to prevent corneal ectasia or in a patient who has ectasia after refractive surgery, or in combination with photo-therapeutic keratectomy, or as LASIK where the inlay is ablated with an excimer laser to correct the refractive power of the eye. This could be a one step or two step procedure after the initial healing has taken place.

In one embodiment, the implant is used in two different procedures. In one embodiment, in a LASIK procedure, the implant is positioned over the exposed corneal stromal surface after creating a cornea stromal flap, and after ablation the inlay corrects the refractive error of the eye by using an excimer laser and wavefront technology that controls the degree and shape of the tissue removed from the inlay during the surgery to eliminate refractive errors of each eye individually during the surgery, then the corneal stromal flap is returned back over the reshaped anterior surface of the inlay and ultimately the reshaped inlay is surrounded on all sides by the corneal stroma and away from the host's Bowman's membrane or corneal epithelium.

In another embodiment, a small incision **852** (e.g., an incision having length of between 4 and 5 millimeters) is created in the cornea **850** of the eye **848** for inlay implantation inside the cornea **850**. Initially a 3-D cavity is created inside the corneal stroma with a femtosecond laser, and through the small incision **852** the folded pieces of the inlay **830**, **832** or **836**, **838**, or preferably a composite lens, are implanted inside the stromal cavity as shown in FIGS. **62** and **63**. A side cross-sectional view of the stromal cavity **854** of the cornea **850** is shown in FIG. **64**, while an implant or inlay **856** is shown disposed in the cavity **854** of the cornea **850** in FIG. **65**.

A first exemplary composite inlay or implant **828** is illustrated in FIG. **61A**. The composite two-piece inlay or implant **828** in FIG. **61A** comprises a central portion **830** and a peripheral portion **832**. A second exemplary composite inlay or implant **834** is illustrated in FIG. **61B**. The composite two-piece inlay or implant **834** in FIG. **61B** is transversely divided into two pieces, and comprises a first section **836** and a second section **838**. A third exemplary composite inlay or implant **840** is illustrated in FIG. **61C**. The composite three-piece inlay or implant **840** in FIG. **61C** is transversely divided into three pieces, and comprises a first section **842**, a second section **844**, and a third section **846**.

In one embodiment, the implant or inlay is reshaped prior to implantation inside a 3-D pocket made with a femtosecond laser inside the corneal stroma, with or without the removing of the stromal tissue, e.g., after removal of a part of the scarred stromal tissue or a stromal lenticule, as performed in known SMILE procedure, to fill the cavity with an implant which has a smooth surface and appropriate refraction matching the needed refraction of the eye, using an excimer laser to ablate the surface of the inlay. In one embodiment, the implant or inlay is folded and implanted inside the corneal stromal cavity. In another embodiment, after correcting the inlay to compensate for the refractive error of the patient's eye by using an excimer laser and wavefront technology, the inlay is cut into various pieces and shapes, thus permitting the pieces of the inlay to be implanted inside the stromal cavity through a small incision (e.g., an incision having length of between 4 and 5 milli-

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meters), and simultaneously eliminating the often seen significant glare after the SMILE procedure that lasts about one month or more. The SMILE surgery involves cutting a small 3-D lenticule inside the corneal stroma with a femtosecond laser, then digging out this lenticule with the help of a bent-tipped needle shaped like a hook, but because the femtosecond laser does not completely separates the lenticule from the rest of the stroma, many attempts are done to separate the lenticule from the stroma in a space where there is almost no room for manipulation, thus many scratches are created in the wall of the cavity, which is left to itself to heal. These surface irregularities with bits of the pieces of the left over stromal tissue refract the incoming light and produce glare in extreme situations like scratched on or inside the reading glasses, etc. However, after a month or more, gradually the tissue heals and the glare is reduced or eliminated.

In one embodiment, the post-operative glare can be significantly reduced by implanting a thin non-refracting one-piece inlay or a composite inlay inside the cavity left in the stroma after SMILE procedure where the inlay has a very smooth surface.

In one embodiment, after the SMILE procedure, the stromal cavity can be injected with riboflavin and the excess solution removed immediately by exerting a minor pressure on the cornea to empty the cavity through the small incision in the cornea, the cornea is radiated for about 1-5 minutes with UV radiation to cross-link the wall of the cavity. In one embodiment, after the SMILE procedure, a cross-linker is mixed with a very dilute collagen, or elastin, or just a saline solution to wash out the cavity and immediately removed under mild pressure to cross-link the area so that the stroma with UV radiation heals and strengthens the corneal resiliency despite removal of a part of the corneal stroma.

In another embodiment, composite lenses of implants are used to simplify insertion of the implant through a small incision **852** (see FIGS. **62** and **63**) using forceps with the folded pieces of the inlay and a lubricating agent, such as low molecular weight heparin or hyaluronic acid, and then open the inlay with a tiny plastic rod inside the corneal cavity.

In one embodiment, the composite lenses are inserted in the stromal cavity with ease using each piece separately then joining the pieces inside the stromal cavity with or without lubricating agents. Because these lenses are organic and/or formed from a human corneal stroma, they join easily together and the smooth cut edges heal easily without producing significant glare in the post-operative period.

In another embodiment, the implant is a hybrid lens made from partially human cross-linked corneal tissue supported by a more resilient polymeric coating such as polyethylene glycol (PEG), chitosan, or another polymer etc. that is water permeable while providing more resiliency to the implant and keeps the inlay surfaces smooth and easy to implant them through a small incision.

In another embodiment, a ribbed, perforated, or sieved polymeric structure (see FIGS. **66-69C**) provides support to the overlying human cross-linked cornea, which drapes over the polymeric structure during the implantation thereof, while the polymeric structure does not prevent the free flow of the fluid from the back of the cornea to the anterior part of the cornea and the holes in this polymeric structure are filled by the surrounding tissue and mostly the fluid in the cornea. Initially, with collective reference to FIGS. **66-68**, the cross-linked donor corneal implant or inlay **858** depicted in FIG. **66** may be placed on the polymeric inlay back support structure **860** of FIG. **67** to form the hybrid lens

implant depicted in FIG. 68. As shown in FIGS. 67 and 68, the polymeric inlay back support structure 860 is provided with a plurality of tiny holes 862 (e.g., having a diameter of 20 microns to 1 millimeter) disposed therethrough to enable free flow of the fluid from the back of the cornea to the anterior part of the cornea and the holes in this polymeric structure. In the illustrative embodiment, the organic cross-linked corneal implant 858 (e.g., formed from donor corneal tissue) has flat parallel anterior and posterior surfaces, but the shape of the organic cross-linked corneal implant 858 can be modified using a specially-shaped polymeric back support structure. For example, as shown in FIGS. 69A-69C, the polymeric back support structure may have a flat structure 864 (see FIG. 69A) for maintaining the flat shape of the organic cross-linked corneal implant 858, a convex structure 866 (see FIG. 69B) for giving the organic cross-linked corneal implant 858 a convex shape, or a concave structure 868 (see FIG. 69C) for giving the organic cross-linked corneal implant 858 a concave shape. Also, these thin-walled transparent polymeric structures can have an index of refraction similar to the cornea (e.g., 1.3).

In one embodiment, the polymeric supporting elements are made from hydrogel with a very high water content between 50-90% of water. This hybrid implant can be used after a LASIK procedure or small incision intrastromal inlay implantation, the structure can remain in place or removed from the eye by forceps.

In one embodiment, all implants used in conjunction with a LASIK procedure have no refractive surfaces initially, except for an index of refraction which is exactly as the corneal tissue, but after positioning over the exposed corneal stroma and modification of its surface with an excimer laser and wavefront technology to correct all refractive errors only the anterior surface of the inlay is changed during the refractive surgery in vivo, before the corneal flap is repositioned over the inlay. This in vivo correction of refractive error has the advantage that the position of inlay over the corneal stroma does not change after the surgery, since the inlay remains in place all the time during the excimer laser ablation and the flap prevents its motion postoperatively, that is why one can achieve very high satisfaction after LASIK surgery to >98% of the patients. Therefore, there will not be misplacement of the direction of the corrected astigmatism postoperatively which can happen if the inlay surface(s) were corrected ex vivo, and then moved on the cornea or inside in the corneal stroma.

In one embodiment, the crosslinker can be any crosslinker, but preferably it is riboflavin which is excited by UV radiation of 350-390 nm or a 370 nm to 390 nm laser wavelength which crosslinks all the proteins present in the implant including the potential pathogens including viral, bacterial, fungal and parasitic ones.

In one embodiment, the crosslinker is in a nanoparticle form conjugated with cell penetrating peptides (CPP) that penetrate all cellular membranes and cross-link the intercellular proteins eliminating the cells of the human corneal inlay, thus preventing an immune response from the tissue surrounding the inlay that can incite an antigenic response toward a corneal transplant.

In one embodiment, the combination of a riboflavin nanoparticle solution with CPP and an antibiotic, antiviral, antifungal, or anti-parasitic medication is used as topical drops applied to the cornea, then irradiated with UV light to cross-link the infected part of the cornea, so as to kill the pathogens and treat the infectious keratitis.

In one embodiment, all implants are prepared as described, but after implantation of the inlay, the riboflavin

is injected in the corneal stromal cavity with a fine needle so that the riboflavin penetrates the wall of the stromal cavity around the implant for a distance of at least 20 microns deep on each side that is cross-linked after termination of the surgery with external UV radiation for short period of time or 1-5 minutes without crosslinking any other part of anterior or posterior stroma not exposed to the riboflavin. The crosslinking of the wall of the cavity kills the stromal cells in that area and creates an immune privileged cavity that prevents an immune response to the implant, the in short situ cross-linking kills potential pathogen that may have contaminated the inlay during the transport.

In one embodiment, medication is present in the riboflavin solution, and the riboflavin nanoparticle solution, and the solution contains free or polymeric slow release nanoparticles of an anti-inflammatory agents such as NSAIDs, Rock inhibitors such as Fasudil, netarsudil, SAR407899, botulinum toxin at 1-10 international units, etc., anti-integrins, resiteganib, Natalizumab at microgram/ml, MLN-00002, Fintrol, IVL745, antagonists of $\alpha v \beta 3$ and/or $\alpha v \beta 5$ integrins, LM609, Vitaxin, Abegrin, CNT095, Cilengitide etc. wnt inhibitors such as Ant1.4Br/Ant 1.4Cl, ivermectin, niclosamide, apicularen and bafilomycin, XAV939, XAV939, G007-LK and G244-LM, NSC668036, SB-216763, gemtuzumab, GSK inhibitors, such as SB-216763, etc., anti-VEGF avastin, lucentis, bevacizumab, aflibercept, antibiotic moxifloxacin etc., or antivirals, valacyclovir etc. antifungal amphotericin B, voriconazole or anti-parasitic agents, such as niclosamide, ivermectin, etc. at non-toxic concentrations alone or in combination. These compounds particularly Rock inhibitors and GSK inhibitors encourage sprouting of the cut corneal nerves and regeneration of nerves, and also can be used in combination with nerve growth factors and Brimonidine as topical or slow release polymeric implants, incorporated in the inlay or placed as polymeric implant, in the anterior chamber of the eye, in the vitreous cavity of the eye, under the conjunctiva, etc. (e.g., such as in the subconjunctival drug delivery implant 722 shown in FIG. 46).

In one embodiment, the donor corneal tissue is cut with a circular trephine of at least 7-8 mm in diameter, separating the circular corneal area from the donor eye and removing the corneal epithelium and corneal endothelial cells from the donor corneal tissue, and cutting in two circular leftover corneal donor portions of two equal dimensions with a keratome or a femtosecond laser, one portion from the anterior surface to a distance of 50-200 microns uniform thickness with Bowman's membrane/stroma 3-D inlay, and another portion from the posterior corneal surface with a 50-200 microns uniform thickness having the Descemet membrane to build a 3-D circle of the corneal stroma and Descemet inlay, and then cross-linking portions with a solution having riboflavin nanoparticles and an antibiotic, and irradiating the portions with UV radiation to kill the stromal keratocytes along with potential pathogens while crosslinking the collagen fibrils in the inlays, then transplanting the cross-linked inlay over the exposed corneal stroma after creation of a LASIK hinged flap to ablate the front surface of the inlay with an excimer laser using wavefront technology, and finally replacing the LASIK flap over the inlay covering it with the corneal stroma from all sides.

In another embodiment, the donor corneal tissue is cut with a circular trephine of at least 7-8 mm in diameter, separating the circular corneal area from the donor eye, and removing the corneal epithelium and corneal endothelial cells from the donor corneal tissue, and cutting in two

circular leftover corneal donor portions into two equal dimensions with a keratome or a femtosecond laser, one portion from the anterior surface to a distance of 50-200 microns uniform thickness with Bowman's membrane/stroma 3-D inlay, and another portion from the posterior corneal surface with a 50-200 microns uniform thickness having the Descemet membrane to build a 3-D circle of the corneal stroma and Descemet inlay, and then cross-linking the portions with a solution having riboflavin nanoparticles and an antibiotic, irradiating the portions with UV radiation to kill the stromal keratocytes along with potential pathogens while crosslinking the collagen fibrils in the inlays, modifying the front surface of the inlays with an excimer laser to a desired shape, and then cutting the inlay with its Descemet membrane in two or three sections to be implanted inside the host corneal stroma. In this embodiment, the Bowman's membrane and/or Descemet's membrane is kept so as to contribute to biomechanical stability of the inlay. A pocket is created with a femtosecond laser, and then the inlay pieces are inserted through a small incision to correct the refractive power of the host eye. The inlay can be cut into 2-3 pieces with a knife prior or after crosslinking the inlay.

In another embodiment, only a single inlay is cut from the donor cornea, rather than the multiple inlay portions described above.

In one embodiment, the corneal endothelium and epithelium of a human donor cornea is removed mechanically or in combination with low concentration of alcohol in water, etc. prior to implantation of the human donor cornea.

In one embodiment, a uniform rectangular or circular piece of human donor stromal tissue, a tissue culture grown cornea, a genetically modified cornea, or an acellular human or animal cornea is cut with either a microkeratome or a femtosecond laser or a trephine to create a 3-D corneal stromal circle with a thickness of 10 to 50 microns, 50 to 150 microns, or 50 to 200 microns under sterile conditions (see FIG. 70). The acellular human or animal cornea may be created using 0.05-0.5% or more sodium dodecyl sulfate (SDS), Triton X-100, benzalkonium chloride (BAK), Igepal, or genipin for 15 minutes to 6 hours or more depending on the concentration used to generate acellular corneal inlays, and the corneal tissue is cleaned in 100% glycerol, and sterilized by killing bacteria, viruses and/or parasites, or the corneal inlay may be re-sterilized by ethylene oxide or during the surgery by crosslinking the donor tissue with the host cornea with UV light and 0.1% of riboflavin nanoparticles.

In one embodiment, the human corneal inlay is made acellular by glycerin and cryopreservation and then thawing it for use in a physiological solution of 0.1% riboflavin nanoparticles, and finally cross-linking it with UV radiation after implantation.

In one embodiment, 0.1% of riboflavin nanoparticles in a physiological solution of pH 6.5-7.5 and osmolality of about 300 milliosmole (mOsm) with or without antibiotics or antivirals, etc. as additives, applied topically on the cornea, the conjunctiva, and/or the lid margin and exposed to UV light or just the sunshine or a white light can sterilize the infected cornea, conjunctiva, or the lid, while repeated daily one time administration can cross-link the anterior part of corneal collagen, etc. preventing corneal ectasia, progression of keratoconus, or high myopia.

In one embodiment, by using a trephine, the donor cornea is cut to create inlay circles of 3 to 9 millimeters (mm) or more in diameter for intrastromal implantation that can be preserved in a tissue culture or a physiological solution

having an antibiotic, antifungal, and a preservative or detergent, such as sodium dodecyl sulfate (SDS), to eliminate the contaminated pathogens. In addition, the donor cornea may be stored in a low concentration of chondroitin sulfate, hyaluronic acid, or albumin to exert an oncotic pressure on the cornea so as to prevent it from swelling during the transport thereof.

In one embodiment, the corneal stromal inlay is decellularized with a solution of 0.1-0.5% sodium dodecyl sulfate (SDS) for simultaneous preservation of the corneal inlay and killing the bacteria, viral, or parasites prior to implantation inside a corneal LASIK flap, then placing the corneal inlay over the exposed corneal stroma after cleaning it in 100% glycerol, correcting the refractive error of the host eye with an excimer laser, then applying a solution of riboflavin/medication over the inlay, replacing the flap, and crosslinking the inlay and the stromal wall with UV radiation.

In one embodiment, the corneal stromal inlay is decellularized with a solution of 0.1-0.5% sodium dodecyl sulfate (SDS) for simultaneous preservation of the corneal inlay and killing the bacteria, viral or parasites prior to implantation inside a corneal pocket created with a femtosecond laser, correcting the refractive error of the inlay prior to implantation in the host eye with the excimer laser or a femtosecond laser, then the corneal inlay is placed inside a corneal pocket, and a solution of 0.1% riboflavin/medication in a hyaluronic acid solution is applied to the inlay and the surrounding tissue, excess hyaluronic acid is removed from the cavity, and the inlay and the surrounding wall of the cornea are cross-linked with UV radiation.

In one embodiment, the stromal inlays are preserved in a hypothermic storage at 2 to 8° C., or stored by freezing, by freeze drying, in glycerol, and/or gamma radiated and placed in hypothermic solution of tissue culture medium or organ culture medium.

In one embodiment, hypothermia reduces the demand for metabolic energy and since the stromal inlay does not have a corneal epithelium and endothelium or keratocytes its demand is very low, therefore it can be preserved longer (e.g., 45 days or more) than a normal cornea can be preserved, which is about one month.

In one embodiment, the preserving solution for the stromal inlay can have chondroitin sulfate to prevent swelling of the tissue or it can have 4-8% dextran 40, or include ascorbic acid, albumin, hyaluronic acid, or low molecular weight heparin, etc.

In one embodiment, the solution can become a gel with the corneal inlay in it by the addition of a crosslinker to it so as to crosslink the albumen or adding trehalose to the solution and to maintain a gel-like structure with proper pH and osmolality of the surrounding inlay.

In one embodiment, the preserving solution can be McCarey-Kaufman medium (M-K medium) containing 5% dextran 40.

In one embodiment, the storage temperature can be 2-4 degrees C. or more, and the storage can be 30-45 days or more.

In one embodiment, the solution can have non-ionic surfactants (Poloxamer 188) for hypothermic storage.

In one embodiment, the intrastromal inlay can be placed in 10-80 ml Eagle's medium containing HEPES buffer, 26 mmol/l, NaHCO₃, 2% FBS, 2 mmol/l L-glutamine, penicillin, streptomycin and amphotericin B, and stored at 4-34° C. for 30-45 days in a sterile container.

In one embodiment, the intrastromal inlays can be stored in a solution of amphotericin B Optisol-GS that is effective against *Candida* species.

In one embodiment, the intrastromal inlay is transplanted over the center of an exposed corneal stroma which is created in a known laser-assisted in situ keratomileusis (LASIK) procedure using a femtosecond laser or a micro-keratome where the flap is created and hinged away to expose the corneal stroma (see FIGS. 71-73).

In one illustrative embodiment, with reference to FIGS. 70-75, a method of corneal implantation with cross-linking will be described. Initially, referring specifically to FIG. 72, a LASIK flap 914 is formed in a cornea 912 of an eye 910 so as to expose a stromal tissue 916 of the cornea 912 underlying the flap 914. Then, the flap 914 is pivoted so as to expose the stromal tissue 916 of the cornea 912 underlying the flap 914 (see FIG. 72). After pivoting the flap 914, an implant or inlay 918 is inserted under the flap 914 so as to overlie the stromal tissue 916 of the cornea 912 (see FIG. 73). Then, as shown in FIG. 73, laser energy 920 (e.g., by means of an excimer laser) and/or microwaves are applied to the implant 918 in the eye 910 so as to modify the refractive power of the implant 918 while being monitored using a Shack-Hartmann wavefront system so as to achieve a desired refractive power for the implant 918. Turning to FIG. 74, after the laser application, a cross-linking solution 922 that includes a photosensitizer is applied to the implant 918 (e.g., by drops or injection). Next, the implant 918 is covered with the flap 914 such that the implant 918 is surrounded entirely by the stromal tissue of the cornea 912. Finally, as shown in FIG. 75, the implant 918 is irradiated (e.g., with ultraviolet (UV) radiation 924) so as to activate cross-linkers in the implant 918, and thereby cross-link the implant 918 to prevent an immune response to the implant 918 and/or rejection of the implant 918 by the patient. The wall of the stroma of the eye 910 surrounding the implant 918 may also be cross-linked together with the implant 918 for a predetermined time duration (e.g., ten minutes).

In the illustrative embodiment of FIGS. 70-75, the implant or inlay 918 may be formed from donor corneal tissue, a tissue culture grown cornea, or a 3-D printed cornea. When the implant or inlay 918 is formed from donor corneal tissue, the donor corneal tissue may initially be cut into a circular implant 926 using a circular trephine (see FIG. 70), where the circular implant has a diameter between 3 millimeters and 10 millimeters. Then, the circular implant 926 may be cut into two circular disks using a keratome or a femtosecond laser, each of the two circular disks having a thickness between 50 microns and 200 microns.

In one embodiment, the stromal inlay size and thickness is chosen depending on the degree of the corrective dioptric power that is needed. In one embodiment, an inlay with a thickness of 50 microns can correct 5 dioptric powers, and with 150 micron thickness can correct 15 dioptric power, etc.

In one embodiment, the stromal inlay has a diameter of about 3 mm and is placed just inferior to the center of the visual axis and corrected to have a dioptric power of $\pm 1-3.00D$ or more for patients with presbyopia for reading that is creating a bifocal cornea, where the patient sees far through the center of the cornea and near, through the small inlay.

In one embodiment, the refractive error of the eye is corrected after the LASIK flap is created and the inlay is positioned over the exposed center of the corneal stroma by the use of an excimer laser and wavefront technology and a Shack-Hartmann sensor to correct myopic, hyperopic, or

astigmatic or other higher order of refractive errors of the eye (see FIG. 73), then a few drops of the solution of 0.1% riboflavin nanoparticles, conjugated with cell penetrating peptides (CPP) or activatable cell penetrating peptides (ACPP) having antibiotic, antifungal, or anti-parasitic medication, etc. or anti-inflammatory agents, such as Rock inhibitors, anti-integrins, GSK inhibitors, etc. at a non-toxic dose of 1 microgram to 100 microgram in 0.1 milliliters (ml) or a concentration for use in intravitreal irrigating solutions, or as slow release biodegradable polymeric polylactic acid or polyglycolic acid or their combinations of porous silicon, micelles or polycaprolactone nanoparticles are administered over the inlay and exposed corneal stroma (see FIG. 74) to treat the inflammatory processes and prevent tissue scarring.

In one embodiment, the riboflavin may penetrate the inlay within 1-3 minutes.

In one embodiment, the hinged corneal flap is repositioned so that the inlay is covered on all sides by the host corneal stroma, which now is also exposed the riboflavin solution.

In one embodiment, the nanoparticles of riboflavin solution can be injected under the corneal flap over the inlay though a small 25-30 gauge needle or smaller needle to expose the inlay and the corneal cavity to the riboflavin solution, then the cornea, including the inlay is irradiated with UV radiation of 3-10 milliW/cm² from outside the cornea for a short period of 3-10 minutes to cross-link only the inlay and its surrounding corneal stroma, thereby killing all remaining cells in the inlay and any potential pathogens that might have contaminated the inlay or the stroma by the combination effects of the UV radiation, riboflavin nanoparticles, CPP, and medication and sterilize the inlay and the stromal cavity. In one embodiment, the corneal cavity and the flap can be irradiated with UV laser brought under the corneal flap through a fiber optic, thus crosslinking the inlay and the surrounding stromal tissue while the tissue close to the corneal endothelium or close to the anterior surface of the cornea are not cross-linked.

In one embodiment, the nanoparticles of riboflavin solution can be injected under the corneal flap over the inlay though a small 25-30 gauge needle or smaller needle to expose the inlay and the corneal cavity to the riboflavin solution, then the cornea, including the inlay is irradiated with UV radiation of 3-10 milliW/cm² for a short period of 5-10 minutes using a fiber optic with an small lens on its tip to focus the laser beam only for a distance of 20-50 microns, and then it diffuses out so that the endothelial cells, or the crystalline lens, or the retina is not exposed to a heavy dose of UV radiation. In one embodiment, the tip of the fiber optic moves all over the front surface of the intrastromal inlay to crosslink only the inlay and its surrounding corneal stroma, killing all remaining cells in the inlay and any potential pathogens that might have contaminated the surgical area by the combination effects of the UV radiation, riboflavin nanoparticles, CPP, and medication and sterilize the inlay and the stromal cavity while eliminating the toxic effect of the UV radiation to the anterior or the posterior corneal tissue.

In one illustrative embodiment, with reference to FIGS. 76-81, another method of corneal implantation with cross-linking will be described. Initially, referring specifically to FIGS. 76 and 77, an intrastromal cavity 932 is formed in the cornea 930 of an eye 928. Then, the intrastromal cavity 932 of the cornea 930 is filled with a shaped donor cornea implant 934 (see FIG. 78). After inserting the shaped donor cornea implant 934 in the intrastromal cavity 932, a cross-linking solution 936 that includes a photosensitizer (e.g.,

riboflavin) is injected into the implant **934** (e.g., by using a needle for injection—see FIG. **79**). Finally, as shown in FIG. **80**, the implant **934** is irradiated (e.g., with ultraviolet (UV) radiation **938**) so as to activate cross-linkers in the implant **934**, and thereby cross-link the implant **934** to prevent an immune response to the implant **934** and/or rejection of the implant **934** by the patient. The wall of the stroma of the eye **928** surrounding the implant **934** may also be cross-linked together with the implant **934** for a predetermined time duration (e.g., ten minutes). In FIG. **80**, the ultraviolet (UV) radiation **938** is shown being externally applied to the cornea **930** of the eye **928** with the implant **934**. Although, rather than being externally applied, the ultraviolet (UV) radiation may also be internally applied to the implant **934** by means of a fiber optic **940** (see FIG. **81**). For example, the UV radiation may be applied inside the tissue with the fiber optic **940**, which may include a small incision lens at its tip to focus the lens in the tissue for a very short distance of about 100 microns and then it diverges, thus preventing damage in the back of the corneal, endothelial cells, the crystalline lens, the retina, the corneal epithelial cells, or the corneal nerves plexus under the epithelium.

In the illustrative embodiments described above, the intrastromal implant or inlay **918**, **934** is not cross-linked, and its refractive power is not corrected prior its transplantation into the recipient cornea **912**, **930**. The implantation procedure may be either a MESICK (Meso=inside, I=inlay, C=Crosslinking, K=Keratomeileusis) procedure, or small incision implant procedure. In one or more other embodiments, the refractive power of the implant or inlay is corrected, but cross-linking is done after implantation inside a stromal pocket.

In one embodiment, microwave radiation may be used to modify the refractive error of the cornea or the inlay rather than laser radiation delivered by an excimer laser. Also, nanojoule pulses of a femtosecond laser may be applied to small lenticles after implantation to change the index of the refraction of the small incision lenses in the stromal cavity. In addition, a combination of cross-linking and microwave application may be used to crosslink the cornea and correct the refractive error while maintaining the corneal transparency.

In one embodiment, subsequently an antibiotic ointment or solution is applied to the operated eye in the post-operative period, and in some embodiments, if needed, the crosslinking process can be repeated inside the stroma and in the inlay to kill potential invading cells as needed and to keep the inlay's transparency.

In one embodiment, after the surgery, the refractive power of the cornea or the inlay can be modified using focally administrated focused microwaves.

In one embodiment, the corneal crosslinking can be combined with microwave application in the same session to achieve synergistic results.

In one embodiment, one can modify the refractive index of the corneal inlay by the use of few nanojoules of femtosecond laser application which does not cut or damage the inlay.

In one embodiment, the crosslinking solution and its components are used to treat an infected wound of the cornea, skin, or mucosa to be irradiated with UV light and kill the bacterial, viral, fungal and/or parasitic infection or the antibiotic resistant bacteria MRSA, or viruses or prions in the infected tissue and the corneal implant, etc.

In one embodiment, the implant is formed from donor corneal tissue, and wherein, prior to the step of inserting the implant under the flap, the method further comprises the

steps of cutting a circular implant from a donor cornea using a circular trephine, the circular implant having a diameter between approximately 3 millimeters and approximately 9 millimeters or more; and cutting the circular implant disks using a trephine or a femtosecond laser, the two circular disks having a thickness between approximately 50 microns and approximately 100 microns and shaping the inlay surface's refractive power with an excimer laser.

In one embodiment, a corneal pocket is created in the host corneal stroma with a diameter of 3-9 mm slightly larger than the diameter of the inlay for ease of implantation, using a femtosecond laser, and marking its center and making a small side incision in the cornea to access the inside of the intrastromal pocket and to separate the stromal pocket using a micro-spatula or a bent needle and hyaluronic acid, and then a pre-shaped corneal inlay and hyaluronic acid is implanted alone or with N-Vinylpyrrolidone and riboflavin for ease of implantation in the corneal stromal pocket, riboflavin or photosensitizer nanoparticles are injected in the corneal pocket to penetrate the inlay and slightly the wall of the pocket, and then the inlay and the wall of the pocket and hyaluronic acid are cross-linked with UV radiation, thus converting the hyaluronic acid into a gel.

In one embodiment, one uses a fluid containing inorganic or organic molecules such as hyaluronic acid, collagen, elastin, chondroitin sulfate or hydrogel, polyethylene glycol, or biotin, polycaprolactone, chitosan, etc. in various combinations in a physiologic solution having in addition a crosslinker such as riboflavin, N-Vinylpyrrolidone, or other medications.

In another embodiment, the implant is formed from donor corneal tissue or a 3-D printed cornea, and prior to the step of inserting the implant in the corneal stroma, one performs the steps of cutting a circular implant from a donor corneal stroma, etc. using a circular trephine, the circular implant having a diameter between approximately 3 millimeters and approximately 9 millimeters and creating a corneal pocket in the host corneal stroma with a diameter of 4-10 mm using a femtosecond laser, or any other laser and making one or two small side incisions in the cornea with the same femtosecond laser or a knife, to access the inside of the intrastromal pocket and to separate the walls of the pocket using a micro-spatula and hyaluronic acid and cutting the circular implant into two concentric circular parts, one circular disk inside the outer doughnut part using a smaller trephine than the diameter of the inlay or a femtosecond laser, the two concentric circular parts having a thickness between approximately 50 microns to approximately 100 microns and their surfaces are shaped to form a desired refractive power with an excimer laser to correct a myopic, hyperopic or astigmatic error of the eye after their separate implantation using forceps or an injector, the annular part first, then the disc part through a small incision that is made from outside to access the inside of the stromal pocket where the two pieces are joined again after their implantation. The pre-shaped corneal inlay is implanted with the annular part first, then the disk inside it second along with hyaluronic or low molecular weight heparin for ease of implantation in the corneal stromal pocket, riboflavin or riboflavin nanoparticles are injected as a 0.1-1% solution or another photosensitizer is injected along with other antibacterial or antifungal medications in the corneal pocket to penetrate the inlay and slightly the wall of the pocket, and finally the inlay and the wall of the pocket are crosslinked with UV radiation and to kill the potential pathogenic contamination associated with the implant.

In one embodiment, the inner disk implant can be organic and the outer annular part of the implant can be made from a synthetic transparent polymer or acrylic that is resilient or flexible and in one piece or multiple separate pieces, which may be implanted through a small incision made in the wall of the cornea after creating a corneal pocket with a larger diameter for the desired implant for ease of implantation.

In one embodiment, one uses a fluid containing inorganic or organic molecules, such as hyaluronic acid, low molecular weight heparin, collagen, elastin, chondroitin sulfate, hydrogel, polyethylene glycol, chitosan, biotin, polycaprolactone, etc. or in various combinations in a physiologic solution having, in addition to a crosslinker such as riboflavin or N-Vinylpyrrolidone, other medications, such as Rock inhibitors or anti-integrins, where the transparent solution has a refractive index of equal to 1.34 or more for correction of a hyperopic error, and is to be injected into a corneal stromal pocket of the recipient cornea prepared using a femtosecond laser or picosecond laser, etc. and to increase the anterior curvature of the stromal cavity by pushing the anterior wall of the cavity forward after injecting more solution, or reducing its curvature by removing an amount of fluid from the stromal cavity, to treat a hyperopic refractive error to the desired degree using a wavefront technology during the processes of the injection and immediately irradiating the cornea with UV radiation to convert the fluid-filled cavity into a cross-linked gel creating an intrastromal transparent lens with an anterior convex surface and an index of refraction of 1.34 or more and cross-linked adjacent corneal stromal tissue to prevent an immune response to the gel and kill potential existing pathogens in the gel and the cornea and prevent the lens from extruding through the small stromal opening made for injection.

In one embodiment, the crosslinking of the cornea and the inlay can be repeated using an internal injection of riboflavin nanoparticles in a physiological solution in the stromal cavity or applying riboflavin/medication drops from outside to penetrate the anterior cornea and the inlay, which are being cross-linked again with externally applied UV radiation to kill potential invading inflammatory cells, or organisms if any sign of rejection or cellular invasion or infection is observed to stop the process, or the implant is emulsified with a femtosecond laser and washed out or removed or replaced.

In one embodiment, a corneal pocket is created inside the corneal stroma using a femtosecond laser and a circular 3-D circular area is cut with a uniform thickness of approximately 60 microns inside the pocket prior to separation of the anterior stromal layer, then a small incision is cut to the side of the cornea to access inside the corneal pocket and remove the circular uniform disc from the stroma near the back of the cavity to create more space inside the corneal pocket and a new corneal inlay is placed inside the pocket, the inlay having a corrected refractive error of the eye for hyperopia, myopia, and astigmatism, and riboflavin nanoparticles along with hyaluronic acid in solution is injected inside the corneal cavity, and the inlay and the surrounding stromal tissue is cross-linked with UV radiation or another wavelength of light if a different crosslinker is used, to prevent corneal rejection or infection.

In another embodiment, a cavity is formed and a uniform disk or oval-shaped portion of the stroma of about 60-100 micron tissue thickness is removed, and the cavity is injected with cross-linkable molecules in a transparent biocompatible solution having a crosslinker to correct the hyperopic, hyperopic, or astigmatic refractive error of the eye, and the cross-linkable fluid is cross-linked by externally applied UV

laser radiation to cross-link the fluidic cavity and convert it into a cross-linked gel and cross-link the corneal pocket wall, while killing potential pathogens infecting the surgical area and preventing the escape of the gel inlay from the small incision that was initially made to access the corneal cavity and preventing glare seen after removal of the stromal tissue.

In one embodiment of corneal implantation with cross-linking, the method comprises the steps of: (i) forming a flap in a cornea of an eye so as to expose a stromal tissue of the cornea underlying the flap; (ii) pivoting the flap so as to expose the stromal tissue of the cornea underlying the flap; (iii) inserting an implant under the flap so as to overlie the stromal tissue of the cornea; (iv) applying laser energy to the implant in the eye so as to modify the refractive power of the implant and the eye while being monitored using a Shack-Hartmann wavefront system so as to achieve a desired refractive power for the implant.

In one embodiment, a cross-linking solution that includes a photosensitizer is applied to the implant and to the stromal tissue of the cornea surrounding the implant, the implant is covered with the flap, the implant being surrounded entirely by the stromal tissue of the cornea; and the implant and the stromal tissue of the cornea surrounding the implant is irradiated so as to activate cross-linkers in the implant and in the stromal tissue of the cornea surrounding the implant, and thereby cross-link the implant and the stromal tissue of the cornea surrounding the implant to prevent an immune response to the implant and/or rejection of the implant by the patient.

In one embodiment, the implant is formed from donor corneal tissue, a tissue culture grown cornea, a 3-D printed cornea, or preferably from human donor corneal tissue that is stored in a sterile container having an antibiotic and a preservative solution without cross-linking the implant prior to implantation, and a corneal LASIK flap is formed in the recipient cornea using one a femtosecond laser or a mechanical keratome.

In one embodiment, the photosensitizer of the cross-linking solution comprises nanoparticles of riboflavin, or another photosensitizer, and the step of irradiating the implant comprises irradiating the implant with ultraviolet light or a chemical crosslinker.

In one embodiment, the cross-linking solution contains one or more medications for treating a medical condition of the patient, the one or more medications selected from a group consisting of a Rock inhibitor, Wnt inhibitor, an integrin inhibitor, a GSK inhibitor, an antibiotic, an antifungal medication, and combinations thereof.

In one embodiment, the correction of the refractive power of the eye is done after placement of the inlay over the exposed corneal stroma using an excimer laser so as to modify the refractive power of the implant and the eye under the control of wavefront technology controlled with a Shack-Hartmann sensor.

In one embodiment, the cross-linking solution includes a photosensitizer in the form of riboflavin nanoparticles and a medication selected from the group consisting of an antibiotic, an antiviral, an antifungal, an anti-parasitic, and combinations thereof; and the cornea is irradiated with ultraviolet light to cross-link an infected part of the cornea so as to kill pathogens and treat infectious keratitis.

In one embodiment, the implant is formed from donor corneal tissue by cutting one or more circular implant disks from a donor cornea using a trephine or a femtosecond laser so as to form the implant, the one or more circular implant disks having a diameter between approximately 3 millime-

ters and approximately 9 millimeters, and the one or more circular implant disks having a thickness between approximately 50 microns and approximately 100 microns; and a surface of the one or more circular implant disks is shaped using an excimer laser prior to implantation under a corneal flap or inside a corneal stromal cavity.

In one embodiment, the implant and the wall of the corneal cavity is cross-linked after implantation using a fiber optic or the implant and the wall of the corneal cavity contains a crosslinker that can be irradiated using an external energy source (e.g., UV radiation) through the surface of the cornea, and the anterior part of the cornea does not have a crosslinker and is not cross-linked during this procedure.

In one embodiment, the cross-linking solution contains nanoparticles of riboflavin, cell penetrating peptides, and an antibiotic, antifungal, anti-viral, and/or antiparasitic medication to treat a bacterial infection of the cornea, skin, or mucosa or the eyelids with ultraviolet radiation to kill the invading pathogens.

In one embodiment, a corneal cavity of 3-9 mm wide or larger is created in the host corneal stroma using a femtosecond laser where a small side incision is made in a recipient cornea of an eye of a patient for insertion of a circular implant into the stromal pocket by injecting the cross linking solution, and then crosslinking the inlay and the wall of the cavity so as to prevent an immune response to the implant and/or prevent rejection of the implant by the patient's tissue.

In one embodiment, the step of forming the small side incision in the recipient cornea of the eye further comprises making the small side incision in the recipient cornea with a femtosecond laser or a knife.

In one embodiment, a cavity is formed in the host stroma and the wall of the cavity is separated using a spatula or a bent needle while injection of a solution of hyaluronic acid, and an implant is inserted along with injection of riboflavin nanoparticles to crosslink the inlay and the wall of the cavity and the remaining hyaluronic acid irradiating the circular implant with ultraviolet light so as to cross-link the circular implant and the portion of the bounding wall of the pocket and the hyaluronic acid, and kill any potential pathogenic contaminants associated with the circular implant and create a clear tissue transition preventing postoperative glare.

In one embodiment, the circular implant is cut with a circular trephine having an outer diameter between approximately 3 millimeters and approximately 9 millimeters; then the circular implant is cut into two concentric circular parts, the first one of the two concentric circular parts comprising an annular disk portion and the second one of the two concentric circular parts comprising a central disk portion inside the annular outer portion, the central disk portion being cut using a femtosecond laser or a trephine having a diameter smaller than the outer diameter of the circular implant, each of the two concentric circular parts having a thickness between approximately 30 microns and approximately 100 microns.

In one embodiment, the surface of the implant is shaped with an excimer laser prior to implantation in the corneal stromal pocket to form a desired refractive power so as to correct a myopic, hyperopic, or astigmatic error of the eye.

In another embodiment, the implant inserted into the pocket of the recipient cornea via a small incision by initially inserting the annular disk portion into a pocket via the small incision using an injector syringe or a forceps, and then subsequently inserting the central disk portion into the pocket via the small incision using the injector syringe or the forceps, and joining the two concentric circular parts

together inside the pocket. Then, a crosslinker with hyaluronic acid is injected inside the pocket over the implant, and the implant and the stromal wall are cross-linked with UV radiation to prevent postoperative glare.

In one embodiment, the central disk portion of the circular implant is organic and the annular disk portion of the circular implant is made from a synthetic transparent polymer, acrylic, or hydrogel, etc. that is resilient or flexible.

In one embodiment, a corneal pocket is formed in the host's posterior corneal stroma using a femtosecond laser, first a circular or oval-shaped three-dimensional disk is cut in the corneal stroma, and a second cut is made in front of the first cut, but the second cut is slightly larger with a substantially uniform thickness or distance from the first cut of approximately 60 microns and 100 microns using the femtosecond laser so as to form the pocket, and the circular or oval-shaped three-dimensional posterior disk is removed from the host corneal stroma to create more space inside the pocket; and the anterior lamella is separated and the posterior lamellar is removed using a micro-spatula or a bent needle, hyaluronic acid is applied for ease of removal of the tissue and insertion of the circular implant into the pocket in the recipient cornea of the eye and to smoothen up the wall of the cavity; riboflavin or other photosensitizer nanoparticles are injected into the pocket in the recipient cornea of the eye so as to penetrate the circular implant and a portion of a bounding wall of the pocket; and the circular implant is irradiated with ultraviolet light or another wavelength of light so as to cross-link the circular implant and the portion of the bounding wall of the pocket, and to prevent corneal rejection of the circular implant and/or prevent corneal infection and postoperative glare.

In one embodiment, a corneal pocket is formed in the host's posterior corneal stroma using a femtosecond laser, first a circular or oval-shaped three-dimensional disk is cut from the corneal stroma, then a second cut is made in front of the first cut, but the second cut is slightly larger with a substantially uniform thickness or distance from the first cut of approximately 60 microns and 100 microns using the femtosecond laser so as to form the pocket, and the circular or oval-shaped three-dimensional posterior disk is removed from the host corneal stroma to create more space inside the pocket by creating two small side incisions opposite each other to gain entry in the pocket from two sides, thus simplifying removal of the stromal tissue or implantation of an inlay afterwards having the two instruments inserted inside the corneal pocket, one having a sharp edge to separate the tissue from the underlying stroma, and the other to hold the edges or two prongs to pull the tissue for ease of manipulation and removal of the circular or oval-shaped three-dimensional disk with a uniform thickness from the host's posterior corneal stroma to create more space inside the pocket; and portions of the pocket are separated using a micro-spatula or a bent needle and hyaluronic acid; and hyaluronic acid is used for ease of insertion of the circular shaped implant into the pocket to correct the refractive error of the eye in the recipient cornea of the eye and smoothen up the wall of the cavity, an riboflavin or other photosensitizer nanoparticles are injected with or without hyaluronic acid into the pocket in the recipient cornea of the eye so as to penetrate the circular implant and a portion of a bounding wall of the pocket; and the circular implant is irradiated with ultraviolet light or another wavelength of light so as to cross-link the circular implant and the portion of the bounding wall of the pocket, and to prevent corneal rejection of the circular implant and/or prevent corneal infection and postoperative glare.

In one embodiment, after creating and removing a circular or oval-shaped three-dimensional disk from the pocket, cross-linkable molecules or monomers are injected in a transparent biocompatible solution containing a crosslinker into the pocket so as to correct a hyperopic, hyperopic or astigmatic refractive error of the eye; and under control of wavefront technology and a Shack-Hartmann sensor, the transparent biocompatible solution is cross-linked by externally applying ultraviolet radiation so as to cross-link the solution inside the pocket and convert the solution to a cross-linked gel, and to further cross-link a bounding wall of the pocket and kill any potential pathogens infecting the surgical area of the eye and preventing postoperative glare.

In one embodiment, one uses a femtosecond laser or picosecond laser to form the pocket in the recipient cornea of the eye, then injects a cross-linking solution into the pocket in the recipient cornea of the eye so as to penetrate the circular implant and a portion of a bounding wall of the pocket, the cross-linking solution comprising a transparent fluid that contains inorganic or organic molecules or monomers that include hyaluronic acid, and its modified derivatives, N-isopropylacrylamide, alginate, thiol, tyramine, collagen, elastin, chondroitin sulfate, hydrogel, photocrosslinked hydrogel, polyethylene glycol, poly(vinyl alcohol) (PVA), biotin, chitosan, and combinations thereof, in addition to a photosensitizer and one or more medications, the transparent fluid having a refractive index equal to 1.34 or more for correction of a hyperopic error; increasing the corneal curvature by pushing an anterior wall of the pocket forward by injecting more of the transparent fluid, or reducing an amount of the transparent fluid to treat a hyperopic refractive error to the desired degree using wavefront technology; and the recipient cornea is irradiated with ultraviolet light to convert the transparent fluid in the pocket into a cross-linked gel, thus creating an intrastromal transparent lens with an anterior convex surface and an index of refraction of 1.34 or more, producing cross-linked adjacent corneal tissue so as to prevent an immune response to the cross-linked gel and kill potential existing pathogens in the cross-linked gel and the recipient cornea, and to prevent the intrastromal transparent lens from extruding through the small side incision in the recipient cornea and prevent postoperative glare.

In one embodiment, the crosslinking preparation containing medications is cross-linked and used as a device that can be placed over or under the conjunctiva, or the lid, so as to be used as a drug delivery device for delivering a non-toxic dose of antibiotic, antiviral, antifungal, anti-parasitic, Rock inhibitors, integrin inhibitors, or Wnt inhibitors after LASIK surgery, corneal transplant, PRK, or after any corneal or conjunctival procedure to treat infection, reduce inflammatory process to reduce scarring, after ocular surgery, encourage nerve growth in the cornea after refractive surgery and treat infective diseases of the cornea, conjunctiva, or the lid.

In one embodiment, the photosensitizer or the cross-linking solution is riboflavin, and the one or more medications are selected from a group consisting of a Rock inhibitor, Wnt inhibitor, an integrin inhibitor, a GSK inhibitor, an antibiotic, an anti-fungal medication, and combinations thereof.

In one embodiment, the recipient cornea and the intrastromal inlay are cross-linked one or more additional times with the crosslinker and a suitable medication is applied if any sign of rejection and/or cellular invasion or infection is observed so as to stop the rejection and/or the cellular invasion.

In one embodiment, one can emulsify the intrastromal transparent gel using a femtosecond laser, washing out or removing the emulsified intrastromal transparent gel, and replacing the emulsified intrastromal transparent gel with a new intrastromal transparent gel-able preparation.

In one or more further embodiments described hereinafter, improved procedures for preserving donor corneas will be explained. Although the transplantation of the cornea was introduced a century ago, it was much later in the 1940s-1950s that corneal preservation for the full thickness transplantation was attempted. Using the so-called moist chamber, initially the entire eye was stored at 2-6° C. as hypothermic. Subsequently, the cornea and some surrounding tissues were kept in organ culture at 31-37° C. Freezing the corneal tissue was discouraged, since it damages the endothelial cells that are useful for regulating the fluid transport in the full thickness corneal graft.

At present, the maximal time that the corneal tissue can be preserved is about 14 days, but, in general, the preserved cornea loses their viability after 3-4 days. These corneas become permeable to water in the culture media, become cloudy and useless as corneal transplant are discarded.

In one embodiment, the corneas are obtained shortly after the donor's death. Biopsies of the eye surfaces are taken for bacteriological evaluation and culture. Corneal storage after corneal or scleral crosslinking does not require any matching, since all proteins, such as collagen are cross-linked during the process of cross-linking using a photosensitizer, such as riboflavin, and light, such as ultraviolet (UV) light and the cross-linked collagen is not immunogenic. Nevertheless, after the initial step of bacteriological biopsy, the eye is placed in a bottle for transfer under aseptic conditions having a solution containing antibiotics and/or antimicrobials, since about 10% of the eyes can be contaminated, but the actual incidence of endophthalmitis is very low.

In one embodiment, the retrieved corneas from the eye bank are prepared by dissecting the soft tissue conjunctiva, muscles, retina, choroid, iris, the ciliary body, and the optic nerve, which are cut away and removed. The endothelial cells and the corneal epithelium are scraped away leaving only the sclera and the corneal stroma. The corneal stroma and scleral tissue are washed and cleansed in a physiological solution or washed with a solution of povidone iodine and then with a physiological solution, and prepared for in vitro crosslinking.

In one embodiment, the advantage of using a photosensitizer and its activation in the tissue with light (e.g. UV light) is that the corneal tissue is cross-linked, and it does not lose its transparency, which is very important after corneal transplantation so that the light reaches the retina without being distorted or absorbed, whereas the cross-linking of tissue with the use of chemical crosslinkers, such as a formaldehyde, interferes with not only the transparency of the cornea, but makes it very stiff, in addition to being very toxic to the surrounding tissue.

In one embodiment, the cornea or the sclera are cut to a specific shape and size as needed.

In one embodiment, the corneal tissue can be cut using a knife, keratome, a femtosecond laser, or excimer laser to the shape and thickness of a lens with defined diameter, or dioptric power creating plus or minus astigmatic lens, or a toric lens.

In one embodiment, the corneal or scleral tissues are initially cross-linked prior to the surgery, and then cut using a knife, keratome, a femtosecond laser, or an excimer laser to the shape and thickness of a lens with defined diameter,

or dioptric power and thickness creating a plus or minus astigmatic lens or a combination thereof.

In one embodiment, the corneas may be cut to be circular, with a central hole of 1-7 millimeters (mm) or rectangular, oval, etc. or any desired shape for implantation inside the host cornea in a pocket created with a femtosecond laser at any depth of the host corneal stroma.

In one embodiment, the lens implant may have one of: (i) a concave surface to correct myopic refractive errors (i.e., a minus lens for correcting nearsightedness), (ii) a convex surface to correct hyperopic refractive errors (i.e., a plus lens for correcting farsightedness), or (iii) a toric shape to correct astigmatic refractive errors.

In one embodiment, the corneas are cross-linked for storage in eye bank by placing them in a crosslinking solution having any photosensitizer, such as riboflavin, at any desired concentration and osmolarity, preferably of 300 mOsm/L at a pH of 4-8, preferably pH 7.4, and having a concentration between about 0.1% to about 100%, for example, of riboflavin therein, etc.

In one embodiment, the cross-linked cornea and sclera can be placed in a sterile vial under vacuum, or in a standard storage solution, such as tissue culture medium 199 (M-199) containing 5% dextran 40 for hypothermia, even cryopreservation, and can be kept frozen under very low temperature after ice-free cryopreservation by vitrification of the cross-linked cornea in a high concentration of propane-1,2-diol or liquid nitrogen by stepwise cooling the tissue, etc.

In one embodiment, the cellular components of the corneal stroma may be initially destroyed using enzymatic agents, such as Trypsin, Nucleases, Phospholipases A₂ (PLA₂), Sera, all have shortcomings of damaging the basement membrane, such as Bowman's or Descemet's membrane. Although destruction of the cellular elements can be beneficial, alone it is useless since the antigenic component of the stroma that is the collagen is not cross-linked.

In one embodiment, one can destroy the cellular component of the corneal or sclera initially using chemical means, such as Ethanol, Glycerol, acids and alkalis, Peracetic acid, Ammonium hydroxide ionic detergents, Sodium dodecyl sulfate, Sodium deoxycholate non-ionic detergents, zwitterionic detergents, (CHAPS), but all damage the Glycosaminoglycans (GAG) component of the stroma of the cornea or the sclera and do not cross-link the collagen, thus the tissue maintains its antigenicity.

In one embodiment, the corneal stroma or the sclera is exposed to hypo- and hypertonic solutions that are basically harmless, but it is not effective in destroying the cellular elements of the cornea or sclera and does not cross-link the collagen, etc. while maintaining the antigenicity of the tissue.

In one embodiment, the cornea or sclera is exposed to Tris-HCL that eliminates cellular elements and also damages the GAG component of the tissue, and it does not cross-link the collagen.

In one embodiment, the physical means, such as freeze-thawing or snap freezing can be used that damages the cellular elements of the tissue, but also damages the extracellular matrix (ECM) of the tissue without crosslinking the collagen.

In another embodiment, the tissue can be exposed to high ultrasonic energy and frequency that can damage the cells when the cornea is immersed in the fluid and the ultrasonic transducer in also immersed in the solution. This technique and/or exposing the cornea or sclera to high hydrostatic pressure can destroy the cells without damaging the collagen in the stroma or in the sclera, but both techniques do not

cross-link the cornea, and the tissues maintain their immunogenicity and create an immune response that encourage immune cell migration and enzymatic destruction of the implant, and the implant is ultimately rejected.

In one embodiment, one can expose the donor cornea or sclera to ultrasonic destruction or high hydrostatic destruction of the cellular elements, then apply a photosensitizer, such as riboflavin, to the tissue immersed in a riboflavin solution that is then radiated with UV radiation or another light to crosslink the cornea or the scleral tissue. Subsequently, the tissue is cleansed in a saline solution to wash out the reactive oxygen species and enzymes that are released by destroying the cells by a simple agitation of the tissue or a low frequency ultrasound. In one embodiment, this ultrasonic technique can be also applied to in vivo corneas or the sclera after their cross-linking with a photosensitizer/UV radiation.

In one embodiment, the cross-linked cornea can be used as an inlay to augment the thickness of the cornea and render it more resilient for refractive surgery procedures.

In one embodiment, the cross-linked cornea is placed over the host cornea using a Shack-Hartmann system or other available instrument to correct the refractive error of the eye with an excimer laser, prior to implantation of the implant inside a corneal pocket.

In one embodiment, the cross-linked cornea or sclera can be used to replace injured lip, nose, or cheek augmentation in cosmetic surgery, chin augmentation, or reconstruction, while the cornea or sclera that is cross-linked maintains its flexibility and softness.

In one embodiment, for eye bank storage, the cornea and/or sclera can be cross-linked (see e.g., FIG. 4B) using a photosensitizer/cross-linker and UV radiation to kill the cellular elements and crosslink the cornea and scleral collagen making them less antigenic while being able to store them for a long time (e.g., greater than 1 year storage) and maintaining the corneal stromal clarity.

In one embodiment, the corneal stroma and/or the sclera can be preserved by the photosensitizer crosslinking technique and kept almost indefinitely if the cross-linked cornea is kept at temperature close to zero, and is sealed in a bottle under vacuum for many years. The cross-linked corneal stroma can be washed with a sterile physiological saline solution to regain its shape and transparency and shaped with a femtosecond laser to any shape lenslet for correction of myopia, hyperopia, and astigmatism while strengthening the cornea by increasing its shape rather than decreasing it as is done with the present photorefractive, or LASIK or SMILE laser procedures. The preserved corneal inlay also can be once again cross-linked or exposed to UV radiation or gamma radiation to completely sterilize it. This procedure prevents post implantation bacteria, fungal parasites, or viral infections seen often after implantation of the partial corneal tissue or inlay where the pathogens can hide in the crevices of the tissue (e.g., inside the cornea) and grow a few days or weeks later producing a difficult to manage endophthalmitis or corneal ulcer.

In one embodiment, the corneal transplant is cross-linked to render the corneal tissue more resilient to expansion as occurs after refractive surgery.

In one embodiment, the photosensitizer (e.g., riboflavin) or other crosslinkers are conjugated with cell penetrating peptides (CPP) or activatable cell penetrating peptides (ACPP) or other cell penetrating agents, such as cyclodextrin, etc. that enhances penetration of the crosslinker in the

tissue, saving time for surgery if the crosslinking is done on a patient, or if the crosslinking is done for the storage of the tissue.

In one embodiment, the photosensitizer formulation permits faster penetration of the crosslinker inside the corneal transplant, thus reducing the operative time less than $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{5}$ or less than $\frac{1}{10}$ of the time that it would take to cross-link the corneal tissue as done by the standard procedure.

In one embodiment, the crosslinkers are made in the form of nanoparticles, such as dendrimers, and can be organic, non-organic, metallic, such as gold, iron or ferric oxide or zinc, silver, silica, graphene, carbon nanotubes, nanowires, quantum dots (QDs), etc. or coated with lipids, or an antibody that encourages the cell penetration, etc.

In one embodiment, the crosslinker is conjugated with CPP and antibody against a bacterium, fungi, viruses, prions, parasites, attached to either metallic or organic nanoparticles, such as a carbon nanospheres or nanotubes, or gold or silver or zinc nanoparticles, etc. and CPP, and radiated with UV light or other wavelengths of light that is absorbed by the nanoparticles and the photosensitizer heats up the nanoparticle/pathogens complex creating thermal damage to the pathogens in addition to being cross-linked by the photosensitizer. The use of UV radiation, in addition, damages the genetic components of the pathogens along with the medication that kills the pathogens or makes them more sensitive to the medication to which the pathogens would not be sensitive normally or were resistant, such as methicillin-resistant *Staphylococcus aureus* (MRSA) bacteria, after administering the crosslinker complex locally or topically.

In one embodiment, the antibody-coated nanoparticles can be attached to the antimicrobial peptides, and the crosslinker is applied topically to the eye, lid, skin or mucosa, or any reachable infection even with a fiber optic (e.g., in the lung, eye, bladder, vagina, uterus, throat, nose, or exposed tumor) as needed and radiated with UV light to crosslink the protein of the bacteria and kill them along with potentially a cancerous lesion or cells (e.g., bladder cancer).

In one embodiment, the process of crosslinking using UV radiation creates singlet oxygen and reactive oxygen species in the presence of the oxygen that cross links the proteins, including those present in the bacteria, fungi, viruses, prions, and parasites, thereby killing them, and in case of the cornea or the sclera, crosslinks the collagens in the tissue and makes then less immunogenic.

In one embodiment, the corneal tissue is cross-linked to be used in refractive surgery as a corneal inlay to change the refractive power of the cornea. In another embodiment, it replaces the damaged corneal tissue. In yet another embodiment, the cross-linked cornea or sclera can be used as a tissue filler or provide a shape for the surrounding tissue structures, such as in cosmetic surgery in the face, replacement for the damaged lids, reconstructing the facial appearance, etc. In still another embodiment, it can serve as a protecting container to prevent immune response to the cells, etc. inside the container or can be used as a reservoir for drug delivery etc.

In one embodiment, the cross-linked cornea or sclera is co-cultured in a medium having nanoparticles/CPP or lipids that penetrates the tissues scaffold, and the nanoparticles are made of polylactic, polyglycolic or polycaprolactone of silicon nanoparticles conjugated with one or more medications (e.g., anti-inflammatory agents, steroids, NSAIDs, anti-proliferative agents, immunosuppressives, such as cyclosporine, mycophenolic acid, antibiotics, anti-glaucoma agents, such as prostaglandin analogues, Rock inhibitors, such as Fasudil, rhopressa, etc., anti-integrins, Wnt inhibi-

tors, or GSK inhibitors, anti-VEGFs, or combinations thereof). The cross-linked cornea or sclera can be cut to any size or thickness (e.g., 10 microns to 1 millimeter and more) be radius shaped, circular, rectangular, etc. to release the medication depending on the bio-degradation of polymeric nanoparticles. The implant can be implanted anywhere in the eye (see e.g., FIG. 4E).

In one embodiment, the cross-linked cornea or sclera is made any desired shape and thickness using a femtosecond laser or an excimer laser, and using the same lasers one can punch partial or full thickness holes in the implant, so that after penetration of the nanoparticles, the release time will depend on the number and the size of the holes created inside the cross-linked cornea ranging from 1 month to few years, the cross-linked cornea or sclera is then co-cultured in a medium having nanoparticles/CPP or lipids that penetrate the tissues scaffold and nanoparticles are made of polylactic, polyglycolic or polycaprolactone of silicon nanoparticles conjugated with medications (e.g., anti-inflammatory agents, steroids, NSAIDs, anti-proliferative agents, immunosuppressives, such as cyclosporine, mycophenolic acid, antibiotics, anti-glaucoma agents, such as prostaglandin analogues, Rock inhibitors, such as Fasudil, rhopressa, etc., anti-integrins, Wnt inhibitors, or GSK inhibitors, anti-VEGFs, or a combination thereof). The cross-linked cornea or sclera can be cut to any size or thickness (e.g., 10 microns to 1 millimeter and more), be radius shaped, circular, rectangular, etc. to release the medication depending on the bio-degradation of polymeric nanoparticles. The implant can be implanted anywhere in the eye or enhanced its release by a focused ultrasound.

In one embodiment, the scleral tissue is cross-linked for long term storage.

In one embodiment, the photosensitizer, such as riboflavin, is in a physiological solution with osmolarities of 200 mOsm/L to 400 mOsm/L, or preferably about 300 mOsm/L.

In one embodiment, the pH of the solution is between 6-8 pH or preferably 7.4 pH.

In one embodiment, the photosensitizer solution contains dextran of 0.1%-3%.

In one embodiment, the photosensitizer/nanoparticle solution can contain low molecular weight heparin or hyaluronic acid or chondroitin sulfate, etc., the low molecular weight heparin enhances the killing of the bacteria in vitro and in vivo after topical application and irradiation with UV light or other light. These preparations can be used for treatment of bacterial, fungi, viral, parasitic keratitis or conjunctivitis.

The crosslinking of the tissue is done when the photosensitizer is radiated with a light wavelength of ultraviolet to infrared depending on the photosensitizer that absorbs the light beam. In one embodiment, the photosensitizer solution can, in addition, contain some additives, such as antibiotics, antifungals, antivirals, and/or anti-parasites.

In one embodiment, the corneal crosslinker is conjugated with CPP, nanoparticles applied to the cornea and irradiated with UV light or other light to kill all cellular elements of the cornea and sclera and crosslink the collagen, and sterilize the corneal implant for storage prior to its use.

In one embodiment, for the cross-linked corneal storage (CCS), the corneal crosslinker is conjugated with CPP, antibody-coated metallic nanoparticles, such as gold nanoparticles, etc. When the implant is completely penetrated by the crosslinker, it is radiated with UV light or other light wavelengths to kill the bacteria by thermal energy, and the crosslinking kills bacteria, fungi, parasites, viruses and prions for corneal storage prior to its use.

In still a further embodiment, the photosensitizer is conjugated with one or more metallic or non-metallic nanoparticles and/or one or more antibody-coated nanoparticles, and the one or more nanoparticles and/or one or more antibody-coated nanoparticles are further conjugated with one or more cell penetrating peptides (CPP) and/or activatable-cell penetrating peptides (ACPP), thereby forming a photosensitizer-nanoparticle-CPP/ACPP complex for facilitating an enhanced penetration of the photosensitizer into a corneal stroma and stromal cells of the host cornea with or without the use of low intensity ultrasound to kill the stromal cells after UV radiation.

In a further embodiment of the present invention, the step of cross-linking the corneal implant prior to implantation of the cornea comprises the substeps of: (a) applying a photosensitizer to the cornea so as to penetrate the entire corneal implant; (b) irradiating the cornea with ultraviolet light so as to activate cross-linkers in the cornea and thereby crosslink the entire corneal implant; (c) cutting the cross-linked cornea to the desired shape or dioptric power; (d) creating a pocket in the recipient cornea; and (e) implanting the cross-linked cornea inside the corneal pocket along with a lubricating substance, such as hyaluronic acid, etc.

In one embodiment, the corneal inlay is implanted as above, then cross-linked with a photosensitizer injected inside the corneal pocket to penetrate the implant and the wall of the corneal pocket, and then the cornea and the inlay are simultaneously cross-linked. Followed by copious washing of the cornea with saline and/or an anti-inflammatory agent, such as steroid, NSAIDs, or ROCK inhibitors, anti-integrins, or GSK inhibitors or Wnt inhibitors.

In still a further embodiment, the photosensitizer is conjugated with one or more nanoparticles and/or one or more antibody-coated nanoparticles, and the one or more nanoparticles and/or one or more antibody-coated nanoparticles are further conjugated with an antimicrobial peptide that damages the bacterial cell membrane while the normal cells are not affected, and one or more cell penetrating peptides (CPP) and/or activatable-cell penetrating peptides (ACPP), and thereby forming a photosensitizer-nanoparticle-CPP/ACPP complex for facilitating an enhanced penetration of the photosensitizer into a corneal stroma and stromal cells of the host cornea to kill the stromal cells. The preparation is administered inside an infected corneal pocket of a large abscess, of nasolacrimal duct, or the lid, or skin or mucosa and irradiated with the light to activate the photosensitizer and kill all organisms while protecting the normal tissue.

In still a further embodiment, the photosensitizer is conjugated with one or more antibody-coated nanoparticles or dendrimers, and the one or more antibody-coated nanoparticles or dendrimers are further conjugated with one or more cell penetrating peptides (CPP) and/or activatable-cell penetrating peptides (ACPP), thereby forming a photosensitizer-antibody-coated nanoparticle or dendrimer-CPP/ACPP complex; and the method further comprises the steps of: (iv) administering the photosensitizer-antibody-coated nanoparticle or dendrimer-CPP/ACPP complex topically or by injection to the host cornea, the photosensitizer being absorbed by bacteria, viruses, fungi, and/or protozoa present in an infected corneal ulcer of the host cornea; and (v) irradiating the host cornea with light so as to kill the bacteria, viruses, fungi, or protozoa by cross-linking the cytoplasmic proteins of the bacteria, viruses, fungi, and/or protozoa by thermal energy and crosslinking of the proteins of the micro-organisms, viruses, and/or prions, and sterilizing the entire cross-linked corneal and scleral implant's thickness and killing all

pathogens and cells in the implant and cleansed with a physiological solution to wash out the toxins and the debris in the tissue.

In one embodiment, the antimicrobial peptides are from the group of chemotactic peptides having inflammatory and antimicrobial activity, such as polybia-CP (Pol-CP—NH₂: Ile-Leu-Gly-Thr-Ile-Leu-Gly-Leu-Leu-Lys-Ser-Leu-NH₂), Pol-CP—NH₂, and protonection and polybia-MPI, linear cationic amphipathic AMPs, or wasp Polybia paulista.

In one embodiment, when the patient suffers from an infected corneal ulcer, or skin ulcer, or mucosa ulcer, the crosslinker is conjugated with CPP, antibiotics, antifungals, anti-parasites, antivirals etc. and the antibody-coated metallic nanoparticles, such as gold nanoparticles, and the ulcer is treated with a topical application of the crosslinker complex to the ulcer, followed by UV radiation or other light wavelengths to kill the bacteria/fungi parasites, or viruses, etc. by thermal energy, medication, and crosslinking the proteins of the bacteria, fungi, parasites, viruses, while crosslinking the cornea or the skin to resist enzymatic degradation and healing the infected ulcer.

In one embodiment, corneal crosslinking is performed with CPP, antibody-coated metallic gold nanoparticles and antibiotics and/or antimicrobial proteins, etc. using UV or other light wavelengths to kill the bacteria by thermal energy and crosslinking. The thermal energy and crosslinking kills bacteria, fungi, parasites, viruses and prions, and their infectious pathogens, such as viral infections, such as hepatitis B, West Nile virus, Chagas disease, and prion agents.

In one embodiment, scleral crosslinking with CPP, nanoparticles using UV light or other wavelengths of light kills all cellular elements of the cornea and/or sclera, and crosslinks the collagen, and sterilizes the implant.

In one embodiment, scleral crosslinking with CPP, antibody-coated metallic gold nanoparticles, etc. using UV light or other light wavelengths kills the bacteria, fungi, parasites, viruses and prions by thermal energy and crosslinking.

In one embodiment, scleral crosslinking with CPP, antibody-coated metallic gold nanoparticles and antibiotics and/or antimicrobial proteins, etc. using UV or other light wavelengths kills the bacteria, fungi, parasites, viruses and prions by thermal energy and crosslinking, and subsequently, the tissue is washed with a physiologic solution and exposed to low frequency ultrasound to cleanse the tissue toxins released by killing the bacteria, fungi, viruses, prions, or parasites.

In one embodiment, after storage of the cross-linked cornea or sclera, if the cross-linked tissue appears swollen, it can be treated with a physiological solution containing antibiotics, deturgescence agents like dextran 4-8%, and chondroitin sulphate, resulting in deswelling of the cornea or the sclera.

In one embodiment, the cross-linked tissue after long term storage can be re-sterilized with gamma radiation.

In one embodiment, the corneas can be stored using standard techniques while suspended in 80 ml Eagle's MEM containing HEPES buffer, 26 mmol/l NaHCO₃, 2% FBS, 2 mmol/l L-glutamine, penicillin, streptomycin and amphotericin B, and stored at 34° C. for 1-7 days, then, if they have not used for corneal transplantation, they can now be prepared as described above for crosslinking or the cornea and sclera, which can be then stored for greater than 1 year, etc.

In one embodiment, the cross-linked cornea is denuded from corneal epithelial cells and endothelial cells while keeping the Descemet's membrane co-cultured with stem cells, or endothelial stem cells, or pre-corneal conjunctival stem cells, or mesenchymal stem cells of the potentially host

patient with polymeric anti-inflammatory nanoparticles of PLA, PLGA having Rock inhibitors, anti-integrins, GSK inhibitors, and/or immunosuppressive agents combined with electrophoresis tissue penetration in the implant and/or followed with medical treatment using topical antibiotics, anti-inflammatory compounds, such as Rock inhibitors, anti-integrins, GSK inhibitors, and/or immunosuppressive agents.

In one embodiment, the cross-linked cornea is denuded from corneal epithelial cells and endothelial cells while keeping the Descemet's membrane co-cultured with stem cells, or endothelial stem cells, or pre-corneal conjunctival stem cells, or mesenchymal stem cells of the potentially host patient and polymeric nanoparticles conjugated with anti-inflammatory compounds, such as Rock inhibitors, anti-integrins, GSK inhibitors, and/or immunosuppressive agents, such as cyclosporine, mycophenolic acid, tacrolimus, combined with electrophoresis tissue penetration in the cross-linked implant etc. to release the medication after implantation for a long time after implantation.

In one embodiment, the cross-linked cornea is denuded from corneal epithelial cells and endothelial cells while keeping the Descemet's membrane co-cultured with stem cells, endothelial stem cells, and transfection with transcription factors such as E2F2, which has been reported to stimulate cell-cycle progression and endothelial cell replication to grow endothelial cells on the Descemet's membrane for transplantation in a patient. The technology can be used as Descemet stripping automated endothelial keratoplasty (DSEK) or Descemet membrane endothelial keratoplasty (DMEK) or deep lamellar keratoplasty (DLK) for partial transplantation of the cornea that is now cross-linked with or without corneal endothelial cells as known in the art, simplifying the 3-D printing of the cornea, for the transplantation, since one has now a resilient cross-linked cornea to work with, and the process can be made simple by manufacturing all the tissues except for growing endothelial cells on the back of a cross-linked cornea with its Descemet's membrane intact, thus reducing the chance of the rejection since the corneal stroma part of the transplant is cross-linked and will not induce an immune response in the host followed after implantation with topical or subconjunctival medical treatment using antibiotics, anti-inflammatory compounds, such as Rock inhibitors, anti-integrins, GSK inhibitors, and/or immunosuppressive agents.

In one embodiment, one can use a combination or cross-linked cornea or cross-linked animal cornea or genetically engineered animal cornea to grow corneal endothelial cells prior to their implantation in DSEK or DMEK, since the cross-linked corneal stromas are not immunogenic. The cross-linked tissue is co-cultured with a medium having nanoparticles/PPP or lipids that penetrates the tissue scaffolds and the nanoparticles are made of polylactic, polyglycolic or polycaprolactone or silicon nanoparticles conjugated with medications, such as anti-inflammatory agents, steroids, NSAID, anti-proliferative agents, immunosuppressives, such as cyclosporine, mycophenolic acid, antibiotics, anti-glaucoma agents, such as prostaglandin analogues, Rock inhibitors, such as Fasudil, rhopressa, etc., anti-integrins, Wnt inhibitors, or GSK inhibitors, anti-VEGFs, or combinations thereof combined with or without electrophoresis tissue penetration in the implant thereof, to release the medication after implantation depending on the bio-degradation of polymeric nanoparticles.

In one embodiment, one can use a combination or cross-linked cornea or cross-linked animal cornea or genetically engineered animal cornea to grow corneal endothelial cells

prior to their implantation in DSEK or DMEK since the cross-linked corneal stromas are not immunogenic. The cross-linked tissue is co-cultured with a medium having nanoparticles/PPP or lipids that penetrates the tissue scaffolds and the nanoparticles are made of polylactic, polyglycolic or polycaprolactone or silicon nanoparticles conjugated with medications, such as anti-inflammatory agents, steroids, NSAIDs, anti-proliferative agents, immunosuppressives, such as cyclosporine, mycophenolic acid, antibiotics, anti-glaucoma agents, such as prostaglandin analogues, Rock inhibitors, such as Fasudil, rhopressa, etc., anti-integrins, Wnt inhibitors, or GSK inhibitors, anti-VEGFs, or combinations thereof and the tissue is exposed to low intensity ultrasound to enhance penetration of the nanoparticles in the implant and to release the medication after implantation slowly, depending on the bio-degradation of polymeric nanoparticles to release the medication after implantation depending on the bio-degradation of polymeric nanoparticles.

In one embodiment, one can use a combination or cross-linked cornea or cross-linked animal cornea or genetically engineered animal cornea to grow corneal endothelial cells prior to their implantation in DSEK or DMEK since the cross-linked corneal stromas are not immunogenic. The cross-linked tissue is co-cultured with a medium having nanoparticles/PPP or lipids that penetrates the tissue scaffolds and nanoparticles are made of polylactic, polyglycolic or polycaprolactone or silicon nanoparticles conjugated with medications, such as anti-inflammatory agents, steroids, NSAIDs, anti-proliferative agents, immunosuppressives, such as cyclosporine, mycophenolic acid, antibiotics, anti-glaucoma agents, such as prostaglandin analogues, Rock inhibitors, such as Fasudil, rhopressa, etc., anti-integrins, Wnt inhibitors, or GSK inhibitors, anti-VEGFs, or their combination, and the tissue is exposed to low intensity ultrasound to enhance penetration of the nanoparticles in the implant and to release the medication after implantation slowly, depending on the bio-degradation of polymeric nanoparticles.

In one embodiment, the cross-linked cornea can be recellularized by either injecting appropriate stem cells inside the cross-linked implant, or after implantation, encouraging stem cells, such as mesenchymal cells migration from the adjacent tissue. In either case, the topical administration of nerve growth factors alone or Rock inhibitors alone, or a combination of them encourage cell migration and migration of nerve axons in the cross-linked graft.

In one embodiment, the epithelial cells can be seeded on the cross-linked Bowman's membrane or the endothelial cells seeded on the Descemet's membrane prior or after implantation.

In one embodiment, the corneal or scleral graft can be built using a standard 3-D printing technique using the cross-linked cornea or a part of it as a base in addition to use of one or the other polymeric materials such as Silk sericin, Poly(vinyl alcohol), Poly(2-hydroxyethyl methacrylate), Polyethylene (glycol) diacrylate, Poly(lactic-co-glycolic acid), Poly(ethylene glycol), Poly(acrylic acid), Poly(glycerol sebacate), Poly(ester urethane) urea, Peptide amphiphiles, naturally/synthetically derived, m Arginine-glycine-aspartic acid, the cellular elements of the cornea can be added subsequently in the tissue culture, such a hybrid cornea can renders itself to modification of its refractive power after implantation or before implantation, using low energy level pulses of a femtosecond laser.

In one embodiment, the corneal stroma is made acellular using 0.05-0.5% or more sodium dodecyl sulfate (SDS),

Triton X-100, benzalkonium chloride (BAK), Igepal, or genipin for 15 minutes to 6 hours or more depending on the concentration used to generate acellular corneal inlays or 100% glycerol for a period of time of 6 to 26 hours then rinsed, then cut into layers of 30-100 microns or more, then using a sized trephine of 3-8 mm, the circular, oval, or rectangular-shaped corneal stroma is reshaped to make concave or convex lenses of +20.00 D to -20.00D of the desired diameter that can be crosslinked with a solution of 0.1-5% riboflavin or any other photosensitizer to crosslink the corneal stroma, prior to the storage.

In one embodiment, the stored cornea is reshaped before or after positioning it over the exposed recipient stroma during the refractive surgery to make additions plus or minus, spherical, or astigmatic correction under the control of a wavefront system to correct a refractive power of the patient eye, an additional crosslinker drop is positioned over the stromal inlay to penetrate the wall of the inlay, the LASIK-like flap is repositioned over the inlay and the corneal tissue surrounding the inlay, and the inlay is crosslinked with UV radiation.

In one embodiment, the donor corneal circles or tissue culture grown corneal stroma are not crosslinked before the surgery and they stored in a medium to preserve the cornea such as benzalkonium chloride (BAK) etc. at a desired concentration and time that makes the corneal stroma acellular tissue, or in one embodiment, the acellular donor corneal tissue or tissue culture grown corneal stroma can be crosslinked, then reshaped into a lens with a predetermined refractive power with a femtosecond laser and/or an excimer laser to any shape lenslet and size for correction of myopia, hyperopia, presbyopia and astigmatism, while strengthening the cornea by increasing the corneal thickness and biomechanics after implantation under a corneal LASIK flap or implanted in the corneal pocket as in the SMILE procedure or presbyopia with a small sized pocket of about 5 mm in diameter laser procedures, or in one embodiment, the crosslinking is done after implantation of the lenslet by injecting a photosensitizer over the lenslet implantation and irradiated.

In one embodiment, the implant works as a drug delivery device to the cornea, conjunctiva, or to the external eye conjunctiva and the lid. In this embodiment, the cornea can be repeatedly penetrated with a femtosecond laser to produce hundreds of holes in the tissue, then the corneal stroma is exposed to drug delivery polymeric nanoparticles of polylactic or glycolic acid, or a combination of polylactic or glycolic acid, or polycaprolactone nanoparticles, chitosan, dendrimers, polyanhydrides, liposomes, or porous silicon nanoparticles that can release slowly the medication over a very long period of time (e.g., 3 to 12 months), at a non-toxic concentration, of antibiotics, anti-virals, antifungals, macrolides such as cyclosporine A or mycophenolic acid, steroids, NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors, and/or GSK-3 inhibitors or riboflavin as long as it is desired for the drug delivery to the anterior or posterior chamber of the eye or enhancing penetration of a crosslinker or medication.

In one embodiment, the inlay may be loaded with anti-VEGF nanoparticles, or riboflavin, anti-VEGFs, Rock inhibitors, or another medication to deliver the medication to the external or internal eye.

In one embodiment, the human corneal stromal lamella or tissue culture grown corneal stroma composed of keratocytes and collagen is made acellular prior to the use as a drug delivery device or as a mechanically supporting inlay or changing the corneal refraction.

In one embodiment, a tube of 1-60 millimeters long and a diameter of 0.02 millimeters to a few millimeters is made from a combination of collagen and chitosan with or without collagen fibers and filled up with the desired medication, cut to the desired size, and crosslinked with riboflavin and UV radiation or another chemical cross-linker for implantation in a crosslinked space or without crosslinking the space since the device is already crosslinked and releases the medication through a diffusion process in the presence of the tissue moisture. In one embodiment, the device can be implanted in the cornea, anterior chamber, in the vitreous cavity, or under the conjunctiva, or mucosa, or the skin as desired.

In one embodiment, the implant is made acellular and shaped as a convex, concave, or astigmatic lens using an excimer laser according to the need of the patient eye to produce emmetropia and implanted in a corneal pocket created with a femtosecond laser sufficiently large to place the inlay of any size with a lubricant, such as hyaluronic acid, or with a preservative such as optosol, etc. through a small incision made into the corneal pocket and the implant is injected with an injector in the cavity, then the riboflavin is injected over the implant to cover the implant and the surrounding stromal wall of the acellular implant and the host cornea.

In one embodiment, the drug delivery device is made of a polymeric compound such as polylactic or glycolic acid, or a combination of polylactic or glycolic acid, or polycaprolactone nanoparticles, chitosan, dendrimers, polyanhydrides, liposomes, or porous silicon nanoparticles, and then encapsulated in a collagen fiber which expands over the core drug delivery device and is subsequently crosslinked, thus creating a device for a long drug delivery and slow release of a medication, such as Rock inhibitors, Wnt inhibitors, GSK inhibitors, or integrin inhibitors, which are effective in picogram to microgram concentrations externally or inside the eye.

In one embodiment, the corneal stromal inlay is decellularized with a solution of 0.1-0.5% sodium dodecyl sulfate (SDS) or glycerin 100% or benzalkonium chloride (BAK) solution at a desired concentration for simultaneous preservation of the corneal inlay and killing the cellular elements, bacteria, viruses, or parasites prior to implantation inside a corneal pocket of 1-5 mm inferior to the center of an eye or in the center of the visual axis created with a femtosecond laser, for correction of presbyopia of 1-3.00 D power, creating a bifocal cornea for seeing in the far or near without glasses.

In another embodiment, the corneal stromal inlay is decellularized with a solution of 0.1-0.5% sodium dodecyl sulfate (SDS) and riboflavin nanoparticles or glycerin 100% or benzalkonium chloride (BAK) solution for simultaneous preservation of the corneal inlay and killing the bacteria, viruses, or parasites, then coated with polyethylene glycol and crosslinked with UV radiation prior to implantation inside a corneal pocket of 1-5 mm inferior to the center of an eye or in the center of the visual axis created with a femtosecond laser, for correction of presbyopia of 1-3.00 D power, creating a bifocal cornea for seeing in the far or near without glasses.

In another embodiment, the corneal inlay is made acellular, shaped with an excimer laser then infused with a riboflavin nanoparticle solution, coated with polyethylene glycol, and crosslinked with UV radiation to crosslink the inlay's stromal collagen or sterilized with ethylene oxide

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prior to the implantation inside the recipient corneal stroma, to correct the refractive error of the eye of to treat a presbyopia.

In a further embodiment, one minimizes the thermal damage to the retinal structure using a technique to use the beam as a paint brush to move the laser spot to avoid the foveal area (refer to FIGS. 84 and 85). In FIG. 82, a posterior portion 1000 of a normal eye is shown with its retinal vessels 1014, optic nerve head 1010, and the circular area 1016 is the macular area, the center of the macula 1016 is the fovea 1012. In FIG. 83, a retinal area is illustrated with subretina neovascular tissue as seen in age-related macular degeneration (ARMD) demonstrated as a kidney-shaped lesion area 1022 behind the retina adjacent to the fovea 1012. In FIG. 83, a standard PDT treatment with a circular laser spot 1016 with a large size is being applied to the macular area 1016 of a posterior eye portion 1020, which unfortunately can damage the fovea 1024 simultaneously.

In one embodiment, the painting photodynamic therapy is combined with foveal sparing that avoids radiation of the 500 micron diameter of the area over the fovea completely (see FIGS. 84 and 85) over the treatment area even if the part of the lesion, in wet form of age-related macular degeneration or sub-retinal neovascular tissue, lies under the fovea, but surprisingly is eliminated by disrupting the network of the abnormal vessels under the fovea by a painting brush PDT (PPDT). For example, as shown in FIGS. 84 and 85, the kidney-shaped lesion 1022 in a posterior eye portion 1030 is being treated with a paint brush technique using a small laser spot 1026 which moves over the lesion 1022 and its border while sparing the fovea 1012 from laser damage (the treatment with the laser can be repeated up to 7 times depending on the thickness of the lesion 1022 without damaging either the fovea 1012 or even the retina overlying the lesion 1022). In addition, the laser also avoids the optic nerve head 1010.

In one embodiment, one can combine the paint brush photodynamic therapy (PPDT) with foveal sparing PDT by eliminating the irradiating of the foveal area directly which has a diameter of about 500 micron (0.5 mm) in size, while the rest of the retina and the sub-retinal neovascular tissue is irradiated with a PPDT.

In one embodiment, the painting PDT also avoids damaging the normal retinal vessels which are not leaky, however the abnormal vessels, not only are fragile and get damaged easily, but have the photosensitizer dye inside their lumens and outside their leaky walls. When the walls of these vessels leak, serum containing the photosensitizer leaks outside the vessel wall and bathes the same vessel also from the outside. Therefore, the walls of these abnormal vessels, after receiving a laser radiation that affects the walls of their endothelial cells both from the outside and inside since they have been exposed as the light passes from the outside to the inside of the vascular lumen.

In one embodiment in PPDT, the leaky capillaries receive a double dose of degradation caused by PPDT/laser radiation, that damages outer wall of the endothelial cells of the capillaries and inner wall of the same endothelial cells that is now sufficient damage to cause platelet aggregation and to block the blood flow in the capillaries and now more difficult to repair itself after the treatment.

In one embodiment, moving the laser spot as a painting brush technique (PPDT), reduces the accumulation of the thermal energy side effect of the laser in one spot, since the laser spot moves avoiding accumulation of the thermal energy in one place and which causes damage to the endothelial cells if kept in one spot, such as 83 seconds

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during the standard PDT, whereas during the PPDT, the flow of blood in fact cools down the heated tissue since the laser beam is mostly not stationary, also avoiding the risk of destruction of the adjacent neurosensory retina surrounding the leaky capillaries, as seen with conventional laser therapy.

In one embodiment, the painting brush PDT is used as monotherapy for age-related macular degeneration (AMD), or in combination with triamcinolone or any other medication, such as pathway or complement inhibitors, that reduce the inflammatory process induced by the photosensitizer.

In another embodiment, anti-vascular endothelial growth factor (VEGF) agents are used before or after PPDT.

In another embodiment, one can administer steroids, such as dexamethasone, or NSAIDs, such as diplophenac, in solution or in slow release format intravitreally, in combination with the PPDT to eliminate the need for the recurrent injection in the eye.

In another embodiment, the anti-VEGF alone or with integrin inhibitors, or GSK inhibitors or Wnt inhibitors can be given intravitreally to counteract the effect of VEGF and inflammatory cytokines, which are known to be increased locally after PDT.

In one embodiment, the combination of PPDT with bevacizumab, or Rock inhibitors or integrin inhibitors or GSK inhibitors decreases the number of bevacizumab treatments by injection that would be needed to stabilize the disease process.

In one embodiment, the Rock inhibitors, or Wnt inhibitors or GSK inhibitors or integrin inhibitors are used alone or in various combinations with anti-VEGF in a solution, or as a slow release delivery wherein the drug delivery implant is formed from nanoparticles, micelles, liposomes, solid lipids, polylactic acid, polyglycolic acid, polycaprolactone, polyester, or another biodegradable polymer, porous silicon, hydrogel, or collagen or alginate or acrylic, methacrylate, hydroxyethyl methacrylate (HEMA), a metallic container, a non-metallic container, or combinations thereof, releasing medications in pictogram, nanogram, or microgram/ml concentrations per day for 1 month to 3 years or more depending on the polymeric delivery component.

In one embodiment, in verteporfin-induced photodynamic effect, the medication can be applied topically, injected, or implanted in various locations of the eye, such as intracorneal, in the anterior chamber of the eye, in the lens capsule after cataract removal, intravitreal, under the retina, in the suprachoroidal space, under the conjunctiva, or in the retrobulbar space, the medication containing Rock inhibitors, Wnt inhibitors, GSK inhibitors or integrin inhibitors alone or in various combinations with anti-VEGF in a solution or polymeric nanoparticles, as a slow release delivery system allowing treatment of retinal, choroidal, optic nerve, or uveal diseases (e.g., uveitis) for an extended period of time, thereby providing a lasting anti-inflammatory effect.

In one embodiment, the medication can be administered prophylactically before the PPDT or after PPDT.

In one embodiment, the photosensitizer is verteporfin (e.g., Visudyne) or any other photosensitizer that is activated by the light and creates a singlet oxygen and reactive species, such as superoxide, hydroxyl radical, singlet oxygen, and alpha-oxygen, etc.

In one embodiment, the PPDT treatment is done for eliminating an infected area of the eye or the skin, lid, Meibomian gland, conjunctiva, cornea, or the mouth, or infected mucosa in the body.

In one embodiment, using PPDT allows more precise and customized treatment of the lesion, while providing flexibility in treatment areas of the pathology without extending

treatment into unaffected tissues, or treating an area of the normal tissue which is affected by a pathology without making a visible appearance of diseased tissue. Furthermore, at the same time, PPDT allows prolonged treatment over the neovascular membrane without damaging the normal vessels.

In one embodiment, the treatment comprises treating a retina or choroidal condition by injecting a photosensitizer intravenously which is activated in the abnormal choroidal or retinal vessels using a wavelength of the light from ultraviolet to infrared laser by focusing the aiming spot of the laser over the diseased areas or adjacent normal areas of the choroid and retina with a spot size of 50 microns to 3-4 mm or more to move slowly over the desired area for a desired time of 1 to 10 minutes or more as needed in a paint brushing fashion to produce a photodynamic effect in the vessels containing the dye inside and outside and preferentially damage the abnormal leaky vessels without coagulating the tissue while damaging the diseased tissue by releasing reactive singlet oxygen species to damage the endothelial vessels and closing them to prevent bleeding and prevent damaging the rest of the normal retina or skin or mucosa, while avoiding the sensitive retinal areas, etc. important for vision, such as the fovea or the optic nerve, etc. from getting damaged.

In one embodiment, a contact lens is used to see a large area of the macula or adjacent areas.

In one embodiment, the laser light can be delivered endoscopically to the peripheral part of the retina or elsewhere in the body such as bladder, mouth, vagina, cervix, nose, ear, infected nail, skin, or mucosa, etc. to specifically damage abnormal tissue of an infection or a tumor, or bacteria, viruses, and fungi, etc. at any location or administer the photosensitizer locally or topically prior to the illuminating the areas with PPDT.

In one embodiment for visualization of the retina, one can choose a contact lens positioned on the cornea to make the retina visible. In one embodiment, the contact lens can reverse the image of the retina, but provide a view of a larger area of the retina, while the use of a contact lens with a direct view of the lesion provides a limited field of view.

In one embodiment, the laser may be delivered at a standard fluence of 600 mW/cm² and a dose of 50 mJ/cm². During the procedure, the operator keeps the fundus contact lens steady while moving the laser beam at 1 to 3 seconds or more for a distance of 0.5 mm or more using a spot size of 0.2 mm-3 mm or more depending on the size of the lesion, and the degree of intra-retinal or sub-retinal fluid. Repeating the full treatment, once or twice or more depending on the thickness of the retina for every 100 micron increase above 180 microns as measured by the OCT.

In one embodiment, all the parameters of PPDT, namely the laser wavelength, duration frequency, power, fluence, etc. or the photosensitizer can be changed as needed as long as the principles of PPDT is kept to provide the specific effect to the damaged cells, such as abnormal endothelial cells, be it traumatic, hypoxically induced or a tumor or an infection or an inflammation caused by a pathogen or an abnormal immune response or auto-immune response of the body, where one can also control the immune response with the described medication or the addition of immunosuppressive agents, such as macrolide cyclosporine A, mycophenolic acid, ascomycin or Humira®, etc. or can be administered with an antibiotic, antibacterial, anti-fungi, antiviral, etc.

In one embodiment, the operator can also achieve moving the laser spot by slightly tilting the contact lens which

diverts the laser beam slowly or fast from one direction to another desired direction, thus creating a continuous painting brush motion (technique) of the laser beam in a circular manner or semi-circular manner around the fovea or around the optic nerve. In one embodiment the operator has the control of choosing any parameter (e.g., size of the laser spot, time, or the power of the laser) according to the size of the lesion or the thickness of the retina, pigmentation of the fundus and desired time of application. Thus applies the principle of one size does not fit all in his or her practice of PPDT.

In one embodiment, PPDT reduces the risk of retinal pigment epithelial damage since it decreases total fluence which itself depends on the speed of the oscillation. By using small spot size and moving it over the treatment area, one can avoid damaging the healthy retina that does not leak the photosensitizer; this advantage is especially useful when there is a marked irregularly shaped lesions.

Example 1

A 67-year-old patient with a history of the wet form of age-related macular degeneration having severe exudation in the sub-retinal space and Swiss cheese fluid filled space in the retina covering the entire macular area and thickened choroid (see FIG. 86A). The visual acuity was reduced to 20/400 and the retinal thickness was 518 microns. The patient had been treated over many years with monthly anti-VEGF without response and unfortunately had gotten worse. The patient was smoking daily. The patient underwent a session of treatment with PPDT covering most of the macula except the foveal area with the parameters described, which treatment was repeated five times without damaging the retina. Over a period of 6-8 weeks, the sub-retinal fluid disappeared and the visual acuity improved to 20/80 despite the thickened choroid and distinct areas outside the fovea that had some remaining sub-retinal fluid (see FIG. 86B). No whitening or damage to the retina or retinal vessels was observed. FIG. 86A depicts an optical coherence tomography (OCT) image 1040 of the central retina showing an elevated sensory retina 1042 by sub-retinal fluid; the retinal area 1042 shows numerous lacunae 1044 filled with fluid 1046 while the structure of the sensory retina 1042 is being pulled and separated by the pressure of the fluid 1046, and the underlying structure shows scarred retinal pigment epithelium (RPE), and beneath it are varicose-like abnormal choroidal vessels which leak fluid. FIG. 86B depicts an optical coherence tomography (OCT) image 1050 after PPDT treatment demonstrating disappearance of most of the fluid 1046 and re-establishment of a somewhat normal relationship of the sensory retina 1042 with its underlying congested choroid.

In one embodiment, the photosensitizer can also be used as a contrast agent for imaging, such as indocyanine green, or can be combined with another photosensitizer such as verteporfin.

In one embodiment, the PPDT can be repeated as described and the number of treatments in one session is decided upon the degree of the increased thickness of the retina, any degree of decrease in visual acuity requires an evaluation of the patient with OCT and/or fluorescein angiography or OCT-angiography for the examination of the retina. In case any subretinal fluid is discovered, the area should be treated with the same principle of therapy (i.e., every 100 increase in thickness should be treated with PPDT once or more as the situation presents itself). In every

situation, foveal PDT is avoided including the standard 83 second radiation with the parameters described.

In one embodiment, a specially recurrent lesion is treated with a combination of PPDT and one or more anti-VEGFs and one or more medications that affect the chronic inflammatory processes or cell migration associated with the basic disease. These medications can include Rock inhibitors, Wnt inhibitors, integrin inhibitors, and GSK-3 inhibitors, or in combinations that work synergistically with PPDT and anti-VEGFs, where the medication can be a solution or as polymeric nanoparticles or an implant.

Example 2

A 70-year-old patient with longstanding CSR A subretinal fluid was seen on OCT with elevation of the sensory retina (refer to FIG. 87A). The visual acuity was reduced to 20/100 and the retinal thickness increased to >300 microns. The patient was treated with one session of PPDT after intravenous injection of verteporfin using, 600 mW/cm², 2400-μm spot size for 83 seconds at a fluence rate of 50 mJ/cm² while avoiding the center of the fovea. Also, a similar concentration of verteporfin or riboflavin or other photosensitizers applied topically as drops, spray, or injection in the body's cavity, such as bladder, could be used and irradiated with UV other wavelength of a laser. No visible whitening or subsequent fibrotic reaction was observed from PPDT application. The serous detachment resolved after treatment and the retinal thickness reduced to 180 microns and the visual acuity improved to 20/40 which remained stable with the one year follow up (see FIGS. 87A and 87B). FIG. 87A depicts an optical coherence tomography (OCT) image **1060** of the retina **1062** and fovea **1064** overlying a separated fluid area **1066** where the fluid caused the separation of the sensory retina from the choroid in central serous retinopathy that prevents proper nutrition of the retinal cells in that area with reduced vision. FIG. 87B depicts an optical coherence tomography (OCT) image **1070** after PPDT treatment of the area, but avoiding the central or foveal area **1064** directly (the retina **1062** is now reattached after therapy and the normal anatomical connection between the retina **1062** and the choroid is re-established which leads to improvement of the vision in the patient).

In one embodiment, where the lesion covers the entire macular area and the fovea, one applies PPDT over the entire lesion and beyond it repeatedly as needed while the foveal area is avoided, surprisingly the treatment of the rest of the lesion affects the neovascular tissue even under the fovea and the subretinal fluid along the abnormal vessels regress.

In one embodiment, this treatment with PPDT can be applied over the pre-papillary (optic disc) areas containing abnormal vessels avoiding irradiation of the optic nerve. In this case, normal retinal vessels do not get damaged by PPDT since the normal retinal vessels do not leak. Therefore, the photosensitizer is not in the intercellular space.

In one embodiment, the thickened choroid specifically behind the macula can be treated by a choroidal bandage or implant from the back of the sclera that pushes over these leaks of enlarged vessels and causes them to collapse like an elastic stalking that collapses a varicose vein prior to the PPDT if needed.

In one embodiment, one administers intravitreally or subconjunctival or retrobulbar injection of a medication prior to irradiation or after irradiation, such as an anti-inflammatory agent, such as one of the steroids, NSAID or Rock inhibitors, Wnt inhibitors, anti-VEGFs, GSK-inhibitors, or integrin inhibitors, or a combination thereof.

In one embodiment, the medication is either administered as a solution or in polymeric nanoparticles, liposomes, micelles, or polylactic, polyglycolic acid, porous silicon, polyester, hydrogel or a combination thereof to prevent damage to the retina and choroid caused by numerous diseases of the retina and the choroid.

In one embodiment, the Rock inhibitors are selected from the group of Fasudil, Ripasudil, RKI-1447, Y-27632, GSK429286A, Y-30141 in a solution or a nanoparticle slow release compound.

In one embodiment, the Rock inhibitors are administered in combination with anti-integrins such as abciximab, Eptifibatide, Tirofiban, αIIbβ3 antagonists, Natalizumab, cyclic RGD (Arg-Gly-Asp) peptide or Lifitegrast, Celengitide, Abegrin, etc.

In one embodiment, the Rock inhibitors are administered in combination with Wnt inhibitors such as FH535, IWP-2, PNU-74654, IWR-1endo, IWR-exo, niclosamide or GSK inhibitors SB-216763 etc.

In one embodiment, the Rock inhibitors are administered in combination with GSK-3 inhibitors, such as SB-216763, etc.

In one embodiment, the Rock inhibitors are administered in combination with anti-VEGFs, such as Bevacizumab, Ranibizumab, Aflibercept, Avastin, pegaptanib, and anti-platelet derived growth factor (PDGF) agents, etc.

In one embodiment, the Rock inhibitors are SAR407899, fasudil, netarsudil, etc. and/or Wnt inhibitors such as FH535, IWP-2, PNU-74654, IWR-1endo, IWR-exo, niclosamide or GSK inhibitors SB-216763, etc., integrin inhibitors such as abciximab, Eptifibatide, Tirofiban, αIIbβ3 antagonists, Natalizumab, in combination with PPDT. In one embodiment, these medications are released at non-toxic concentrations of 1 picogram to nanogram/ml or microgram/ml concentrations daily.

In one embodiment, the Wnt/β-catenin pathway can be inhibited indirectly by inhibiting GSK-3 glycogen synthesizing kinase, such as lithium chloride, a simple salt, or its Phosphoaminophosphonic form at an extremely low concentration of 1-3 pico or nano-micromolar that can only be delivered using antibody coated pluralities of nanoparticles conjugated with polymers, such as chitosan, etc.

In one embodiment, peripapillary choroidal neovascularization (CNV) from presumed ocular histoplasmosis (OHS) or idiopathic chorioretinal neovascularization or occult choroidal neovascularization or any other vascularized lesion, can be treated with PPDT or in combination with anti-VEGF.

In one embodiment, in diabetic retinopathy with severe capillary leakage and accumulation of subretinal fluid, the disease can be treated with PPDT alone or in combination with Rock inhibitors, anti-integrins, GSK inhibitors or intravitreal NSAIDs or steroids, etc. or in combination with a mild millipulse panretinal laser coagulation.

In one embodiment, one administers intravitreal injections of bevacizumab (0.5 mg-1.25 mg or more) and dexamethasone (200 microgram-1 mg or more) or in combination with Rock inhibitors, Wnt inhibitors, GSK inhibitors and integrin inhibitors, such as vitronectin receptor inhibitor-type integrin at microgram concentrations.

In one embodiment, recalcitrant idiopathic CNV and a possible episode of post-anti-VEGF stroke, intravitreal dexamethasone (360 mcg) and triamcinolone acetate (400 mcg) in combination with PPDT can be performed or with addition or of Rock inhibitors that do not constrict the small vessels, and in fact dilate them.

In one embodiment of retinal pigment epithelial (RPE) detachment with accumulation of sub-RPE fluid, the PPDT seems to be protective of the RPE and this procedure can be combined with anti-VEGFs, Rock inhibitors, Diamox, or probenecid to enhance removal of sub-RPE fluid.

In one embodiment, post-operative funduscopy, fluorescein angiography, and OCT is performed to confirm re-absorption of the fluid as well as resolution of hemorrhage and subretinal fluid in these patients.

In one embodiment, combining the PPDT with anti-VEGF can also treat the recalcitrant cases that have not responded previously to anti-VEGF. In the examples described, one of the patients was treated for years with anti-VEGF therapy without achieving a desired response while the condition got worse, and after treatment with PPDT the treatment caused the sub-retinal fluid to disappear and the visual acuity improved indicating that one should consider these cases as not lost since saving the vision or improving the vision to any degree is valuable for the patients suffering from loss of sight in age-related macular degeneration.

In one embodiment, the PPDT causes less damage to the retina than the standard therapy as evidence of not observing scar formation after therapy or any burning or whitening effect on the retina after PPDT.

In one embodiment, adding the anti-VEGF or Rock inhibitors or integrin inhibitors or GSK-inhibitors or Wnt inhibitors work as anti-inflammatory in these cases where the disease is associated with chronic not fulminant inflammation and slow release medication eliminates the need for reinjection of these medications inside the eye.

In one embodiment, PPDT treatment for CSR or ARMD causes short-term choriocapillaris hypoperfusion and long-term choroidal vascular remodeling, leading to reduction in choroidal congestion, vascular hyperpermeability, and extravascular leakage.

In one embodiment, PPDT offers the choice of early treatment for CSR, in contrast to the present concept of waiting until the fluid reabsorbs by itself that often may lead to further atrophy of the retina and affecting long term vision of the patient.

In one embodiment, foveal sparing PPDT is an effective treatment for CSR, wet ARMD, OHPS, idiopathic choroidal neovascularization and neovascularization after the trauma of infection while treating the pathogens that produce the inflammatory response, the effect of PPDT is enhanced by concomitant intravitreal or local or systemic administration of Rock inhibitors, anti-integrins, GSK inhibitors or Wnt inhibitors and contributing to faster recovery of the eye and contributing to neuronal regeneration of the damaged neurons because of the diseases process.

In one further embodiment, lamellar corneal transplantation is performed using a circular lamellar cornea similar to the shape of the optical lenses as a circle to replace or add a circular implant inside the corneal stroma (refer to the cross-sectional views illustrated in FIGS. 88A, 88B, 88C, 88D, and 88E). More specifically, in one illustrative embodiment, as shown in FIG. 88A, a double cut 1102 is formed in the cornea 1100 using a femtosecond laser. Then, a cut portion 1104 of the corneal stroma of the cornea 1100 is removed (see FIG. 88B). The removed cut portion of the corneal stroma of the cornea 1100 forms a corneal graft 1106 (see FIG. 88C), which is then ablated using an excimer laser 1108 so as to modify the shape of the corneal graft 1106. Finally, the modified corneal graft 1106 is re-implanted in the cornea 1100 (see FIG. 88E).

In FIG. 89A, a top view of the cornea 1100 in FIG. 88A is shown, which illustrates the intrastromal cut 1102 being formed in the cornea 1100 from the top of the eye. FIG. 89B illustrates a top view of the cornea 1100 in FIG. 88B with the cut portion 1104 of the corneal stroma indicated. In FIG. 89C, a top view of the removed corneal graft 1106 of the cornea stroma of FIGS. 88C and 88D is shown being ablated by using the excimer laser 1108. In FIG. 89D, a top view of the cornea 1100 in FIG. 88E is shown with the rotation of the corneal graft 1106 being diagrammatically indicated using the curved arrow 1112. FIGS. 89A and 89B diagrammatically depict four (4) rotational reference points 1110 of the corneal graft 1106 as a 3 o'clock position, 6 o'clock position, 9 o'clock position, and 12 o'clock position. In FIG. 89E, a top view of the cornea in FIG. 88E is illustrated after the modified corneal graft 1106 has been re-implanted and positioned in the cornea 1100.

In one embodiment, the homograft or allograft is created from the eye bank. The eyes are washed in a solution of povidone iodide to sterilize them. The corneal epithelium and endothelium are removed by mechanical abrasion, etc. Using a femtosecond laser, or a Moria microkeratome, multiple parallel lamellar sections of 50-200 micron thickness are cut inside the corneal stroma, from the back area to the front, subsequently a rectangular section is cut through the full thickness cornea using either a femtosecond laser or a vibrating knife or blade creating 3 to 5 rectangular lamellar tissue, which are then separated for preparation of the inlay in refractive surgery.

In another embodiment, the lamellar corneal tissues are curved out of the eye bank eye corneas in a rectangular shape with the desired thickness to start with, using a mechanical microkeratome or a femtosecond laser then separated from the whole eye to be used as an inlay.

In one embodiment, the lamellar allograft corneas are de-cellularized using a solution of benzalkonium chloride or an organic acid, Triton X-100, SDS, and trypsin-EDTA, etc., to eliminate an immune response prior to the implantation which simultaneously kills the bacteria and other pathogens and reduces the immune response by the host tissue.

In one embodiment, the graft is rectangular, squared, diamond-shaped, or circular with a non-refractive sloped peripheral edge, etc.

In one embodiment, the lamellar cuts are done in the eye bank eye creating 2-4 lamella of 100 micron thickness which are not separated from each other at one side, the cornea is removed and placed on a flat or curved surface, then using a knife or a vibrating knife, one cuts a rectangular or square-shaped lamellae that now are separated from each other.

In one embodiment, the inlays are treated only in their central 3-8 mm area limited by a circle, bordering to the sides of the rectangular inlay.

In one embodiment, inlay transplantation is performed so that the inlay has a refractive section located centrally surrounded with a non-refractive section.

The disadvantage of a circular organic or polymeric or a mixture of organic and polymeric lens is that there is a significant misalignment of the correction of the astigmatic error. Since the lens or the inlay can rotate either before, during, or after implantation for 1-20 degrees which creates a post-surgical visual disturbance which requires another surgery for its adjustment because the patient notices the visual problem after surgery when the graft cannot be easily manipulated to be surgically re-adjusted (see the top views in FIGS. 89A, 89B, 89C, 89D, 89E, 89F, and 89G).

FIG. 89F depicts a top view of a cornea 1114 of an eye illustrating a circular implant 1116 with a rotatable astigmatic axis 1118. FIG. 89G depicts a top view of the cornea 1114 with the rotatable circular implant 1116 and astigmatic axis 1118. The rotation of the implant 1116 in FIG. 89G is indicated by the curved arrow 1120. In FIG. 89H, a top view of a cornea 1122 of an eye is illustrated, wherein the cornea 1122 is provided with a rectangular implant 1124 with a non-movable graft and astigmatic axis 1126. In FIG. 89I, another top view of the cornea 1122 with the rectangular implant 1124 with the non-movable graft and astigmatic axis 1126 is illustrated to represent the unchanged position of the implant 1124.

In one embodiment, the graft can be crosslinked with riboflavin and ultraviolet (UV) radiation either before or after the transplantation.

In one embodiment, the allogeneic or homologous graft can be decellularized with solution of 0.1% riboflavin or more and Benzalkonium chloride 0.05%-2% or more, and subsequently crosslinked with ultraviolet (UV) radiation to enhance its killing effect of UV radiation on the keratocytes and decellularize the inlay preventing graft rejection without affecting the transparency of the inlay before the transplantation. In one embodiment, the decellularized corneal tissue is modified with an excimer laser to the desired shape or refractive power to eliminate refractive error of the recipient eye.

In one embodiment, the allogeneic or homologous graft can be decellularized with solution of 0.1% riboflavin or more and Benzalkonium chloride 0.05%-2% or more, and subsequently crosslinked with ultraviolet (UV) radiation to enhance its killing effect of UV radiation on the keratocytes and decellularize the inlay preventing graft rejection without affecting the transparency of the inlay and simultaneously sterilize the inlay by killing all pathogens.

In one embodiment, one can destroy the cellular components of the graft using chemical means, such as Ethanol, Glycerol, acids and alkalis, Peracetic acid, Ammonium hydroxide ionic detergents, Sodium dodecyl sulfate, Sodium deoxycholate non-ionic detergents, zwitterionic detergents, (CHAPS), but all damage the Glycosaminoglycans (GAG) component of graft and do not cross-link the collagen, thus the tissue maintains its antigenicity.

In one embodiment, the acellular graft may be created using 0.05-0.5% or more sodium dodecyl sulfate (SDS), Triton X-100, benzalkonium chloride (BAK), Igepal, or genipin for 15 minutes to 6 hours or more depending on the concentration used to generate the acellular graft, and the graft is cleaned in 100% glycerol, and sterilized by killing bacteria, viruses and/or parasites, or the graft may be re-sterilized by ethylene oxide or during the surgery by cross-linking the graft with UV light and 0.1% of riboflavin nanoparticles.

In one embodiment, the allogeneic or homologous graft can be crosslinked with solution of 0.1% riboflavin and Benzalkonium chloride 0.05%-2% or more and ultraviolet (UV) radiation to enhance killing effect on the keratocytes and decellularize the inlay preventing graft rejection either before the transplantation.

In one embodiment, autologous corneal transplantation is used as an adjunct to the refractive surgery to modify the corneal shape outside the eye to be implanted in the corneal pocket in treatment of myopia and hyperopia, presbyopia, or astigmatism, etc. (refer to FIGS. 89A, 89B, 89C, 89D, 89E, 89F, 89G, 89H, and 89I; also refer to FIGS. 90A, 90B, 90C, 90D, and 90E).

In one embodiment, using an autologous graft saves a number of the eye bank corneas for the patients in need of a corneal transplant and eliminates the reservation that some religious people might have about a homologous transplant from another person, and also would eliminate the potential transmission of the prion and viral diseases to the patient.

In one embodiment, the autologous corneal lamellar transplant can be used as an inlay removed from the eye so as to modify its refractive power outside the eye, the central part of the implant located in the visual axis is modified with an excimer laser, and subsequently implanted in the corneal pocket in treatment of myopia and hyperopia, or astigmatism, etc. in refractive surgery (see FIGS. 90A, 90B, 90C, 90D, and 90E).

The implantation of lamellar corneal stroma that is cross-linked or without crosslinking is described in U.S. Pat. No. 10,314,690 has the potential of expanding the refractive surgery including for patients with high myopia or hyperopia. However, it has the disadvantage of making a larger corneal LASIK flap for its implantation.

In one embodiment, the corneal tissue implanted uses allograft corneal tissue and crosslinking of the tissue can expand the refractive correction without thinning the cornea, which increases the biomechanical properties of the cornea, and eliminates an immune response from the host.

In one embodiment, inlay transplantation is performed so that the inlay has a refractive section located centrally surrounded with a non-refractive section.

In one embodiment, one can use either a homologous graft from an eye bank or autologous corneal graft taken from the same eye of the patient. It differentiates itself from the other inlays by choosing a rectangular-like structure so that it can be implanted inside a prepared rectangular corneal pocket with the same shape or replace a part of the cornea without being able to change its direction or orientation since a rectangular implant cannot rotate in its place inside a rectangular pocket, while a circular implant can rotate in its place.

In one embodiment, using a rectangular implant, the location of the axis of astigmatism cannot change since the implant cannot move in position; therefore, the astigmatic axis of the modified inlay is stable at the desired axis to compensate for the astigmatic error of the eye at the exact axis (refer to FIG. 89H).

In one embodiment, the graft can be taken from a corneal stroma, homologous or autologous, the graft's surface would be modified to the desired refractive power needed to correct the refractive power of the eye after its removal and subsequently re-implanted in the patient's eye through a small incision in the patient's cornea having a corneal pocket created by a femtosecond laser. If the implant originates from the patient, it would eliminate the need for a cornea from the eye bank for refractive surgery in developing countries or when there is a shortage of the allograft corneas.

Now, the preparation of the patient cornea for modifying the refractive error of the eye will be described. In one embodiment, the refractive error of the cornea is measured with wavefront technology, such as refractometer or a wavefront technology equipped with a Shack-Hartmann system with tunable fluidic lenses or any other instrument indicating lower order aberrations such as spherical, cylindrical, or higher order aberration, coma, spherical aberration, trifol or quadrafoil, etc. recognized by mathematical expressions of Zernike polynomials causing glare, blurred vision, headache, or double vision.

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In one embodiment, the patient eye is disinfected using a solution of povidone iodide prior to the surgery to prevent infection.

In one embodiment, the refractive error is modified using a femtosecond laser of any wavelength from UV to infrared light and a processor available from the present companies, such as Zeiss, Alcon, Bausch and Lomb, Ziemer companies with a modified algorithm, etc.

In one embodiment, one can cut a three-dimensional desired part of the corneal stroma of 5-250 micron thickness in a shape of a circle of 2 mm to 12 mm in diameter, preferentially of 100-200 microns and a diameter of 3-11 mm, in either a circular or rectangular shape, or diamond shape, triangle, trapezoidal, oval, or any other shape, such as a doughnut or strip, etc. of the stromal tissue.

In one embodiment, the femtosecond cuts are made parallel to each other in the cornea. The deeper incision is made first, then the more superficial cut is made to prevent the small bubbles produced with the femtosecond laser from obstructing the field of view for the laser application by the operator.

Another illustrative corneal graft embodiment is depicted in FIGS. 90A-90E. Initially, as shown in FIG. 90A, two parallel intrastromal cuts 1130 are formed using a femtosecond laser in a cornea 1128 of an eye. In FIG. 90B, two parallel corneal access incisions 1134, which are outward from the intrastromal cuts 1132, are shown being formed in the cornea 1128. The two parallel corneal access incisions 1134 enable the intrastromal cut portion 1132 of the eye to be accessed. In FIG. 90C, the removed rectangular graft portion 1136 from the cornea 1128 is shown being ablated by an excimer laser 1138. The excimer laser 1138 creates a modified surface area 1140 of the rectangular graft portion 1136. FIG. 90D illustrates the re-implantation of the modified rectangular graft portion 1136 in the cornea 1128. FIG. 90E is another top view of the cornea 1128 illustrating the modified circular intrastromal area 1142 of the graft portion 1136.

In one embodiment, both surfaces of the implant are smoothened with an excimer laser to remove the irregularities produced by previous laser spots without changing the refractive power of the implant that has been modified.

In one embodiment, the incision has the shape of a circular 3-dimensional cut and in another preferential technique, the cuts are done in a rectangular fashion in the corneal stroma and the side of the borders are connected by a series of vertical cuts connecting the two cuts with each other so that now a rectangular-shaped corneal stroma is separated further with a moving spatula or a vibrating spatula or an ultrasonic spatula as known in the art with a tip that travels vertically or orthogonally for about 0.05 to 0.125 mm and easily separates the attachment of the femtosecond flap from the rest of the stroma.

In one embodiment, a small incision is made at the corneal periphery, not exceeding $\frac{1}{4}$ of the corneal thickness, to create a path inside the corneal stroma to reach the lenticule inside the stroma for its removal or implantation. In one embodiment, two small incisions are done opposite to each other, to support separation of the explant from the sclera with a spatula or a vibrating spatula and/or forceps.

The circular correction of the ocular aberration is done so that the central area of the explant surface is now corrected using an excimer laser with wavefront technology ablating or correcting the main ocular aberration, spheric or astigmatic, etc. while the position of the explant has an up or down or north or south direction so that after implantation

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the four angles of the explant fits exactly inside the rectangle that has been removed from the eye (refer to FIGS. 90A, 90B, 90C, 90D, and 90E).

In one embodiment, the explant can be implanted and crosslinked inside the corneal pocket by injecting 0.1% ml of riboflavin and subsequently irradiated with UV radiation from the outside using 3-10 mW/cm² for a predetermined time depending on the laser power and/or concentration of the photosensitizer.

In one embodiment, the rectangular implant is crosslinked before or after modification of its surface, and then implanted inside the corneal stroma.

In one embodiment, after creating this pathway inside the cornea, a small amount of hyaluronic acid is injected into the space between the superior corneal flap and inferior intact stroma.

In one embodiment, a blunt curved spatula or one with vibration that is controlled ultrasonically by creating a vibration in the spatula's edge is moved in this virtual space between the cut stromal part and the surrounding the tissue to shell out the corneal stroma from its surrounding tissue.

In one embodiment, the separation of the corneal stroma graft can be aided by inserting a similar small instrument from the opposite side of the cornea inside the outlined corneal pocket, assisting its separation, which is then removed by a forceps non-traumatically.

In one embodiment, if the explant is circular, the four clock hours of the tissue explant is marked with a dye at its 12, 3, 6, and 9 o'clock hours so that it can maintain its position properly after its surface modification and implantation. However, often circular implants can be rotated during the implantation or after its implantation changing the axis of the corrected astigmatism.

In one embodiment, the explanted cornea is placed on a flat surface and the refractive errors of the eye that have been measured previously are corrected with an excimer laser to correct a hyperopic eye prior to re-implantation in the cornea.

In one embodiment, the explanted cornea is placed on a flat surface and the refractive errors of the eye that have been measured previously are corrected with an excimer laser to correct a presbyopic eye prior to re-implantation in the cornea.

In one embodiment, the explanted cornea is placed on a flat surface and the refractive errors of the eye that have been measured previously are corrected with an excimer laser to correct a myopic eye prior to re-implantation in the cornea.

In one embodiment, the explanted cornea is placed on a flat surface and the refractive errors of the eye that has been measured previously are corrected with an excimer laser to correct an astigmatic eye prior to re-implantation in the cornea.

In one embodiment, the explanted and modified cornea is replaced in the corneal pocket along with hyaluronic acid, or combined alginates, chitosan, or polyvinyl alcohol to smoothen up its implantation and prevent glare in the post-operative period by slight up and down bumps produced by a femtosecond laser that causes glare.

In an alternative embodiment, the explanted corneal stroma is modified with an excimer laser using one of the wavefront technologies to correct the refractive power of the recipient eye, then flooded with riboflavin and crosslinked prior to or after its transplantation inside the corneal pocket then crosslinked with UV radiation.

In an embodiment, the explanted corneal stroma is flooded with riboflavin and antibiotic solution and cross-

linked with UV radiation prior or after its transplantation inside the corneal pocket to kill potential pathogens of a contaminated explant.

In one embodiment, the riboflavin is combined with an anti-bacterial or anti-viral or anti-fungal in from of nanoparticles to kill all pathogens after UV radiation including the prions that can subsequently affect the cerebrospinal system.

In one embodiment, the cornea is treated with a topical anti-inflammatory agent, such as steroids, NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors, GSK inhibitors alone or in combination to release the medication in the cornea for a long time after implantation.

In one embodiment, the cornea is treated with topical anti-inflammatory agents, such as steroids, NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors, and/or GSK inhibitors alone or in combination with slow release nanoparticle polymers, such as polylactic or polyglycolic acid, micelles, liposomes, porous silicon, polyester, or an ointment, etc. to release the medication in the cornea, for a long time after implantation.

In one embodiment, the corneal explant is treated after its modification with the excimer laser, and is impregnated with topical anti-inflammatory agents such as steroids, NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors, and/or GSK inhibitors alone or in combination in slow release nanoparticle polymers, such as polylactic or polyglycolic acid, micelles, liposomes, porous silicon, polyesters, or any other slow release compound, etc. to release the medication in the cornea for a long time after implantation.

In one embodiment, the entire procedure can be performed using a human corneal stroma from the eye bank as homologous tissue, modified with an excimer laser prior to its implantation inside a pocket of the corneal cavity after removal of a corneal stoma to create space for the implantation for the newly formed implant to act as a lens in the patient's eye as needed to fully correct its refractive errors of the eye with or without crosslinking it.

In one embodiment, topical anti-inflammatory agents such as steroids, NSAIDs, Rock inhibitors, Wnt inhibitors, integrin inhibitors, and/or GSK inhibitors alone or in combination with slow release nanoparticle polymers, such as polylactic or polyglycolic acid, micelles, liposomes, porous silicon, polyesters, or any other slow release compound, etc. or injected subconjunctivally to release the medication in the cornea for a long time after implantation.

In one embodiment, the above refractive procedure is done in a human patient or in an animal, such as pets, dogs, cats, horses or other animals to correct the refractive errors of their eyes, after using a handheld phoropter to measure the refractive errors of these animals.

In one embodiment, a lamellar corneal graft implantation is performed on a human patient, a lamellar corneal graft is decellularized and crosslinked, and the lamellar corneal graft implanted into the human patient is formed from an animal cornea. The cornea is a transparent tissue, and has no blood vessels, etc. This permits a cornea of an animal, either a genetically modified or non-genetically modified, to be transplanted into a human since the corneal cells are eliminated and the corneal proteins or collagens are permanently modified by crosslinking, which means the cornea is not immunogenic. Also, the transplanted corneal stroma may be modified to the desired thickness or size to correct refractive errors of the eye in the animal or human.

In one embodiment, a lamellar corneal graft implantation is performed on an animal, a lamellar corneal graft is

decellularized and crosslinked, and the lamellar corneal graft implanted into the animal is formed from a human cornea.

In one embodiment, the aforescribed procedures can be performed on animals (e.g., horses, cats, dogs, etc.) after measuring their refractive errors.

Any of the features, attributes, or steps of the above described embodiments and variations can be used in combination with any of the other features, attributes, and steps of the above described embodiments and variations as desired.

Although the invention has been shown and described with respect to a certain embodiment or embodiments, it is apparent that this invention can be embodied in many different forms and that many other modifications and variations are possible without departing from the spirit and scope of this invention.

Moreover, while exemplary embodiments have been described herein, one of ordinary skill in the art will readily appreciate that the exemplary embodiments set forth above are merely illustrative in nature and should not be construed as to limit the claims in any manner. Rather, the scope of the invention is defined only by the appended claims and their equivalents, and not, by the preceding description.

REFERENCES

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The invention claimed is:

1. A method of lamellar corneal graft implantation, said method comprising the steps of:

- forming one or more intrastromal cuts in a cornea of an eye of a patient;

- removing a cut portion of the cornea defined by the one or more intrastromal cuts from the eye of the patient so as to form a pocket in the cornea, the pocket being bounded on all sides by corneal tissue of the cornea;

- forming one or more corneal access incisions in the cornea of the eye of the patient, the pocket being accessible through the one or more corneal access incisions in the cornea;

- applying laser energy to a lamellar corneal graft using an excimer laser so as to modify the refractive power of the lamellar corneal graft, the lamellar corneal graft being in the form of an autologous corneal graft or a homologous corneal graft; and

- implanting the lamellar corneal graft into the pocket in the cornea of the eye of the patient in a location previously occupied by the cut portion of the cornea so as to correct the refractive power of the eye of the patient.

2. The method according to claim 1, wherein the step of forming one or more intrastromal cuts in the cornea of the eye of the patient further comprises:

- forming two parallel cuts in the cornea of the eye using a femtosecond laser; and wherein the step of forming the one or more corneal access incisions in the cornea of the eye of the patient further comprises:

- forming two parallel corneal access incisions in the cornea outwardly from the two parallel cuts.

3. The method according to claim 1, wherein the lamellar corneal graft is in the form of an autologous corneal graft, and wherein the cut portion of the cornea that is removed from the eye of the patient forms the autologous corneal graft.

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4. The method according to claim 1, wherein the lamellar corneal graft is in the form of an homologous corneal graft, and wherein a donor corneal portion that is removed from an eye bank cornea forms the homologous corneal graft.

5. The method according to claim 4, wherein, prior to implantation of the homologous corneal graft or after implantation of the homologous corneal graft, the method further comprises the steps of:

treating the homologous corneal graft with a solution of riboflavin or benzalkonium chloride (BAK) for making the homologous corneal graft acellular, and for killing any bacteria, viruses, or parasites prior to implantation; and

irradiating the homologous corneal graft so as to cross-link the homologous corneal graft, thereby killing keratocytes and decellularizing the homologous corneal graft and preventing graft rejection without affecting the transparency of the homologous corneal graft.

6. The method according to claim 1, wherein the lamellar corneal graft is circular in shape such that the lamellar corneal graft is capable of being rotated about an astigmatic axis of the eye in order to permit adjustment after the lamellar corneal graft has been implanted into the eye.

7. The method according to claim 1, wherein the lamellar corneal graft is rectangular in shape such that the lamellar corneal graft is not able to be rotated about an astigmatic axis of the eye, thus preventing movement of the lamellar corneal graft after the lamellar corneal graft has been implanted into the eye.

8. The method according to claim 7, wherein the rectangular lamellar corneal graft comprises a central refractive section surrounded by a peripheral non-refractive section.

9. The method according to claim 1, wherein the method further comprises the step of:

disinfecting the eye of the patient using a solution of povidone iodine prior to surgery to prevent infection.

10. The method according to claim 1, wherein the step of forming the one or more intrastromal cuts in the cornea of the eye of the patient further comprises cutting out a three-dimensional portion of stromal tissue from the cornea of the eye using a femtosecond laser; and

wherein the step of removing the cut portion of the cornea further comprises removing the three-dimensional cut portion of the cornea using forceps so as to create the pocket for receiving the lamellar corneal graft.

11. The method according to claim 1, wherein the method further comprises the step of:

treating the cornea of the patient with one or more topical anti-inflammatory agents in combination with one or more slow release nanoparticle polymers so as to

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release the one or more anti-inflammatory agents in the cornea for a long time after implantation of the lamellar corneal graft.

12. The method according to claim 11, wherein the one or more topical anti-inflammatory agents are selected from the group consisting of steroids, NSAIDs, ROCK inhibitors, Wnt inhibitors, integrin inhibitors, GSK inhibitors, and combinations thereof, and wherein the slow release nanoparticle polymers are selected from the group consisting of polylactic or polyglycolic acid, micelles, liposomes, porous silicon, polyester, and combinations thereof.

13. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on a human patient, and the lamellar corneal graft implanted into the human patient is formed from a human cornea.

14. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on an animal, and the lamellar corneal graft implanted into the animal is formed from an animal cornea.

15. The method according to claim 14, wherein the method further comprises the step of:

measuring the refractive error of the animal cornea with a handheld phoropter.

16. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on an animal, and the lamellar corneal graft implanted into the animal is formed from a human cornea.

17. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on a human patient, and the lamellar corneal graft implanted into the human patient is formed from an animal cornea.

18. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on a human patient, the lamellar corneal graft is decellularized and crosslinked, and the lamellar corneal graft implanted into the human patient is formed from an animal cornea.

19. The method according to claim 1, wherein the lamellar corneal graft implantation is performed on an animal, the lamellar corneal graft is decellularized and crosslinked, and the lamellar corneal graft implanted into the animal is formed from a human cornea.

20. The method according to claim 19, wherein the lamellar corneal graft is decellularized using chemical means, the chemical means for destroying the cellular elements in the lamellar corneal graft are selected from the group consisting of ethanol, glycerol, acids, alkalis, peracetic acid, ammonium hydroxide ionic detergents, sodium dodecyl sulfate, sodium deoxycholate non-ionic detergents, zwitterionic detergents, Triton X-100, benzalkonium chloride, Igepal, genipin, and combinations thereof.

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